Algebraic Geometry

Dhruv Goel

Contents

Preface

Chapter 1

Fundamentals

1.1 Definitions and Examples

1.2 The Zariski Base

Let X be a locally ringed space.An open subset $D \subset X$ is said to be *distinguished* if there is an $f \in \mathcal{O}(X)$ such that $D = X \smallsetminus \mathbb{V}(f) = \{p \in X : f(p) \neq 0\}$; in general, there are many such f for which this holds–for instance, every power of f gives the same D.

A scheme X is affine iff the natural morphism $X \to \operatorname{Spec} \mathcal{O}(X)$ is an isomorphism. In this case, if $D \subset X$ is a distinguished open subset, then $(D, \mathcal{O}|_D)$ is also an affine scheme, almost by construction of the sheaf $\mathcal{O}$. The following results will be used repeatedly.

Lemma 1.2.1.

(a) If $\pi : X \to Y$ is a morphism of locally ringed spaces and $D \subset Y$ a distinguished open, then so is $\pi^{-1}(D) \subset X$.
(b) The intersection of two distinguished opens in a locally ringed space is again distinguished.
(c) Suppose X is a locally ringed space and $D' \subset D \subset X$ open subsets. If D' is distinguished in X, then it is also in D. Conversely, if X is an affine scheme, D distinguished in X, and D' distinguished in D, then D' is distinguished in X.
(d) Distinguished open subsets are a basis for the topology on an affine scheme. Affine open subsets are a basis for the topology on any scheme.
(e) Suppose X is an affine scheme, $\{f_i\} \subset \mathcal{O}(X)$ a family of elements, and for each i the set $D_i := X \smallsetminus \mathbb{V}(f_i)$. Then $X = \bigcup_i D_i$ iff $(f_i) = (1)$ in $\mathcal{O}(X)$. In particular, any affine scheme is quasicompact.

Proof. Exercise. ∎

Example 1.2.2. On a general locally ringed space, distinguised opens do not have to be a basis for the topology. A simple example is obtained by taking a non-affine scheme like $X = \mathbb{P}^1_{\mathbb{C}}$, in which the only distinguished opens as $\emptyset$ and X.

The first fundamental result we need is

Lemma 1.2.3. Let X be a scheme and $U, V \subset X$ affine opens. Then $U \cap V$ can be covered by affine opens which are simultaneously distinguished in both U and V.

Proof. Let $p \in U \cap V$. Using 1.2.1(d), pick a distinguished open D in U such that $p \in D \subset U \cap V$. Next, again using 1.2.1(d), pick a distinguished open D' in V such that $p \in D' \subset D$. Then D' is distinguished in D and hence in U by 1.2.1(c). ∎

The above observations allow us to construct sheaves very explicitly on a scheme.

Theorem 1.2.4. Let X be a scheme. Suppose $\mathcal{F}$ is a sheaf on $X^{\mathrm{dist}}_{\mathrm{Zar}}$, i.e., an assignment to each affine open $U \subset X$ a set[1] $\mathcal{F}(U)$, and to each distinguished inclusion $D \subset U$ a restriction morphism $\mathcal{F}(U) \to \mathcal{F}(D)$, satisfying the usual presheaf axioms, such that for every affine open $U \subset X$ and finite cover $U = \bigcup_\alpha D_\alpha$ of U by distinguished affine opens, the associated sequence

$$* \to \mathcal{F}(U) \to \prod_\alpha \mathcal{F}(D_\alpha) \rightrightarrows \prod_{\alpha,\beta} \mathcal{F}(D_{\alpha\beta})$$

is an equalizer diagram, where $D_{\alpha\beta} = D_\alpha \cap D_\beta$. Then there is a sheaf $\mathcal{F}^+$ on X along with an isomorphism $\mathcal{F} \to \mathcal{F}^+_{\mathrm{Zar}} := \mathcal{F}^+|_{X^{\mathrm{dist}}_{\mathrm{Zar}}}$, and this pair is unique up to unique isomorphism.

Proof.

[1] Or a group, or ring, or a module over a fixed ring, etc.

Existence. For $x \in X$, define $\mathcal{F}_x := \varinjlim_{U \ni x} \mathcal{F}(U)$, where the colimit is taken over affine opens U containing x and distinguished inclusions. For *any* open $V \subset X$, define $\mathcal{F}^+(V) \subset \prod_{x \in V} \mathcal{F}_x$ via compatible germs, namely those $(\sigma_x)_{x \in V}$ such that for each $x \in V$ there is an affine open U with $x \in U \subset V$ and an element $\sigma_U \in \mathcal{F}(U)$ such that for all $y \in U$, we have $\sigma_y = \sigma_U|_y$. Because of 1.2.1(d), the natural projections make $\mathcal{F}^+$ into a presheaf on X, which is automatically a sheaf by construction. There is clearly a natural morphism $\mathcal{F} \to \mathcal{F}^+|_{X^{\text{dist}}_{\text{Zar}}}$. To check that it is an isomorphism, first note that injectivity follows from 1.2.1(de) and the separatedness axiom for $\mathcal{F}$: if $U \subset X$ is an affine open and $\sigma, \tau \in \mathcal{F}(U)$ map to the same element of $\mathcal{F}^+(U)$, then there is a finite cover $U = \bigcup_\alpha D_\alpha$ of U by distinguished affine opens such that for all α we have $\sigma|_{D_\alpha} = \tau|_{D_\alpha}$, and so $\sigma = \tau$ by the injectivity of $\mathcal{F}(U) \to \prod_\alpha \mathcal{F}(D_\alpha)$. Similarly, we show surjectivity: given $(\sigma_x) \in \mathcal{F}^+(U)$, pick a finite cover $U = \bigcup_\alpha D_\alpha$ of U by distinguished affine opens and elements $\sigma_\alpha \in \mathcal{F}(D_\alpha)$ such that for all α and all $x \in D_\alpha$, we have $\sigma_x = \sigma_\alpha|_x$. Then on each intersection $D_{\alpha\beta}$, the restrictions $\sigma_\alpha|_{D_{\alpha\beta}}$ and $\sigma_\beta|_{D_{\alpha\beta}}$ map to the same element of $\mathcal{F}^+(D_{\alpha\beta})$ and hence agree by the injectivity proven above. Therefore, by the gluing axiom for $\mathcal{F}$, there is a $\sigma \in \mathcal{F}(U)$ such that for all α, we have that $\sigma_\alpha = \sigma|_{D_\alpha}$, and then σ maps to $(\sigma_x) \in \mathcal{F}^+(U)$.

Uniqueness. Given any sheaf $\mathcal{G}$ on X, it is easy to see that given any morphism $\varphi : \mathcal{F} \to \mathcal{G}_{\text{Zar}}$ gives rise to a unique morphism $\varphi^+ : \mathcal{F}^+ \to \mathcal{G}$ whose restriction to $X^{\text{dist}}_{\text{Zar}}$ precomposed with the isomorphism $\mathcal{F} \to \mathcal{F}^+|_{\text{Zar}}$ is φ; this combined with the fact that if φ is an isomorphism, then so is $\varphi_x : \mathcal{F}_x \to (\mathcal{G}_{\text{Zar}})_x \xrightarrow{\sim} \mathcal{G}_x$ finishes the proof. Indeed, one shows existence by the following construction: given an open $V \subset X$ and $\sigma = (\sigma_x) \in \mathcal{F}^+(V)$, pick an affine open cover $V = \bigcup U_\alpha$ and $\sigma_\alpha \in \mathcal{F}(U_\alpha)$ such that for all α and all $x \in U_\alpha$ we have $\sigma_x = \sigma_\alpha|_x$. Consider the elements $\varphi(\sigma_\alpha) \in \mathcal{G}(U_\alpha)$. We claim that for all α, β we have $\varphi(\sigma_\alpha)|_{U_{\alpha\beta}} = \varphi(\sigma_\beta)|_{U_{\alpha\beta}}$, and this follows from the sheaf axiom for $\mathcal{G}$ and using 1.2.3 to produce an open cover $U_{\alpha\beta} = \bigcup_\gamma U_{\alpha\beta\gamma}$ where $\sigma_\alpha|_{U_{\alpha\beta\gamma}} = \sigma_\beta|_{U_{\alpha\beta\gamma}}$ and $U_{\alpha\beta\gamma}$ is simultaneously distinguised in both U_α and U_β, for all α, β, γ. Therefore, the $\varphi(\sigma_\alpha)$ glue to produce a unique element of $\mathcal{G}(V)$ which we call $\varphi^+(\sigma)$. The verification that this is well-defined, restricts to φ, and is uniquely determined by this condition is left to the reader.

■

Remark 1.2.5. In fancy language, the above results say that for any scheme X, the site $X^{\text{dist}}_{\text{Zar}}$ consisting of affine opens and distinguished inclusions is enough to capture all (Zariski-) sheafy business on X. Precisely, the continuous inclusion $X^{\text{dist}}_{\text{Zar}} \hookrightarrow X_{\text{Zar}}$ gives rise to an adjoint equivalent of categories $\mathsf{Shv}(X^{\text{dist}}_{\text{Zar}}, \mathcal{O}) \rightleftharpoons \mathsf{Shv}(X_{\text{Zar}}, \mathcal{O})$, where $\mathcal{O}$ denotes either a constant sheaf of rings or the structure sheaf ([1, Theorem 6.2.2]). This adjoint equivalence preserves various structures on these categories such as (quasi)coherence (1.3.3).

1.3 Quasicoherent Sheaves

Theorem/Definition 1.3.1 (QCoh(Affine))**.** Let X be an affine scheme with $A := \mathcal{O}(X)$.

(a) The global sections functor $\Gamma : \mathcal{O}_X\text{-}\mathsf{Mod} \to A\text{-}\mathsf{Mod}$ admits a left adjoint $\widetilde{\cdot}$.
(b) This left adjoint $\widetilde{\cdot}$ is a fully faithful and faithfully exact additive monoidal functor.

The full subcategory on the essential image of $\widetilde{\cdot}$ is called the subcategory of *quasicoherent sheaves* on X, and denoted $\mathsf{QCoh}(X)$.

(c) The subcategory $\mathsf{QCoh}(X) \subset \mathcal{O}_X\text{-}\mathsf{Mod}$ is closed under arbitrary direct sums, taking (co)kernels, and taking tensor products, and is hence a full abelian symmetric monoidal subcategory of $\mathcal{O}_X\text{-}\mathsf{Mod}$.
(d) An $\mathcal{O}_X$-module $\mathcal{F}$ lies in $\mathsf{QCoh}(X)$ iff the co-unit $\varepsilon_{\mathcal{F}} : \widetilde{\mathcal{F}(X)} \to \mathcal{F}$ is an isomorphism.
(e) The functors $\widetilde{\cdot} : A\text{-}\mathsf{Mod} \to \mathsf{QCoh}(X) : \Gamma$ are inverse equivalences of abelian symmetric monoidal categories.

Proof.

(a) We give two constructions of $\widetilde{\cdot}$.[2]

1. If $\tau : (X, \mathcal{O}_X) \to (X, \underline{A}_X)$ is the change of structure morphism, then we have the adjunctions

$$\begin{tikzcd} A\text{-}\mathsf{Mod} \arrow[r, bend left, "\underline{[\cdot]}"] \arrow[r, phantom, "\perp"] & \underline{A}_X\text{-}\mathsf{PMod} \arrow[l, bend left, "\Gamma"] \arrow[r, bend left, "\mathrm{sh}"] \arrow[r, phantom, "\perp"] & \underline{A}_X\text{-}\mathsf{Mod} \arrow[l, bend left, "\mathrm{incl}"] \arrow[r, bend left, "\tau^*"] \arrow[r, phantom, "\perp"] & \mathcal{O}_X\text{-}\mathsf{Mod}. \arrow[l, bend left, "\mathrm{Res}^{\mathcal{O}_X}_{\underline{A}_X}=\tau_*"] \end{tikzcd}$$

Therefore, the adjoint is given by the composite $M \mapsto \widetilde{M} = \mathcal{O}_X \otimes_{\underline{A}_X} \underline{M}_X$.

2. We factor via the adjunctions

$$\begin{tikzcd} A\text{-}\mathsf{Mod} \arrow[r, bend left, "\widetilde{[\cdot]}_{\mathrm{Zar}}"] \arrow[r, phantom, "\perp"] & \mathcal{O}_{X^{\mathrm{dist}}_{\mathrm{Zar}}}\text{-}\mathsf{Mod} \arrow[l, bend left, "\Gamma"] \arrow[r, bend left, "+"] \arrow[r, phantom, "\perp"] & \mathcal{O}_X\text{-}\mathsf{Mod}. \arrow[l, bend left, "{[\cdot]_{\mathrm{Zar}}=[\cdot]|_{X^{\mathrm{dist}}_{\mathrm{Zar}}}}"] \end{tikzcd}$$

Explicitly, given $M \in A\text{-}\mathsf{Mod}$, consider the $\mathcal{O}_{X^{\mathrm{dist}}_{\mathrm{Zar}}}$-premodule $\widetilde{M}_{\mathrm{Zar}}$ given by $U \mapsto \mathcal{O}(U) \otimes_A M$. Check, as in the definition of $\mathcal{O}$ [TODO], that this is actually an $\mathcal{O}_{X^{\mathrm{dist}}_{\mathrm{Zar}}}$-*module*: if $U = \bigcup D_\alpha$ is a finite cover of U by distinguished affine opens, then the sequence

$$0 \to \mathcal{O}(U) \otimes_A M \to \bigoplus_\alpha \mathcal{O}(D_\alpha) \otimes_A M \to \bigoplus_{\alpha,\beta} \mathcal{O}(D_{\alpha\beta}) \otimes_A M$$

of A-modules is exact. The proof 1.2.4 (or its slightly strengthened version 1.2.5), produces an $\mathcal{O}_X$-module $\widetilde{M}$ such that for every affine $U \subset X$, the map $\mathcal{O}(U) \otimes_A M \to \widetilde{M}(U)$ is an isomorphism, i.e., $\widetilde{M}_{\mathrm{Zar}} \xrightarrow{\sim} \widetilde{M}|_{X^{\mathrm{dist}}_{\mathrm{Zar}}}$. Then for any $\mathcal{F} \in \mathcal{O}_X\text{-}\mathsf{Mod}$, we have the natural bijections

$$\mathrm{Hom}_{\mathcal{O}_X\text{-}\mathsf{Mod}}(\widetilde{M}, \mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{dist}}_{\mathrm{Zar}}}}(\widetilde{M}_{\mathrm{Zar}}, \mathcal{F}_{\mathrm{Zar}}) \xrightarrow{\sim} \mathrm{Hom}_{A\text{-}\mathsf{Mod}}(M, \mathcal{F}(X)),$$

where the first map is given by restriction (1.2.5), and the second by taking global sections (check!).

[2] These two constructions are not *obviously* the same, but they are a posteriori by the uniqueness of adjoints.

(b) To show that $\widetilde{\cdot}$ is fully faithful, it suffices to show that for each $M \in A\text{-}\mathsf{Mod}$, the unit $\eta_M : M \to \widetilde{M}(X)$ is an A-module isomorphism, but this is clear from the second construction above. For $M \in A\text{-}\mathsf{Mod}$ and $p \in X$, there is a natural isomorphism $M_p \to \widetilde{M}_p$ of modules over $\mathcal{O}_{X,p} = A_p$, so faithful exactness follows from the corresponding fact in commutative algebra about the localizations of a module at primes. This functor is clearly additive and preserves tensor products, for instance, because all the functors it is a composite of in the first construction do.

The statements in (c), (d), and (e) follow immediately from (a) and (b). ■

Example 1.3.2. Let $A \to B$ be a ring homomorphism corresponding to the morphism of affine schemes $f : \operatorname{Spec} B \to \operatorname{Spec} A$. For any A-module M, we have a natural isomorphism of $\mathcal{O}_{\operatorname{Spec} B}$-$\mathsf{Mod}$-modules $f^*\widetilde{M} \cong \widetilde{B \otimes_A M}$. This follows from the theory of adjunctions: for any $\mathcal{G} \in \mathcal{O}_{\operatorname{Spec} B}$-$\mathsf{Mod}$, we have the natural bijections

$$\begin{aligned}
\operatorname{Hom}_{\operatorname{Spec} B}(f^*\widetilde{M}, \mathcal{G}) &\cong \operatorname{Hom}_{\operatorname{Spec} A}(\widetilde{M}, \pi_*\mathcal{G}) \\
&\cong \operatorname{Hom}_{A\text{-}\mathsf{Mod}}(M, \operatorname{Res}^B_A \mathcal{G}(\operatorname{Spec} B)) \\
&\cong \operatorname{Hom}_{B\text{-}\mathsf{Mod}}(B \otimes_A M, \mathcal{G}(\operatorname{Spec} B)) \\
&\cong \operatorname{Hom}_{\operatorname{Spec} B}(\widetilde{B \otimes_A M}, \mathcal{G}).
\end{aligned}$$

Theorem/Definition 1.3.3 (QCoh(Scheme))**.** Let X be a scheme and $\mathcal{F} \in \mathcal{O}_X$-$\mathsf{Mod}$. The following conditions on $\mathcal{F}$ are equivalent.

(a) For every affine open $U \subset X$, we have $\mathcal{F}|_U \in \mathsf{QCoh}(U)$.
(b) For every $x \in X$ and affine open neighborhood U of x in X, the natural morphism $\mathcal{O}_{X,x} \otimes_{\mathcal{O}_X(U)} \mathcal{F}(U) \to \mathcal{F}_x$ is an isomorphism.
(c) For every chain of affine opens $V \subset U \subset X$ in X, the natural $\mathcal{O}(V)$-module morphism $\mathcal{O}(V) \otimes_{\mathcal{O}(U)} \mathcal{F}(U) \to \mathcal{F}(V)$ is an isomorphism.
(d) For every chain of affine opens $V \subset U \subset X$ in X with V distinguished in U, the natural $\mathcal{O}(V)$-module morphism $\mathcal{O}(V) \otimes_{\mathcal{O}(U)} \mathcal{F}(U) \to \mathcal{F}(V)$ is an isomorphism.
(e) There is an affine open cover $\mathscr{U}$ of X such that for all $U \in \mathscr{U}$, we have $\mathcal{F}|_U \in \mathsf{QCoh}(U)$.
(f) For each $x \in X$, there is a neighborhood U of x in X, sets I and J and an exact sequnece of $\mathcal{O}_U$-modules[3] of the form $\mathcal{O}_U^{\oplus J} \to \mathcal{O}_U^{\oplus I} \to \mathcal{F} \to 0$.

An $\mathcal{F} \in \mathcal{O}_X$-$\mathsf{Mod}$ is said to be *quasicoherent* if it satisfies these equivalent conditions. Further:

(f) The full subcategory $\mathsf{QCoh}(X)$ of $\mathcal{O}_X$-Mod consisting of quasicoherent $\mathcal{O}_X$-modules is closed under arbitrary direct sums, taking (co)kernels, and taking tensor products, and is hence a full abelian symmetric monoidal subcategory of $\mathcal{O}_X$-Mod.

[TODO]

Proof. The implications (b) ⇒ (c) and (a) ⇒ (d) are clear (1.2.1(d)).

(a) ⇒ (b) Follows from 1.3.2.
(c) ⇒ (a) Follows from 1.3.1(d).
(d) ⇒ (e) Modules admit presentations, and so the result follows from 1.3.1(b).
(e) ⇒ (a) Without loss of generality, X is affine. If there were a global such exact sequence, then $\mathcal{F}$ would be quasicoherent because $\mathsf{QCoh}(X)$ is closed under taking cokernels and free sheaves are obviously quasicoherent. Therefore, the result follows immediately from the following lemma (1.3.4).

The statement in (f) then follows from 1.3.1(c). ■

[3]Here $\mathcal{O}_U := \mathcal{O}_X|_U$.

Lemma 1.3.4. Let X be an affine scheme and $\mathcal{F} \in \mathcal{O}_X\text{-}\mathsf{Mod}$. Suppose there is a finite cover $X = \bigcup_\alpha D_\alpha$ by distinguished open affines such that for all α, we have $\mathcal{F}|_{D_\alpha} \in \mathsf{QCoh}(D_\alpha)$. Then $\mathcal{F} \in \mathsf{QCoh}(X)$.

Proof. We want to show that for all distinguished opens $D \subset X$, the natural morphism $\mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{F}(U) \to \mathcal{F}(D)$ is an isomorphism. For this, consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{F}(X) \arrow[r] \arrow[d] & \bigoplus_\alpha \mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{F}(D_\alpha) \arrow[r] \arrow[d] & \bigoplus_{\alpha,\beta} \mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{F}(D_{\alpha\beta}) \arrow[d] \\
0 \arrow[r] & \mathcal{F}(D) \arrow[r] & \bigoplus_\alpha \mathcal{F}(D \cap D_\alpha) \arrow[r] & \bigoplus_{\alpha,\beta} \mathcal{F}(D \cap D_{\alpha\beta}),
\end{tikzcd}$$

where the top row comes from the sheaf axiom for $\mathcal{F}$, the finiteness of the cover, and the fact that the $\mathcal{O}(X)$-algebra $\mathcal{O}(D)$ is flat. Consider the middle column: for each α, we have

$$\mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{F}(D_\alpha) \cong \mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{O}(D_\alpha) \otimes_{\mathcal{O}(D_\alpha)} \mathcal{F}(D_\alpha) \cong \mathcal{O}(D \cap D_\alpha) \otimes_{\mathcal{O}(D_\alpha)} \mathcal{F}(D_\alpha),$$

and under this isomorphism, the middle map is the obvious one; a similar observation holds for the last column. Since for all α, we have $\mathcal{F}|_{D_\alpha} \in \mathsf{QCoh}(D_\alpha)$, the middle map is an isomorphism; since for all α and β, we have $\mathcal{F}|_{D_{\alpha\beta}} \in \mathsf{QCoh}(D_{\alpha\beta})$ (which follows because the restriction of a quasicoherent sheaf on affine scheme to a distinguished open is clearly quasicoherent, and $D_{\alpha\beta}$ is distinguished in both D_α and D_β), the last column is an isomorphism. Then the five-lemma tells us that $\mathcal{O}(D) \otimes_{\mathcal{O}(X)} \mathcal{F}(X) \to \mathcal{F}(D)$ is also an isomorphism, as needed. ■

Lemma 1.3.5. Let X be an affine scheme and $\mathcal{F} \in \mathsf{QCoh}(X)$.

(a) Let $f \in \mathcal{O}(X)$ and $D = D(f) \subset X$ be the corresponding distinguished open.
  (i) If $\sigma \in \mathcal{F}(X)$ is such that $\sigma|_D = 0$, then there is an $n \gg 1$ such that $f^n\sigma = 0 \in \mathcal{F}(X)$.
  (ii) Given a section $\sigma \in \mathcal{F}(D)$, there is an $n \gg 1$ such that $f^n\sigma$ comes from an element of $\mathcal{F}(X)$.

(b) Supose we are given a finite cover $X = \bigcup_\alpha D_\alpha$ of X by distinguished affine opens, and suppose for all α, β, we are given $\sigma_{\alpha\beta} \in \mathcal{F}(D_{\alpha\beta})$ such that for all α, β, γ we have $\sigma_{\beta\gamma} - \sigma_{\alpha\gamma} + \sigma_{\alpha\gamma} = 0$ in $\mathcal{F}(D_{\alpha\beta\gamma})$.[4] Then there are elements $\tau_\alpha \in \mathcal{F}(D_\alpha)$ for all α such that for all α, β, we have $\tau_\beta - \tau_\alpha = \sigma_{\alpha\beta} \in \mathcal{F}(D_{\alpha\beta})$.

Proof. The statement in (a) is clear. For (b), first pick $f_\alpha \in \mathcal{O}(X)$ such that for all α we have $D_\alpha = D(f_\alpha)$. Next, fix a γ. Using (a)(ii), produce an $n \gg 1$ and for all α a section $\tau_\alpha^\gamma \in \mathcal{F}(D_\alpha)$ that restricts to $f_\gamma^n \cdot \sigma_{\gamma\alpha} \in \mathcal{F}(D_{\alpha\gamma})$. Use (a)(i) and the cocycle condition to increase n if needed to assume further that for all α, β we have $\tau_\beta^\gamma - \tau_\alpha^\gamma = f_\gamma^n \sigma_{\alpha\beta} \in \mathcal{F}(D_{\alpha\beta})$. Since $(f_\gamma^n) = (1)$ (1.2.1(e)), there are $c_\gamma \in \mathcal{O}(X)$ such that $\sum_\gamma c_\gamma f_\gamma^n = 1$. For each α, let $\tau_\alpha := \sum_\gamma c_\gamma \tau_\alpha^\gamma$. This works. ■

The statement of 1.3.5(a) will be generalized in 3.3.

Theorem 1.3.6. Let X be a scheme and $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be an exact sequence of $\mathcal{O}_X$-modules.

(a) If $\mathcal{F}' \in \mathsf{QCoh}(X)$, then for any affine open $U \subset X$, the sequence

$$0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U) \to 0$$

of sections on U is exact.

[4] Here, and in what follows, restrictions are left implicit.

(b) If $\mathcal{F}', \mathcal{F}'' \in \mathsf{QCoh}(X)$, then also $\mathcal{F} \in \mathsf{QCoh}(X)$. In other words, $\mathsf{QCoh}(X)$ is stable under extensions.

Proof.

(a) This is a formal consequence of 1.3.5(b).
(b) The result follows from 1.3.3(a), our (a), and the five lemma.

∎

Remark 1.3.7. 1.3.5(b) is saying, of course, that for any affine scheme X, cover $\mathcal{D}$ by finitely many distinguished affines, and $\mathcal{F} \in \mathsf{QCoh}(X)$, the Čech cohomology $\check{\mathrm{H}}^1(\mathcal{D}, \mathcal{F})$ vanishes. This is an important idea that will come back repeatedly.

1.4 The Proj Construction

Let A be a ring and S a $\mathbb{Z}_{\geq 0}$-graded A-algebra which is generated in degree 1, i.e., such that $\operatorname{Sym}_A^\bullet S_1 \twoheadrightarrow S$. Consider the functor $\operatorname{Proj}_A S : \mathsf{Sch}^{\mathrm{op}} \to \mathsf{Set}$ given as follows.

(a) On objects, for a $T \in \mathsf{Sch}$, the set $(\operatorname{Proj}_A S)(T)$ is the set of isomorphism classes of triples $(\pi, \mathcal{L}, \psi)$, where
  (1) π is morphism $\pi : T \to \operatorname{Spec} A$,
  (2) $\mathcal{L}$ is a line bundle on T, and
  (3) $\psi : S \to \Gamma_\bullet(T, \mathcal{L})$ is a graded A-algebra homomorphism such that $\mathcal{L}$ is generated by the global sections $\psi(S_1)$.

  Here two such triples $(\pi_i, \mathcal{L}_i, \psi_i)$ for $i = 1, 2$ are isomorphic iff $\pi_1 = \pi_2 := \pi$ say, and there is an isomorphism $\phi : \mathcal{L}_1 \to \mathcal{L}_2$ of line bundles on the A-scheme $\pi : T \to A$ such that $\psi_2 = \Gamma_\bullet(T, \phi) \circ \psi_1$.

(b) On morphisms $f : T' \to T$, the map $(\operatorname{Proj}_A S)(f) : (\operatorname{Proj}_A S)(T) \to (\operatorname{Proj}_A S)(T')$ is given by sending the isomorphism class of a triple $(\pi, \mathcal{L}, \psi)$ to that of $(\pi \circ f, f^*\mathcal{L}, \Gamma_\bullet(f^*) \circ \psi)$.

Theorem/Definition 1.4.1. In the above setting, the functor $\operatorname{Proj}_A S$ is representable by a scheme.

The representing scheme $\operatorname{Proj}_A S$ comes with a morphism $\pi : \operatorname{Proj}_A S \to \operatorname{Spec} A$, a line bundle called $\mathcal{O}(1)$, and a graded A-algebra homomorphism $\psi : S \to \Gamma_\bullet(\operatorname{Proj}_A S, \mathcal{O}(1))$ called *saturation* such that $\mathcal{O}(1)$ is generated by the global sections $\psi(S_1)$.

Proof. The functor $\operatorname{Proj}_A S$ is evidently a Zariski sheaf, so it remains to cover it with representable open subfunctors. For each $f \in S_1$, consider the open subfunctor $D_+(f)$ consisting of triples $(\pi, \mathcal{L}, \psi)$ such that $\psi(f) \in \Gamma(T, \mathcal{L})$ is everywhere nonzero (and hence defines an isomorphism $\mathcal{O}_T \xrightarrow{\sim} \mathcal{L}$). These cover $\operatorname{Proj}_A S$ by the hypothesis that S is generated in degree 1. It remains to note that $D_+(f)$ is representable by the scheme $\operatorname{Spec} S[f^{-1}]_0$ with structure morphism to A clear, the universal line bundle $\mathcal{O}_{\operatorname{Spec} S[f^{-1}]_0}$ trivial, and $\psi : S \to S[f^{-1}]_0[t]$ given by sending a $g \in S_n$ to $(gf^{-n})t^n$ for $n \in \mathbb{Z}_{\geq 0}$. ∎

Lemma 1.4.2. The scheme $\operatorname{Proj}_A S$ constructed in 1.4.1 is

(a) separated, and
(b) of finite type over A if S_1 is a finite A-module.

Proof.

(a) We use 3.6.3(b). For $f, g \in S_1$, the intersection $D_+(f) \cap D_+(g)$ is affine because, say, it is the distinguished open $D_+(g/f)$ on the affine scheme $D_+(f)$. The morphism $\mathcal{O}(D_+(f)) \otimes \mathcal{O}(D_+(g)) \to \mathcal{O}(D_+(f) \cap D_+(g))$ is surjective because $1 \otimes f/g \mapsto (g/f)^{-1}$.
(b) Clear.

∎

Remark 1.4.3. Can define more generally. Universaly property is more complicated, but still separated and of finite type if it is a finitely generated graded ring.

Let X be a scheme and $\mathcal{S}$ a $\mathbb{Z}_{\geq 0}$-graded quasicoherent $\mathcal{O}_X$-algebra that is generated in degree 1, i.e., such that $\underline{\operatorname{Sym}}^\bullet_{\mathcal{O}_X} \mathcal{S}_1 \twoheadrightarrow \mathcal{S}$. Consider the functor $\underline{\operatorname{Proj}}_X \mathcal{S} : \mathsf{Sch}^{\mathrm{op}} \to \mathsf{Set}$ given as follows.

(a) On objects, for a $T \in \mathsf{Sch}$, the set $(\underline{\operatorname{Proj}}_S \mathcal{S})(T)$ is the set of isomorphism classes of triples $(\pi, \mathcal{L}, \psi)$, where

(1) π is morphism $\pi : T \to X$,
(2) $\mathcal{L}$ is a line bundle on T, and
(3) $\psi : \mathcal{S} \to \bigoplus_{n\geq 0} \pi_* \mathcal{L}^{\otimes n}$ is a graded $\mathcal{O}_X$-algebra morphism such that $\psi^1 : \pi^* \mathcal{S}_1 \to \mathcal{L}$ is surjective.[5]

Here two such triples $(\pi_i, \mathcal{L}_i, \psi_i)$ for $i = 1, 2$ are isomorphic iff $\pi_1 = \pi_2 := \pi$ say, and there is an isomorphism $\phi : \mathcal{L}_1 \to \mathcal{L}_2$ of line bundles on the X-scheme $\pi : T \to X$ such that $\psi_2 = (\bigoplus_{n\geq 0} \pi_* \phi^{\otimes n}) \circ \psi_1$.

(b) On morphisms $f : T' \to T$, the map $(\underline{\mathrm{Proj}}_X \mathcal{S})(f) : (\underline{\mathrm{Proj}}_X \mathcal{S})(T) \to (\underline{\mathrm{Proj}}_X \mathcal{S})(T')$ is given by sending the isomorphism class of a triple $(\pi, \mathcal{L}, \psi)$ to that of $(\pi \circ f, f^* \mathcal{L}, (\bigoplus_{n\geq 0} \pi_* f^{*\otimes n}) \circ \psi)$.

As in the absolute case, we have

Theorem/Definition 1.4.4. In the above setting, the functor $\underline{\mathrm{Proj}}_X \mathcal{S}$ is representable by a scheme.

The representing scheme $\mathrm{Proj}_X \mathcal{S}$ comes with a morphism $\pi : \underline{\mathrm{Proj}}_X \mathcal{S} \to X$, a line bundle called $\mathcal{O}(1)$, and a morphism $\mathcal{S} \to \bigoplus_{n\geq 0} \pi_* \mathcal{O}(n)$ of graded $\mathcal{O}_X$-algebras (where $\mathcal{O}(n) := \mathcal{O}(1)^{\otimes n}$) such that $\pi^* \mathcal{S}_1 \twoheadrightarrow \mathcal{O}(1)$.

Proof. As above, it suffices to check that $\underline{\mathrm{Proj}}_X \mathcal{S}$ is a Zariski sheaf and that it can be covered by representable open subfunctors. In this case, for each affine open $U \subset X$, consider the open subfunctor F_U sending T to triples $(\pi, \mathcal{L}, \psi)$ with $\pi(T) \subset U$. These clearly cover $\underline{\mathrm{Proj}}_X \mathcal{S}$, and finally each F_U is isomorphic to $\mathrm{Proj}_{\mathcal{O}_X(U)} \mathcal{S}(U)$. ■

Corollary 1.4.5. Let X be a scheme and $\mathcal{E} \in \mathsf{QCoh}(X)$. The scheme $\underline{\mathrm{Proj}}_X \mathrm{Sym}^\bullet \mathcal{E}$ represents the functor taking a scheme T to isomorphism classes of triples $(\pi, \mathcal{L}, \psi)$ such that $\pi : T \to X$ is a morphism, $\mathcal{L}$ a line bundle on T, and a surjection $\psi : \pi^* \mathcal{E} \twoheadrightarrow \mathcal{L}$ of $\mathcal{O}_T$-modules.

[5] Here we are using the adjunction $\pi^* \dashv \pi_*$ to treat $\psi^1 : \mathcal{S}_1 \to \pi_* \mathcal{L}$ as an $\mathcal{O}_T$-module morphism $\pi^* \mathcal{S}_1 \to \mathcal{L}$.

1.5 Exercises

Exercise 1.1. Let $f, g : X \to Y$ be morphisms of schemes. Show that $f = g$ iff for all $p \in X$, the restrictions of f and g to $\operatorname{Spec} \mathcal{O}_{X,p}$ agree.

Exercise 1.2. Let X be a scheme and $p \in X$.

(a) Produce a natural bijection between the set of irreducible components of X containing p and the minimal primes of the ring $\mathcal{O}_{X,p}$. (The morphism $\operatorname{Spec} \mathcal{O}_{X,p} \to X$ "sees" all the irreducible components of p in X.)
(b) Show that if $\mathcal{O}_{X,p}$ is a domain, then there is a unique irreducible component of X passing through p. In particular, if this is true of all $p \in X$, then the irreducible components of X are disjoint.

Exercise 1.3. Let $X \to Z \leftarrow Y$ be morphisms of schemes.

(a) For each $p \in Z$ with residue field $\kappa(p)$, produce a natural isomorphism between the fiber $(X \times_Z Y)_p$ and $X_p \times_{\kappa(p)} Y_p$. (In this sense, the fiber product of schemes is the "fiber-wise product".)
(b) Let $x \in X$ and $y \in Y$. Show that there is a point $q \in X \times_Z Y$ mapping simultaneously to x and y iff x and y map to the same point, say p, in Z. In this case, produce a morphism $\operatorname{Spec} \kappa(x) \otimes_{\kappa(p)} \kappa(y) \to X \times_Z Y$ that induces a homeomorphism of the first space with the subset of all such points q in $X \times_Z Y$.

Exercise 1.4. Let $\pi : X \to Y$ be a surjective morphism of schemes with Y reduced. Show that $\pi^\sharp : \mathcal{O}_Y \to \pi_* \mathcal{O}_X$ is injective.

Exercise 1.5. Show that a direct summand of a quasicoherent module is quasicoherent.

Exercise 1.6. Let X be a scheme, $\mathcal{E}, \mathcal{E}'$ be quasicoherent sheaves on X with $\mathcal{E}'$ of finite type, and $\phi : \mathcal{E} \to \mathcal{E}'$ a morphism of $\mathcal{O}_X$-modules. Consider the following conditions on ϕ.

(a) The map ϕ is an isomorphism.
(b) The map ϕ is an epimorphism.
(c) The induced map on fibers is a surjection: for all points $p \in X$, the induced map $\phi(p) : \mathcal{E}(p) \to \mathcal{E}'(p)$ is a full rank linear map of $\kappa(p)$-vector spaces.
(d) The induced map on geometric fibers is a surjection: for all geometric points $\overline{p} \to X$, the induced map $\phi(p) : \mathcal{E}(\overline{p}) \to \mathcal{E}'(\overline{p})$ is a full-rank linear map of $\kappa(\overline{p})$-vector spaces.

Show that (a) $\Rightarrow$ (b) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d), and all conditions are equivalent when $\mathcal{E}$ and $\mathcal{E}'$ are line bundles, i.e., locally free of rank 1.

Exercise 1.7. Let R be a Dedekind domain with $K := \operatorname{Frac} R$, and let $X := \operatorname{Spec} R$ have generic point η so that $\mathcal{O}_{X,\eta} \xrightarrow{\sim} K$.

(a) Produce a natural bijection between the set of nonzero fractional ideals of R and the set of pairs $(\mathcal{L}, \varphi)$, where $\mathcal{L} \to X$ is a line bundle and $\varphi : \mathcal{L}_\eta \xrightarrow{\sim} K$ is an $R^\times$-equivalence class of K-vector-space trivializations of $\mathcal{L}$ at the generic point.
(b) Use (a) to give a group isomorphism $\operatorname{Pic}(R) \xrightarrow{\sim} \operatorname{Pic}(X)$.

Exercise 1.8. Let X be a scheme. Suppose we are given the following data.

(a) For each affine open $U \subset X$ a U-scheme $\pi_U : Z_U \to U$.
(b) For each distinguished inclusion $V \subset U \subset X$ of affine opens, a morphism $\rho^U_V : Z_V \to Z_U$.

Suppose further that the following conditions are satisfied.

(c) For each triple $W \subset V \subset U$ of distinguished inclusions, we have $\rho^U_W = \rho^U_V \circ \rho^V_W$.
(d) For each $V \subset U$ the following diagram is commutative and Cartesian.

$$\begin{tikzcd} Z_V \arrow[r, "\rho_V^U"] \arrow[d, "\pi_V"'] \arrow[dr, phantom, "\lrcorner"] & Z_U \arrow[d, "\pi_U"] \\ V \arrow[r, hook] & U \end{tikzcd}$$

Show that there is an X-scheme $\pi : Z \to X$ and morphisms $\rho_U : Z_U \to Z$ for each affine open $U \subset X$ such that for all distinguished inclusions $V \subset U$ we have $\rho_V = \rho_U \circ \rho_V^U$, and such that for all $U \subset X$ the following diagram is commutative and Cartesian.

$$\begin{tikzcd} Z_U \arrow[r, "\rho_U"] \arrow[d, "\pi_U"'] \arrow[dr, phantom, "\lrcorner"] & Z \arrow[d, "\pi"] \\ U \arrow[r, hook] & X \end{tikzcd}$$

Further, this data $(Z, \pi, \{\rho_U\})$ is unique upto unique isomorphism.[6]

Exercise 1.9. Continuing the notation of 1.8, suppose we are given the following data.

(a) For each affine open $U \subset X$, an $\mathcal{F}_U \in \mathcal{O}_{Z_U}\text{-}\mathsf{Mod}$.
(b) For each distinguished inclusion $V \subset U \subset X$ of affine opens of X, an isomorphism $\phi_V^U : \mathcal{F}_V \to (\rho_V^U)^*\mathcal{F}_U$ of $\mathcal{O}_{Z_V}$-modules.

Suppose further that the following conditions are satisfied.

(c) For each triple $W \subset V \subset U$ of distinguished inclusions, we have $\phi_W^U = (\rho_W^V)^*\phi_V^U \circ \phi_W^V$.

Show that there is an $\mathcal{F} \in \mathcal{O}_Z\text{-}\mathsf{Mod}$ and isomorphisms $\phi_U : \mathcal{F}_U \to \rho_U^*\mathcal{F}$ of $\mathcal{O}_{Z_U}$-modules such that for all distinguished inclusions $V \subset U$, we have $\phi_V = (\rho_V^U)^*\phi_U \circ \phi_V^U$.[7] Further, this data $(\mathcal{F}, \{\phi_U\})$ is again unique up to unique isomorphism.[8]

Exercise 1.10. Let X be a scheme and $\mathcal{E}$ be a finite rank vector bundle on X. Let $P := \underline{\mathrm{Proj}}_X \operatorname{Sym}^\bullet \mathcal{E}$ and let $\pi : P \to X$ be the structure morphism.

(a) Show that the adjoint to morphism $\pi^*\mathcal{E} \twoheadrightarrow \mathcal{O}_P(1)$ obtained from the definition of π is an isomorphism $\mathcal{E} \xrightarrow{\sim} \pi_*\mathcal{O}_P(1)$.
(b) Show that the isomorphism of (a) is functorial in $\mathcal{E}$ in the following sense. Let $\mathcal{E} \twoheadrightarrow \mathcal{F}$ be a surjection of finite rank vector bundles on X, and let $Q := \underline{\mathrm{Proj}}_X \operatorname{Sym}^\bullet \mathcal{F}$ with structure morphism $\rho : Q \to X$. Produce a morphism $j : Q \to P$ such that $\pi \circ j = \rho$ and $j^*\mathcal{O}_P(1) \cong \mathcal{O}_Q(1)$, check that j is a closed embedding, produce a morphism $\pi_*\mathcal{O}_P(1) \to \rho_*\mathcal{O}_Q(1)$ of vector bundles on X, and check that the following diagram commutes:

$$\begin{tikzcd} \mathcal{E} \arrow[r, "\sim"] \arrow[d, two heads] & \pi_*\mathcal{O}_P(1) \arrow[d] \\ \mathcal{F} \arrow[r, "\sim"] & \rho_*\mathcal{O}_Q(1). \end{tikzcd}$$

[6] We remark also that the same result works when, in stead of all affine opens $U \subset X$, we are given this data only over "sufficiently small" affine opens, e.g., those whose images under a given morphism $\pi : X \to Y$ are contained in an affine open, or those on which a given locally free sheaf $\mathcal{F} \to X$ is trivial, etc.

[7] In particular, if each $\mathcal{F}_U$ is quasicoherent (resp. locally free, resp. a line bundle), then so is $\mathcal{F}$.

[8] A similar remark to the previous exercise about "sufficiently small" affine opens also holds.

Chapter 2

Topology for Algebraic Geometers

2.1 Irreducible, Kolmogorov, and Sober Spaces

In this section, we introduce irreducible, Kolmogorov, and sober spaces. These concepts are absolutely fundamental to understanding the topology of schemes. To motivate the definition, we first recall the following facts about connected spaces.

Proposition/Definition 2.1.1 (Connected Spaces)**.**

- The following conditions on a topological space X are equivalent.
  (a) The space X cannot be expressed as a union of two proper clopen subspaces.
  (b) Any nonempty clopen subspace must be all of X.
  (c) The only subspaces of X with empty boundary are $\emptyset$ and X.
  (d) There does not exist a surjective continuous function $X \to \{0,1\}$, where $\{0,1\}$ is given the discrete topology.

  A topological space is said to be *connected* iff it satisfies these equivalent conditions.
- A *connected component* of a space is a maximal connected subspace.

Proof. Exercise. ■

Recall the following basic properies.

Lemma 2.1.2.

(a) If $f : X \to Y$ is a continuous map with X connected, then $f(X) \subset Y$ is connected.
(b) If $Y \subset X$ is a subspace and Y is connected, then so is $\overline{Y}$.[1] In particular, any connected component of X is closed in X.[2]
(c) A topological space that can be covered by a family $\{A_i\}$ of connected subspaces such that the total intersection $\bigcap_i A_i$ is nonempty is connected.
(d) Every connected subset of a topological space is contained in a unique connected component. In particular, distinct connected components of a space are disjoint, and every space is the (disjoint) union of its connected components.
(e) Any clopen subset of a topological space is a disjoint union of some of its connected components.

Proof. Exercise. We note only that (d) uses Zorn's Lemma for existence. ■

Often in algebraic geometry, we need a slightly stronger topological property than being connected. This is provided by

Proposition/Definition 2.1.3 (Irreducible Spaces)**.**

- The following conditions on a nonempty topological space are equivalent.
  (a) The space cannot be expressed as a union of two proper closed subspaces.
  (b) Any two nonempty open subspaces have nonempty intersection.
  (c) Every nonempty open subspace is dense.
  (d) A subspace is not dense iff it is nowhere dense.
  (e) Every nonempty open subspace is connected.

  A topological space is said to be *irreducible* iff it is nonempty and it satisfies these conditions.
- An *irreducible component* of a space is a maximal irreducible subspace.
- Given a space X, a point $\eta \in X$ is said to be a *generic point* of X iff $X = \overline{\{\eta\}}$.

[1] The converse is clearly not true.
[2] It is not true in general that a connected component is also necessarily open. A counterexample is given by taking the product topology on $2^{\mathbb{N}}$, or if you prefer $\operatorname{Spec} \mathbb{F}_2^{\mathbb{N}}$.

Proof. Exercise. ■

It is immediate that any irreducible space is connected.

Example 2.1.4. The space $\mathbb{C}$ with the Euclidean topology is connected but not irreducible, whereas the affine line $\mathbb{A}^1_{\mathbb{C}} \cong \operatorname{Spec} \mathbb{C}[x]$ with the Zariski topology is irreducible.

The analogous basic properties for irreducibility are given in

Lemma 2.1.5.

(a) If $f : X \to Y$ is a continuous map with X irreducible, then $f(X) \subset Y$ is irreducible.
(b) If X is any topological space and $Y \subset X$ a subspace, then Y is irreducible iff $\overline{Y}$ is. In particular, any irreducible component of X is closed in X, and a dense subspace of an irreducible space is irreducible.
(c) Every nonempty open subspace of an irreducible space is irreducible.
(d) A topological space that admits an open cover by irreducible open subspaces with nonempty pairwise intersections is irreducible.
(e) Every irreducible subspace of a topological space is contained in an irreducible component. In particular, every topological space is the union of its irreducible components.[3]
(f) If $x \in X$ is a point in a space, then the closure $\overline{\{x\}} \subset X$ is an irreducible closed subspace of X. In particular, a space admitting a generic point is irreducible, and the set of generic points of such a space is precisely the intersection of all its nonempty open subsets.
(g) If X is any topological space and $U \subset X$ a nonempty open subspace, then there is an inclusion-preserving bijection

$$\{\text{irreducible closed } Y \subset U\} \leftrightarrow \{\text{irreducible closed } Z \subset X \text{ with } Z \cap U \neq \emptyset\} \text{ given by}$$
$$Y \mapsto \overline{Y},$$
$$Z \cap U \leftarrow\!\shortmid Z.$$

This bijection restricts to one between irreducible components of U and those of X meeting U (or equivalently those irreducible subsets $Z \subset X$ such that the intersection $Z \cap U \subset Z$ is dense).

Proof. Exercise; the proofs are very similar to those in 2.1.2. We prove only (b) to give a flavor of the proofs. Note that a subspace $Y \subset X$ is irreducible iff for any two open sets U, V of X meeting Y, the intersection $U \cap V$ meets Y. The result follows from noting that an open subspace of X meets Y iff it meets $\overline{Y}$. ■

A relationship between these two notions is provided in

Lemma 2.1.6.

(a) If X is a topological space, $Y \subset X$ a connected component and $Z \subset X$ an irreducible subspace, then $Y \cap Z \neq \emptyset$ implies $Z \subset Y$. In particular, every connected component of a topological space is the union of some of its irreducible components.
(b) A connected space is irreducible iff it is nonempty and it admits a cover by irreducible open subspaces.

Proof. Exercise. ■

[3] It is evident that not every topological space is the *disjoint* union of its irreducible components.

Next, we recall some point-set topology before introducing sober spaces.

Proposition/Definition 2.1.7 (Kolmogorov Spaces)**.** The following conditions on a topological space X are equivalent.

(a) If $x \neq y \in X$, then there is an open subspace $U \subset X$ containing exactly one of x and y.
(b) The map $X \to \mathrm{Irred}(X)$ given by $x \mapsto \overline{\{x\}}$ is an injection, where $\mathrm{Irred}(X)$ is the set of irreducible closed subsets of X.
(c) The specialization relation on X given by $x \rightsquigarrow y$ iff $y \in \overline{\{x\}}$ is a partial order.

A topological space is said to be *Kolmogorov* or T_0 if it satisfies these equivalent conditions.

Proof. Clear. ■

In a Kolmogorov space, the minimal points for the specialization preorder are the closed points, and the maximal points are the generic points of irreducible components (when they exist).

Definition 2.1.8 (Sober Spaces)**.** A topological space X is said to be *sober* if the map

$$X \to \mathrm{Irred}(X), \quad x \mapsto \overline{\{x\}}$$

is a bijection, i.e., every irreducible closed subspace of X has a unique generic point.

Remark 2.1.9.

(a) A sober space is, in particular, Kolmogorov. In [2], a Noetherian sober space is called a Zariski space.
(b) In an irreducible sober space, the intersection of all nonempty open subsets consists of a single point–*the* generic point.

Lemma 2.1.10.

(a) A topological space admitting an open cover by sober spaces is itself sober. In particular, the underlying topological space of any scheme is sober.
(b) A constructible subspace[4] of a sober space is sober.

Proof.

(a) This is clear for Kolmogorov spaces, and generic points can be produced locally (using 2.1.5(g)). The second statement follows from the first because the underlying topological space of an affine scheme is clearly sober.
(b) The statement is clearly true for locally closed subspaces; for constructible subspaces it follows from 2.3.3(b).

■

Remark 2.1.11. If X is a sober space and $Y \subset X$ an arbitrary subspace, it is not necessarily true that Y is also sober. For a simple counterexample, take $X = \mathbb{A}^1_{\mathbb{C}}$ with generic point $\eta \in X$ and let $Y := X \smallsetminus \{\eta\}$.

Here are a few of applications of these concepts that are often useful (and help provide clean solutions to the exercises in, say, [2, §II.3])

[4]See 2.3.3 below if needed.

Proposition 2.1.12. Let $f : X \to Y$ be an open continuous map of topological spaces and suppose that Y is irreducible and has a generic point η. The following conditions are equivalent.

(a) The generic fiber $X_\eta = f^{-1}(\eta)$ is irreducible.
(b) There is a dense subset of points $y \in Y$ such that the fiber $X_y = f^{-1}(y)$ is irreducible.
(c) The space X is irreducible.

Proof.

(a) ⇒ (b) Consider the subset $\{\eta\}$.

(b) ⇒ (c) Suppose $X = Z_1 \cup Z_2$ for proper closed $Z_1, Z_2 \subset X$. Since f is open, for $i = 1, 2$, the subset $f(X \smallsetminus Z_i) \subset Y$ is open; since Y is irreducible, then intersection $f(X \smallsetminus Z_1) \cap f(X \smallsetminus Z_2)$ is nonempty and open; by the density hypothesis, there is a $y \in f(X \smallsetminus Z_1) \cap f(X \smallsetminus Z_2)$ such that X_y is irreducible. Then $X_y \cap Z_i$ for $i = 1, 2$ are proper closed subspaces of X_y whose union is X_y, which is a contradiction.

(c) ⇔ (a) For any subspace $Z \subset Y$, continuity of f tells us that $\overline{f^{-1}(Z)} \subset f^{-1}(\overline{Z})$ with equality when f is open. Applying this to $Z = \{\eta\}$ gives us that $\overline{X_\eta} = f^{-1}(\overline{\{\eta\}}) = f^{-1}(Y) = X$, and so the result follows from 2.1.5(b).

■

Proposition/Definition 2.1.13. Let $f : X \to Y$ be a continuous map of irreducible sober spaces, and let ξ (resp. η) be the generic point of X (resp. Y). The following are equivalent:

(a) $f(\xi) = \eta$.
(b) $\eta \in f(X)$.
(c) $\overline{f(X)} = Y$.
(d) If $\emptyset \neq U \subset Y$ is open, then $f^{-1}(U) \neq \emptyset$.

A map f satisfying these equivalent properties is said to be *dominant*, and we say that X *dominates* Y (via f).

Proof. It is clear that the (c) ⇔ (d) holds very generally (i.e., does not need the irreducible or sober hypotheses). The implications (a) ⇒ (b) ⇒ (c) ⇒ (d) are clear. For (d) ⇒ (a), note that if $\emptyset \neq U \subset Y$ is open, then $f^{-1}(U) \ni \xi$, whence $f(\xi) \in U$, and so the result follows from 2.1.9(b). ■

Definition 2.1.14. When X and Y are sober (but not necessarily irreducible) topological spaces, we say that a continuous map $f : X \to Y$ is *dominant* iff each irreducible component of Y is dominated by an irreducible component of X.

Remark 2.1.15. In algebraic geometry, this is a slightly better definition than "dense image" (conditions (c) and (d) above), which is not as well-behaved. In particular, our definition of dominant implies dense image, but is in general stronger.[5] Besides, in most cases of interest (e.g., when the morphism f is quasicompact and X, Y are schemes; see 3.3.3(a)), the two definitions are equivalent.

Proposition 2.1.16. Let $f : X \to Y$ be a continuous map of sober spaces, and suppose that Y is irreducible. Let $\eta \in Y$ be the generic point and $X_\eta = f^{-1}(\eta)$ the generic fiber. There is

[5] Note that this convention differs from the one in [3]. For an example where these definitions are not equivalent, consider the natural morphism $\coprod_{\mathbb{Z}} \operatorname{Spec} \mathbb{C} \to \mathbb{A}^1_{\mathbb{C}}$ given by the inclusion of the integers.

an inclusion preserving bijection

$$\{\text{irreducible closed } Z \subset X_\eta\} \leftrightarrow \{\text{irreducible closed } W \subset X \text{ dominating } Y\} \text{ given by}$$
$$Z \mapsto \overline{Z},$$
$$W_\eta = W \cap X_\eta \leftarrow\!\shortmid W.$$

Further, under this bijection, irreducible components of X_η correspond to irreducible components of X dominating Y.

Proof. Clear from 2.1.13. (Note that we are also using 2.1.10(b).) ∎

2.1.1 Applications to Geometric Properties

Theorem/Definition 2.1.17. (Geometrically Irreducible Schemes) Let k be a field, and fix separable and algebraic closures $k \to k^{\mathrm{s}} \to k^{\mathrm{a}}$. The following conditions on a k-scheme X are equivalent.

(a) For all field extensions K/k, the K-scheme X_K is irreducible.
(b) The k^{a}-scheme $X_{k^{\mathrm{a}}}$ is irreducible.
(c) The k^{s}-scheme $X_{k^{\mathrm{s}}}$ is irreducible.
(d) The scheme X is irreducible, and if $K := \kappa(\eta)$ is the function field of the generic point $\eta \in X$, then $\operatorname{Spec} k^{\mathrm{s}} \otimes_k K$ is irreducible.
(e) The scheme X is irreducible, and with K as in (d), the fields k^{s} and K are linearly disjoint over k.
(f) The scheme X is irreducible, and with K as in (d), the field k is separably closed in K (i.e., under compatible k-embeddings, we have $k^{\mathrm{s}} \cap K = k$ in K^{a}.)

[TODO: Cleanup]

Proof. The implication (a) ⇒ (b) is clear, and the equivalence of (d)-(f) is field theory ([4, Exercise 5.7]). Note also that if K/k is any field extension and X_K irreducible, then so is X; this follows from 2.1.5(a) and the fact that $X_K \to X$ is surjective, since it is the base-change of the surjective $\operatorname{Spec} K \to \operatorname{Spec} k$; this proves (b) ⇒ (c).

(b) ⇒ (a) By the observation above and embedding every field in sight compatibly in an algebraic closure of K, we are reduced to showing: if k is an algebraically closed field and X an irreducible k-scheme, then for any field K/k, the scheme X_K is also irreducible. Since the reduction morphism is a universal homeomorphism, we may assume that X is integral. By 2.1.12, we may replace X by the spectrum of its function field, and at this point we finish by invoking [1, 10.5.O].

(c) ⇒ (b) The field extension $k^{\mathrm{a}}/k^{\mathrm{s}}$ is purely inseparable and so the morphism $\operatorname{Spec} k^{\mathrm{a}} \to \operatorname{Spec} k^{\mathrm{s}}$ is a universal homeomorphism ([1, 10.5.7]).

(c) ⇔ (d) By the observation above, if $X_{k^{\mathrm{s}}}$ is irreducible, then so is X. The rest is the equivalence of (a) and (c) in 2.1.12, using [1, 10.5.6] for the openness of $X_{k^{\mathrm{s}}} \to X$.

∎

Theorem/Definition 2.1.18. (Geometrically Reduced Schemes) Let k be a field, and fix perfect closure and algebraic closures $k \to k^{\mathrm{p}} \to k^{\mathrm{a}}$ of k. The following conditions on a k-scheme X are equivalent.

(a) For all field extensions K/k, the K-scheme X_K is reduced.
(b) The k^{a}-scheme $X_{k^{\mathrm{a}}}$ is reduced.
(c) The k^{p}-scheme $X_{k^{\mathrm{p}}}$ is reduced.

(d) The scheme X is reduced, and if $\eta \in X$ is the generic point of an irreducible component, then the field extension $K := \kappa(\eta)$ of k is separable, i.e., k^{p} and K are linearly disjoint over k.

In particular, if k is a perfect field, then every reduced k-scheme is geometrically reduced.

Note that if K is a separable field extension of a field k, then necessarily k is its own perfect closure in K (i.e., under compatible embeddings, we have $k^{\mathrm{p}} \cap K = k$ in K^{a}), but the converse is not always true ([4, Ex. 5.9]).

[TODO: Cleanup]

Proof. The implication (a) $\Rightarrow$ (b) is clear. Note that if K/k is any field extension and X_K reduced, then so is X: this is a local question, but for any k-algebra A we have $A \hookrightarrow A_K = K \otimes_k A$; this proves (b) $\Rightarrow$ (c).

(c) $\Rightarrow$ (a) By the observation above, we have to show that if k is a perfect field and X a reduced k-scheme, then for any field extension K/k, the K-scheme X_K is also reduced. This is true; see [1, 10.5.22].

(a) $\Leftrightarrow$ (d) ([5, Prop. 5.49]) The question is local, so we may assume $X = \operatorname{Spec} A$ for some reduced k-algebra A. Since A is reduced, for each generic point say $\eta_i \in \operatorname{Spec} A$, the ring $\kappa(\eta_i)$ is a localization of A, and further, the canonical homomorphim $A \to \prod_i \kappa(\eta_i)$ is injective. If K is a field extension of k and A_K is reduced, the so it its localization $\kappa(\eta_i) \otimes_k K$; if this is true for all K, then each $\kappa(\eta_i)$ is separable. Conversely, if every $\kappa(\eta_i)$ is separable, then for every field K and i, we have that $K \otimes_k \kappa(\eta_i)$ is reduced, and hence so is A_K because

$$A_K \hookrightarrow K \otimes_k \prod_i \kappa(\eta_i) \hookrightarrow \prod_i (K \otimes_k \kappa(\eta_i)).$$

■

Corollary 2.1.19. (Geometrically Integral Schemes) Let k be a field, and fix an algebraic closure $k \to k^{\mathrm{a}}$ of k. The following conditions on a k-scheme X are equivalent.

(a) For all field extensions K/k, the K-scheme X_K is integral.
(b) The k^{a}-scheme $X_{k^{\mathrm{a}}}$ is integral.
(c) The scheme X is integral, and if $K = \kappa(X)$ denotes its fraction field, then k^{a} and K are linearly disjoint over k.
(d) The scheme X is integral, and if K denotes its fractional field, then the extension K/k is separable, and k is separably closed in K.

In particular, if k is perfect, then this is also equivalent to: the scheme X is integral, and if K is its fraction field, then k is algebraically closed in K.

Proof. The implications (c) $\Rightarrow$ (d) $\Rightarrow$ (a) $\Leftrightarrow$ (b) are clear from 2.1.17 and 2.1.18. For (b) $\Rightarrow$ (c), this is a local question, so suppose that $X = \operatorname{Spec} A$ for a k-algebra A, with $A_{k^{\mathrm{a}}} = k^{\mathrm{a}} \otimes_k A$ a domain. Then A is also a domain because $A \hookrightarrow A_{k^{\mathrm{a}}}$, and the localization $k^{\mathrm{a}} \otimes_k K$ of $k^{\mathrm{a}} \otimes_k A$ is also a domain, which says precisely that k^{a} and K are linearly disjoint over k (c.f. [4, Ch. 5]). ■

For an example of a field k and an integral scheme X over k with fraction field K such that k is algebraically closed in K but X is not geometrically integral over k, see again [4, Ex. 5.9].

Corollary 2.1.20. Let k be a field and X be a geometrically reduced integral (e.g., geometrically integral) k-scheme locally of finite type. The function field $K = \kappa(X)$ of X is a separably generated extension of k.

Proof. The extension K/k is both separable (2.1.18) and finitely generated. ■

2.2 Closure-Complete and Noetherian Spaces

In this section, we describe two fundamental finiteness conditions on topological spaces: closure-completeness and the Noetherian property.

Definition 2.2.1. A topological space is *closure-complete* if every point has a closed point in its closure.

Remark 2.2.2. A nonempty closed subspace of a closure-complete space contains a closed point, and in particular, every nonempty closure-complete space has a closed point. For a Kolmogorov space, closure-completeness is equivalent to saying that every point specializes to a minimal point for the specialization partial order.

Lemma 2.2.3. A Kolmogorov space admitting a finite open cover by closure-complete spaces is closure-complete. In particular, the underlying topological space of any quasicompact scheme is closure-complete and hence contains a closed point.

Proof. By induction on the size on the open cover, we are reduced to the case of two open sets, so say $X = U \cup V$ with U, V closure-complete open subspaces. It suffices to show that if $x \in U$, then x has a closed point in its closure. Since U is closure-complete, there is a $y \in \overline{\{x\}} \cap U$ which is closed in U. If $y \notin V$, then y is also closed in X. Else, $y \in V$ and so since V is closure-complete, there is a $z \in \overline{\{y\}} \cap V$ which is closed in V. Then z is closed in X. Indeed, if $w \in \overline{\{z\}} \smallsetminus \{z\}$, then $w \notin V$ and so $w \in \overline{\{z\}} \cap U \subset \overline{\{y\}} \cap U = \{y\}$, but $y \notin \overline{\{z\}}$ by the Kolmogorov hypothesis. The second result follows from the first, since the underyling space of an affine scheme is closure-complete (by Zorn's Lemma!). ■

For a scheme that is not closure-complete (and, in fact, has no closed points at all), see [6, Exercise 3.3.27].

Proposition/Definition 2.2.4 (Noetherian Spaces)**.** The following conditions on a topological space are equivalent.

(a) Every descending chain of closed subspaces is eventually stationary.
(b) Every non-empty collection of closed subspaces has a minimal element.[6]
(c) Every ascending chain of open subspaces is eventually stationary.
(d) Every non-empty collection of open subspaces has a maximal element.
(e) Every open subspace is quasicompact.

A topological space is *Noetherian* if it satisfies the above equivalent conditions.

Proof. The equivalence of (a)-(d) is left to the reader.

(d) ⇒ (e) Let X satisfy (d) and $U \subset X$ be an open subspace. Let $\{U_i\}$ be an open cover of U. The collection of all finite unions of the U_i is a non-empty collection of open subspaces, and thus has a maximal element, which must then be U.

(e) ⇒ (c) Suppose X is a space such that every open subspace of X is quasicompact, and let $U_1 \subset U_2 \subset \cdots$ be an ascending chain of open subspaces in X. Let $U := \bigcup_{i=1}^{\infty} U_i$, which is an open subset of X. Since $\{U_i\}$ is an open cover of U and U is quasicompact, this forces $U = U_i$ for some i as needed.

■

[6] Here "minimal" or "maximal" element always means the same with respect to the inclusion partial order.

Example 2.2.5. Let R be a Noetherian ring. Then the underlying topological space of $\operatorname{Spec} R$ is Noetherian. This follows from the inclusion-reversing bijection between closed subsets of $\operatorname{Spec} R$ and radical ideals of R.

Example 2.2.6. The space $\mathbb{C}$ equipped with the analytic topology is not Noetherian.

Here are some basic properties.

Lemma 2.2.7.

(a) A topological space that admits a finite cover by Noetherian spaces is Noetherian.
(b) Every subspace of a Noetherian space is Noetherian, and hence quasicompact.
(c) A Noetherian space has only finitely many irreducible components, and hence only finitely many connected components.
(d) A connected component of a Noetherian space is open. In particular, the union of any collection of connected components is clopen.

Proof.

(a) Follows from 2.2.4(a).
(b) If X is Noetherian, $Y \subset X$ a subspace, and (Z_j) a descending chain of closed subspaces in Y, then the chain $(\overline{Z}_j)$ in X is eventually stationary, and hence we are done by $Z_j = Y \cap \overline{Z}_j$.
(c) Apply 2.2.4(b) to the collection of closed subspaces which cannot be written as a finite union of irreducible subspaces. The second result follows from the first, thanks to 2.1.6(a).
(d) By 2.1.6(a), a connected component of a Noetherian space is the complement of the union of some of its irreducible components (namely those disjoint from it); therefore, the first statement follows from (c). The second result follows from the first combined (c).

■

Remark 2.2.8.

(a) Note that if X is Noetherian scheme, then the underlying topological space of X is Noetherian, but the converse is false in general (e.g., take $X = \operatorname{Spec} k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$.) "Noetherian" (along with the corresponding "locally Noetherian" below) is one of the very few adjectives for which this remark holds.
(b) A scheme morphism out of a Noetherian scheme is automatically qcqs.

One of the key reasons we like Noetherian spaces is

Proposition 2.2.9 (Irreducible Decomposition of Noetherian Space)**.** Let X be a Noetherian space.

(a) There is an integer $r \geq 1$ and closed irreducible subspaces $Z_1, \dots, Z_r \subset X$ such that $X = Z_1 \cup \cdots \cup Z_r$ such that for all i, j with $1 \leq i \neq j \leq r$, we have $Z_i \not\subset Z_j$, and such that if $Z \subset X$ is any irreducible subspace, then there is an $i = 1, \dots, r$ such that $Z \subset Z_i$.
(b) The integer r and decomposition as in (a) are uniquely determibned (up to re-ordering).

The decomposition $X = Z_1 \cup \cdots \cup Z_r$ is called the *irreducible decomposition* of X.

Proof. Take Z_j to be the irreducible components of X, using 2.2.7(c). ■

Another reason we like Noetherian spaces is that we can do Noetherian induction on them, as in

Proposition 2.2.10 (Noetherian Induction)**.** Let X be a Noetherian topological space, and let $\mathscr{P}$ be a property satisfied by certain closed subsets of X. Suppose that for each closed subset $Z \subset X$, if $\mathscr{P}$ holds for each proper $Z' \subsetneq Z$, then it holds for Z. Then the property $\mathscr{P}$ holds for all closed subsets of X, and in particular for X itself.

Proof. This is just transfinite induction on the collection of closed subsets of X ordered by inclusion; as stated, this follows immediately from 2.2.4(b). ∎

One kind of space that comes up often in algebraic geometry is a locally Noetherian space.

Definition 2.2.11. A topological space X is said to be *locally Noetherian* if for each $x \in X$, there is an open neighborhood U of x in X such that U is Noetherian as a topological space.

Note that if X is a locally Noetherian topological space and $U \subset X$ any open subspace, then U is also locally Noetherian. Further, a topological space is Noetherian iff it is locally Noetherian and quasicompact. A few of the results from this section can be generalized to the locally Noetherian setting. Here's one example.

Proposition 2.2.12. Let X be a locally Noetherian space. An open subset $U \subset X$ is dense iff it meets all irreducible components of X.

Proof. If U meets all irreducible components of X, then it is dense in X by 2.1.3(c) and 2.1.5(d). To show the converse, suppose we are given a dense open subset $U \subset X$ and an irreducible component $Z \subset X$ such that $U \cap Z = \emptyset$. Since Z is irreducible, it is nonempty; pick any $z \in Z$ and a Noetherian open neighborhood V of z in X. Using 2.1.5(g), we may replace (X, U, Z) by $(V, U \cap V, Z \cap V)$ to assume that in fact X is Noetherian. Let $X = Z_1 \cup \cdots \cup Z_r$ for some $r \in \mathbb{Z}_{\geq 1}$ and (Z_i) be the irreducible decomposition of X as in 2.2.9(a), numbered so that $Z = Z_r$. Then check that $W = X \smallsetminus (Z_1 \cup \cdots \cup Z_{r-1})$ is a nonempty open subset of X contained in Z; this cannot happen, since then U must both meet W (since it is dense) and not (since $W \subset Z$ and $U \cap Z = \emptyset$). ∎

It is an easy and instructive exercise, left to the reader, to come up with a space X (necessarily non-locally-Noetherian) and a dense open subset $U \subset X$ that does not meet some irreducible components of X.

2.3 Locally Closed, Constructible, and Very Dense Subspaces

Now we would like to review some point-set topology about locally closed, constructible, and very dense subspaces.

Proposition/Definition 2.3.1 (Locally Closed Subspaces)**.** Let X be a topological space. The following conditions on a subspace $Y \subset X$ are equivalent.

(a) For each $y \in Y$, there is an open subspace $U \subset X$ containing y such that $Y \cap U$ is closed in U.
(b) There is an open subspace $U \subset X$ such that $Y \subset U$ and Y is closed in U.
(c) There is a closed subspace $Z \subset X$ such that $Y \subset Z$ and Y is open in Z.
(d) There is an open subspace $U \subset X$ and a closed subspace $Z \subset X$ such that $Y = U \cap Z$.
(e) There are open subspaces $V \subset U \subset X$ such that $Y = U \smallsetminus V$.
(f) There are closed subspaces $W \subset Z \subset X$ such that $Y = Z \smallsetminus W$.
(g) Y is an open subspace of its closure $\overline{Y}$ in X.
(h) $\overline{Y} \smallsetminus Y$ is a closed subspace of X.

A subspace Y satisfying these equivalent conditions is said to be a *locally closed subspace* of X. In this case, the largest open subspace U of X as in (b), i.e., containing Y and in which Y is closed, is $U = X \smallsetminus (\overline{Y} \smallsetminus Y)$.

Proof. The implications (a) $\Leftarrow$ (b) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d) $\Leftrightarrow$ (e) $\Leftrightarrow$ (f) and (c) $\Leftrightarrow$ (g) $\Leftrightarrow$ (h) are clear. For (a) $\Rightarrow$ (b), for each $y \in Y$ pick a U_y, and let $U := \bigcup_{y \in Y} U_y$. The last claim is clear. ■

Corollary 2.3.2.

(a) The preimage of a locally closed subspace is locally closed.
(b) If $Z \subset Y \subset X$ are such that Z is locally closed in Y and Y locally closed in X, then Z is locally closed in X.
(c) The intersection of two locally closed subspaces is locally closed.
(d) The complement of a locally closed subspace is the disjoint union of an open subspace and a closed subspace.

Proof. The statements (a) and (b) follow from 2.3.1(d), the statement (c) follows from (a) and (b), and (d) follows from either (e) or (f) of 2.3.1. ■

Proposition/Definition 2.3.3 (Constructible Subspaces)**.** Let X be a topological space. The following conditions on a subset $Y \subset X$ are equivalent.

(a) Y belongs to the algebra of subsets of X generated by the closed sets, i.e., the smallest collection of subsets containing the closed subsets which is closed under taking finite intersections and complements (and hence also under finite unions).
(b) Y can be written as a finite disjoint union of locally closed subsets of X.
(c) There is an integer $n \in \mathbb{Z}_{\geq 1}$ and a nested sequence $G_1 \supset G_2 \supset \cdots \supset G_n$ of closed subsets of X such that $Y = G_1 \smallsetminus (G_2 \smallsetminus (G_3 \smallsetminus (\cdots \smallsetminus G_n)) \cdots)$.

If X is Noetherian, these properties are also equivalent to

(d) For every closed irreducible $Z \subset X$, the intersection $Y \cap Z$ either contains a nonempty open subset of Z or is nowhere dense in Z.

A subspace satisfying equivalent conditions (a)-(c) is said to be a *constructible* subspace of X.

Proof. Here we prove the equivalence of (a), (b), and (c). The proof of equivalence with (d) in the Noetherian case is deferred to 2.3.5.

(a) $\Leftrightarrow$ (b) It suffices to show that the collection of subsets described by (b) is closed under taking finite intersections and complements. The first of these is clear from 2.3.2(c), and the second follows from the first and 2.3.2(d).

(c) $\Rightarrow$ (b) Clear, since $G_1 \smallsetminus (G_2 \smallsetminus (G_3 \smallsetminus (\cdots \smallsetminus G_n)) \cdots) = (G_1 \smallsetminus G_2) \amalg (G_3 \smallsetminus G_4) \amalg \cdots$.

(a) $\Rightarrow$ (c) This is somewhat trickier than it looks, and may be skipped on a first reading. One possible approach is to show directly that the collection of subsets described by (c) is closed under taking finite intersections and complements. The latter is clear, but the former involves some nontrivial combinatorics and Boolean algebra. Here's a different approach.[7]

Given a topological space X, an $n \in \mathbb{Z}_{\geq 1}$, and a collection $\mathscr{Z} = \{Z_1, \ldots, Z_n\}$ of closed subsets of X, consider the Boolean subalgebra $\mathcal{S}(\mathscr{Z})$ of subsets of X generated by the Z_i. This consists of finite unions of subsets of the form $\bigcap_{i=1}^n Z_i^*$ where each Z_i^* is either Z_i or $X \smallsetminus Z_i$; in particular, $\mathcal{S}(\mathcal{Z})$ has size at most 2^{2^n}, and is therefore finite.[8] The subset $\mathcal{S}(\mathscr{Z}) \subset 2^X$ inherits a natural partial order by inclusion, and we call a element $A \in \mathcal{S}(\mathscr{Z})$ an *atom* iff $A \neq \emptyset$ and A is minimal with respect to this partial order. Each atom A is locally closed (since it must itself be of the form $\bigcap_{i=1}^n Z_i^*$ as above by minimality), distinct atoms are disjoint, and every element of $\mathcal{S}(\mathscr{Z})$ is a (disjoint) union of atoms.

Suppose we are given a Y as in (a). Then there is a finite set $\mathscr{Z}$ of closed subsets of X such that $Y \in \mathcal{S}(\mathscr{Z})$; fix such a $\mathscr{Z}$. Write $Y \in \mathcal{S}(\mathscr{Z})$ as a disjoint union of atoms, say $Y = \coprod_{k=1}^m A_k$ for some $m \in \mathbb{Z}_{\geq 1}$, with the A_k pairwise distinct. Further express each A_k as $A_k = S_k \smallsetminus T_k$ for some closed sets $T_k \subset S_k \subset X$. We claim that it suffices to take $n = 2m$, with

$$G_{2k-1} = \bigcup_{j=k}^{r} S_j \text{ and } G_{2k} = T_k \cup \bigcup_{j=k+1}^{r} S_j$$

for $1 \leq k \leq m$. Since the inclusions $G_1 \supset G_2 \supset \cdots \supset G_n$ are clear, it suffices to show that for each $k = 1, \ldots, m$, we have $G_{2k-1} \smallsetminus G_{2k} = A_k$. The inclusion $G_{2k-1} \smallsetminus G_{2k} \subset A_k$ is clear. For the reverse inclusion, it suffices to show that if $1 \leq k < k' \leq m$, then $A_k \cap S_{k'} = \emptyset$, but this follows from the fact that distinct atoms are disjoint.[9]

■

Lemma 2.3.4.

(a) The preimage of a constructible set under a continuous map is constructible. If $Z \subset Y \subset X$ are such that Z is constructible in Y and Y is constructible in X, then Z is constructible in X.

(b) Let X be an irreducible topological space and $C \subset X$ a constructible subspace. Then the following are equivalent:

  (i) C contains a nonempty open subspace of X.

  (ii) C is dense.

  If, further, X has a generic point η, then (i) and (ii) are also equivalent to

  (iii) C contains η.

(c) In a locally Noetherian sober space, a subspace is closed (resp. open) iff it is constructible and stable under specialization (resp. generization).

Proof.

[7] Collaboration acknowledgment: this solution was inspired by some conversations with ChatGPT on 09/20/25.

[8] Another way to see this is to note that a subalgebra of $\mathbb{F}_2^X$ generated by n elements, say $\chi_1, \ldots, \chi_n$ has dimension at most 2^n over $\mathbb{F}_2$, since it is spanned as a vector space over $\mathbb{F}_2$ by the elements $\chi_I = \prod_{i \in I} \chi_i$ as $I \subset 2^{\{1,\ldots,n\}}$ ranges over all subsets of $\{1, \ldots, n\}$.

[9] If the reader finds an easier proof of this implication, please let me know!

(a) Follows from 2.3.2 and 2.3.3 (check!).
(b) The implication (i) $\Rightarrow$ (ii) follows from irreducibility. For (ii) $\Rightarrow$(i), use 2.3.3 to write C as a union of locally closed subspaces and use the irreducibility of X to reduce to the case where C is locally closed; then apply 2.3.1(g). When X has a generic point η, the implications (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii) are clear.
(c) Noting that the question is local, and by taking complements for the assertion in parentheses, it suffices to show that if X is Noetherian sober, $C \subset X$ constructible, and $y \in \overline{C}$, then there is a $x \in C$ such that $x \rightsquigarrow y$. Use 2.2.7(b) and 2.2.9 to write an irreducible decomposition of C as $C = Z_1 \cup \cdots \cup Z_r$ for some $r \in \mathbb{Z}_{\geq 1}$ and closed irreducible $Z_1, \ldots, Z_r \subset C$. For each i, the closure $\overline{Z}_i$ of Z_i in X is irreducible (by 2.1.5(c)); since X is sober, $\overline{Z}_i$ has a generic point, say η_i. The intersection $C \cap \overline{Z}_i$ is a constructible set of $\overline{Z}_i$ by (a) and contains Z_i, so it is dense; therefore, by (b), we have $\eta_i \in C$. Now if $y \in \overline{C}$, then there is an i such that $y \in \overline{Z}_i$, and then taking $x = \eta_i$ suffices.

■

Corollary 2.3.5. Let X be a Noetherian topological space. A subset $Y \subset X$ is constructible iff for every closed irreducible $Z \subset X$, the intersection $Y \cap Z$ either contains a nonempty open subspace of Z or is nowhere dense in Z.

The following proof has been taken from [5, Proposition 10.14].

Proof. If Y is constructible, then for each such Z, so is $Y \cap Z \subset Z$ by 2.3.4(a). In this case, either $Y \cap Z$ is dense in Z, in which case it contains a nonempty open subspace of Z by 2.3.4(b), or it is not dense, in which case it is nowhere dense in Z by 2.1.3(d).

Conversely, by Noetherian Induction (2.2.10) applied to property of meeting Y in constructible subset of X, we are reduced to the case in which for every proper closed subset $W \subset X$, the intersection $W \cap Y$ is constructible in X (or equivalently in W, by 2.3.4(a)). If X is not irreducible, then for each irreducible component $X' \subset X$, the intersection $X' \cap W$ is constructible in X by hypothesis; this suffices, since constructible sets are closed under finite unions and X has only finitely many irreducible components (2.2.7(c)). Suppose now that X is irreducible. If Y contains a nonempty open subspace U of X, then the hypothesis applied to $W = X \setminus U$ tells us that $W \cap Y = Y \setminus U$ is constructible in X; then so is $Y = (Y \setminus U) \cup U$. If Y is nowhere dense in X, then in particular the closure $\overline{Y} \subset X$ is a proper closed subset; then the hypothesis applied to $W = \overline{Y}$ tells us that $Y = W \cap Y$ is constructible in X. ■

The final notion we will need is that of *very dense subspaces.* To motivate this, note that if X is a topological space and $Y \subset X$ a subspace, then the map $W \mapsto W \cap Y$ taking open (resp. closed) subspaces $W \subset X$ to open (resp. closed) subspaces of Y is always a surjection.

Proposition/Definition 2.3.6 (Very Dense Subspaces)**.** Let X be a topological space. The following conditions on a subspace $Y \subset X$ are equivalent:

(a) The map $U \mapsto U \cap Y$ is a bijection from the set of open (resp. closed, resp. locally closed, resp. constructible) subspaces of X to those of Y. Equivalently, if $U, V \subset X$ are open (resp. closed, resp. locally closed, resp. constructible) subspaces such that $U \cap Y = V \cap Y$, then $U = V$.
(b) For every closed subspace $Z \subset X$, we have $Z = \overline{Z \cap Y}$.
(c) Every nonempty locally closed (resp. constructible) subset of X meets Y.

A subspace $Y \subset X$ satisfying these equivalent conditions is said to be *very dense* in X.

Proof.

(a) $\Rightarrow$ (b) The property (a) for constructible subsets implies it for locally closed subsets, which in turn implies it for open and closed subsets, and the properties for open and closed subsets are equivalent by taking complements. The property (a) for closed subsets then implies (b); indeed, if $Y \subset X$ is any subset and $Z \subset X$ closed, then $Z \cap Y = \overline{Z \cap Y} \cap Y$.

(b) $\Rightarrow$ (c) The two properties are equivalent by 2.3.3(b), and (b) implies (c) for locally closed subsets by 2.3.1(f).

(c) $\Rightarrow$ (a) The property (c) for constructible sets implies the property (a) for constructible sets: if $C, D \subset X$ are constructible subsets such that $C \cap Y = D \cap Y$, then $C \setminus D$ and $D \setminus C$ are constructible subsets of X not meeting Y, and hence are both empty.

■

Example 2.3.7. The subset $\mathbb{Q} \subset \mathbb{R}$ is dense but not very dense.

Corollary 2.3.8. Let X be a topological space and $Y \subset X$ a subspace.

(a) If $C \subset X$ is a constructible subspace and Y very dense in X, then $Y \cap C$ is very dense in C.
(b) If $X = \bigcup_\alpha X_\alpha$ for some $X_\alpha \subset X$ such that for each α, the intersection $Y \cap X_\alpha$ is very dense in X_α, then Y is very dense in X.

Proof.

(a) If $D \subset C$ is a nonempty constructible subspace, then D is constructible in X by 2.3.4(a) and hence meets Y (and so $Y \cap C$) by 2.3.6(c).
(b) Given a nonempty constructible $C \subset X$, pick a point $x \in C$ and an α such that $x \in X_\alpha$. Then $C \cap X_\alpha$ is a nonempty constructible subspace of X_α by 2.3.4(a), and so meets $Y \cap X_\alpha$ by hypothesis and 2.3.6(c). Therefore, C meeets Y, and we are done.

■

The exercise [6, Exercise 3.3.27] cited in the previous section gives an example of a pathological scheme which does not have *any* closed points. This, however, cannot happen for algebraic varieties. Indeed, the key example of very dense subspaces, and the reason we care, is

Theorem 2.3.9. Let k be a field and X be a scheme locally of finite type over k.

(a) The subspace of closed points of X is very dense in X. In particular, X is closure-complete, so that if it is nonempty, then it has a closed point.
(b) Suppose Y is another scheme locally of finite type over k, and $f, g : X \to Y$ two k-morphisms such that for some algebraic closure $\overline{k} \supset k$ we have $f(\overline{k}) = g(\overline{k}) : X(\overline{k}) \to Y(\overline{k})$. If X is geometrically reduced, then $f = g$.

Proof.

(a) Let $X^{\mathrm{cl}} \subset X$ denote the subset of closed points. If $W \subset X$ is any (locally closed) subset, then $W^{\mathrm{cl}} = W \cap X^{\mathrm{cl}}$, since closed points admit an intrinsic characterization (namely that their residue field is a finite extension of k). Therefore, by 2.3.8(b), it suffices to show the result when X is affine, so suppose X is an affine scheme of finite type over k and $Y \subset X$ a nonempty locally closed subset. We want to show $Y \cap X^{\mathrm{cl}} \neq \emptyset$. Replacing X by $(\overline{Y})^{\mathrm{red}}$, we may assume Y is open in X (using 2.3.1(g) and that every closed subscheme of affine scheme is affine), and by shrinking Y we may assume it is a nonempty principal open subset, say $Y = D(f)$ for some $f \in \mathcal{O}(X)$. Since Y is nonempty, f lies in some prime ideal, and then f lies in some maximal ideal; the corresponding point of X then belongs to $Y \cap X^{\mathrm{cl}}$.

(b) Base change to $\overline{k}$ to assume k is algebraically closed, using that $f_{\overline{k}} = g_{\overline{k}}$ implies $f = g$ (exercise; see [1, 10.2.J]) In this case, $X(k)$ can be identified with the set of closed points, which is very dense by (a). The (underlying subspace) of the equalizer $\mathrm{Eq}(f, g) \subset X$ is locally closed, since the diagonal morphism for schemes is always a locally closed immersion, and so its complement is constructible; if it were to be nonempty, then it would meet $X(k)$ by 2.3.6(e), which it does not. Therefore, $\mathrm{Eq}(f, g) = X$ as topological spaces, and then as schemes because X is reduced, i.e., $f = g$.

■

Informally, for geometrically reduced schemes locally of finite type over a field k, equality of morphisms can be checked at the level of $\overline{k}$-points for an(y choice of an) algebraic closure $\overline{k}$ of k. The hypothesis of geometric reduceness cannot be removed; for a counterexample, consider the ring $\mathbb{C}[\varepsilon] := \mathbb{C}[x]/(x^2)$, $X = Y = \operatorname{Spec} \mathbb{C}[\varepsilon]$, and $f, g : X \to Y$ the two morphisms corresponding to $\varepsilon \mapsto 0, \varepsilon$. It also does not suffice to work with k points when k is not algebraically closed, since a scheme locally of finite type over k need not have any k-points at all (e.g. when $k = \mathbb{R}$ and $X = \operatorname{Spec} \mathbb{C}$).

Remark 2.3.10. The result of the previous proposition holds more generally in other contexts, e.g., for schemes locally of finite type over $\mathbb{Z}$. A scheme is called a *Jacobson scheme* if its subset of closed points is very dense; then 2.3.9 is saying that a scheme locally of finite type over a field is Jacobson. The affine version of a Jacobson scheme is a Jacobson ring, which admits many different characterizations (see [5, Exercises 10.15-16] and [4, §6.2.2]). It is easy to see that $\mathbb{Z}$ is a Jacobson ring, and hence that $\operatorname{Spec} \mathbb{Z}$ is a Jacobson scheme; also, a field k is trivially seen to be a Jacobson ring. The final result needed is that if S is a Jacobson scheme and $f : X \to S$ a morphism locally of finite type, then X is a Jacobson scheme ([5, Exercise 10.16]).

2.4 Separatedness and Properness from a Topological Perspective

In this final section, we provide a few topological results that might help motivate the algebro-geometric definitions of separated and proper morphisms. The first is the classic

Proposition/Definition 2.4.1. (Hausdorff Spaces) Let X be a topological space. The following conditions on X are equivalent:

(a) For any two distinct points $x, y \in X$, there are disjoint open subsets $U, V \subset X$ such that $x \in U$ and $y \in V$.
(b) Limits of nets in X, when they exist, are unique.
(c) Every point $x \in X$ is the intersection of all its closed neighborhoods (where by a closed neighborhood of x we meant a closed subset of X containing x in its interior).
(d) The diagonal $\Delta(X) \subset X \times X$ is a closed subset when $X \times X$ is given the product topology.

A topological space satisfying these equivalent conditions is called *Hausdorff*.

Proof. Exercise. ∎

Schemes are usually not Hausdorff, and so the above definition needs to be modified to obtain the usual definition for separated morphisms. One can also define, just as in the scheme case, a continuous map $X \to Y$ to be separated if the diagonal $\Delta_{X/Y} : X \to X \times_Y X$ is a closed topological embedding, and this notion can then be characterized in terms of the fibers being relatively Hausdorff ([7, Proposition 1.25]), but to me that seems a little artificial; besides, it's less useful to algebraic geometers because it does not translate over to the schemes setting, precisely because the Zariski topology on $X \times_Y X$ is not the usual (subspace-of-) product topology.

There is another related topological notion that is used in scheme theory, namely the notion of quasiseparated topological spaces. I believe this is best handled in a scheme-theoretic manner ([1, §11.1.4]), so I will content myself with giving the definition and one lemma which is sometimes useful (as in 3.3.5 below and in the solution to [1, Exercise 5.1.H]).

Definition 2.4.2. A topological space X is said to be *quasiseparated* if the intersection of any two quasicompact[10] open subsets of X is again quasicompact.

Lemma 2.4.3 ([8])**.** Let X be a topological space admitting a basis of quasicompact open subsets, and $\mathscr{U}$ a cover of X by quasiseparated open subsets, all of whose pairwise intersections are quasicompact. If $W \subset X$ is any quasicompact open subset, then for all $U \in \mathscr{U}$, the intersection $W \cap U$ is quasicompact.

Proof. Cover W by finitely many basic quasicompact open subsets V_i each contained in some element say U_i of the cover $\mathscr{U}$. It suffices to show that for each $U \in \mathscr{U}$, the intersection $U \cap V_i$ is quasicompact, but $U \cap V_i = V_i \cap (U \cap U_i)$ is an intersection of two quasicompact open subsets inside the quasiseparated U_i. ∎

Au contraire, it is easier and more direct to relate the topological notion of compactness to the algebro-geometric notion of properness. The following exposition has been taken from [7, §1.5]. Recall the definition of quasicompact spaces.

[10] See 2.4.4 below if needed.

Proposition/Definition 2.4.4 (Quasicompact Spaces)**.** The following conditions on a topological space X are equivalent:

(a) Every open cover has a finite subcover.
(b) Every collection of closed subspaces with the finite intersection property (i.e., such that every finite intersection of which is nonempty) has nonempty intersection.
(c) Every net in X has a convergent subnet.
(d) The final morphism $X \to \{*\}$ is universally closed, i.e., for every topological space Z, the projection $\pi : Z \times X \to Z$ is a closed map.

A topological space satisfying these equivalent properties is said to be *quasicompact*. A topological space is said to be *compact* if it is both quasicompact and Hausdorff. Further:

(e) A closed subspace of a quasicompact space is quasicompact.
(f) The continuous image of a quasicompact space is quasicompact.
(g) A quasicompact subspace of a Hausdorff space is closed.

Proof. The implications (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) are standard and left as an exercise, as are the statements (e), (f), (g). Since a proof of the equivalence with (d) is trickier (and harder to locate), we reproduce it here.

(a) $\Rightarrow$ (d) Let Z be given, $C \subset Z \times X$ closed, and $z \in Z \smallsetminus \pi(C)$. The following lemma (2.4.5)[11] applied to $\pi^{-1}(z) = \{z\} \times X \subset (Z \times X) \smallsetminus C \subset Z \times X$ tells us that there is an open neighborhood V of z in Z such that $V \times X \subset (Z \times X) \smallsetminus C$. This shows that $V \subset Z \smallsetminus \pi(C)$. Since z was arbitrary, this proves that $Z \smallsetminus \pi(C)$ is open as needed.

(d) $\Rightarrow$ (c) An argument can be found in [7, Prop. 1.29], but I prefer the following argument adapted from [9, Lemma 4.3]. Suppose X is a topological space and $\mathscr{U}$ an open cover of X with no finite subcover. Replace $\mathscr{U}$ by the collection of open subspaces consisting of all finite unions of elements of $\mathscr{U}$ to assume without loss of generality that $\mathscr{U}$ is directed. We put a topology on $Z = \mathscr{T}(X) \subset 2^X$, the collection of open subsets of X, such that the projection map $\pi : Z \times X \to Z$ is not closed. Define a subset $V \subset Z$ to be open iff both (i) and (ii) below hold.

(i) The collection V is upward closed, i.e., for all $U \in V$, if $U' \subset X$ is open and $U \subset U'$, then $U' \in V$.

(ii) If $X \in V$, then there is a $U \in \mathscr{U}$ such that $U \in V$.

It follows from the fact that $\mathscr{U}$ is directed that the above defines a topology on Z, and further that if $U \subset X$ is an open subspace of some element of $\mathscr{U}$, then the upward closure $U^{\uparrow} := \{U' \in Z : U' \supset U\} \subset Z$ is open. It remains to show that the subset $C \subset Z \times X$ defined by $C := \{(U, x) : U \not\ni x\}$ is a closed subset such that $\pi(C) = Z \smallsetminus \{X\}$ is not closed. The latter part follows from the fact that $X \notin \mathscr{U}$. For the former part, we need to show that if $(U, x) \in Z \times X$ is such that $U \ni x$, then there is an open neighborhood of (U, x) in $Z \times X$ contained in $Z \times X \smallsetminus C$. But now since $\mathscr{U}$ covers X, there is a $U' \in \mathscr{U}$ such that $U' \ni x$, and then $(U, x) \in (U \cap U')^{\uparrow} \times (U \cap U') \subset Z \times X \smallsetminus C$ as needed.

■

Lemma 2.4.5. Let X, Y be topological spaces and $K \subset X$ and $L \subset Y$ quasicompact subspaces. Then for any open neighborhood U of $K \times L$ in $X \times Y$, there are open neighborhoods V of K in X and W of L in Y such that $V \times W \subset U$.

Proof. Exercise. ■

[11] In order to avoid circular arguments, the reader is encouraged to prove the Lemma 2.4.5 using definition (a) of compactness given above.

Proposition/Definition 2.4.6 (Proper Maps)**.** The following conditions on a continuous map $f : X \to S$ between topological spaces are equivalent:

(a) The map f is universally closed in the topological category, i.e., for every continuous map $T \to S$, the projection map $f_T : X_T := T \times_S X \to T$ after base-change to T is closed.
(b) For every topological space T, the map $f_{T\times S} : T \times X \to T \times S$ is closed.
(c) The map f is closed and for every quasicompact subspace $K \subset S$, the preimage $f^{-1}(K) \subset X$ is quasicompact.
(d) The map f is closed and for all $s \in S$, the fiber $X_s = f^{-1}(s) \subset X$ is quasicompact.

Further, when S is locally (quasi)compact Hausdorff, these are also equivalent to:

(e) For every (quasi)compact subset $K \subset S$, the preimage $f^{-1}(K) \subset X$ is quasicompact.

A continuous map $f : X \to S$ is said to be *proper* iff it is satisfies equivalent conditions (a)-(d).

Note that as a consequence, we have that a space X is quasicompact iff the final morphism $X \to \{*\}$ is proper.

Proof.

(a) $\Rightarrow$ (b) Base-change along the projection $T \times S \to T$.
(b) $\Rightarrow$ (c) Taking T to be a point shows f is closed. Let $K \subset S$ be a quasicompact subspace and $T = Z$ an arbitrary space. Since $f_Z : Z \times X \to Z \times S$ is closed, so is the restriction $f_Z|_{Z\times f^{-1}(K)} : Z \times f^{-1}(K) \to Z \times K$.[12] By 2.4.4(d), the projection $Z \times K \to Z$ is also closed. Therefore, the composition $Z \times f^{-1}(K) \to Z \to K \to Z$, which is just the projection onto Z, is also closed. Since this is true for all Z, another application of 2.4.4(d) tells us that $f^{-1}(K)$ is quasicompact.
(c) $\Rightarrow$ (d) The one point subspace $\{s\} \subset S$ is quasicompact.
(d) $\Rightarrow$ (a) Given $g : T \to S$, the image $f_T(X_T) = g^{-1}(f(X))$ is closed because f is. Given a closed $Z \subset X_T$, we have to show that $T \smallsetminus f_T(Z)$ is open, i.e., for all $t \in T \smallsetminus f_T(Z)$, there is an open neighborhood U of t in T contained in $T \smallsetminus f_T(Z)$. When $t \notin f_T(X_T)$, we can take $U := T \smallsetminus f_T(X_T)$. When $t \in f_T(X_T)$, note that $f_T^{-1}(t) = \{t\} \times X_{g(t)} \subset X_T \smallsetminus Z$. Since $X_T \subset T \times X$ is given the subspace topology, there is a closed subset $\overline{Z}$ of $T \times X$ such that $Z = \overline{Z} \cap X_T$; then applying 2.4.5 to $\{t\} \times X_{g(t)} \subset T \times X \smallsetminus \overline{Z}$ gives us a neighborhood V of t in T and a neighborhood W of $X_{g(t)}$ in X such that $V \times W \subset T \times X \smallsetminus \overline{Z}$. It then suffices to take $U := V \cap g^{-1}(S \smallsetminus f(X \smallsetminus W))$.

Suppose now that S is locally (quasi)compact Hausdorff. It only remains to prove

(e) $\Rightarrow$ (c) We have to show that f is closed. Let $C \subset X$ be closed; we have to show that for all $s \in S \smallsetminus f(C)$, there is an open neighborhood U of s in S contained in $S \smallsetminus f(C)$. Pick an open neighborhood V of s in S whose closure $\overline{V} \subset S$ is (quasi)compact. Then $f^{-1}(\overline{V}) \subset X$ is quasicompact, and hence so is $C \cap f^{-1}(\overline{V})$ by 2.4.4(e). By 2.4.4(f), the subspace $f(C \cap f^{-1}(\overline{V})) = f(C) \cap \overline{V} \subset S$ is also quasicompact, and hence closed by 2.4.4(g). It suffices to take $U := V \smallsetminus (f(C) \cap \overline{V}) = V \smallsetminus f(C)$.

■

For an example where (e) is strictly weaker than (a)-(d), see [7, Problem 1.21]. We end this section by stating without proof the following wonderful result relating these modern and classical notions. For this recall that when X is scheme of finite type over $\mathbb{C}$, then we can associate to it a complex analytic space X^{an} whose underlying topological space is the set $X(\mathbb{C})$

[12] This uses the following elementary fact in point-set topology: if $p : A \to B$ is a continuous closed map and $C \subset B$ any subset, then the restriction $p|_{p^{-1}(C)} : p^{-1}(C) \to C$ is also closed. Indeed, any closed subspace of $p^{-1}(C)$ is of the form $p^{-1}(C) \cap D$ for some closed $D \subset A$; then it is easy to check that $p(p^{-1}(C) \cap D) = C \cap p(D)$.

of complex points of X equipped with the classical (i.e., analytic) topology (c.f. [2, Appendix B]).

Theorem 2.4.7. Let X be a scheme of finite type over $\mathbb{C}$.

(a) The scheme X is separated iff $X(\mathbb{C})$ with the classical topology is Hausdorff.
(b) The scheme X is complete (i.e., proper over $Y = \operatorname{Spec}\mathbb{C}$) iff $X(\mathbb{C})$ with the classical topology is compact (i.e., quasicompact Hausdorff).

TOCITE. ■

2.5 Exercises

Exercise 2.1. Let X be an irreducible topological space. Show that for any open subset $i_U : U \hookrightarrow X$ and set S, the presheaf on X given by

$$V \mapsto \begin{cases} S, & \text{if } \emptyset \neq V \subset U \text{ and} \\ *, & \text{else} \end{cases}$$

is a sheaf. In particular, if $S = \mathbb{Z}$, then this is the sheaf of abelian groups $i_{U,!}\mathbb{Z}_U$ on X.

Exercise 2.2. Show that openness of continuous maps is local on the source and target. Explicitly, show that the following conditions on a continuous map $\pi : X \to Y$ are equivalent.

(a) The map π is open.
(b) There is an open cover $\{V_j\}$ of Y such that for each j, if we let $X_j := \pi^{-1}(V_j)$ and $\pi_j : X_j \to V_j$ be the induced maps, then each $\pi_j : X_j \to V_j$ is open.
(c) There is an open cover $\{U_i\}$ of X such that for each i, the map $\pi|_{U_i} : U_i \to Y$ is open.
(d) There is a basis $\mathcal{B}$ for the topology on X such that for each $U \in \mathcal{B}$, the image $\pi(U) \subset Y$ is open.

Exercise 2.3 (Local Homeomorphisms)**.** A map $\pi : E \to X$ of topological spaces is a *local homeomorphism* if for each $e \in E$, there is a neighborhood $\widetilde{U}$ of e in E such that $U := \pi(\widetilde{U}) \subset X$ is open and $\pi|_{\widetilde{U}} : \widetilde{U} \to U$ is a homeomorphism.

(a) A continuous map $\pi : E \to X$ is a local homeomorphism iff π and the diagonal $\Delta_\pi : E \to E \times_X E$ are both open.
(b) If $\pi : E \to X$ is a local homeomorphism, then E has a basis of opens that map homeomorphically under π to opens in X. Further, for each $x \in X$, the preimage $\pi^{-1}(x) \subseteq E$ has the discrete topology.
(c) If two local sections of a local homeomorphism agree at a point, then they agree in a neighborhood.
(d) If $\pi : E \to Y$ is a local homeomorphism, and $f : X \to Y$ is any continuous map, then the map $f^*\pi : X \times_Y E \to E$ is a local homeomorphism, and for each $x \in X$, we have that the map $\mathrm{pr}_E|_{(f^*\pi)^{-1}x} : (f^*\pi)^{-1}x \to \pi^{-1}(f(x))$ is a homeomorphism.
(e) If $F \xrightarrow{\tau} E \xrightarrow{\pi} X$ is any sequence of maps, then
 - if τ and π are local homeomorphisms then so is $\pi \circ \tau$,
 - if π and $\pi \circ \tau$ are local homeomorphisms then so is τ, and
 - if τ and $\pi \circ \tau$ are local homeomorphisms then so is π, if τ is surjective. Give a counterexample otherwise.
(f) In particular, every local section of a local homeomorphism is open.

Exercise 2.4. ([5, Exercise 5.23]) Let k be a field and X a connected k-scheme.

(a) Suppose there is a nonempty geometrically connected k-scheme P and a k-morphism $P \to X$. Show that X is geometrically connected.
(b) Assume there is a point $p \in X$ such that k is separably closed in $\kappa(p)$. Show that X is geometrically connected. In particular, if X is a connected k-scheme with $X(k) \neq \emptyset$, then X is geometrically connected.

Exercise 2.5. Let $n \in \mathbb{Z}_{\geq 0}$. Let X be a topological space admitting a cover by open subsets, each of which only has finitely many irreducible components (e.g., if X is locally Noetherian). Show that X is equidimensional of dimension n iff for all $x \in X$ we have $\dim_x X = n$.

Exercise 2.6. Let X be a scheme and $\mathcal{G}$ be a finitely presented $\mathcal{O}_X$-module.

(a) For each $r \in \mathbb{Z}_{\geq 0}$, the locus $\{p \in X : \dim_{\kappa(p)} \mathcal{G}(p) \geq r\}$ is a closed subset. The locus

$\{p \in X : \dim_{\kappa(p)} \mathcal{G}(p) = r\}$ is locally closed.

(b) Let $\mathcal{F}$ be a finitely generated $\mathcal{O}_X$-module and $\varphi : \mathcal{F} \to \mathcal{G}$ a morphism of $\mathcal{O}_X$-modules. Suppose for simplicity that X is Noetherian.[13] Show that for each $r \in \mathbb{Z}_{\geq 0}$, the locus $\{p \in X : \operatorname{rank} \varphi(p) \leq r\}$ is constructible.

[13] This is only because there is some disagreement about the definition of constructible subsets, and our convention does not agree with that of the Stacks project ([3, 04ZC]), although it does when the underlying topological space is Noetherian. It should be clear to the reader from the proof what exactly can be shown in general.

Chapter 3

Affine Communication Lemma and Properties of Morphisms

3.1 The Affine Communication Lemma and Reasonable Classes of Morphisms

The wonderful theory-building tool we will repeatedly use is

Lemma 3.1.1 (Affine Communication Lemma)**.** Let X be a scheme and $\mathscr{P}$ be a property satisfied by some affine open subsets of X such that for any affine open $U \subset X$, the following two conditions hold.

(a) For any distinguished open $D \subset U$, if U has $\mathscr{P}$ then so does D.
(b) Suppose $\{D_\alpha\}$ is a finite collection of distinguished opens in U that cover U (i.e., such that $U = \bigcup_\alpha D_\alpha$). If each D_α has $\mathscr{P}$, then so does U.

In this case, if X admits some affine open cover all of whose elements have $\mathscr{P}$, then every affine open $U \subset X$ has $\mathscr{P}$.

Proof. Suppose we are given an affine open cover $\{V_\beta\}$ of X such that each V_β has $\mathscr{P}$. Using 1.2.3, cover U by distinguished opens D_α that are simultaneously also distinguished in *some* $V_{\beta(\alpha)}$. Since each V_β has $\mathscr{P}$, property (a) tells us that each D_α has $\mathscr{P}$. To apply (b), it remains to only note that finitely many D_α cover U, and this is true because U is quasicompact (1.2.1(e)). ■

The Affine Communication Lemma is best used in conjuction with the following two.

Lemma 3.1.2. Let $\mathscr{P}$ be a property of morphisms of schemes that is local on the target.[1] If $\mathscr{P}$ is stable under affine base change[2], then $\mathscr{P}$ is stable under base change.

Proof. Let $X \to S$ have $\mathscr{P}$ and let $T \to S$ be a morphism. Since $\mathscr{P}$ is local on the target, to show that the pullback $X_T \to T$ has $\mathscr{P}$, it suffices to produce an affine open cover $\mathscr{U}$ of T such that for each $U \in \mathscr{U}$, the morphism $X_U \to U$ has $\mathscr{P}$. For this, let $\mathscr{U}$ be the cover of T consisting of the affine opens $U \subset T$ whose image in S lies in some affine open of S; we show this $\mathscr{U}$ works. Given a $U \in \mathscr{U}$, pick an affine open V of S such that U maps to V in S. Since $\mathscr{P}$ is local on the target, the basechange $X_V \to V$ has $\mathscr{P}$. Then the morphism $X_U \to U$ is the affine base change of $X_V \to V$ along $U \to V$ and hence also has $\mathscr{P}$. ■

Lemma 3.1.3. Let $\mathscr{P}$ be a property of morphisms of schemes that is local on the target. If $\mathscr{P}$ is further local on the source[3] and stable under affine composition,[4] then it is stable under arbitrary composition.

Proof. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms having $\mathscr{P}$; we want to show that so does $gf : X \to Z$. Since $\mathscr{P}$ is local on the target, we may assume without loss of generality that Z is affine. Since $\mathscr{P}$ is local on the source, to show that gf has $\mathscr{P}$, it suffices to produce an affine open cover $\mathscr{U}$ of X such that for each $U \in \mathscr{U}$, the composite $U \hookrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z$ has $\mathscr{P}$. For this, let $\mathscr{U}$ be the cover of X consisting of affine opens $U \subset X$ whose image in Y under f lies in some affine open of Y; we show that this $\mathscr{U}$ works. Given $U \in \mathscr{U}$, pick an affine open V of Y such that $U \subset f^{-1}(V)$. Since $\mathscr{P}$ is local on the source, the restriction $g|_V : V \to Z$ has $\mathscr{P}$. Since $\mathscr{P}$ is local on the target, $f|_{f^{-1}(V)} : f^{-1}(V) \to V$ has $\mathscr{P}$. Since $\mathscr{P}$ is local on the source, the

[1] In other words, if $\pi : X \to Y$ is a morphism of schemes, then π has the property $\mathscr{P}$ iff for each $y \in Y$, there is an open $V \subset Y$ such that the restriction $\pi^{-1}(V) \to V$ has $\mathscr{P}$.

[2] An *affine base change* is a base change in which the source and target bases are affine.

[3] In other words, if $\pi : X \to Y$ is a morphism of schemes, then π has the property $\mathscr{P}$ iff for each $x \in X$, there is an open $U \subset X$ such that the restriction $\pi|_U : U \to Y$ has $\mathscr{P}$.

[4] An *affine composition* is a composition of two morphisms, both of whose sources and targets are affine.

further restriction $f|_U : U \to V$ has $\mathscr{P}$. Finally, since $\mathscr{P}$ is stable under affine composition, the composite $g|_V \circ f|_U : U \to Z$ has $\mathscr{P}$, as needed. ■

The following definition is surprisingly useful.

Definition 3.1.4. A class of morphisms $\mathscr{P}$ of schemes is said to be *reasonable* iff it contains isomorphisms, is local on the target, and is stable under composition and base-change.

Remark 3.1.5. The same definition can also, of course, be applied in other contexts–e.g., of locally ringed spaces (and hence of topological spaces, manifolds, etc.). For instance, injections, embeddings, open embeddings, closed embeddings, and surjections are all reasonable classes of continuous maps.

Lemma 3.1.6. Let $\mathscr{P}$ be a property of morphisms of schemes stable under composition and base-change. Let S be a scheme, and let $f : X \to Y$ and $g : X' \to Y'$ be two morphisms of S-schemes with property $\mathscr{P}$. Then $f \times g : X \times_S X' \to Y \times_S Y'$ also has $\mathscr{P}$.

Proof. Since $\mathscr{P}$ is stable under base-change, the product $f \times 1_{X'} : X \times_S X' \to Y \times_S X'$ (resp. $1_Y \times g : Y \times_S X' \to Y \times_S Y'$) has $\mathscr{P}$, because it is the base-change of f (resp. g) along the projection $Y \times_S X' \to Y$ (resp. $Y \times_S Y' \to Y'$). Since $\mathscr{P}$ is stable under composition, the morphism $f \times g = (1_Y \times g) \circ (f \times 1_{X'})$ also has $\mathscr{P}$. ■

In this chapter, we will meet a huge collection of reasonable morphisms. Before we do that, let's give one general way of producing more

Lemma 3.1.7.

Lemma 3.1.8.

3.2 (Locally) Noetherian Schemes and (Locally) Finite Type and Presentation Morphisms

In this section we discuss some fundamental finiteness properties of schemes and morphisms. For this we start with a general lemma.

Lemma 3.2.1. Let A be a ring, $n \in \mathbb{Z}_{\geq 1}$, and $f_1, \ldots, f_n \in A$ such that $(f_1, \ldots, f_n) = A$.

(a) Let M be an A-module. If for each $i = 1, \ldots, n$, the $A[f_i^{-1}]$-module $A[f_i^{-1}] \otimes_A M$ is finitely generated, then so is M.
(b) Let $\mathfrak{a} \subset A$ be an ideal. If for each $i = 1, \ldots, n$, the ideal $\mathfrak{a}A[f_i^{-1}] \subset A[f_i^{-1}]$ is finitely generated, then so is $\mathfrak{a}$.

Proof.

(a) For each $i = 1, \ldots, n$, pick finitely many elements $m_{ij} \in M$ whose images generate $A[f_i^{-1}] \otimes_A M$ as an $A[f_i^{-1}]$-module. Check that the finite set $\{m_{ij}\}_{i,j}$ generates M as an A-module.
(b) Follows from applying (a) to $M = \mathfrak{a}$, and noting that for each i, we have $A[f_i^{-1}] \otimes_A \mathfrak{a} \xrightarrow{\sim} \mathfrak{a}A[f_i^{-1}]$ because localization is exact (i.e., $A[f_i^{-1}]$ is a flat A-algebra).

■

Proposition/Definition 3.2.2 (Locally Noetherian Schemes)**.** The following conditions on a scheme X are equivalent:

(a) For every affine open $U \subset X$, the ring $\mathcal{O}(U)$ is Noetherian.
(b) There is an affine open cover $X = \bigcup_i U_i$ such that for each i, the ring $\mathcal{O}(U)$ is Noetherian.

A scheme X satisfying these equivalent conditions is said to be *locally Noetherian.* A scheme X is said to be *Noetherian* iff it is locally Noetherian and quasicompact. Further:

(c) If X is a (locally) Noetherian scheme, then the underlying topological space of X is also (locally) Noetherian. Further, any open subscheme $U \subset X$ is also (locally) Noetherian.

Proof. By 3.1.1 it suffices to show the following.

(a) If A is a Noetherian ring, then for each $f \in A$, so is $A[f^{-1}]$. This follows, for instance, from the Hilbert Basis Theorem, since $A[f^{-1}] \cong A[X]/(fX - 1)$.
(b) Suppose A is a ring, $n \in \mathbb{Z}_{\geq 1}$, and $f_1, \ldots, f_n \in A$ such that $(f_1, \ldots, f_n) = (1)$. If for each i the ring $A[f_i^{-1}]$ is Noetherian, then so is A; this follows immediately from 3.2.1(b).

The first statement in (c) follows from 2.2.5 and 2.2.7(a). The second statement for the locally Noetherian case follows immediately from the equivalence of (a) and (b), while the in the Noetherian case follows from the locally Noetherian case coupled with the first statement and 2.2.7(b). ■

Let A be a ring. Recall that an A-algebra B is said to be of *finite type*[5] (resp. *finite presentation*) if B if there is an $n \in \mathbb{Z}_{\geq 1}$ and a surjective A-algebra morphism $A[X_1, \ldots, X_n] \twoheadrightarrow B$ (resp. there is an $n \in \mathbb{Z}_{\geq 1}$ and a surjective A-algebra morphism $A[X_1, \ldots, X_n] \twoheadrightarrow B$ with the kernel a finitely generated ideal). Recall that algebras of finite presentation satisfy the following persistence property.

[5] Also known as being "finitely generated".

Lemma 3.2.3. Let A be a ring and B be an algebra of finite presentation. For any $m \in \mathbb{Z}_{\geq 1}$ and surjective A-algebra morphism $\rho : A[X_1, \dots, X_m] \twoheadrightarrow B$, the kernel is a finitely generated ideal.

Proof. Since B is of finite presentation, there is an $n \in \mathbb{Z}_{\geq 1}$ and a surjection $\pi : A[Y_1, \dots, Y_n] \twoheadrightarrow B$ with the kernel a finitely generated ideal, say generated by the polynomials $f_1, \dots, f_r \in A[Y_1, \dots, Y_n]$ for some $r \in \mathbb{Z}_{\geq 0}$. Let $A[X] := A[X_1, \dots, X_m]$ and $A[Y] := A[Y_1, \dots, Y_n]$ for notational simplicity. Pick A-algebra homomorphisms $\widetilde{\pi} : A[Y] \to A[X]$ and $\widetilde{\rho} : A[X] \to A[Y]$ lifting π and ρ respectively. Check that the kernel of ρ is generated by the $r + m$ elements $\widetilde{\pi}(f_1), \dots, \widetilde{\pi}(f_r), X_1 - \widetilde{\pi}\widetilde{\rho}(X_1), \dots, X_m - \widetilde{\pi}\widetilde{\rho}(X_m)$. ■

With this, we are now ready to define morphisms locally of finite type/presentation.

Proposition/Definition 3.2.4 (Locally Finite Type/Presentations Schemes)**.** Let A be a ring and X be an A-scheme. Then X is said to be *locally of finite type*[6] (resp. *locally of finite presentation*[7]) *over* A if the following equivalent conditions hold:

(a) For every affine open $U \subset X$, the A-algebra $\mathcal{O}(U)$ is of finite type (resp. presentation).
(b) There is an affine open cover $X = \bigcup_i U_i$ such that for each i, the A-algebra $\mathcal{O}(U_i)$ is of finite type (resp. presentation).

Further:

(c) If X is lft (resp. lfp) over A, then so is any open subscheme $U \subset X$. Conversely, if X admits an open cover by lft (resp. lfp) A-schemes, then X is lft over A.

Proof. By 3.1.1, it suffices to show the following. Suppose B is an A-algebra.

(i) If B is of finite type (resp. presentation) over A, then for each $f \in B$, so is $B[f^{-1}] \cong B[T]/(Tf - 1)$; this is clear.
(ii) If there is an $n \in \mathbb{Z}_{\geq 1}$ and $f_1, \dots, f_n \in B$ such that $(f_1, \dots, f_n) = (1)$ and for each i, the localization $B[f_i^{-1}]$ is of finite type (resp. presentation) over A, then so is B.
  "ft". There is an $N \gg 1$ such that for each i, there is a finite set of $g_{ij} \in B$ such that $B[f_i^{-1}]$ is generated as an A-algebra by the $g_{ij} \cdot f_i^{-N}$. If we pick $h_1, \dots, h_n \in B$ such that $\sum_{i=1}^n h_i f_i = 1$, then $B = A[f_i, g_{ij}, h_i]$.
  "fp". By what we have already shown, B is of finite type, and so we may pick an $m \in \mathbb{Z}_{\geq 1}$ and surjection $\pi : A[X_1, \dots, X_m] \twoheadrightarrow B$ of A-algebras. Let $\mathfrak{a} := \ker \pi$. For each f_i, since localization is exact, the kernel of the localized map $\pi_i : A[f_i^{-1}][X_1, \dots, X_m] \to B[f_i^{-1}]$ is exactly $\mathfrak{a}A[f_i^{-1}][X_1, \dots, X_m]$. By 3.2.3, each $\mathfrak{a}A[f_i^{-1}][X_1, \dots, X_m]$ is finitely generated, so we conclude by 3.2.1.

The statement (c) follows immediately from the equivalence of (a) and (b). ■

The relative version of the above property is given in

Proposition/Definition 3.2.5 (Locally Finite Type/Presentations Morphisms)**.** A morphism $\pi : X \to Y$ of schemes is said to be *locally of finite type* (resp. *locally of finite presentation*[8]) if the following equivalent conditions hold:

(a) For every affine open $V \subset Y$, the $\mathcal{O}(V)$-scheme $\pi^{-1}(V)$ is lft (resp. lfp) over $\mathcal{O}(V)$.
(b) There is an affine open cover $Y = \bigcup_i V_i$ such that for each i, the $\mathcal{O}(V_i)$-scheme $\pi^{-1}(V_i)$ is lft (resp. lfp) over $\mathcal{O}(V_i)$.

[6] Abbreviated "lft".
[7] Abbreviated "lfp".
[8] We use the same abbreviations as for schemes.

Further:

(c) Lft (resp. lfp) morphisms are affine local on both the source and target. In particular, an open immersion is lfp (and hence lft).
(d) Lft (resp. lfp) morphisms form a reasonable class.
(e) If $\pi : X \to Y$ is lft and Y locally Noetherian, then so is X.

If A is a ring and X an A-scheme, then X is lft (resp. lfp) over A iff the structure morphism $X \to \operatorname{Spec} A$ is lft (resp. lfp).

Proof. By 3.1.1, it suffices to show the following. Let A be a ring.

(i) If X is an A-scheme lft (resp. lfp) over A, then for each $f \in A$, the scheme $\varphi^{-1}(D_f)$ is lft (resp. lfp) over $\mathcal{O}(D_f) = A[f^{-1}]$. Indeed, for each affine open $U \subset \varphi^{-1}(D_f)$, we know that $\mathcal{O}(U)$ is a finite type (resp. presentation) A-algebra and hence also a finite type (resp. presentation) $A[f^{-1}]$-algebra.
(ii) If $n \in \mathbb{Z}_{\geq 1}$ and $f_1, \ldots, f_n \in A$ are such that $(f_1, \ldots, f_n) = 1$ and for each i, the preimage $\varphi^{-1}(D_{f_i})$ is lft (resp. lfp) over $A[f_i^{-1}]$, then X is lft (resp. lfp) over A. Indeed, for each i, there is an affine open cover $\varphi^{-1}(D_{f_i}) = \bigcup_j U_{ij}$ such that for each j, the $A[f_i^{-1}]$-algebra $\mathcal{O}(U_{ij})$ is of finite type (resp. presentation). Then $X = \bigcup_i \bigcup_j U_{ij}$ is an affine open cover such that for each (i, j), the A-algebra $\mathcal{O}(U_{ij})$ is of finite type (resp. presentation) over $A[f_i^{-1}]$ and hence also over A, so we are done.

Now we show the remaining properties.

(c) Affine-locality on the target follows from the equivalence of (a) and (b). To prove affine-locality on the source, use affine-locality on the target to reduce the statement to 3.2.4(c).
(d) Stability under base-change follows from 3.1.2 and stability under composition follows from 3.1.3.
(e) Clear.

∎

3.3 Quasicompact and Quasiseparated Morphisms

The first order of business is to understand quasicompact morphisms.

Proposition/Definition 3.3.1 (Quasicompact Morphisms)**.** Let $\pi : X \to Y$ be a morphism of schemes. The following conditions on π are equivalent.

(a) For every quasicompact open $V \subset Y$, the preimage $\pi^{-1}(V) \subset X$ is quasicompact.
(b) For every affine open $V \subset Y$, the preimage $\pi^{-1}(V)$ is quasicompact.
(c) There is an affine open cover $Y = \bigcup_i V_i$ such that for each i, the preimage $\pi^{-1}(V_i) \subset X$ is quasicompact.

A morphism π of schemes satisfying these equivalent conditions is said to be *quasicompact*. Further:

(d) Quasicompact morphisms form a reasonable class.
(e) Every morphism out of a Noetherian scheme is quasicompact.

Proof.

(a) $\Rightarrow$ (b) Every affine open subset is quasicompact.
(b) $\Rightarrow$ (a) Every quasicompact open subset is a finite union of affine opens.
(b) $\Leftrightarrow$ (c) By 3.1.1, it remains to check the following in the case Y is affine.

(i) If X is quasicompact, then for any distinguished $D \subset Y$ the preimage $\pi^{-1}(D)$ is quasicompact. Since X is quasicompact, it is a finite union of affines; this reduces us to the case where X is also affine, but then we can invoke 1.2.1(a).

(ii) If $\{D_\alpha\}$ is a finite cover of Y by distinguished opens and each preimage $\pi^{-1}(D_\alpha) \subset X$ is quasicompact, then so is X. This is clear because $X = \pi^{-1}(Y) = \bigcup_\alpha \pi^{-1}(D_\alpha)$.

(d) Stability under composition follows from (a), and the rest is clear from 3.1.2.
(e) This follows from 2.2.7.

■

Next, we discuss a few properties of quasicompact morphisms that make them a very helpful class of morphisms. For this, we need a preparatory lemma.

Lemma 3.3.2. Let $\varphi : A \to B$ be a ring homorphism. The following conditions on φ are equivalent.

(a) The map φ takes non-nilpotent elements of A to non-nilpotent elements of B.
(b) We have $\ker \varphi \subset \mathrm{Nil}(A)$.
(c) Every minimal prime of A is contracted from B.
(d) The induced morphism $\operatorname{Spec} \varphi : \operatorname{Spec} B \to \operatorname{Spec} A$ has dense image.

In particular, if A is domain, then the above equivalent conditions hold iff φ is injective.

Proof.

(a) $\Leftrightarrow$ (b) Clear.
(b) $\Rightarrow$ (c) In general, every prime minimal over $\ker \varphi$ is contracted from B. This follows from the fact that if $A \subset B$ is a *subring*, then every minimal prime of A is contracted from B; this itself follows from the fact that if $\mathfrak{p} \subset A$ is a prime, then $B_\mathfrak{p} \neq 0$, so there is a prime $\mathfrak{q}$ of B such that $A \cap \mathfrak{q} \subset \mathfrak{p}$.
(c) $\Rightarrow$ (d) The closure of the image of $\operatorname{Spec} \varphi$ contains all irreducible components of $\operatorname{Spec} A$.
(d) $\Rightarrow$ (a) For $f \in A$, we have $(\operatorname{Spec} \varphi)^{-1} D(f) = D(\varphi(f))$; finish by using that f is nilpotent iff $D(f) = \emptyset$.

■

As a consequence, we obtain

Theorem 3.3.3. Let $\pi : X \to Y$ be a quasicompact morphism of schemes.

(a) The morphism π is dominant (2.1.14) iff it has dense image, i.e., $\overline{\pi(X)} = Y$.
(b) The image $\pi(X) \subset Y$ is closed iff it is stable under specialization.

Proof.

(a) One direction is 2.1.15. Suppose $\overline{\pi(X)} = Y$, and let $Z \subset Y$ be an irreducible component. Let $V \subset Y$ be an affine open subset such that $Z \cap V \neq \emptyset$; then $Z \cap V$ is an irreducible component of V by 2.1.5(g). The preimage $\pi^{-1}(V)$ is quasicompact, and hence covered by finitely many affine opens, say $U_1, \dots, U_n$ for some $n \in \mathbb{Z}_{\geq 1}$. The morphism $\coprod_{i=1}^n U_i \to \pi^{-1}(V) \to V$ also has dense image. Combining with 2.1.13 reduces us to 3.3.2.
(b) One direction is clear, so suppose that $\pi(X)$ is stable under specialization; we want to show it is closed. Replacing Y by the schematic image of X (3.5.12) and using (a), we may assume π is dominant; then we want to show that π is surjective. It follows from the definition 2.1.14 that $\pi(X)$ contains the generic points of all irreducible component of Y; then the stability of $\pi(X)$ under specialization gives the result.

■

We end with a wonderful fact that is not as well-known as it should be.

Theorem 3.3.4. A universally closed morphism of schemes is quasicompact.

Proof. (Taken from [10].) By the implication (d) $\Rightarrow$ (c) of 2.4.6, we reduce to showing: if k is a field, and $\pi : X \to \operatorname{Spec} k$ a universally closed morphism of schemes, then X is quasicompact. Let $X = \bigcup_{i \in I} U_i$ be an open affine cover of X, and let $A = k[X_i]_{i \in I}$ so that $T := \operatorname{Spec} A = \mathbb{A}_k^I$. Let

$$Z := X \times_k T - \bigcup_{i \in I} U_i \times_k D(X_i).$$

Pick a nonzero $f \in A$ such that $D(f) \subset T \smallsetminus \pi_T(Z)$; this is possible because π_T is closed and $\pi_T(Z) \neq T$, since say $(1, 1, \dots) \in T \smallsetminus \pi_T(Z)$. Pick a finite subset $J \subset I$ such that $f \in k[X_j]_{j \in J}$; we will show that $X = \bigcup_{j \in J} U_j$. Indeed, if there were an $x \in X \smallsetminus \bigcup_{j \in J} U_j$, and we picked a $y \in D(f) \subset \mathbb{A}_k^J \hookrightarrow T$ (where the last closed immersion is given by setting all other variables X_i for $i \notin J$ to zero) and a point $z \in X \times_k T$ above (x, y), then z would have to lie in Z. However, we would also conclude that $f(\pi_T(z)) = f(y) \neq 0$, which would contradict the fact that $\pi_T(Z) \subset \mathbb{V}(f)$. ■

The next order of business is to understand quasiseparated morphisms, but for that we need to understand quasiseparated schemes first.

Proposition/Definition 3.3.5 (Quasiseparated Schemes)**.** The following conditions on a scheme X are equivalent:

(a) If $U, V \subset X$ are quasicompact opens, then so is $U \cap V$.
(b) If $U, V \subset X$ are affine opens, then $U \cap V$ is a finite union of affine opens in X.
(c) There is an affine open cover $\mathscr{U}$ of X all of whose pairwise intersections are quasicompact.

A scheme X satisfying these equivalent conditions is said to be *quasiseparated*. Further:

(d) An open subscheme of an quasiseparated scheme is quasiseparated.

(e) A locally Noetherian scheme is quasiseparated.

Proof.

Step 1. The implications (a) $\Leftrightarrow$ (b) $\Rightarrow$ (c) follow from the fact that an open subset of X is quasicompact iff it can be written as a finite union of affine opens of X.

Step 2. We show that an affine scheme X satisfies (b). Indeed, let X be affine and $U, V \subset X$ be affine opens. By 1.2.3, U (resp. V) can be written as the finite union of affine opens U_i (resp. V_j) which are simultaneously distinguished in U and X (resp. in V and X). Then $U \cap V = \bigcup_{i,j} U_i \cap V_j$, with each intersection $U_i \cap V_j$ being distinguished in X by 1.2.1(b).

Step 3. Now we show (c) $\Rightarrow$ (b). Let $\mathscr{U}$ and $U, V \subset X$ be as given. By 2.4.3 applied to $\mathscr{U}$ and $W = U$, we conclude that for each $U' \in \mathscr{U}$, the intersection $U \cap U'$ is quasicompact. Then 2.4.3 applied to $\mathscr{U} \cup \{U\}$ and $W = V$ shows that $U \cap V$ is quasicompact.

The statement (d) is clear from (a). The statement (e) is clear from (b) combined with (b) and (e) of 2.2.7. ∎

Corollary 3.3.6 (Qcqs Schemes)**.** A scheme X is quasicompact and quasiseparated (abbreviated *qcqs*) iff it admits a finite affine open cover, all of whose pairwise intersections are also covered by finitely many affine opens.

In particular, affine schemes are qcqs. In many ways, qcqs schemes behave like affine schemes. Here is one example.

Lemma 3.3.7. Let X be a qcqs scheme and $\mathcal{F} \in \mathsf{QCoh}(X)$. Then for any $f \in \mathcal{O}(X)$, the natural morphism $\mathcal{O}(X)[f^{-1}] \otimes_{\mathcal{O}(X)} \mathcal{F}(X) \to \mathcal{F}(D(f))$ is an isomorphism.

Proof. Note that if $U \subset X$ is any affine, then $U \cap D(f) = D(f|_U)$ is a distinguished open in U. Now cover X by finitely many affines U_i such that each pairwise intersection $U_i \cap U_j$ is the finite union of affines U_{ijk}. Then for any sheaf $\mathcal{F}$ on X, we have an exact sequence

$$0 \to \mathcal{F}(X) \to \bigoplus_i \mathcal{F}(U_i) \to \bigoplus_{i,j,k} \mathcal{F}(U_{ijk}). \tag{3.3.8}$$

Tensoring with the flat $\mathcal{O}(X)$-algebra $\mathcal{O}(X)[f^{-1}]$ and using that tensor products commute with direct sums, along with some natural identifications, yields the analogous sequence

$$0 \to \mathcal{O}(X)[f^{-1}] \otimes_{\mathcal{O}(X)} \mathcal{F}(X) \to \bigoplus_i \mathcal{O}(D(f|_{U_i})) \otimes_{\mathcal{O}(U_i)} \mathcal{F}(U_i) \to \bigoplus_{i,j,k} \mathcal{O}(D(f|_{U_{ijk}})) \otimes_{\mathcal{O}(U_{ijk})} \mathcal{F}(U_{ijk}).$$

Map this to the sequence corresponding to 3.3.8 for the cover $\{D(f|_{U_i})\}_i$ of $D(f)$, and use that the middle and last maps are then isomorphisms (because $\mathcal{F} \in \mathsf{QCoh}(X)$) along with the five lemma to finish the proof. ∎

Most proofs involving qcqs schemes proceed in this way. We will give another example of such an rgument at the end of this section.

Now we are ready to define quasiseparated morphisms.

Proposition/Definition 3.3.9 (Quasiseparated Morphisms)**.** Let $\pi : X \to Y$ be a morphism of schemes. The following conditions on π are equivalent.

(a) For every quasiseparated open $V \subset Y$, the preimage $\pi^{-1}(V) \subset X$ is quasiseparated.
(b) For every affine open $V \subset Y$, the preimage $\pi^{-1}(V)$ is quasiseparated.
(c) There is an affine open cover $Y = \bigcup_i V_i$ such that for each i, the preimage $\pi^{-1}(V_i)$ is quasiseparated.

(d) The diagonal morphism $\Delta_\pi : X \to X \times_Y X$ is quasicompact.

A morphism π of schemes satisfying these equivalent conditions is said to be *quasiseparated*. Further:

(e) Quasiseparated morphisms form a resaonable class.
(f) Any morphism out of a quasiseparated scheme is quasiseparated.
(g) Any monomorphism (e.g., immersion) is quasiseparated.

Proof. The implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) are clear.

(c) $\Rightarrow$ (d) We produce an affine open cover of $X \times_Y X$ such that for each element of this cover, its preimage under Δ_π is quasicompact. For each i, pick an affine open cover $\pi^{-1}(V_i) = \bigcup_j U_{ij}$ of $\pi^{-1}(U_i)$. Then we claim that the affine open cover $\{U_{ij} \times_{V_i} U_{ij'}\}_{i,j,j'}$ of $X \times_Y X$ has this property; indeed, we have $\Delta_\pi^{-1}(U_{ij} \times_{V_i} U_{ij'}) = U_{ij} \cap U_{ij'}$, which is an intersection of affines in the quasiseparated $\pi^{-1}(V_i)$ and hence quasicompact.

(d) $\Rightarrow$ (b) Let $U, U' \subset \pi^{-1}(V)$ be affine opens; we want to show that $U \cap U'$ is quasicompact. Well, $U \times_V U' \xrightarrow{\sim} U \times_Y U' \subset X \times_Y X$ is an affine open and hence quasicompact, therefore its preimage under Δ_π, namely $U \cap U'$, is also quasicompact.

At this point, we have shown the equivalence of (b), (c), and (d), and it is clear that a scheme X is quasiseparated in the sense of 3.3.5 iff the final morphism $X \to \operatorname{Spec} \mathbb{Z}$ satisfies conditions (d). Next, we note that, with definition (d) of quasiseparatedness, it is clear from the abstract formalism (3.1.8) that quasiseparated morphisms are stable under composition. Given this, we are ready to show

(d) $\Rightarrow$ (a) Let $V \subset Y$ be any open subset and let $X_V = \pi^{-1}(V)$. Then the diagram

$$\begin{tikzcd} X_V \arrow[r, hook] \arrow[d, "\Delta"] & X \arrow[d, "\Delta"] \\ X_V \times_V X_V \arrow[r, hook] & X \times_Y X \end{tikzcd}$$

is a pullback, so stability of quasicompact morphisms under basechange (3.3.1(d)), we conclude that $X_V \to V$ is also quasiseparated in the sense of (d). Since V is quasiseparated, we know that the final morphism $V \to \operatorname{Spec} \mathbb{Z}$ is quasiseparated in the sense of (d). Since quasiseparated morphisms in the sense of (d) are stable under composition, we conclude that the composite $X_V \to V \to \operatorname{Spec} \mathbb{Z}$ is quasiseparated in the sense of (d), but this is just saying that X_V is a quasiseparated scheme.

That quasiseparated morphisms are preserved by basechange follows from 3.1.8, and (f) is clear from 3.3.5(d). Finally, (g) follows from (d) because a morphism $\pi : X \to Y$ is a monomorphism iff $\Delta_\pi : X \to X \times_Y X$ is an isomorphism. ■

Here's another proof of (c) $\Rightarrow$ (b) using the Affine Communication Lemma.

Proof 2 of (c) $\Rightarrow$ (b). By 3.1.1, it remains to check the following in the case Y is affine:

(i) If X is quasiseparated, then for any distinguished $D \subset Y$, so is the preimage $\pi^{-1}(D)$; this follows from 3.3.5(d).
(ii) If $\{D_\alpha\}$ is a finite cover of Y by distinguished opens and each $\pi^{-1}(D_\alpha)$ is quasiseparated, then so is X. Indeed, suppose we are given affines $U, V \subset X$. Covering U by finitely many distinguished opens in U, each of which lies in $\pi^{-1}(D_\alpha)$ for some α, we reduce to the case in which $U \subset \pi^{-1}(D_\alpha)$ for some α. Then

$$U \cap V = U \cap \pi^{-1}(D_\alpha) \cap V = U \cap \pi|_V^{-1}(D_\alpha).$$

Here U is affine and hence quasicompact, and $\pi|_V^{-1}(D_\alpha)$ is distinguished in V (by 1.2.1(a) applied to $\pi|_V$) and hence quasicompact; since this intersection takes place in $\pi^{-1}(D_\alpha)$, which is quasiseparated, it follows that $U \cap V$ is quasicompact.

■

A morphism is said to be *qcqs* if it is both quasicompact and quasiseparated. We like such morphisms because of the corresponding analog of the qcqs lemma:

Lemma 3.3.10. Let $\pi : X \to Y$ be a qcqs morphism and $\mathcal{F} \in \mathsf{QCoh}(X)$. Then $\pi_*\mathcal{F} \in \mathsf{QCoh}(Y)$.

Proof. Since qcqs morphisms are affine-local on the target and the question of quasicoherence is likewise local on Y, we may assume that Y is affine; then X is a qcqs scheme. We have to show that if $D \subset Y$ is distinguished, then the natural morphism $\mathcal{O}(D) \otimes_{\mathcal{O}(Y)} \pi_*\mathcal{F}(Y) \to \pi_*\mathcal{F}(D)$ is an isomorphism. If $g \in \mathcal{O}(Y)$ is such that $D = D(g)$ and $f := \mathcal{O}(\pi)(g) \in \mathcal{O}(X)$ the image of g under the ring homomorphism $\mathcal{O}(\pi) : \mathcal{O}(Y) \to \mathcal{O}(X)$, then $\pi^{-1}(D) = X_f$. Since g acts by f on $\mathcal{F}(X)$, in light of 3.3.7, it remains to note only that the natural morphism $\mathcal{O}(Y)[g^{-1}] \otimes_{\mathcal{O}(Y)} \mathcal{F}(X) \to \mathcal{O}(X)[f^{-1}] \otimes_{\mathcal{O}(X)} \mathcal{F}(X)$ is an isomorphism. ■

We end this section with an application of the above result that we will later need.

Lemma 3.3.11. Let X be a qcqs scheme and $\mathcal{F} \in \mathsf{QCoh}(X)$.

(a) If $U \subset X$ is a quasicompact open, $\mathcal{G}$ a finite-type quasicoherent $\mathcal{O}_U$-module and $\mathcal{G} \to \mathcal{F}|_U$ an $\mathcal{O}_U$-module morphism, then there is a finite-type quasicoherent $\mathcal{O}_X$-module $\overline{\mathcal{G}}$ and an $\mathcal{O}_X$-module morphism $\overline{\mathcal{G}} \to \mathcal{F}$ whose restriction to U is (isomorphic to) $\mathcal{G} \to \mathcal{F}|_U$.

(b) The sheaf $\mathcal{F}$ is the union of its finite-type quasicoherent subsheaves. In other words, for any $x \in X$ and $s \in \mathcal{F}_x$, there is a finite-type quasicoherent subsheaf $\mathcal{F}' \subset \mathcal{F}$ such that in fact $s \in \mathcal{F}'_x \subset \mathcal{F}_x$.

Proof.

(a) First suppose that we have shown the result for affine X; given this, we show how to do it for arbitrary qcqs X. Using that X is quasicompact and inducting on the minimal $n \in \mathbb{Z}_{\geq 1}$ such that there is an open cover of X of the form $X = U \cup \bigcup_{i=1}^n U_i$ with each U_i an affine open, we may first suppose that $X = U \cup V$ for some affine open V. Since X is quasiseparated, the intersection $U \cap V$ is quasicompact; by the affine case, there is an extension of $\mathcal{G}|_{U\cap V} \to \mathcal{F}|_{U\cap V}$ to an $\mathcal{O}_V$-module morphism $\overline{\mathcal{G}}_1 \to \mathcal{F}|_V$ from a finite-type quasicoherent $\mathcal{O}_V$-module $\overline{\mathcal{G}}_1$ whose restriction to $U \cap V$ is isomorphic to $\mathcal{G}|_{U\cap V} \to \mathcal{F}|_{U\cap V}$. Gluing this with our given morphism $\mathcal{G} \to \mathcal{F}|_U$ then produces the required extension.
Hence suppose that X is affine. Now note that the inclusion $\iota : U \hookrightarrow X$ is quasicompact by hypothesis and quasiseparated (3.3.9(g)), so by 3.3.10, we have $\iota_*\mathcal{G}$ is a quasicoherent $\mathcal{O}_X$-submodule of the quasicoherent $\mathcal{O}_X$-module $\iota_*(\mathcal{F}|_U) = \iota_*\iota^*\mathcal{F}$. Next, take

$$\overline{\mathcal{G}} := \ker(\mathcal{F} \oplus \iota_*\mathcal{G} \to \iota_*\iota^*\mathcal{F})$$

with its obvious map to $\mathcal{F}$. This is almost the required extension, except it is not clear that it finite-type. To finish the proof, it suffices to replace $\overline{\mathcal{G}}$ by a finite-type quasicoherent subsheaf which still restricts to $\mathcal{G}$ on U.
Let $M := \overline{\mathcal{G}}(X)$. Since $\mathcal{G}$ is finite-type, for each $x \in U$, there is an $f \in \mathcal{O}(X) =: A$ such that $x \in D(f) \subset U$ and such that $A[f^{-1}] \otimes_A M$ is finitely generated, say by the images of finitely many $m_{f,j} \in M$. Since U is quasicompact, we cover it by finitely many such $D(f_i)$'s, and let M' be the submodule generated by all the $m_{f_i,j}$'s. Replacing $\overline{\mathcal{G}}$ by the subsheaf $\widetilde{M'}$ finishes the proof.

(b) Pick an affine open neighborhood $U \subset X$ of x on which s lifts to a section $\mathcal{O}_U \to \mathcal{F}|_U$. By (a), there is a finite-type quasicoherent sheaf $\overline{\mathcal{G}}$ on X and an $\mathcal{O}_X$-module morphism $\overline{\mathcal{G}} \to \mathcal{F}$ whose restriction to U is (isomorphic to) $\mathcal{O}_U \to \mathcal{F}|_U$. Take $\mathcal{F}'$ to be the image of $\overline{\mathcal{G}} \to \mathcal{F}$.

■

Finally, we can combine the above definitions to arrive at

Definition 3.3.12. A morphism of schemes is said to be

(a) *of finite type* iff it is locally of finite type and quasicompact, and
(b) *of finite presentation* iff it is locally of finite presentation and qcqs.

[TODO: Reason for qcqs.] Both of these are reasonable classes of morphisms. Clearly, a finite presentation morphism is of finite type; the converse holds in Noetherian circumstances: if $\pi : X \to Y$ is a morphim of schemes with Y locally Noetherian, then π is of finite type iff it is of finite presentation.

3.3.1 Chevalley's Theorem

Theorem 3.3.13. Let $\pi : X \to Y$ be a finite-type morphism of Noetherian schemes. If $C \subset X$ is constructible, then so is $\pi(C) \subset Y$.

[TODO: More general situations. Retrocompact, etc. TOCITE: Lei Fu]

Proof. We leave to the reader ■

3.4 Affine, Integral, and Finite Morphisms

Before we define affine morphisms, we use the following consequence of the qcqs lemma taken from [2, Exercise 2.17].

Lemma 3.4.1. A scheme X is affine iff there is an $n \in \mathbb{Z}_{\geq 1}$ and $f_1, \ldots, f_n \in \mathcal{O}(X)$ generating the unit ideal such that each X_{f_i} is affine.

Proof. If X is affine, it suffices to take $n = 1$ and $f_1 = 1$. Conversely, the hypothesis implies that X is qcqs (3.3.6), so for each i, 3.3.7 applied to $\mathcal{F} = \mathcal{O}_X$ and $f = f_i$ tells us $\mathcal{O}(X)[f_i^{-1}] \xrightarrow{\sim} \mathcal{O}(X_{f_i})$. Since X_{f_i} is affine, we get a series of isomorphisms

$$X_{f_i} \xrightarrow{\sim} \operatorname{Spec} \mathcal{O}(X_{f_i}) \xrightarrow{\sim} \operatorname{Spec} \mathcal{O}(X)[f_i^{-1}] \cong D(f_i) \subset \operatorname{Spec} \mathcal{O}(X).$$

This says exactly that the natural morphism $X \to \operatorname{Spec} \mathcal{O}(X)$ is an isomorphism over the cover $\{D(f_i)\}_i$ of $\operatorname{Spec} \mathcal{O}(X)$, so by the locality on the target of isomorphisms we conclude that $X \to \operatorname{Spec} \mathcal{O}(X)$ is also an isomorphism. ∎

Proposition/Definition 3.4.2 (Affine Morphisms)**.** Let $\pi : X \to Y$ be a morphism of schemes. The following conditions on π are equivalent.

(a) For every affine open $V \subset Y$, the preimage $\pi^{-1}(V) \subset X$ is affine.
(b) There is an open affine cover $Y = \bigcup_i V_i$ such that for each i, the preimage $\pi^{-1}(V_i)$ is affine.

A morphism satisfying these equivalent conditions is said to be *affine*. Further:

(c) Affine morphisms form a reasonable class.

Proof. Using 3.1.1, reduce to 1.2.1(a) and 3.4.1. For (c), use 3.1.2. ∎

The locality on the target of affine morphisms has the following unexpected consequence.

Corollary 3.4.3. If $\pi : X \to Y$ is a morphism with Y an affine scheme such that for each $y \in Y$, there is an affine open $V \subset Y$ containing y with $\pi^{-1}(V)$ affine, then X is an affine scheme. In particular, the complement of a locally principal closed subscheme (e.g., an effective Cartier divisor) in an affine scheme is affine.

Proof. As mentioned above, this follows from the fact that affine morphisms are local on the target (3.4.2(c)). ∎

Recall that an algebra $A \to B$ is said to be *finite* if B is a finitely generated A-module. The globalization of this definition is

Proposition/Definition 3.4.4 (Integral and Finite Morphisms)**.** Let $\pi : X \to Y$ be a morphism of schemes. The following conditions on π are equivalent.

(a) open $V \subset Y$, the preimage $\pi^{-1}(V) \subset X$ is affine and $\mathcal{O}(\pi^{-1}(V))$ is an integral (resp. a finite) $\mathcal{O}(V)$-algebra.
(b) There is an open affine cover $Y = \bigcup_i V_i$ such that for each i, the preimage $\pi^{-1}(V_i)$ is affine and $\mathcal{O}(\pi^{-1}(V_i))$ is an integral (resp. a finite) $\mathcal{O}(V_i)$-algebra.

A morphism satisfying these equivalent conditions is said to be *finite*. Further:

(c) Integral (resp. finite) morphisms form a reasonable class.

(d) Integral morphisms have zero-dimensional fibers. Finite morphisms have finite discrete fibers.

Proof. Using that a ring homomorphism $A \to B$ is finite iff it is integral and of finite type along with the already proven results (3.2.4), it suffices to show the integral case. Using 3.1.1, reduce to showing the following.

(i) If $\varphi : A \to B$ is an integral algebra, then for any $f \in A$, the localization $A[f^{-1}] \to B[\varphi(f)^{-1}]$ is also. This is clear.

(ii) If $\varphi : A \to B$ is an A-algebra, $n \in \mathbb{Z}_{\geq 1}$, and $f_1, \ldots, f_n \in A$ with $(f_1, \ldots, f_n) = 1$ such that for each i, the algebra $A[f_i^{-1}] \to B[\varphi(f_i)^{-1}]$ is integral, then $A \to B$ is finite. To show this, replace A by $\varphi(A)$ in B to assume that φ is an inclusion. Then for each $b \in B$, pick $N, M \gg 1$ such that for all $i = 1, \ldots, n$, we have $f_i^N b^M \in \sum_{i=0}^{M-1} Ab^i$. Then use a partition-of-unity argument to check that in fact $b^M \in \sum_{i=0}^{M-1} Ab^i$ as needed.

For (c), use 3.1.2 as usual. For (d), we use (c) to reduce to the case of the target being $\operatorname{Spec} k$ for a field k; in the first case, use incomparability, and in the second, use that X is an Artinian scheme over k. ∎

3.5 Closed Embeddings

Let X be a locally ringed space and $\mathcal{I} \subset \mathcal{O}_X$ an ideal sheaf. We want to define a closed sub-locally-ringed-space $Z = \mathbb{V}(\mathcal{I})$ of X corresponding to the ideal sheaf $\mathcal{I}$. For this, as a topological space we take $Z := \operatorname{Supp}(\mathcal{O}_X/\mathcal{I}) \subset X$, which is a closed subset since $\mathcal{O}_X/\mathcal{I}$ is a sheaf of rings on X. Let $\iota : Z \hookrightarrow X$ denote the inclusion map. Define a sheaf of rings on Z by taking $\mathcal{O}_Z := \iota^{-1}(\mathcal{O}_X/\mathcal{I})$, so that at a point $z \in Z$, the stalk $\mathcal{O}_{Z,z} = \mathcal{O}_{X,z}/\mathcal{I}_z$ is a nonzero quotient of a local ring $\mathcal{O}_{X,z}$ and hence local. This construction therefore gives us a locally ringed space Z. Define a morphism $\iota : Z \to X$ of locally ringed spaces by requiring the corresponding morphism of sheaves to arise from the natural surjection $\iota^\sharp : \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I} \cong \iota_*\mathcal{O}_Z$.[9] In particular, there is an exact sequence of abelian sheaves on X of the form

$$0 \to \mathcal{I} \to \mathcal{O}_X \to \iota_*\mathcal{O}_Z \to 0 \tag{3.5.1}$$

called the *ideal sheaf exact sequence*.[10] With this construction, it is clear that the morphism $\iota : Z \to X$ is a monomorphism in the category of locally ringed spaces.

As with ideals in a ring, this construction satisfies the expected universal property.

Proposition 3.5.2. Let X be a locally ringed space and $\mathcal{I}$ an ideal sheaf on X. Let Z and ι be as above. If Y is any other locally ringed space and $f : Y \to X$ a morphism such that $\mathcal{I} \subset \ker(f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y)$, then f factors uniquely through Z, i.e., there is a unique morphism of locally ringed spaces $\widetilde{f} : Y \to Z$ such that $f = \iota \circ \widetilde{f}$.

We could take this universal property as a *definition* of $\mathbb{V}(\mathcal{I})$; then the above paragraph would amount to giving a *construction*. Note that conversely if a morphism $f : Y \to X$ admits a factorization of the form $f = \iota \circ \widetilde{f}$ for some $\widetilde{f} : Y \to Z$, then automatically we have that $\mathcal{I} \subset \ker(f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y)$.

Proof. The uniqueness of such an f follows from ι being a monomorphism. To show existence, first note that $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ admits a factorization as a morphism of sheaves of rings of the form

$$\mathcal{O}_X \twoheadrightarrow \mathcal{O}_X/\mathcal{I} \to f_*\mathcal{O}_Y. \tag{3.5.3}$$

This implies that set-theoretically $f(Y) \subset Z$; indeed, if $y \in Y$, then we have a factorization $\mathcal{O}_{X,f(y)} \to (\mathcal{O}_X/\mathcal{I})_{f(y)} \to \mathcal{O}_{Y,y}$ of $f_y^\sharp$; since the composite is a nonzero homomorphism of rings between nonzero rings, the middle ring cannot be zero. Therefore, by definition of the subspace topology, there is a unique continuous map $\tilde{f} : Y \to Z$ such that $f = \iota \circ \tilde{f}$ as continuous maps. To upgrade this to a morphism of locally ringed spaces, we need to give a morphism $\widetilde{f}^\sharp : \mathcal{O}_Z \to \widetilde{f}_*\mathcal{O}_Y$ of sheaves of rings on Z such that the resulting composition $f = \iota \circ \widetilde{f}$ holds as morphisms of locally ringed spaces. Since $\iota_* : \mathsf{Ab}(Z) \to \mathsf{Ab}(X)$ is fully faithful (and preserves ring sheaves), it suffices to produce a morphism $\iota_*\widetilde{f}^\sharp : \iota_*\mathcal{O}_Z \to \iota_*\widetilde{f}_*\mathcal{O}_Y$ of sheaves on X; take this to be the composition

$$\iota_*\mathcal{O}_Z = \iota_*\iota^{-1}(\mathcal{O}_X/\mathcal{I}) \xrightarrow{\eta^{-1}_{\mathcal{O}_X/\mathcal{I}}} \mathcal{O}_X/\mathcal{I} \to f_*\mathcal{O}_Y \cong \iota_*\widetilde{f}_*\mathcal{O}_Y,$$

[9] Here we are using the following fact: if $\pi : Z \to X$ is a continuous map, then we have an adjunction $\pi^{-1} \dashv \pi_*$ between $\mathsf{Ab}(Z)$ and $\mathsf{Ab}(X)$. Further, if $\pi = \iota$ is a topological closed embedding (i.e., up to identifications, the inclusion of a closed subset), then the counit $\epsilon : \iota^{-1}\iota_* \to 1$ is an isomorphism, which tells us that $\iota_* : \mathsf{Ab}(Z) \to \mathsf{Ab}(X)$ is fully faithful with essential image consisting of the abelian sheaves on X supported on Z. In particular, if $\mathcal{F}$ is an abelian sheaf on X supported on Z, then the unit $\eta_{\mathcal{F}} : \mathcal{F} \to \iota_*\iota^{-1}\mathcal{F}$ is an isomorphism. This follows also from the fact that if $j : U \hookrightarrow X$ denotes the open complement of Z, then we have a natural exact sequence $0 \to j_!j^{-1}\mathcal{F} \to \mathcal{F} \to \iota_*\iota^{-1}\mathcal{F} \to 0$ on $\mathcal{F}$.

[10] Since the functor ι_* is fully faithful, some texts drop the ι_* from 3.5.1, writing it instead as $0 \to \mathcal{I} \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$, but this means exactly the same thing.

where the second-to-last morphism comes from the factorization (3.5.3). Then the identity $f = \iota \circ \widetilde{f}$ as morphisms of locally ringed spaces holds by construction using (3.5.3). ■

As in the case of abelian sheaves, we can identify the essential image of ι_*. For this we start with a simple lemma.

Lemma/Definition 3.5.4. Let X be a locally ringed space and $\mathcal{I}$ an ideal sheaf on X. For an $\mathcal{F} \in \mathcal{O}_X\text{-}\mathsf{Mod}$, the following conditions are equivalent.

(a) For every open subset $U \subset X$, the ideal $\mathcal{I}(U) \subset \mathcal{O}_X(U)$ annihilates the $\mathcal{O}_X(U)$-module $\mathcal{F}(U)$.
(b) For every open subset $U \subset X$, the ideal $\mathcal{I}(U) \subset \mathcal{O}_X(U)$ annihilates $\mathcal{F}|_U$ (i.e., every section of $\mathcal{F}|_U$ on every open subset contained in U).
(c) For every $x \in X$, the ideal $\mathcal{I}_x \subset \mathcal{O}_{X,x}$ annihilates the $\mathcal{O}_{X,x}$-module $\mathcal{F}_x$.

When $\mathcal{F}$ satisfies these equivalent conditions, we say that $\mathcal{I}$ annihilates $\mathcal{F}$.

Proof. Exercise. ■

Let $\mathcal{O}_X/\mathcal{I}\text{-}\mathsf{Mod} \subset \mathcal{O}_X\text{-}\mathsf{Mod}$ be the full subcategory consisting of all $\mathcal{O}_X$-modules annihilated by $\mathcal{I}$.

Theorem 3.5.5. Let X be a locally ringed space and $\mathcal{I}$ and $\iota : Z \hookrightarrow X$ be as above. The functor $\iota_* : \mathcal{O}_Z\text{-}\mathsf{Mod} \to \mathcal{O}_X\text{-}\mathsf{Mod}$ is fully faithful with essential image $\mathcal{O}_X/\mathcal{I}\text{-}\mathsf{Mod}$. A quasi-inverse on this essential image is given by the functor $\pi^{-1} \cong \pi^*$.

Proof. There is not much to say here. The pushforward $\iota_* : \mathcal{O}_Z\text{-}\mathsf{Mod} \to \mathcal{O}_X\text{-}\mathsf{Mod}$ factors through $\mathcal{O}_X/\mathcal{I}\text{-}\mathsf{Mod}$. If $\mathcal{G} \in \mathcal{O}_X/\mathcal{I}\text{-}\mathsf{Mod}$, then $\pi^{-1}\mathcal{G} \cong_{\mathsf{Ab}(Z)} \pi^*\mathcal{G}$, and this defines a functor $\pi^{-1} : \mathcal{O}_X/\mathcal{I}\text{-}\mathsf{Mod} \to \mathcal{O}_X\text{-}\mathsf{Mod}$, with the adjunction $\pi^{-1} \dashv \pi_*$ giving an adjoint equivalence between $\mathcal{O}_Z\text{-}\mathsf{Mod}$ and $\mathcal{O}_X/\mathcal{I}\text{-}\mathsf{Mod}$. ■

That this is the right construction for schemes follows from

Proposition 3.5.6. Let X be a scheme and $\mathcal{I}$ a *quasicoherent* ideal sheaf on X. Then the locally ringed space Z constructed above is a scheme and $\iota : Z \to X$ is a morphism of schemes. Further, the quotient $\iota_*\mathcal{O}_Z$ is also quasicoherent on X, i.e., the ideal sheaf exact sequence (3.5.1) is an exact sequence in $\mathsf{QCoh}(X)$.

Proof. The question is local on X, so we may assume without loss of generality that X is affine. Let $A := \mathcal{O}(X)$, and let $I := \mathcal{I}(X) \subset A$. The quasicoherence of $\mathcal{I}$ tells us that there is an isomorphism $\mathcal{O}_X/\mathcal{I} \cong \widetilde{A/I}$ of sheaves of X, from which it follows that as a topological space $Z = \operatorname{Supp}(\widetilde{A/I}) = \operatorname{Supp}(A/I) = \mathbb{V}(I) \subset \operatorname{Spec} A$. Therefore, the underlying topological space of Z is homeomorphic to $\operatorname{Spec}(A/I)$, and so admits the sheaf $\mathcal{O}_{\operatorname{Spec}(A/I)}$ also. Now consider the adjunctions (and inclusion)

$$\begin{aligned}\operatorname{Hom}_{A\text{-}\mathsf{Mod}}\left(A/I, \Gamma(Z, \mathcal{O}_{\operatorname{Spec}(A/I)})\right) &\cong \operatorname{Hom}_{\mathcal{O}_X\text{-}\mathsf{Mod}}\left(\widetilde{A/I}, \iota_*\mathcal{O}_{\operatorname{Spec}(A/I)}\right)\\ &\subset \operatorname{Hom}_{\mathsf{Ab}(X)}\left(\widetilde{A/I}, \iota_*\mathcal{O}_{\operatorname{Spec}(A/I)}\right)\\ &\cong \operatorname{Hom}_{\mathsf{Ab}(Z)}\left(\iota^{-1}(\widetilde{A/I}), \mathcal{O}_{\operatorname{Spec}(A/I)}\right).\end{aligned}$$

The obvious isomorphism in the first term above therefore gives rise to an abelian sheaf morphism $\iota^{-1}\widetilde{A/I} \to \mathcal{O}_{\operatorname{Spec}(A/I)}$, which is easily seen to be an isomorphism on stalks and hence an isomorphism. In particular, Z is a(n) (affine) scheme. Further, considering global sections shows that the morphism $i : Z \to X$ constructed above corresponds exactly to the natural morphism $\operatorname{Spec} \pi$ arising from $\pi : A \twoheadrightarrow A/I$. ■

Remark 3.5.7.

(a) This statement is *not* true in general for a scheme X if we do not require $\mathcal{I}$ to be a quasicoherent. Here's a standard counterexample: let $X = \mathbb{A}^1_{\mathbb{C}}$ and $U = \mathbb{A}^1_{\mathbb{C}} \smallsetminus \{0\} \subset X$, with $j : U \hookrightarrow X$ the inclusion. Then $j_! \mathcal{O}_U \subset \mathcal{O}_X$ is a sheaf of ideals. Applying the above construction to this sheaf yields $Z = \{0\}$ equipped with the structure sheaf $\mathcal{O}_{X,0} \cong \mathbb{C}[t]_{(t)}$. This Z is certainly a locally ringed space, but not a scheme.

(b) Different quasicoherent ideal sheaves $\mathcal{I}$ can give rise to the same underlying topological space Z (e.g., the sheaves (t) and (t^2) on $X = \mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[t]$), although the *scheme* Z along with the morphism $\iota : Z \to X$ determines $\mathcal{I}$ uniquely, namely as the kernel of the map of sheaves $\iota^\sharp : \mathcal{O}_X \to \iota_* \mathcal{O}_Z$.

Proposition/Definition 3.5.8 (Closed Immersion)**.** The following conditions on a morphism $\iota : Z \to X$ of schemes are equivalent:

(a) For each open affine $U \subset X$, the preimage $\iota^{-1}(U)$ is affine, and $\mathcal{O}(U) \twoheadrightarrow \mathcal{O}(\iota^{-1}(U))$.

(b) There is an affine open cover $X = \bigcup U_i$ of X such that for each i, the preimage $\iota^{-1}(U_i)$ is affine and $\mathcal{O}(U_i) \twoheadrightarrow \mathcal{O}(\iota^{-1}(U_i))$.

(c) The continuous map $|\iota|$ is a topological closed embedding (i.e., a homeomorphism onto a closed subset of X), and further the sheaf morphism $\iota^\sharp : \mathcal{O}_X \to \iota_* \mathcal{O}_Z$ is surjective.

(d) There is a quasicoherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ such that the map ι factors as an isomorphism $Z \xrightarrow{\sim} \mathbb{V}(\mathcal{I})$ followed by the inclusion morphism $\mathbb{V}(\mathcal{I}) \hookrightarrow Z$ constructed above.

A morphism ι satisfying these equivalent conditions is said to be a *closed embedding* or a *closed immersion*. In this case, the ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ and the isomorphism $Z \to \mathbb{V}(\mathcal{I})$ in (d) are determined uniquely by ι. Further:

(e) Closed immersions form a reasonable class.

Proof.

(a) ⇒ (b) Clear.

(b) ⇒ (c) Topological closed embeddings are a reasonable class of continuous maps (3.1.5), and the surjectivity of $\iota^\sharp$ is a local question.

(c) ⇒ (d) The conditions of (c) imply that ι is qcqs; indeed, a topological closed embedding is a quasicompact, and (c) implies that ι is a monomorphism in the category of locally ringed spaces, so that the property 3.3.9(d) for ι is obvious. By 3.3.10, the sheaf $\iota_* \mathcal{O}_Z$ on X is quasicoherent, and hence so is the kernel $\mathcal{I} := \ker(\iota^\sharp)$ by 1.3.3(e). We claim that this $\mathcal{I}$ works. By the universal property (3.5.2), there is a unique factorization of ι as $Z \xrightarrow{\widetilde{\iota}} \mathbb{V}(\mathcal{I}) \hookrightarrow X$ as a morphism of schemes, and we have to show that $\widetilde{\iota} : Z \to \mathbb{V}(\mathcal{I})$ is an isomorphism. Topologically, since ι and $\mathbb{V}(\mathcal{I}) \hookrightarrow X$ are closed embeddings, so is $\widetilde{\iota}$; it is then a homeomorphism because it is also surjective: the sheaf $\iota_* \mathcal{O}_Z$ is supported exactly on $\iota(Z)$. To check that $\widetilde{\iota}^\sharp : \mathcal{O}_{\mathbb{V}(\mathcal{I})} \to \widetilde{\iota}_* \mathcal{O}_Z$ is an isomorphism, we may check it after pushing forward via the inclusion to X, at which point we have to check that the natural morphism $\mathcal{O}_X / \mathcal{I} \to \iota_* \mathcal{O}_Z$ induced by $\iota^\sharp$ is an isomorphism, but indeed this is true by definition of $\mathcal{I}$ since we assume that $\iota^\sharp$ is surjective as a morphism of sheaves.

(d) ⇒ (a) Follows from (the proof of) 3.5.6.

Affine-locality on the target follows from the equivalence of (a) and (b), stability under composition is clear from (a), and stability under base change follows from 3.1.2. ■

We could also prove (b) ⇒ (a) above directly using 3.1.1; this proof is left to the reader as an exercise.

Definition 3.5.9 (Closed Subschemes)**.** Let X be a scheme and $\iota : Z \to X$ and $\iota' : Z' \to X$ denote two closed embeddings.

(a) We say that Z' *majorizes*[11] Z, written $Z \preceq Z'$, iff there is a morphism $\widetilde{\iota} : Z \to Z'$ such that $\iota = \iota' \circ \widetilde{\iota}$.

By 3.5.2, this happens iff the corresponding ideal sheaves $\mathcal{I}$ and $\mathcal{I}'$ satisfy $\mathcal{I}' \subset \mathcal{I}$, and in this case $\widetilde{\iota}$ is uniquely determined and also necessarily a closed embedding.

(b) We say that Z' and Z are *equivalent* if both $Z \preceq Z'$ and $Z' \preceq Z$.

This notion of equivalence is an equivalence relation on closed embeddings (check!).

(c) A *closed subscheme* of X is an equivalence class of closed embeddings mapping to X.

Therefore, closed subschemes of X correspond to quasicoherent ideal sheaves on X, with the correspondence given by taking a quasicoherent ideal sheaf $\mathcal{I}$ to the (equivalence class of) the morphism $\mathbb{V}(\mathcal{I}) \hookrightarrow X$ constructed above.

Remark 3.5.10. On a locally Noetherian scheme (e.g. a scheme locally of finite type over a field k), every quasicoherent ideal sheaf is automatically coherent; therefore, in this case, closed subschemes correspond to coherent ideal sheaves.

Definition 3.5.11 (Schematic Image)**.**

Theorem 3.5.12. Let $\pi : X \to Y$ be a morphism of schemes. If π is qc or X is reduced, then the schematic image $\mathrm{Im}(\pi)$ of π can be computed affine-locally on Y: for any affine open $V \subset Y$, the intersection $V \cap \mathrm{Im}(\pi)$ is the closed subscheme of V corresponding to the ideal $\ker \pi^\sharp_V : \mathcal{O}_Y(V) \to \mathcal{O}_X(\pi^{-1}(V))$. In particular, the underlying subspace of Y corresponding to $\mathrm{Im}(\pi)$ is exactly $\overline{\pi(X)}$.

Corollary 3.5.13. Let S be a scheme and $\pi : X \to Y$ and $\rho_1, \rho_2 : Y \to Z$ be morphisms of S-schemes. Let $j : W \hookrightarrow Y$ be the schematic image of π. If $\rho_1 j = \rho_2 j$, then $\rho_1 \pi = \rho_2 \pi$; and the converse holds if Z is a separated S-scheme.

[11] Being more pedantic, one could insist on saying that ι' majorizes ι instead, since this notion does depend on the morphisms ι and ι' and not just the abstract isomorphism classes of Z and Z', but we leave this level of pedantry to the pedants.

3.6 Separated and Proper Morphisms

Definition 3.6.1. A morphism $\pi : X \to Y$ of schemes is said to be

(a) *separated* iff the diagonal $\Delta_\pi : X \to X \times_Y X$ is a closed embedding.
(b) *proper* iff it is separated, universally closed, and locally of finite type.

Recall that a universally closed morphism is quasicompact (3.3.4), so a morphism is proper iff it is separated, universally closed, and of finite type.

Remark 3.6.2.

(a) A morphism $\pi : X \to Y$ is separated iff $\Delta_\pi(X) \subset X \times_Y X$ is a closed subset; this follows from the fact that Δ_π is a locally closed immersion.
(b) Since closed immersions form a reasonable class (3.5.8), it follows from the general theory (3.1.8) that separated morphisms and proper morphisms form reasonable classes.
(c) Affine morphisms and monomorphisms of schemes are separated, and separated morphisms are quasiseparated.
(d) A scheme X is said to be *separated* iff the final morphism $X \to \operatorname{Spec}\mathbb{Z}$ is separated. If X is a separated scheme, then every morphism from X to an affine scheme is separated; conversely, if X admits a separated morphism to an affine scheme, then X is itself separated.
(e) Since closed immersion $\Rightarrow$ affine $\Rightarrow$ quasicompact, we have separated $\Rightarrow$ affine diagonal $\Rightarrow$ quasiseparated. These are all distinct, by considering the gluing of two $\mathbb{A}^n_k$'s along the origin for $n \in \{1, 2, \infty\}$. [TODO]

Lemma 3.6.3. Let $\pi : X \to Y$ be a morphism of schemes.

(a) If π is separated, then for every pair of affine opens $U, V \subset X$ which map to a common affine open of S, (a) the intersection $U \cap V$ is affine and (b) the ring map $\mathcal{O}_X(U) \otimes_{\mathbb{Z}} \mathcal{O}_X(V) \to \mathcal{O}_X(U \cap V)$ is surjective.
(b) If for any $x, y \in X$ such that $\pi(x) = \pi(y)$, there exist affine opens $U, V \subset X$ with $x \in U$ and $y \in V$ such that U and V map to a common affine open of S such that (a) and (b) hold, then π is separated.

Proof. (Stacks) ∎

In the following, we let A be a valuation ring with maximal ideal $\mathfrak{m}$, and $K := \operatorname{Frac} A$. We let $\Delta := \operatorname{Spec} A$, and $\Delta^* := \operatorname{Spec} K$, with the inclusion $A \to K$ corresponding to the open embedding $\Delta^* \hookrightarrow \Delta$. For any morphism $\pi : X \to Y$, we can ask about the extension property of morphisms of the form

$$\begin{tikzcd} \Delta^* \arrow[r] \arrow[d, hook] & X \arrow[d, "\pi"] \\ \Delta \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd} \tag{3.6.4}$$

In other words, give morphisms $\Delta^* \to X$ and $\Delta \to Y$ such that the solid diagram (3.6.4) commutes, we can ask about the existence and uniqueness of dashed lifts $\Delta \to X$ making the diagram commute. We say the morphism π *satisfies the uniqueness (resp. existence, resp. existence and uniqueness) part(s) of the valuative criterion (with respect to all valuation rings)* iff for all valuation rings A and solid diagrams 3.6.4, there is at most one (resp. at least one, resp. exactly one) lift $\Delta \to X$ making the diagram 3.6.4 commute.

Theorem 3.6.5 (Valuative Criteria)**.**

(a) A morphism is separated iff it is qs and satisfies the uniqueness part of the valuative criterion.
(b) A morphism is universally closed iff it is qc and satisfies the existence part of the valuative criterion.
(c) A morphism is separated and universally closed iff it is qcqs and satisfies the existence and uniqueness parts of the valuative criterion.

In particular, a morphism is proper iff it is qs and finite-type, and satisfies the existence and uniqueness parts of the valuative criterion.

Proof. Clearly, (c) follows from combining (a) and (b). Note that in the above notation, we have a natural bijection of $X(A)$ with triples (η, s, φ), where $\eta, s \in X$ are points such that $\eta \rightsquigarrow s$ and $\varphi : \kappa(\eta) \to K$ is an field homomorphism such that under this inclusion, the local ring $\mathcal{O}_{\overline{\eta},s}$ of $\kappa(\eta)$ is dominated by the local ring A of K.[12]

(a) A separated morphism is qs (3.6.2(c)), and satisfies the uniqueness part of the valuative criterion: if π is separated, the equalizer of two lifts $\Delta \to X$ is a closed subscheme of the integral scheme Δ, and so if it contains Δ^*, it must be all of Δ. For the converse, under the given hypotheses, it suffices to show using 3.6.2(a) and 3.3.3(b) that $\Delta_\pi(X) \subset X \times_Y X$ is stable under specialization, so suppose that $\eta \in \Delta_\pi(X)$ specializes to some $s \in X \times_Y X$. By Zorn's Lemma, there is a valuation ring A of $\kappa(\eta)$ dominating $\mathcal{O}_{\overline{\eta},s}$. As noted above, this yields a morphism $f : \operatorname{Spec} A \to X \times_Y X$ such that $f([\mathfrak{m}]) = s$ and which restricts to the natural morphism $\{\eta\} \to X \times_Y X$ on $\Delta^* = \operatorname{Spec} \kappa(\eta)$. Since $\eta \in \Delta_\pi(X)$, composing f with the canonical projections gives two morphisms $f_1, f_2 : \Delta \to X$ and a solid diagram (3.6.4) in which these two morphisms fit as lifts. By hypothesis, this forces $f_1 = f_2$, i.e., that f factors through Δ_π, and hence that $s \in \Delta_\pi(X)$ as well.
(b) A universally closed morphism is quasicompact (3.3.4). To show that it satisfies the existence part of the valuative criterion, suppose we are given a solid diagram (3.6.4), and consider the corresponding fiber product $X \times_Y \Delta$ to reduce to the following problem: if $\pi : X \to \Delta$ is a closed morphism, then every section $\sigma : \Delta^* \to X$ of π extends to a section over Δ. A given section σ corresponds to a point $\eta \in X$ as well as a field *iso*morphism $\varphi : \kappa(\eta) \xrightarrow{\sim} K$. The restriction $\pi|_{\overline{\eta}}$ has closed image, so there is an $s \in \overline{\eta}$ such that $s \mapsto [\mathfrak{m}]$. This gives us a local homomorphism $A \to \mathcal{O}_{\overline{\eta},s}$ which extends to the inverse isomorphism $\varphi^{-1} : K \to \kappa(\eta)$. Since A is a valuation ring, it is maximal with respect to dominance in its fraction field K, so that the induced map $A \to \mathcal{O}_{\overline{\eta},s}$ is an isomorphism. Therefore, the triple (η, s, φ) correspnds to a morphism $\Delta \to X$ which is the extension of the section σ. Conversely, suppose that $\pi : X \to Y$ is qc and satisfies the existence part of the valuative criterion. Then every base-change of π also satisfies these properties; therefore, it remains to show only that π is closed. If $Z \subset X$ is closed, then giving it the reduced closed subscheme structure, the composite $Z \hookrightarrow X \to Y$ is also qc and satisfies the existence part; therefore, using 3.3.3(b), we have reduced to showing that $\pi : X \to Y$ is qc and satisfies the existence part, then $\pi(X) \subset Y$ is stable under specialization. For this, suppose that $\eta \in \pi(X)$ specializes to some $s \in Y$. Pick a point $\theta \in X$ mapping to η; then $\mathcal{O}_{\overline{\eta},s} \subset \kappa(\eta) \overset{\varphi}{\hookrightarrow} \kappa(\theta) =: K$. As above, let A be a valuation ring of K dominating $\mathcal{O}_{\overline{\eta},s}$. At this point, we have enough data to produce a solid diagram 3.6.4 where the bottom arrow $\Delta \to Y$ corresponds to (η, s, φ). The existence of a lift $\Delta \to X$ then tells us that $s \in \pi(X)$ as needed.

∎

Example 3.6.6. For any scheme S and integer $n \in \mathbb{Z}_{\geq 0}$, the morphism $\mathbb{P}^n_S \to S$ is proper. Since proper morphisms form a reasonable class, it suffices to do the case of $S = \operatorname{Spec} \mathbb{Z}$. Then

[12] Here $\overline{\eta} \subset X$ is given the reduced closed subscheme structure.

3.6.5 tells us we only have to check the existence and uniqueness part of the valuative criterion for $\mathbb{P}^n_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$ with respect to valuation rings A, but existence is the standard argument on clearing denominators, whereas uniqueness follows from noting that any two points of $\mathbb{P}^n_{\mathbb{Z}}$ lie in a common distinguished affine open of the form $D_+(X_0)$ (up to automorphisms of $\mathbb{P}^n_{\mathbb{Z}}$ when $n \geq 1$–with the case $n = 0$ being trivial).

When we allow mild finiteness hypotheses, we can in fact check only on *discrete* valuation rings.

Theorem 3.6.7 (Noetherian Valuative Criteria)**.** Let $\pi : X \to Y$ be a finite-type morphism of locally Noetherian schemes. Then π is separated (resp. universally closed, resp. proper) iff it satisfies the uniqueness (resp. existence, resp. existence and uniqueness) part(s) of the valuative criterion with respect to all *discrete* valuation rings.

Proof. ([2, Ex. II.4.11(a)]) By (the proof of) 3.6.5, it only remains to quote the following result in commutative algebra: if $(\mathcal{O}, \mathfrak{m}, k)$ is a Noetherian local domain that is not a field, with fraction field κ, and K a finitely generated field extension of κ, then there is a *discrete* valuation ring A of K dominating $\mathcal{O}$. This is [4, 7.1.10]. ■

It is clear from the proof that we have proven slightly more generally that if $\pi : X \to Y$ is an *lft* morphism of locally Noetherian schemes, then π is separated iff it safises the uniqueness part of the valuative criterion with respect to all DVRs. Even in non-Noetherian settings, 3.6.7 is often helpful, because we can sometimes reduce to the Noetherian case by a limiting or base-change argument.

Example 3.6.8. Suppose X is an irreducible Noetherian regular curve proper over a field k (resp. over $\mathbb{Z}$). There is a bijection between discrete valuations on $k(X)$ trivial on k (resp. with no conditions) and the closed points of X. For instance, for any field k, we recover a classification of the discrete valuations on a field $k(t)$ which are trivial on k (by applying this to $X = \mathbb{P}^1_k$), and similarly a classification of discrete valuations on any number field K (by applying this to $X = \operatorname{Spec} \mathcal{O}_K$).

The valuative criterion gives us an important result about extending morphisms from curves to proper schemes.

Theorem 3.6.9 (Curve to Proper Extension)**.** Let C be a curve (i.e., a locally Noetherian equidimensional scheme with $\dim C = 1$) and $P \in C$ a regular closed point. Suppose we have a solid diagram of schemes of the form

$$\begin{tikzcd} C \smallsetminus \{P\} \arrow[r] \arrow[d, hook] & Y \arrow[d] \\ C \arrow[r] \arrow[ur, dashed, "\exists!"] & S. \end{tikzcd}$$

If $Y \to S$ is proper, then there is a unique dashed arrow making the diagram commute.

Proof. Without loss of generality, we may replace C by a small neighborhood of P to assume that $C = \operatorname{Spec} A$ is affine, with A a Noetherian domain of dimension one, and that there is a $t \in A$ such that P corresponds to the maximal ideal (t), where t maps to a uniformizer of the DVR $A_{(t)}$. At this point, uniqueness is clear from the separatedness of $Y \to S$. For existence, let $K := \operatorname{Frac} A$ and use 3.6.5 to produce a morphism $\operatorname{Spec} A_{(t)} \to Y$ such that the composite $\operatorname{Spec} K \to \operatorname{Spec} A_{(t)} \to Y$ agrees with the map $\operatorname{Spec} K \to C \smallsetminus \{P\} \to Y$. At this point, we have to solve a gluing problem in which the properness of $Y \to S$ is no longer needed. Replacing

Y by a small affine neighborhood of the image of the closed point of $\operatorname{Spec} A_{(t)}$ and shrinking C further as necessary, it remains to show: if B is a ring, and $B \to A[1/t]$ and $B \to A_{(t)}$ ring homomorphisms which "agree" on the generic point, i.e., such that the two induced maps $B \to K$ are the same, then both factor through A. But this is clear, since $A = A_{(t)} \cap A[1/t] \subset K$, which follows immediately from the primality of (t) in A (check!). ■

Of course, 3.6.5 tells us that if $Y \to S$ is qs and finite-type, then it is proper iff such extensions always exist.

We end this section by characterizing finite morphisms.

Theorem 3.6.10. A morphism is finite iff it is affine and proper.

Proof. (TOCITE: Lovely Little Lemmas) A finite morphism is affine by definition. It is also proper: it is separated since it is affine 3.6.2(c), it is universally closed because it is integral and hence closed [TODO], and finiteness is stable under base-change, and it is clearly locally of finite type.

Conversely, since finite, affine, and proper morphisms are all local on the target, we are reduced to showing that if $\varphi : A \to B$ a ring homomorphism such that the induced map $\operatorname{Spec}\varphi : \operatorname{Spec} B \to \operatorname{Spec} A$ is proper, then φ is finite. By the Cancellation Theorem, we may replace A by $A/\ker\varphi$ to assume that φ is injective. Since φ is finite-type, it suffices to show that φ is integral. Replacing A by $\varphi(A)$, we may assume that φ is the inclusion of a subring and suppress it from the notation.

Therefore, we have reduced to showing: if $A \subset B$ is a subring such that the induced morphism $\mathbb{A}^1_B \to \mathbb{A}^1_A$ is closed, then the extension $A \subset B$ is integral. Suppose the hypotheses hold, and let $b \in B$. Let $\mathfrak{b} = (bX - 1) \subset B[X]$ and $\mathfrak{a} := A[X] \cap \mathfrak{b}$, so that $A[X]/\mathfrak{a} \hookrightarrow B[X]/\mathfrak{b}$. The induced map on spectra for this ring extension is the restriction of the map $\mathbb{A}^1_B \to \mathbb{A}^1_A$ to the closed subset $\mathbb{V}(\mathfrak{b})$ and is hence a closed morphism. Since it is also dominant (3.3.2), we conclude that it is surjective. This means that $\overline{X} \in A[X]/\mathfrak{a}$ is a unit: if it were not, it would be contained in some prime $\mathfrak{p} \subset A[X]/\mathfrak{a}$, which by surjectivity would be pulled back from a prime $\mathfrak{q} \subset B[X]/\mathfrak{b}$ containing $\overline{X}$, but $\overline{X}$ is a unit there since it corresponds to b^{-1} under the isomorphism $B[X]/J \cong B[b^{-1}]$. This means that there is an $n \in \mathbb{Z}_{\geq 0}$ and $a_0, \dots, a_n \in A$ such that

$$X(a_0 X^n + a_1 X^{n-1} + \cdots + a_n) - 1 \in I.$$

This relation in $B[X]/\mathfrak{b} \cong B[b^{-1}]$ says that $b^{-1}(a_0 b^{-n} + \cdots + a_n) = 1$, which after clearing denominators says precisely that b is integral over A. ■

3.7 Exercises

Exercise 3.1. Let A be a DVR with fraction field K, and let $X := \operatorname{Spec} A$.

(a) Give an equivalence of categories between $\mathcal{O}_X$-Mod and triples (M, V, φ), where $M \in A$-Mod, $V \in K$-Mod, and $\varphi : M \to \operatorname{Res}_A^K V$ is an A-linear map, with morphisms $(f_A, f_K) : (M_1, V_1, \varphi_1) \to (M_2, V_2, \varphi_2)$ given by pairs (f_A, f_K) where $f_A : M_1 \to M_2$ is an A-module homomorphism and $f_K : V_1 \to V_2$ a K-linear map such that $f_K \varphi_1 = \varphi_2 f_A$. Under this equivalence, quasicoherent $\mathcal{O}_X$-modules correspond to triples (M, V, φ) where φ induces an isomorphism $K \otimes_A M \to V$, and coherent (i.e., finite-type, i.e., finite-presentation) $\mathcal{O}_X$-modules correspond to triples (M, V, φ) where in addition M is finitely generated.

(b) Under the identification of (a), suppose $\mathcal{F}_i \in \mathcal{O}_X$-$\mathsf{Mod}$ corresponds to (M_i, V_i, φ_i) for $i = 1, 2$. Show that $\underline{\operatorname{Hom}}_{\mathcal{O}_X}(\mathcal{F}_1, \mathcal{F}_2) \in \mathcal{O}_X$-$\mathsf{Mod}$ corresponds to the triple

$$(\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{F}_1, \mathcal{F}_2), \operatorname{Hom}_{K\text{-}\mathsf{Mod}}(V_1, V_2), \operatorname{pr}),$$

where pr is the projection map which takes in a pair (f_A, f_K) and remembers only the map $f_K : V_1 \to V_2$.

Exercise 3.2. Let p be a prime, $k := \mathbb{F}_p(t)$, and $C := \operatorname{Spec} k[X, Y]/(Y^2 - X^p + t)$, and P the point corresponding to the prime $(Y, X^p - t)$.

(a) Check that C is an integral curve and that $P \in C$ is a regular closed point, even though C is not smooth over k at P.

(b) The k-morphism $C \smallsetminus \{P\} \to \mathbb{P}^1_k$ given by $(X, Y) \mapsto [X^p - t, Y]$ extends to a k-morphism $C \to \mathbb{P}^1_k$ by 3.6.9. What is the image of P under this extension?

Exercise 3.3. ([1, 15.4.N]) Let X be a scheme, and let $\mathcal{L}$ be a line bundle on X. Let

$$\Gamma_*(X; \mathcal{L}) := \bigoplus_{n=0}^{\infty} \Gamma_n(X, \mathcal{L}) := \bigoplus_{n=0}^{\infty} \Gamma(X, \mathcal{L}^{\otimes n}),$$

which is naturally an $\mathbb{N}$-graded algebra over $\Gamma_0(X; \mathcal{L}) = \mathcal{O}(X)$. Similarly, if $\mathcal{F} \in \mathcal{O}_X$-$\mathsf{Mod}$, then for $n \in \mathbb{N}$, let $\mathcal{F}(n) := \mathcal{L}^{\otimes n} \otimes_{\mathcal{O}_X} \mathcal{F}$; then $\Gamma_*(X; \mathcal{L}, \mathcal{F}) := \bigoplus_{n=0}^{\infty} \Gamma(X, \mathcal{F}(n))$ is naturally a graded $\Gamma_*(X, \mathcal{L})$-module.

(a) Let $s \in \Gamma(X, \mathcal{L})$. Describe a natural morphism $\Gamma_*(X; \mathcal{L}, \mathcal{F})[s^{-1}]_0 \to \Gamma(X_s, \mathcal{F})$.

(b) Show that if X is qcqs and $\mathcal{F} \in \mathsf{QCoh}(X)$, then the above morphism is an isomorphism.

(c) Reinterpret (b) as saying that, in this setting, for every $\sigma \in \Gamma(D(s), \mathcal{F})$, there is an $n \in \mathbb{Z}_{>0}$ such that $s^n \sigma \in \Gamma(D(s), \mathcal{F}(n))$ extends to a global section in $\Gamma(X, \mathcal{F}(n))$, and similarly if $\sigma, \tau \in \mathcal{F}(X, \mathcal{F})$ are such that $\sigma|_{D(s)} = \tau|_{D(s)} \in \Gamma(D(s), \mathcal{F})$, then there is an $n \in \mathbb{Z}_{>0}$ such that $s^n \sigma = s^n \tau \in \Gamma(X, \mathcal{F}(n))$.

Exercise 3.4. Suppose we have a commutative square of morphisms of schemes as follows:

$$\begin{tikzcd} W \arrow[r, "f"] \arrow[d, "i"] & X \arrow[d, "g"] \\ Y \arrow[r, "h"] & Z. \end{tikzcd}$$

Suppose that i is a closed embedding and g is separated. Show that $(f, i) : W \to X \times_Z Y$ is also a closed embedding.

Exercise 3.5. Let S be a scheme, X be a proper S-scheme, and Y be a separated S-scheme. Let $f : X \to Y$ be an S-morphism. Show that $f(S) \subset Y$ is closed. Give a counterexample

(a) when X is not proper but Y is separated, and
(b) when X is proper but Y is not separated.

Exercise 3.6. Let $\pi : X \to Y$ be a finite dominant morphism of integral schemes. By 3.3.10, the sheaf $\pi_*\mathcal{O}_X$ is a finite-type quasicoherent sheaf on Y. Show that its rank (i.e., the dimension of its fiber at the generic point) is the degree of the finite extension $\pi^* : K(Y) \to K(X)$ of fields.

Exercise 3.7. Let X be a Noetherian scheme.

(a) Let $\mathcal{I}, \mathcal{J}$ be quasicoherent ideal sheaves[13] such that the underlying subsets of the closed subschemes cut out by $\mathcal{I}$ and $\mathcal{J}$ are identical, i.e., such that set-theoretically we have $\mathrm{Supp}(\mathcal{O}_X/\mathcal{I}) = \mathrm{Supp}(\mathcal{O}_X/\mathcal{J})$. Show that $\mathcal{I}$ and $\mathcal{J}$ are comparable in the sense that there are $m, n \in \mathbb{Z}_{\geq 0}$ such that $\mathcal{I}^m \subset \mathcal{J}$ and $\mathcal{J}^n \subset \mathcal{I}$.
(b) Taking $\mathcal{I} = \mathrm{Nil}(X)$ to be the sheaf cutting out the reduction X^{red} and $\mathcal{J} = 0$, show that $\mathcal{I}$ is nilpotent, i.e., that there is an $n \in \mathbb{Z}_{>0}$ such that $\mathcal{I}^n = 0$.
(c) Give counterexamples to show that the above results are not true if X is only locally Noetherian but not quasicompact, or quasicompact but not locally Noetherian.

Exercise 3.8. Let X be a scheme and $Z \to X$ a closed immersion.

(a) If $Z \to X$ is an isomorphism, then the underlying map of sets is surjective; show the the converse holds if X is reduced.
(b) Produce an example to show that the converse in (a) need not hold in general if X is not reduced.

In this sense, the reduced scheme structure is the "smallest" scheme structure supported on the same topological space.

Exercise 3.9. Let k be a field, $r \in \mathbb{Z}_{\geq 1}$, and $X_1, \ldots, X_r$ be irreducible finite type k-schemes. Show that if $U \subset X_1 \times_k X_2 \times_k \cdots \times_k X_r$ is a dense open subset, then for each i, the image of the map $U \to X_i$ contains a dense open subset.

Exercise 3.10. Let $\varphi : X \to Y$ be a morphism and Z the closed schematic image of φ. Suppose that $j_W : W \hookrightarrow Y$ is a closed subscheme and φ factors through W in the sense that there is a morphism $\psi : X \to W$ such that $\varphi = j_W\psi$. Let Z' be the closed schematic image of ψ. Show that there is a unique isomorphism $Z' \xrightarrow{\sim} Z$ compatible with the morphism from X.

Exercise 3.11. Let k be an algebraically closed field (for simplicity), X_1, X_2, Z integral proper k-schemes, and $\phi_i : Z \to X_i$ for $i = 1, 2$ two closed immersions. Let $X := X_1 \cup_Z X_2$ be the result of gluing X_1 and X_2 along $\phi_i(Z)$, with corresponding closed immersions $j_i : X_i \to X$ for $i = 1, 2$. Show that if $\mathcal{L}$ is a line bundle on X such that $j_i^*\mathcal{L} \cong \mathcal{O}_{X_i}$ for $i = 1, 2$, then $\mathcal{L}$ is trivial.

Exercise 3.12. (Adapted from [1, 19.1.2])

(a) Let $\pi : X \to Y$ be a finite morphism. Suppose that for each $y \in Y$, the degree of π at y is zero or one. Show that π is a closed embedding.
(b) Suppose further that Y is reduced and that for each $y \in Y$, the degree is actually one and not zero. Show that π is an isomorphism.
(c) Suppose in (a) and (b) that Y is locally of finite type over a field; show that it suffices to check on closed points $y \in Y$ to get the same conclusion.

Exercise 3.13.

(a) Let $j : Z \hookrightarrow X$ be an lfp locally closed embedding. Show that for a $z \in Z$, the following conditions are equivalent:
  (i) The morphism j is a local isomorphism, i.e., open embedding, in a neighborhood of

[13] Recall that quasicoherent ideal sheaves on locally Noetherian schemes are automatically coherent.

Z. In other words, there is an open neighborhood U of z in Z such that $j(U)$ is an open neighborhood of z in X and the induced morphism $U \to j(U)$ is an isomorphism of schemes.

(ii) The conormal bundle $\mathcal{N}_{Z/X}^{\vee}$ has stalk zero at z, i.e., $z \notin \operatorname{Supp} \mathcal{N}_{Z/X}^{\vee}$.

In particular, j is an open embedding iff $\mathcal{N}_{Z/X}^{\vee} = 0$.

(b) Show that if $\pi : X \to Y$ is an lft morphism, then the diagonal $\Delta_\pi : X \to X \times_Y X$ is lfp.

3.7.1 PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 3.14. If $j : X \hookrightarrow Y$ is a locally closed embedding of schemes and the underlying topological subspace $j(X) \subset Y$ is closed (resp. open), then j is a closed (resp. open) embedding of schemes.

Exercise 3.15. Let $\pi : X \to Y$ be a morphism of schemes, and let $\iota : X^{\mathrm{red}} \hookrightarrow X$ denote the reduction. Then π is separated (resp. universally closed, resp. proper) iff $\pi \circ \iota$ is.

Chapter 4

Dimension

4.1 Basics

Next, we discuss *dimension*, a purely topological notion that is very important in algebraic geometry.

Definition 4.1.1 (Dimension)**.** Let X be a topological space.

(a) A *(finite) chain of irreducible closed subsets of* X consists of an integer $n \in \mathbb{Z}_{\geq 0}$ and irreducible closed subsets
$$X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n \subset X.$$
Here n is called the *length* of the chain.

(b) The *dimension* of X, denoted $\dim X$, is the supremum of all $n \in \mathbb{Z}_{\geq 0}$ such that there is a chain of irreducible closed subsets of X of length n. We say that X is *equidimensional* or *pure of dimension* n if every irreducible component of X has dimension $n \in \mathbb{Z}_{\geq 0}$. We say that X is equidimensional if it is equidimensional of dimension n for some $n \in \mathbb{Z}_{\geq 0}$; in particular, an equidimensional topological space has finite dimension.

(c) For a point $x \in X$, we define the *local dimension of* X *at* x, denoted $\dim_x X$, to be the infimum of $\dim U$, where U ranges over all open neighborhoods of x in X.[1]

(d) For an irreducible subset $Y \subset X$, we define the *codimension of* Y *in* X, denoted $\operatorname{codim}_X Y$, to be the supremum of all $n \in \mathbb{Z}_{\geq 0}$ such that there is a chain $X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n \subset X$ of irreducible closed subsets of X of length n containing Y, i.e., such that $Y \subset X_0$.

(e) For any arbitrary nonempty closed subset $Y \subset X$, we define the *codimension of* Y *in* X to be $\operatorname{codim}_X Y := \inf_{Z \in \mathrm{Irred}(Y)} \operatorname{codim}_X Z$. We say that Y is *pure of codimension* r *in* X for some $r \in \mathbb{Z}_{\geq 0}$ if all irreducible components of Y have codimension r in X.

The dimension of a scheme is the defined to be the dimension of its underlying topological space; the codimension of a closed subscheme is the codimension of its underlying closed subspace.

Remark 4.1.2.

(a) For any space X, we have $\dim X \in \mathbb{Z}_{\geq 0} \cup \{\pm\infty\}$, with $\dim X = -\infty$ iff $X = \emptyset$.

(b) The local dimension is a local property in the sense that if X is a space, $x \in X$, and $U \subset X$ an open neighborhood of x in X, then $\dim_x X = \dim_x U$. Further, if X has only finitely many irreducible components and $x \in X$ is contained in all of them, then $\dim_x X = \dim X$. In particular, if X is irreducible, then for all $x \in X$ we have $\dim_x X = \dim X$.

(c) When X is a sober space, the above definition(s) can be made using point specializations instead.

(d) If $X = \operatorname{Spec} R$ is an affine scheme, then it is clear by definition that the dimension of X is the Krull dimension $\dim R$ of R.

(e) If $Y \subset X$ is an irreducible subspace of a topological space, then by definition we have that $\operatorname{codim}_X Y = \operatorname{codim}_X \overline{Y}$, and that $\operatorname{codim}_X Y + \dim Y \leq \dim X$. Unfortunately, equality does not always hold. A simple counterexample is given as follows: let R be a DVR with uniformizer π, and let $X = \mathbb{A}^1_R = \operatorname{Spec} R[x]$, so that $\dim X = 2$. If $Y := \mathbb{V}(\pi x - 1)$, then Y is a closed subscheme with $\operatorname{codim}_X Y = 1$ (by Krull's Hauptidealsatz) and $\dim Y = 0$. However, we will show below (4.1.8(a)) that equality holds in reasonable circumstances (e.g., for varieties).

Lemma 4.1.3.

(a) Let X be a space and $Y \subset X$ a subspace. Then $\dim Y \leq \dim X$. If X is irreducible and finite dimensional, and $Y \subsetneq X$ is a proper closed subset, then the inequality is strict.

[1] This definition disagrees with that given in [2, Prop. III.9.5], which is poorly behaved, and agrees with the definition in EGA or [11, Definition 3.3.6]. For the relationship between these definitions (at least for schemes locally of finite type over a field), see 4.1.8(b).

(b) For any space X, we have
$$\dim X = \sup_{Y\in\mathrm{Irred}(X)} \dim Y = \sup_{x\in X} \dim_x X.$$
(c) If $\{U_\alpha\}$ is an open cover of X, then $\dim X = \sup_\alpha \dim U_\alpha$.
(d) If a topological space X is the union of finitely many closed subspaces $Z_i \subset X$, then $\dim X = \max\{\dim Z_i\}$. If each Z_i is pure-dimensional of dimension $n \in \mathbb{Z}_{\geq 0}$, then X is also pure-dimensional of dimension n.
(e) If X is a closure-complete space (e.g., the underlying space of a quasicompact scheme), then $\dim X = \sup_{x\in X^{\mathrm{cl}}} \dim_x X$, where X^{cl} is the subset of closed points.
(f) If $\pi : X \to Y$ is dominant integral morphism of integral schemes, then $\dim X = \dim Y$.
(g) If there is an $n \in \mathbb{Z}_{\geq 0}$ such that X admits an open cover by subspaces, each equidimensional of dimension n, then also X is equidimensional of dimension n.

Proof.

(a) Follows from 2.1.5(g) and the observation that if $Z \subset Y$ is any closed subset with closure $\overline{Z}$ in X, then $Z = \overline{Z} \cap Y$. The second statement follows from appending X to a chain of irreducible closed subsets of Y.
(b) By (a), these suprema are at most $\dim X$. The first result is clear. For the second, if $X_0 \subsetneq \cdots \subsetneq X_n \subset X$ is a chain of irreducible closed subsets, then there is an $x \in X_0$; the result follows from applying 2.1.5(g) to any open neighborhood U of x in X.
(c) Follows from (b).
(d) Follows from (b) (exercise!).
(e) In any chain as above, the set X_0 contains a closed point.
(f) The case of affine Y (and hence X) is an immediate consequence of Cohen-Seidenberg Theory ([4, Cor. 4.2.6(a)]) combined with 3.3.2. The general case then follows from covering Y with affines and using (c).
(g) This follows (b) and 2.1.5(g).

■

Let's first see an algebraic interpretation of the (co)dimension for (possibly non-affine) schemes.

Lemma 4.1.4. Let X be a scheme.

(a) If $Y \subset X$ is an irreducible subset, then $\operatorname{codim}_X Y = \dim \mathcal{O}_{X,Y}$, where $\mathcal{O}_{X,Y}$ is the local ring of X at the generic point of $\overline{Y}$.
(b) In particular, we have $\dim X = \sup_{x\in X} \dim \mathcal{O}_{X,x}$.

Proof.

(a) By replacing Y by its generic point, we are immediately reduced to showing that if X is a scheme and $x \in X$, then $\dim \mathcal{O}_{X,x} = \operatorname{codim}_X\{x\}$. By 2.1.5(d), we are then reduced to the case of affine X, where the result is clear by the inverse correspondence between prime ideals and irreducible subsets.
(b) Clear from (a).

■

Now let us focus on the case of schemes locally of finite type over a field k; in this section, we follow [5, §5.6]. By 4.1.3(b), we may restrict to considering irreducible schemes.[2]

[2]Since dimension is a topological property, we may pass to the underlying reduced subscheme to restrict ourselves to considering integral schemes. The added generality in considering irreducible and not just integral schemes is somewhat superficial.

The standard result from commutative algebra we need is

Theorem 4.1.5 (Noether Normalization)**.** Let k be a field and X be an affine scheme of finite type over k. Then $\dim X < \infty$, and there is a finite surjective[3] morphism $X \to \mathbb{A}_k^{\dim X}$. In addition, if X is integral, then $\dim X = \operatorname{trdeg}_k k(X)$.

Proof. The usual (algebraic) way of stating the Noether Normalization Lemma (say [4, Thm. 6.2.7(a)]) produces an integer $r \in \mathbb{Z}_{\geq 0}$ and a finite surjective morphism $X \to \mathbb{A}_k^r$ (using [4, Rmk. 3.2.2] if needed). Using this, from Cohen-Seidenberg Theory, ([4, Cor. 3.2.6]), we conclude that $\dim X = r$. If X is also integral, the above map expresses $k(X)$ as a finite algebraic extension of $k(\mathbb{A}_k^r)$, so that $\operatorname{trdeg}_k k(X) = \operatorname{trdeg}_k k(\mathbb{A}_k^r) = r$ as well. ■

Theorem 4.1.6. Let X be an irreducible scheme lft over a field k, and let $\eta \in X$ be the generic point of X. Let $k(X) := \kappa(\eta)$.

(a) We have $\dim X = \operatorname{trdeg}_k k(X)$. In particular, $\dim X < \infty$.
(b) If $U \subset X$ is any nonempty open subscheme, then $\dim U = \dim X$.
(c) More generally, let $f : Y \to X$ be a dominant lft morphism.[4] Then $\dim Y \geq \dim X$.
(d) If $x \in X$ is any closed point, then $\dim X = \dim \mathcal{O}_{X,x}$.
(e) If $f : Y \to X$ is a quasi-finite morphism,[5] then $\dim Y \leq \dim X$.

In particular, if $f : Y \to X$ is a dominant quasi-finite morphism, then $\dim Y = \dim X$.

Proof.

(a) Replacing X by X_{red}, we may assume that X is integral. Using 4.1.3(c), we are reduced to the case of affine X; this reduces us to 4.1.5.
(b) Clear from (a).
(c) By 2.1.13, there is a $\theta \in Y$ such that $f(\theta) = \eta$. Then f induces a k-embedding $\kappa(\eta) \hookrightarrow \kappa(\theta)$, and so
$$\dim X = \operatorname{trdeg}_k \kappa(\eta) \leq \operatorname{trdeg}_k \kappa(\theta) = \dim \overline{\{\theta\}} \leq \dim Y,$$
where we are repeatedly using (a) and 4.1.3(a).
(d) Using (b), we may assume that X is integral and affine. This reduces us to: if R is a domain finitely generated over a field and $\mathfrak{m} \subset R$ a maximal ideal, then $\dim R = \operatorname{ht} \mathfrak{m}$; this is a standard consequence of Noether Normalization as well ([4, Theorem 6.2.7(d)]).
(e) By 4.1.3(b), we may assume Y to be irreducible. Replacing X and Y by their reductions, we can assume that both X and Y are integral. By (b), we may further reduce to the case when X and Y are affine. This reduces us to: if R and S are domains finitely generated over a field k and $f : R \to S$ a k-algebra homomorphism such that for all primes $\mathfrak{p} \subset R$, the $\kappa(\mathfrak{p})$-algebra $\kappa(\mathfrak{p}) \otimes_R S$ has only finitely many primes, then $\dim S \leq \dim R$. To show this, take $\mathfrak{p} := \ker \varphi$. Then $\operatorname{Spec}(\kappa(\mathfrak{p}) \otimes_R S)$ is a nonempty finite-type $\kappa(\mathfrak{p})$-scheme whose underlying topological space is *finite*. The subset of non-closed points is then open, and hence by 2.3.9, must then be empty, i.e., every point is closed. In particular, applying this to the prime corresponding to $(0) \subset S$ shows us that $\operatorname{Frac} S$ is a *finite* extension of $\kappa(\mathfrak{p})$, and hence that $\dim S = \operatorname{trdeg}_k S = \operatorname{trdeg}_k \kappa(\mathfrak{p}) = \operatorname{coht} \mathfrak{p} \leq \dim R$ as needed.

■

Corollary 4.1.7. Let X be a scheme of finite type over a field k. Then $\dim X < \infty$.

[3]In this case, "finite surjective" is the same as "finite dominant" because 2.4.4 tells us that, in this context, dominance is the same as having dense image, and finite morphisms are closed [TOCITE].
[4]Recall that when f is quasicompact, our definition of dominant agrees with the usual one; see 2.1.15.
[5]Our definition of "quasi-finite" includes "finite-type".

Proof. Follows from 2.2.7(c), 4.1.3(b), and 4.1.6(a). ■

Parts (b) and (d) of 4.1.6 combine to tell us that, in the above setting, if $x \in X$ is any closed point, then $\dim_x X = \dim \mathcal{O}_{X,x}$. When x is not necessarily closed, the correct generalization is

Theorem 4.1.8. Let X be a scheme that is locally of finite type over a field k.

(a) Let $Y \subset X$ be a closed subset. Suppose that either X is equidimensional, or that X has finitely many components and Y lies in each of them. Then

$$\operatorname{codim}_X Y + \dim Y = \dim X < \infty.$$

(b) In general, for any $x \in X$, we have $\dim \mathcal{O}_{X,x} + \operatorname{trdeg}_k \kappa(x) = \dim_x X$.

Note that some sort of hypothesis as in (a) is clearly necessary, as easily constructed counterexamples otherwise show.

Proof.

(a) The hypotheses allow us to reduce to the case of affine integral X and Y (exercise). Let $R := \mathcal{O}(X)$. If Y corresponds to the prime ideal $\mathfrak{p} \subset R$, we have $\operatorname{codim}_X Y = \dim R_{\mathfrak{p}} = \operatorname{ht} \mathfrak{p}$ by 4.1.4, and $\dim Y = \dim R/\mathfrak{p} = \operatorname{coht} \mathfrak{p}$. This reduces us to the following standard result in commutative algebra: if R is a domain that is finitely generated over a field k and $\mathfrak{p} \subset R$ a prime ideal, then $\dim R = \operatorname{ht} \mathfrak{p} + \operatorname{coht} \mathfrak{p}$ ([4, Theorem 6.2.7(d)]).
(b) We will reduce to (a). First, replacing X by an affine open containing x, we may assume that X is of finite type over k. In particular, X is Noetherian and so has only finitely many irreducible components. Removing all irreducible components of X which do not contain x, we may assume that x is contained in every irreducible component of X. In this case, we have by 4.1.2(b) that $\dim_x X = \dim X$. Applying (a) to $Y := \overline{\{x\}}$, and using 4.1.4(a) and 4.1.6(a) gives the result.

■

Next, we check that the dimension of a scheme locally of finite type over a field behaves well under changing the base field.

Theorem 4.1.9. Let k be a field, X a k-scheme, and $k \subset K$ be a field extension. If either

(a) X is affine over k and K/k algebraic, or
(b) X is lft over k (and K/k arbitrary), then

the scheme X is equidimensional iff the base-change X_K of X to K is, and then $\dim X_K = \dim X$.

This result is (I think) more subtle than is often given credit for, and the (only detailed) proof (I know of it) is somewhat tricky. Some sort of hypothesis as in (a) or (b) is clearly necessary; a simple counterexample otherwise is given by taking $X = \operatorname{Spec} k(t)$ and $K = k(s)$, both purely transcendental; then X_K is (the spectrum of) a suitable localization of (the coordinate ring of) $\mathbb{A}^1_{k(s)}$ and can be easily shown to be one-dimensional.[6] The following proof strategy has been taken from [1, Chapter 12]. For the proof, let's isolate two small results.

Lemma 4.1.10. Let X be an affine integral scheme over a field k, and K/k a field extension.

(a) Every irreducible component of X_K dominates X.

[6] I suspect similar counterexamples exist when we drop *any* of the hypotheses of the theorem.

(b) If K/k is purely transcendental, then X_K is also integral.

Proof. Let $X = \operatorname{Spec} R$ for a k-domain R.

(a) For any minimal prime $\mathfrak{p}$ of $R_K = K \otimes_k R$, the natural map $R \to R_K \to R_K/\mathfrak{p}$ is injective. This follows from two facts: (i) that R_K is a free R-module and R is a domain, and (ii) if S is any ring and $\mathfrak{p} \subset S$ a minimal prime, then any element of $\mathfrak{p}$ is a zero divisor.[7]

(b) If $K = k(e_i)_{i \in I}$, then $X_K = \operatorname{Spec} R_K$ where

$$R_K = K \otimes_k R \cong k(e_i) \otimes_{k[e_i]} (k[e_i] \otimes_k R) \cong (k[e_i] \smallsetminus \{0\})^{-1} R[e_i]$$

is a localization of the domain $R[e_i]$ and hence also a domain.

■

Proof of 4.1.9. First we prove the forward statement; suppose that X is equidimensional. In both (a) and (b), we can reduce to the case of integral X. Indeed, we reduce to the case of irreducible X by noting that an irreducible component of X_K must be an irreducible component of Y_K for some irreducible component Y of X. Also, the reduction map $X^{\text{red}} \to X$ is a surjective closed immersion and hence a universal homeomorphism; in particular, the natural map $(X^{\text{red}})_K \to X_K$ is a homeomorphism. Hence suppose that X is integral.

(a) Since K/k is algebraic, every component of X_K maps to X via an integral morphism. Since X is affine and integral, the result follows from 4.1.10(a) combined with 4.1.3(e).

(b) By 4.3 and 4.1.3(f), it suffices to deal with the case of affine X. Since any K/k can be written as a composite of a purely transcendental extension followed by an algebraic extension, by part (a), it suffices to treat the case when K/k is purely transcendental. Using 4.1.10(b) and 4.1.6, it remains only to justify that $\operatorname{trdeg}_k k(R) = \operatorname{trdeg}_K K(R_K)$, but indeed $K(R_K) = k(R)(e_i)$, so the result is clear by, say, [4, Excercise 5.10].

For the converse, suppose now that X_K is equidimensional. The morphism $X_K \to X$ is affine and hence quasicompact, and it is surjective because it is the base-change of the surjective morphism $\operatorname{Spec} K \to \operatorname{Spec} k$. Therefore, it follows from 3.3.3(a) that $X_K \to X$ is dominant (2.1.14); in particular, if $Y \subset X$ is any irreducible component, then there is an irreducible component $Z \subset X_K$ dominating Y.

(a) Giving Z and Y the reduced subscheme structures, they are both integral. Since X is affine, so are Y and Z, and the result follows from 4.1.3(e) applied to the induced morphism $Z \to Y$.

(b) By 4.1.3(f) and 4.3, it suffices to deal with affine X. As in the forward implication in (b) above, we may reduce to the case of K/k purely transcendental. If Y and Z are as above, then we may once again give Y and Z the reduced subscheme structure to make them integral. Then $Z \subset Y_K$, but Y_K is integral by 4.1.10(b), so $Z = Y_K$ by maximality. By the forward implication, we get $\dim Y = \dim Y_K = \dim Z = \dim X_K$ as needed.

■

Corollary 4.1.11. Let k be a field and X be a k-scheme. Let K/k be a field extension. Suppose either that K/k is algebraic or that X is lft over k. Then for any $p \in X_K$ over $q \in X$, we have $\dim_p X_K = \dim_q X$. In particular, we have $\dim X_K = \dim X$.z

Proof. Follows from the above theorem, and is left as an exercise. ■

Finally, we deal with the dimension of the product of two k-schemes.

[7] Indeed, since $\mathfrak{p}$ is minimal, the ring $S_\mathfrak{p}$ has a unique prime ideal $\mathfrak{p}S_\mathfrak{p}$. Therefore, $\operatorname{Nil}(S_\mathfrak{p}) = \mathfrak{p}S_\mathfrak{p}$, so that any element of $\mathfrak{p}$ is nilpotent in $S_\mathfrak{p}$ and hence a zero-divisor in S.

Theorem 4.1.12. Let k be a field, and X, Y be nonempty equidimensional schemes, lft over k. Then $X \times_k Y$ is also nonempty equidimensional and lft over k, with $\dim X \times_k Y = \dim X + \dim Y$.

Proof 1. That the product $X \times_k Y$ is nonempty and lft over k is clear, so we only have to show that it is equidimensional of dimension $\dim X + \dim Y$. We proceed in several steps.

Step 1. Suppose that k is algebraically closed and X, Y affine integral. Then $X \times_k Y$ is also affine and integral ([1, 10.4.H]), and the result follows from 4.1.5: pick finite surjective morphisms $X \to \mathbb{A}_k^{\dim X}$ and $Y \to \mathbb{A}_k^{\dim Y}$, and observe that their product $X \times_k Y \to \mathbb{A}_k^{\dim X + \dim Y}$ is also finite surjective, whence the result follows from 3.3.3(a) and 4.1.3(e).

Step 2. Suppose that k is algebraically closed and X, Y affine. Replacing X and Y by their reductions by using that $X^{\mathrm{red}} \times_k Y^{\mathrm{red}} \to X \times_k Y$ is a homeomorphism, we may assume that X and Y are reduced. The result follows from noting that an irreducible component of $X \times_k Y$ must be (contained in, and hence) of the form $X' \times_k Y'$ for some irreducible components $X' \subset X$ and $Y' \subset Y$ (given the reduced subscheme structure).

Step 3. Suppose that X and Y are affine (and k arbitrary). By 4.1.9(a) applied to an algebraic closure $K = \overline{k}$ of k, we get that X_K (resp. Y_K)is equidimensional of the same dimension as X (resp. Y). By Step 2, $X_K \times_K Y_K$ is equidimensional of dimension $\dim X_K + \dim Y_K = \dim X + \dim Y$. We conclude by invoking 4.1.9(a) once again by noting that $X_K \times_K Y_K = (X \times_k Y)_K$.

Step 4. Suppose that X and Y are irreducible and k arbitrary. Let $Z \subset X \times_k Y$ be an irreducible component. Pick affine opens $U \subset X$ and $V \subset Y$ such that $Z \cap (U \times_k V) \neq \emptyset$. From 4.1.6(b), we conclude that $\dim Z = \dim Z \cap (U \times_k V)$. Now $Z \cap (U \times_k V)$ is an irreducible component of $U \times_k V$ by 2.1.5(g), and so by Step 3 we have $\dim Z \cap (U \times_k V) = \dim U + \dim V = \dim X + \dim Y$, where in the last step we are again using 4.1.6(b).

Step 5. The main theorem in full generality then follows easily from Step 4 and the fact (as in Step 2) that every irreducible component $Z \subset X \times_k Y$ is an irreducible component of $X' \times_k Y'$ for some irreducible components $X' \subset X$ and $Y' \subset Y$.

■

Proof 2. Since $X \to \operatorname{Spec} k$ is flat of relative dimension $\dim X$, the base-change $X \times_k Y \to Y$ is also flat of relative dimension $\dim X$ by [1, 24.5.K]; the result then follows from [1, 24.5.L]. ■

4.2 Krull's Theorems

Corollary 4.2.1. Let k be a field, X a scheme locally of finite type over k. Let $n, r \in \mathbb{Z}_{\geq 1}$ with $r \leq n$. Let $Z \subset X$ be a regular embedding of codimension r (e.g., for $r = 1$ this means that Z is an effective Cartier divisor). If X has dimension at most n, then Z has dimension at most $n - r$. If X is equidimensional of dimension n, then Z is equidimensional of dimension $n - r$.

4.3 Smoothness

Definition 4.3.1 (Smoothness over a Field)**.** Let k be a field and X an lft k-scheme.

(a) If X is irreducible, we say that X is *smooth* iff Jacobian corank function of X is constant with value $\dim X$.
(b) For arbitrary X, we say that X is smooth iff the irreducible components of X are disjoint, and each component is smooth.

Lemma 4.3.2. Let X be lft and equidimensional. Then X is smooth iff the Jacobian corank function of X is constant with value $\dim X$.

Proof. When k is algebraically closed (or more generally perfect), this follows from the fact that smooth implies regular at closed points, and regular local rings are domains. To do the general case, base-change to an algebraic closure of k, using 4.1.9, stability of the Jacobian corank function under field extensions, and 4.15. ■

Definition 4.3.3 (Smooth Morphisms I)**.** Let $n \in \mathbb{Z}_{\geq 0}$ and let A be a ring.

- An A-scheme X is *smooth of relative dimension n over A at a point $x \in X$* iff there is an open neighborhood U of x in X, an integer $r \in \mathbb{Z}_{\geq 0}$, polynomials $f_1, \ldots, f_r \in A[x_1, \ldots, x_{n+r}]$, and an isomorphism of A-schemes of the form

$$U \xrightarrow{\sim} \operatorname{Spec} A[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r)[1/\det \mathrm{Jac}_{\leq r}] \tag{4.3.4}$$

where

$$\mathrm{Jac}_{\leq r} = \left[\frac{\partial f_i}{\partial x_j}\right]_{i,j=1,\ldots,r}.$$

An A-scheme X is *smooth of relative dimension n* or a *smooth A-scheme of relative dimension n* iff it is smooth of relative dimension n over A at all $x \in X$.
- An A-scheme X is *étale* over A (at $x \in X$) iff it is smooth of relative dimension 0 (at $x \in X$).
- An A-scheme X is *smooth* iff for all $x \in X$, there is an integer $n \in \mathbb{Z}_{\geq 0}$ (possibly depending on x) such that X is smooth of relative dimension n over A at x.

Remark 4.3.5. Let $n \in \mathbb{Z}_{\geq 0}$, and let A be a ring.

(a) Let X be an A-scheme smooth of relative dimension n over A (at $x \in X$). For any ring homomorphism $A \to B$, the base-change X_B is a B-scheme smooth of relative dimension n over B (at any x_B over x).
(b) If the structure morphism $a_X : X \to \operatorname{Spec} A$ is an open embedding, then X is étale over A. This follows immediately from 1.2.1(d). The same proof shows that if X is a smooth A-scheme of relative dimension n over A (at $x \in X$), then for any open subset $U \subset X$ (containing x), the A-scheme U is smooth of relative dimension n over A (at $x \in U$). Conversely, if X is an A-scheme that admits an open cover by smooth A-schemes of relative dimension n over A, then X itself is a smooth A-scheme of relative dimension n over A.
(c) If $A = k$ is a field, then X is a smooth k-scheme of relative dimension n iff X is a lft k-scheme which is equidimensional of dimension n and smooth in the sense of 4.3.1.[8]

Definition 4.3.6 (Smooth Morphisms II)**.** Let $n \in \mathbb{Z}_{\geq 0}$.

[8] To note that a k-scheme that is smooth of relative dimension n is equidimensional, it might be helpful to use Krull's theorems [TODO] to show that every irreducible component of X has dimension at least n, and then use [1, 13.1.M] along with 4.1.3(b) to prove $\dim X \leq n$.

- A morphism $\pi : X \to Y$ of schemes is said to be *smooth of relative dimension* n *at a point* $x \in X$ iff there is an open affine neighborhood V of $\pi(x) \in Y$ such that the restriction of π to the preimage $X_V = \pi^{-1}(V)$ makes the $\mathcal{O}(V)$-scheme X_V smooth of relative dimension n over $\mathcal{O}(V)$ at the point $x \in X_V$. A morphism $\pi : X \to Y$ is said to be *smooth of relative dimension* n iff it is smooth of relative dimension n at each $x \in X$.
- A morphism $\pi : X \to Y$ is said to be *étale* (at $x \in X$) iff it is smooth of relative dimension zero (at $x \in X$).
- A morphism $\pi : X \to Y$ is said to be *smooth* iff for all $x \in X$, there is an integer $n \in \mathbb{Z}_{\geq 0}$ (possibly depending on x) such that π is smooth of relative dimension n at x.

Remark 4.3.7. If $n \in \mathbb{Z}_{\geq 0}$ and $\pi : X \to Y$ is a smooth morphism of relative dimension n at $x \in X$, then there is an open neighborhood U of x in X such that the restriction $\pi_U : U \to Y$ can be factored as an étale map $U \to \mathbb{A}^n_Y$ followed by the projection $\mathbb{A}^n_Y \to Y$.

Lemma 4.3.8.

(a) (Affine Communication) Let $n \in \mathbb{Z}_{\geq 0}$, let A be a ring, let $a_X : X \to \operatorname{Spec} A$ be an A-scheme, and let $x \in X$. Then X is smooth of relative dimension n over A (at x) iff the structure morphism a_X is smooth of relative dimension n (at x).
(b) For any morphism $\pi : X \to Y$ and integer $n \in \mathbb{Z}_{\geq 0}$, the locus $X_n \subset X$ of points of X at which π is smooth of relative dimension n is open.
(c) Smooth (and hence étale) morphisms are lfp (and hence lft). For any $n \in \mathbb{Z}_{\geq 0}$, smooth morphisms of relative dimension n are affine local on both the source and target, and stable under base-change: if $\pi : X \to Y$ is smooth of relative dimension n (at $x \in X$), then for any morphism $Y' \to Y$, the basechange $\pi' : X' = X \times_Y Y' \to Y'$ is smooth of relative dimension n (at any x' over x).
(d) Given integers $m, n \in \mathbb{Z}_{\geq 0}$, if $\pi : X \to Y$ is smooth of relative dimension n at $x \in X$, and $\rho : Y \to Z$ is smooth of relative dimension m at $y := \pi(x)$, then the composite $\rho \circ \pi : X \to Z$ is smooth of relative dimension $m + n$ at x. In particular, étale morphisms are stable under composition, and hence form a reasonable class of morphisms.

Proof.

(a) Follows from clearing denominators, and left as an exercise (c.f. 4.3.5(b)).
(b) Clear from the definition.
(c) Affine-locality on the target follows from the definition; affine-locality on the source follows from 4.3.5(b). Finally, stability under base-change follows at once from (a), 4.3.5(a), and 3.1.2.
(d) By a minor variant of the proof of 3.1.3 (left to the reader), it suffices to do the case when X, Y, Z are all affine and π and ρ are globally standard smooth (i.e., of the form (4.3.4)); then the result is clear.

$\blacksquare$

Combining 4.3.8(c) with 4.3.5(c) tells us that smooth morphisms have smooth fibers in the sense of 4.3.1.

Example 4.3.9.

(a) Open embeddings of schemes are étale.
(b) For any $m \in \mathbb{Z}_{\geq 2}$, the morphism $\mathbb{A}^1_{\mathbb{Z}} \to \mathbb{A}^1_{\mathbb{Z}}$ corresponding to the ring homomorphism $\mathbb{Z}[x] \to \mathbb{Z}[y]$ given by $x \mapsto y^m$ is étale over the subset $\operatorname{Spec} \mathbb{Z}[x, (mx)^{-1}] \cong D(mx) \subset \mathbb{A}^1_{\mathbb{Z}}$.
(c) For a number field K with ring of integers $\mathcal{O}_K$, the structure morphism $\operatorname{Spec} \mathcal{O}_K \to \operatorname{Spec} \mathbb{Z}$ is étale over the subset $\operatorname{Spec} \mathbb{Z}[1/\operatorname{disc}(K)] \cong D(\operatorname{disc} K) \subset \operatorname{Spec} \mathbb{Z}$.

(d) Let k be a field, X, Y be lft k-schemes, and $\pi : X \to Y$ a k-morphism. Let x be a closed point with residue field $\kappa(x)$ a finite separable extension of k (e.g., any closed point when k is perfect), and let $y := \pi(x)$. Suppose for simplicity that the natural morphism $\kappa(y) \to \kappa(x)$ is an isomorphism, which happens, e.g., when k is algebraically closed. Then π is étale at x iff the differential $\mathrm{d}\pi_x : \mathrm{T}_x X \to \mathrm{T}_y Y$ is an isomorphism of $\kappa(x)$-vector spaces. For example, if $k = \mathbb{C}$ and X and Y are smooth separated finite-type $\mathbb{C}$-schemes, and $\pi : X \to Y$ a $\mathbb{C}$-morphism, then π is étale iff the induced map $X(\mathbb{C}) \to Y(\mathbb{C})$ of the corresponding complex manifolds is a local biholomorphism.

(e) Let k be a field. A morphism $\pi : X \to \operatorname{Spec} k$ is étale iff X is discrete, and each point of X is the spectrum of a finite separable field extension of k.

(f) For any integer $n \in \mathbb{Z}_{\geq 0}$ and scheme S, the structure morphisms of the S-schemes $\mathbb{A}^n_S$ and $\mathbb{P}^n_S$ are smooth of relative dimension n.

Theorem 4.3.10. Let k be a field and X an lft k-scheme. The following conditions are equivalent.

(a) The k-scheme X is generically geometrically reduced, i.e., for each generic point η of an irreducible component of X, the local ring $\mathcal{O}_{X,\eta}$ is a separable field extension of k.
(b) There is a dense open subset $U \subset X$ such that U is smooth over k.
(c) The smooth locus $X^{\mathrm{sm}} \subset X$, i.e., the open locus of points at which X is smooth over k, is dense.

Further, if X is reduced, then these conditions are also equivalent to

(d) The k-scheme X is geometrically reduced.

Proof. The equivalence of (b) and (c) is clear. For the equivalence with (a), using 2.2.12, we are immediately reduced to the case in which X is affine and irreducible, which we assume hence. Let $n := \dim X \in \mathbb{Z}_{\geq 0}$. Let $\eta \in X$ be the generic point and $K = K(X) = \kappa(\eta)$ be the function field of X, so that $n = \operatorname{trdeg}_k K$.

(a) ⇒ (b) It suffices to show that the rank of the coherent sheaf $\Omega_{X/k}$ at η is n; by upper semicontinuity, it would then follow that it is n on an open subset U containing η, which would then be smooth over k. The stalk of $\Omega_{X/k}$ at η is precisely $\Omega_{K/k}$. Since K/k is separable and finitely generated, we conclude from [4, 4.2.8] [unstable ref].

(b) ⇒ (a) We may shrink X to assume that it is smooth over k. In this case, X is geometrically reduced, and in particular reduced. It follows that for each such point η, the ring $\mathcal{O}_{X,\eta}$ is a reduced Artinian local ring and hence a field extension of k. This field extension is just $\kappa(\eta)$, which then has to be separable over k thanks to 2.1.18.

The equivalence of (a)-(c) with (d) when X is reduced follows immediately from 2.1.18. ■

Corollary 4.3.11. Let k be a perfect field and X reduced lft k-scheme. There is a dense open subset $U \subset X$ such that U is smooth over k.

Proof. This follows from the equivalence of (b) and (d) in 4.3.10, and the fact that a reduced lft k-scheme if automatically geometrically reduced when k is perfect (2.1.18). ■

Theorem 4.3.12 (Generic Smoothness)**.** Let $\pi : X \to Y$ be a morphism of finite presentation of schemes with Y integral and the fraction field $K(Y)$ perfect (e.g., if $\operatorname{char} K(Y) = 0$, or if $Y = \operatorname{Spec} k$ for a perfect field k).

(a) If X is integral and π dominant, then there is a dense open $U \subset X$ such that $\pi|_U : U \to Y$ is smooth of relative dimension $\operatorname{trdeg}_{K(Y)} K(X)$.
(b) Suppose that X and Y are Noetherian and that X is regular. There is a dense open $V \subset Y$ such that the pullback $\pi|_{X_V} : X_V = \pi^{-1}(V) \to V$ is smooth.

Compare this with [1, 21.6.4-6] and [5, Exercise 10.40]; as is explained in (b) of the second source, slightly weaker hypotheses suffice in (b).

Proof. In (b), we may further assume also that π is dominant; if not, the image of π is constructible by Chevalley's Theorem ([1, 8.4.2]), but does not contain the generic point, and hence the closure of $\pi(X)$ is a proper closed subset of Y (2.3.4(b)). Therefore, it suffices to take a dense open subset $V \subset Y$ such that the preimage X_V of V is empty, proving (b). Similarly, since a Noetherian regular scheme is the disjoint union of finitely many integral Noetherian regular schemes, we can assume also that X is integral. Therefore, in proving both (a) and (b), we may assume that X is integral and π is dominant, which we do from now on.

The problem is easily reduced to the case of $Y = \operatorname{Spec} A$ for some domain A and $X = \operatorname{Spec} B$ for some domain B which is a finitely presented A-algebra such that $A \to B$ is injective. Let $K := \operatorname{Frac} A$ and $L := \operatorname{Frac} B$, and let $n := \operatorname{trdeg}_K L$. The generic fiber corresponds to the K-algebra $K \otimes_A B \subset L$, which is an integral finite-type K-algebra.

(a) By 4.3.11, there is a nonzero $f \in B$ such that $K \otimes_A B[1/f]$ is smooth over K of dimension n. This reduces the statement (a) to a statement in commutative algebra, namely 4.3.13.
(b) We are given that B is regular. In this case, $K \otimes_A B$, being a localization of B, is also regular. Since K is perfect, this means that $K \otimes_A B$ is smooth over K of dimension n (unstable [4, 6.5.2]).

■

Lemma 4.3.13. (Spreading out Smoothness, V1) Let A be a domain and B a domain which is a finitely presented A-algebra such that $A \to B$ is injective. Let $K := \operatorname{Frac} A$. Suppose that $K \otimes_A B$ is smooth of dimension $n \in \mathbb{Z}_{\geq 0}$ over K. Then there is a nonzero $f \in B$ such that $B[1/f]$ is smooth of relative dimension n over A.

Proof. Let $L := \operatorname{Frac} B$. Pick a presentation of B of the form $B = A[x_1, \ldots, x_s]/(f_1, \ldots, f_s)$ for some $s, t \in \mathbb{Z}_{\geq 0}$ and $f_1, \ldots, f_t \in A[x_1, \ldots, x_s]$. Since B is a domain, the ideal $\mathfrak{P} := (f_1, \ldots, f_s) \subset A[x_1, \ldots, x_s]$ is prime. Since $A \to B$ is injective, $\mathfrak{P} \cap A = (0)$; in other words, there is a prime $\mathfrak{P}' \subset K[x_1, \ldots, x_s]$ such that $\mathfrak{P} = \mathfrak{P}' \cap A[x_1, \ldots, x_s]$. For use in a moment, we compute also $\operatorname{coht} \mathfrak{P}'$: this is $\dim K[x_1, \ldots, x_s]/\mathfrak{P}'$, which is the transcendence degree of the fraction field of $K[x_1, \ldots, x_s]/\mathfrak{P}'$ over K. This fraction field is just L; indeed, the injection $B \hookrightarrow K[x_1, \ldots, x_s]/\mathfrak{P}'$ is easily seen to extend to an isomorphism of the corresponding fraction fields. Therefore, $\operatorname{coht} \mathfrak{P}' = \dim K[x_1, \ldots, x_s]/\mathfrak{P}' = \operatorname{trdeg}_K L = n$.

The above choice of presentation of B as an A-algebra yields a presentation of $\Omega_{B/A}$ as a B-module of the form

$$B^{\oplus t} \xrightarrow{[\partial f_j/\partial x_i]} B^{\oplus s} \to \Omega_{B/A} \to 0.$$

Apply the right-exact functor $L \otimes_B -$ to get the exact sequence

$$L^{\oplus t} \xrightarrow{[\partial f_j/\partial x_i]} L^{\oplus s} \to L \otimes_B \Omega_{B/A} \to 0.$$

Now $L \otimes_B \Omega_{B/A} \cong \Omega_{L/K}$ as L-vector spaces ([1, 21.2.L]). Since $K \otimes_A B$ is smooth over K, it is in particular geometrically reduced; in particular, its fraction field L is a separable extension of K. Therefore, $\dim_L \Omega_{L/K} = \operatorname{trdeg}_K L = \dim K \otimes_A B = n$. Let $r := \operatorname{rank}_L[\partial f_j/\partial x_i]$, so that $r \leq t$ and $s = n + r$. Reorder the f_j to assume that the rank of the first r columns is already r, i.e., the matrix $J := [\partial f_j/\partial x_i]_{\substack{1 \leq i \leq s \\ 1 \leq j \leq r}}$ has rank r over L. Let $m \in A[x_1, \ldots, x_s]$ be the $r \times r$ minor of J which is not in $\mathfrak{P}$.

Let $\overline{\mathfrak{P}}$ be the prime of $A[x_1, \ldots, x_s]/(f_1, \ldots, f_r)$ corresponding to $\mathfrak{P}$, and consider the

$$(A[x_1, \ldots, x_s]/(f_1, \ldots, f_r))_{\overline{\mathfrak{P}}} \twoheadrightarrow (A[x_1, \ldots, x_s]/\mathfrak{P})_{(0)} = L. \qquad (4.3.14)$$

If $\overline{\mathfrak{P}}'$ is the prime of $K[x_1, \ldots, x_s]/(f_1, \ldots, f_r)$ corresponding to $\mathfrak{P}$, then it is easy to see that

$$(A[x_1, \ldots, x_s]/(f_1, \ldots, f_s))_{\overline{\mathfrak{P}}} \cong (K[x_1, \ldots, x_s]/(f_1, \ldots, f_r))_{\overline{\mathfrak{P}}'}$$

as A-algebras. By the Jacobian criterion, the ring $K[x_1, \ldots, x_s]/(f_1, \ldots, f_r)$ is smooth of relative dimension n over K at $\overline{\mathfrak{P}}'$; therefore, the corresponding localization is a regular local ring of dimension $\operatorname{ht}\mathfrak{P}' - r = (s - \operatorname{coht}\mathfrak{P}') - r = 0$, i.e, it is a field (c.f. [4, 6.5.1]). Therefore, the map 4.3.14 is an isomorphism. This proves that $(f_1, \ldots, f_r)A[x_1, \ldots, x_s]_{\mathfrak{P}} = (f_1, \ldots, f_t)A[x_1, \ldots, x_s]_{\mathfrak{P}} = \mathfrak{P}A[x_1, \ldots, x_s]_{\mathfrak{P}}$. It follows that there is a $g \in A[x_1, \ldots, x_s] \smallsetminus \mathfrak{P}$ such that

$$(f_1, \ldots, f_r)A[x_1, \ldots, x_s][1/g] = (f_1, \ldots, f_t)A[x_1, \ldots, x_s][1/g].$$

It suffices to take f to be the image of gm in B. ■

Smoothness is equivalent to diagonal being regular [TODO].

4.4 Bertini's Theorems

[TODO: Estimate dimensions of cones.]

Convention 4.4.1. Let k be a field, $d, n \in \mathbb{Z}_{\geq 0}$, and $X \subset \mathbb{P}^n_k$ be a closed subscheme containing a dense open subset V which is smooth of dimension d over k.[9] Let $Z := X \smallsetminus V$.

Theorem 4.4.2 (Bertini)**.** Assume the hypotheses 4.4.1.

(a) For any $r \in \mathbb{Z}_{\geq 0}$, there is a nonempty open subset $U \subset (\mathbb{P}^{n\vee}_k)^{\times_k r}$ such that for all $p \in U$ with corresponding hyperplanes $H_1, \ldots, H_r \subset \mathbb{P}^n_{\kappa(p)}$, the schematic intersection

$$X_{\kappa(p)} \cap H_1 \cap \cdots \cap H_r \subset \mathbb{P}^n_{\kappa(p)}$$

is smooth of dimension $d - r$ away from $Z_{\kappa(p)} \cap H_1 \cap \cdots H_r$, and in particular contains a dense open subset which is smooth of dimension $d - r$ over $\kappa(p)$.

(b) In particular, for $r = d$, there is a nonempty open subset $U \subset (\mathbb{P}^{n\vee}_k)^{\times_k d}$ such that for all $p \in U$ with corresponding hyperplane $H_1, \ldots, H_r \subset \mathbb{P}^n_{\kappa(p)}$, the intersection $X_{\kappa(p)} \cap H_1 \cap \ldots H_r$ is smooth of dimension 0 over $\kappa(p)$, i.e., consists of finitely many reduced closed points q_j with residues fields finite separable extensions of $\kappa(p)$, say q_j, and $\sum_j [\kappa(q_j) : \kappa(p)] = \deg X$.

TODO. Use $k \to \kappa(p)$ universally open. For the last part of (b), by generic flatness and constancy of Hilbert polynomial in flat families, we may assume that the sum is constant; to evaluate this constant, we may then pass to the algebraic closure of k to assume k is algebraically closed.

Find a nonempty open subset $V_1 \subset \mathbb{P}^{n\vee}_k$ such that U meets the fiber over each $q_1 \in V_1$ (3.9), for each closed $p_1 \in V_1$ corresponding to hyperplane $H_1 \subset \mathbb{P}^n_k$, the intersection $X \cap H_1$ is smooth of dimension $d - 1$ away from $Z \cap H_1$, and H_1 avoids all the associated points of X. Then $\deg(X \cap H_1) = \deg X$. Now the fiber of the projection $(\mathbb{P}^{n\vee}_k)^{\times_k r} \to \mathbb{P}^{n\vee}_k$ over p_1 meets U is a nonempty open subset of $(\mathbb{P}^{n\vee}_k)^{\times_k (r-1)}$. We are done by induction on r. ∎

Sometimes when X is smooth, can use Miracle Flatness to say more precisely what this locus is.

Generalized to intersection of hypersurfaces of arbitrary degree.

Theorem 4.4.3. Let k be a field of characteristic zero, X a reduced finite-type k-scheme, and V a (finite dimensional) linear series on X; in other words, suppose there is a line bundle $\mathcal{L}$ on X, and that $V \subset \mathrm{H}^0(X, \mathcal{L})$ a finite-dimensional subspace. The general element of V, when considered as a closed subscheme of X, is smooth away from the base locus of V and the singular locus of X. Specifically, there is a nonempty open subset $U \subset \mathbb{P}V^\vee$ such that for each $\sigma \in U(k)$, the correspondiding vanishing locus $\mathbb{V}(\sigma) \subset X$ is smooth after removing its intersection with the base locus of V and the singular locus of X.

[9]This condition is automatic when X is reduced and k is perfect by generic smoothness. [TODO]

4.5 Unramified Morphisms

Lemma/Definition 4.5.1. Let k be a field and X be an lft k-scheme.

- Let $x \in X$. The following conditions are equivalent.
  (a) The diagonal $\Delta : X \to X \times_k X$ is a local isomorphism, i.e., open embedding, at $x \in X$.
  (b) We have $\Omega_{X/k,x} = 0$. Equivalently, by Nakayama, $\Omega_{X/k}(x) := \kappa(x) \otimes_{\mathcal{O}_{X,x}} \Omega_{X/k,x} = 0$.
  (c) The point $x \in X$ is an isolated reduced point with $\kappa(x)/k$ a finite separable extension.

  A k-scheme X is said to be *unramified* at $x \in X$ if it is lft and the equivalent conditions (a)-(c) hold.
- The following conditions are equivalent.
  (d) The k-scheme X is unramified at each $x \in X$.
  (e) The diagonal morphism $\Delta : X \to X \times_k X$ is an open embedding.
  (f) We have $\Omega_{X/k} = 0$.
  (g) We have $\dim X = 0$, the scheme X is reduced, and for each $x \in X$, the extension $\kappa(x)/k$ is finite separable.
  (h) The scheme X is the schematic disjoint union of points of the form $\operatorname{Spec} K$ for finite separable extensions K/k.
  (i) For any algebraically (or separably) closed field extension Ω/k, the base-change X_Ω is the schematic disjoint union of points of the form $\operatorname{Spec} \Omega$.
  (j) For every k-algebra B and ideal $I \subset B$ with $I^2 = 0$, the map $X(B) \to X(B/I)$ is injective.

  A k-scheme X is said to be *unramified* over k if it is lft and it satisfies the equivalent conditions (d)-(j).

Proof.

(a) $\Leftrightarrow$ (b) Follows from 3.13 and that $\Omega_{X/k} \cong_{\mathcal{O}_X\text{-}\mathsf{Mod}} \mathcal{N}_\Delta^\vee$.

(b) $\Rightarrow$ (c) For any $x \in X$, we know $\dim_x X \le \dim_{\kappa(x)} \Omega_{X/k}(x)$, so that if $\Omega_{X/k}(x) = 0$, then $\dim_x X = 0$. It then follows from 4.1.1(d), 4.1.8(b), and 4.1.6(a) that $\overline{\{x\}}$ is an irreducible component of X of dimension zero, which along with 2.3.9(a) implies that x is isolated in X, with $\kappa(x)/k$ finite. At this point, we quote 4.5.2 to finish.

(c) $\Rightarrow$ (b) Follows immediately from the fact that if K/k is finite separable, then $\Omega_{K/k} = 0$.

The proof of the equivalence of (d)-(i) is left to the reader.

(f) $\Leftrightarrow$ (j) The question is local on X, so we may reduce to the case of $X = \operatorname{Spec} A$ for A some finitely generated k-algebra. At this point, we finish by quoting the following result from commutative algebra (4.5.3)

■

Lemma 4.5.2. Let k be a field, A finitely generated k-algebra which is also an Artinian local ring. If $\Omega_{A/k} = 0$, then A is a finite separable field extension of k.

Proof. Exercise; c.f. [4, 6.3].[10] ■

Lemma 4.5.3. Let k be a ring and A a k-algebra. Then $\Omega_{A/k} = 0$ iff for every k-algebra B and ideal $I \subset B$ with $I^2 = 0$, for every k-morphism $A \to B/I$, there is at most one lift to a k-morphism $A \to B$.

[10] Unstable ref

Proof. Exercise; c.f. [4, 4.1].[11]

■

Theorem/Definition 4.5.4 (Unramified Morphisms)**.** Let $\pi : X \to Y$ be an lft morphism of schemes.

- Let $x \in X$ with $y := \pi(x) \in Y$. The following conditions are equivalent.
  (a) The diagonal $\Delta_\pi : X \to X \times_Y X$ is a local isomorphism, i.e., open embedding, at $x \in X$.
  (b) We have $\Omega_{\pi,x} = 0$. Equivalently (by Nakayama), we have $\Omega_\pi(x) = 0$.
  (c) The lft $\kappa(y)$-scheme $X_y = \pi^{-1}(y)$ is unramified (4.5.1) at x.
  (d) We have $\mathfrak{m}_{X,x} = \mathfrak{m}_{Y,y}\mathcal{O}_{X,x}$, and $\kappa(x)/\kappa(y)$ is finite separable.

  A morphism $\pi : X \to Y$ is said to be *unramified at* $x \in X$ if π is lft and the equivalent conditions (a)-(d) hold.
- The following conditions are equivalent.
  (e) The morphism π is unramified at each $x \in X$.
  (f) The diagonal morphism $\Delta_\pi : X \to X \times_Y X$ is an open embedding.
  (g) For each $y \in Y$, the $\kappa(y)$-scheme X_y is the schematic disjoint union of points of the form $\operatorname{Spec} K$ for $K/\kappa(y)$ finite separable.
  (h) For each geometric point $\overline{y} \to X$, the $\kappa(\overline{y})$-scheme $X_{\overline{y}}$ is the schematic disjoint union of points $\operatorname{Spec} \kappa(\overline{y})$.
  (i) We have $\Omega_\pi = 0$.
  (j) For all Y-schemes Z and for all closed subschemes $Z_0 \subset Z$ defined by a square-zero ideal sheaf, the map $\mathrm{Hom}_Y(Z, X) \to \mathrm{Hom}_Y(Z_0, X)$ is injective. In other words, given a solid diagram

$$\begin{tikzcd} Z_0 \arrow[r] \arrow[d, hook] & X \arrow[d, "\pi"] \\ Z \arrow[r] \arrow[ur, dashed] & Y, \end{tikzcd} \tag{4.5.5}$$

  there is at most one lift $Z \to X$ making the diagram commute.

  A morphism $\pi : X \to Y$ is said to be *unramified* iff it is lft and the equivalent conditions (e)-(j) hold.

Proof.

(a) $\Leftrightarrow$ (b) Follows from 3.13.

For use in the other implications, we note that the formation of Ω commutes with base change; specifically, $\mathcal{O}_{X_y,x} \cong \mathcal{O}_{X,x}/\mathfrak{m}_{Y,y}\mathcal{O}_{X,x}$ as $\mathcal{O}_{X,x}$-algebras, and

$$\Omega_{X_y/\kappa(y),x} \cong \mathcal{O}_{X_y,x} \otimes_{\mathcal{O}_{X,x}} \Omega_{\pi,x} \cong \Omega_{\pi,x}/\mathfrak{m}_{Y,y}\Omega_{\pi,x} \tag{4.5.6}$$

as $\mathcal{O}_{X_y,x}$-modules.

(b) $\Rightarrow$ (c) Follows immediately from (4.5.6).

(c) $\Rightarrow$ (d) Since $\mathcal{O}_{X_y,x}$ is a field, $\mathfrak{m}_{Y,y}\mathcal{O}_{X,x}$ is maximal and hence $\mathfrak{m}_{Y,y}\mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$. The residue fields of X and X_y at x are the same, so the separability of $\kappa(x)/\kappa(y)$ follows from 4.5.1(c).

(d) $\Rightarrow$ (b) The hypotheses imply that $\mathcal{O}_{X_y,x} \xrightarrow{\sim} \kappa(x)$, so that x is an isolated reduced point of X_y and $\Omega_{X_y/\kappa(y),x} = 0$. It follows from 4.5.6 that $\Omega_{\pi,x} = \mathfrak{m}_{Y,y}\Omega_{\pi,x} = \mathfrak{m}_{X,x}\Omega_{\pi,x}$; at this point, Nakayama's Lemma tells us that $\Omega_{\pi,x} = 0$.

The equivalences (e)-(i) are left as an exercise,, and that of (i) and (j) is a local question which again reduces to 4.5.3. ■

[11] Unstable ref

Remark 4.5.7.

(a) It is clear from the proof that it suffices to check (j) for all affine Z, or even only those Z which are the spectra of local rings (using 1.1). Similarly, it suffices to check those Z which are of finite type over Y.

(b) A morphism $\pi : X \to Y$ satisfying equivalent conditions (i) and (j) (the equivalence of which did not need π to be lft) is said to be *formally unramified*; a morphism is unramified iff it is lft and formally unramified. For instance, for any ring A and multiplicative subset $S \subset A$, the morphism $A \to S^{-1}A$ is formally unramified, but not always of finite type (e.g., consider $k[x] \to k(x)$ for any field k).

4.6 Exercises

Exercise 4.1.

(a) Show that an irreducible topological space of dimension zero is indiscrete, and conversely.
(b) Show that a locally Noetherian sober topological space of dimension zero is discrete, and conversely. Is the result true without either the "locally Noetherian" or "sober" hypotheses?

In particular, for a Noetherian scheme, being zero-dimensional is equivalent to having a finite discrete underlying topological space, and an irreducible Noetherian scheme of dimension zero is the spectrum of an Artinian local ring.

Exercise 4.2. Let X be a nonempty scheme of finite type over a field. Show that the following conditions are equivalent.

(a) We have $\dim X = 0$.
(b) The underlying topological space of X is finite and discrete.
(c) The underlying topological space of X is finite.
(d) There are no curves on X, i.e., there does not exist an irreducible closed subspace of X of dimension one.

Show by counterexample that (c) does not imply (a) for an arbitrary scheme X.

Exercise 4.3.

(a) Give an example of a topological space X and a nonempty open subset $U \subset X$ such that X is equidimensional but $\dim U < \dim X$.
(b) Show that (a) cannot happen if X is (the underlying topological space of) a scheme locally of finite type over some field. In fact, show that, in this case, every nonempty open subset $U \subset X$ is equidimensional of dimension $\dim X$.

Exercise 4.4. Let X be a topological space and $Y \subset X$ a subspace with closure $\overline{Y}$.

(a) Show that $\dim Y \leq \dim \overline{Y}$.
(b) Suppose that X is (the underlying topological space of) a scheme locally of finite type over a field, and $Y \subset X$ a constructible subset. Show that equality holds in (a).
(c) Give examples to show that the result does not hold in general if we drop either hypothesis. Specifically, give an example of a integral Noetherian scheme X and an open subset $Y \subset X$ such that $\dim Y < \dim \overline{Y}$, and of an integral scheme X of finite type over $\mathbb{C}$ and a subset $Y \subset X$ such that $\dim Y < \dim \overline{Y}$.

Exercise 4.5. Let k be a field, $n \in \mathbb{Z}_{\geq 0}$, and $\mathfrak{a} \subset k[x_0, \ldots, x_n]$ a homogeneous ideal. Let $X := \operatorname{Proj} k[x_0, \ldots, x_n]/\mathfrak{a} \subset \mathbb{P}^n_k$ and $CX := \operatorname{Spec} k[x_0, \ldots, x_n]/\mathfrak{a} \subset \mathbb{A}^{n+1}_k$.

(a) Show that every prime of $k[x_0, \ldots, x_n]$ minimal over $\mathfrak{a}$ is homogeneous.
(b) Suppose that X is nonempty. Give a bijection between the following three sets.
  (i) The irreducible components of CX.
  (ii) The irreducible components of X.
  (iii) The primes of $k[x_0, \ldots, x_n]$ minimal over $\mathfrak{a}$.
  What goes wrong when X is empty?
(c) Show that if X is nonempty and pure of dimension $d \in \mathbb{Z}_{\geq 0}$, then the cone CX is also pure of dimension $d + 1$.

Exercise 4.6. Let k be a field, $n \in \mathbb{Z}_{\geq 1}$, and $X \subset \mathbb{P}^n_k$ a closed subscheme of codimension $d \in \mathbb{Z}_{\geq 0}$. Show that the following conditions on X are equivalent.

(a) There is a k-isomorphism $\mathbb{P}^{n-d}_k \xrightarrow{\sim} X$ such that $\mathcal{O}_X(1) := \mathcal{O}_{\mathbb{P}^n_k}(1)|_X$ pulls back to $\mathcal{O}_{\mathbb{P}^{n-d}_k}(1)$.

(b) There are d hyperplanes $H_1, \ldots, H_d \subset \mathbb{P}^n_k$ such that $X = H_1 \cap \cdots \cap H_d$.
(c) The subscheme X is schematically cut out by hyperplanes, i.e., there is some family $H_\lambda \subset \mathbb{P}^n_k$ of hyperplanes such that $X = \bigcap_\lambda H_\lambda$. Equivalently, the saturated ideal $I \subset k[x_0, \ldots, x_n]$ is generated by some elements $h_\lambda \in k[x_0, \ldots, x_n]_1$.
(d) The subscheme X is the schematic intersection of all hyperplanes containing it. Equivalently, the saturated ideal $I \subset k[x_0, \ldots, x_n]$ of X satisfies $I = (I_1)$.

A closed subscheme X of $\mathbb{P}^n_k$ of this form is said to be a *linear subspace of* $\mathbb{P}^n_k$ *of codimension* d.

(e) In the above set-up, suppose that $Y \subset \mathbb{P}^n_k$ is a closed subscheme, and that there is a field extension $k \to k'$ such that the base-change $Y' := Y_{k'} \subset \mathbb{P}^n_{k'}$ is a linear subspace of $\mathbb{P}^n_{k'}$ of codimension d. Show that Y is itself a linear subspace of $\mathbb{P}^n_k$ of codimension d.

The converse to (e) is hopefully clear.

Exercise 4.7. Let k and n be as in 4.6, and let $X \subset \mathbb{P}^n_k$ be a closed subscheme.

(a) Show that there is a unique smallest linear subspace containing X as a closed subscheme. This is called the *(schematic) linear span* of X, and is denoted by $\langle X \rangle$. Show that $\langle X \rangle = \bigcap_{H \supset X} H = \bigcap_{h \in I_1} \mathbb{V}(h)$, where $I = \Gamma_*(\mathcal{I}_X)$, and that X is linear iff $X = \langle X \rangle$. On the other extreme, we say that X is *nondegenerate* iff $\langle X \rangle = \mathbb{P}^n_k$.
(b) Show that if X is reduced, then $\langle X \rangle$ is the smallest linear subspace whose underlying set contains the underlying subset of X in $\mathbb{P}^n_k$.
(c) Show that the schematic linear span of $X = \mathbb{V}(x^2) \subset \mathbb{P}^1_k = \operatorname{Proj} k[x, y]$ is all of $\mathbb{P}^1_k$, i.e., X is nondegenerate. Conclude that the result of (b) is false when X is not required to be reduced.
(d) Show that the formation of linear span commutes with extension of scalars: if $k \to k'$ is a field extension, then $\langle X \rangle_{k'} = \langle X_{k'} \rangle$.

Exercise 4.8. (Adapted from [12, Ex. 11.6]) Let k be a field, $n \in \mathbb{Z}_{\geq 1}$, and $X \subset \mathbb{P}^n_k$ a locally closed subset. Show that if $Z \subset \mathbb{P}^n_k$ is a hypersurface not containing an irreducible component of X, then
$$\dim(Z \cap X) = \dim X - 1.$$

Exercise 4.9. (Adapted from [2, Ex. 3.22].) Let $\pi : X \to Y$ be a dominant morphism of integral schemes of finite type over a field k.

(a) Let $W \subset Y$ be an irreducible closed subset and $Z \subset \pi^{-1}(W)$ an irreducible component dominating W. Show that $\operatorname{codim}_X(Z) \leq \operatorname{codim}_Y(W)$. Show by example that the result is false in general if we do not require Z to dominate W.
(b) Let $r := \dim X - \dim Y = \operatorname{trdeg}_{K(Y)} K(X) \in \mathbb{N}$ denote the *relative dimension* of X over Y. Show that for every $y \in Y$, every irreducible component of the fiber $X_y = \pi^{-1}(y)$ has dimension at least r.
(c) Show that there is a dense open subset $V \subset Y$ such that for all $y \in V$, the fiber X_y is equidimensional of dimension r.
(d) (Upper Semicontinuity of Fiber Dimension) Show that the function $X \to \mathbb{N}$ defined by $x \mapsto \dim_x X_{\pi(x)}$ is upper semicontinuous, i.e., for each $n \in \mathbb{N}$, show that
$$D_n := \{x \in X : \dim_x X_{\pi(x)} \geq n\} \subset X$$
is closed. Show that $D_r = X$.
(e) (Chevalley) For each $n \in \mathbb{N}$, let $C_n := \{y \in Y : \dim X_y = n\}$. Show that each $C_n \subset Y$ is a constructible subset, and that C_r contains an open dense subset of Y.
(f) (Upper Semicontinuity in the Base) Suppose further that π is closed. Show that the function $Y \to \mathbb{N}$ defined by $y \mapsto \dim X_y$ is upper semicontinuous. Show by example that the result is false in general if we do not require π to be closed.

(g) Show that the result of (d) holds when we only assume X and Y to be finite type k-schemes (with no conditions on the k-morphism π); similarly, the result of (f) holds when, in addition, π is closed.

Exercise 4.10. Let k be a field, X and Y be equidimensional lft k-schemes, and $\pi : X \to Y$ a k-morphism of finite type. There is a dense open subset $V \subset Y$ such that for all $y \in V$, the fiber X_y is equidimensional of dimension $\dim X - \dim Y$ over $\kappa(y)$ (or empty).

Exercise 4.11. (Adapted from [1, 12.4.D] and [12, Theorem 11.14]) Let k be a field, X, Y be finite type k-schemes with Y irreducible, and let $\pi : X \to Y$ be a closed k-morphism. Show that if there is an $n \in \mathbb{N}$ such that for every $y \in Y$, the fiber X_y is irreducible of dimension n, then X is irreducible with $\dim X = \dim Y + n$.

Exercise 4.12. (Adapted from [12, Exercise 11.15].) Let k be a field, X, Y, Z be integral finite-type k-schemes, and let $f : X \to Z$ and $g : Y \to Z$ be dominant k-morphisms.

(a) Show that $X \times_Z Y$ is nonempty, and that

$$\dim X \times_Z Y \geq \dim X + \dim Y - \dim Z.$$

(b) Give examples to show that the inequality can be strict, and that $X \times_Z Y$ can, in general, have irreducible components of smaller dimension than the right side, although this cannot happen if $Z = \mathbb{A}^n_k$.

(c) Is the result true when we don't require integrality or dominance in the hypotheses? Produce at least one counterexample to the result without at least one of these hypotheses, and one (nontrivial) situation where these hypotheses do not hold but the result still does. Open ended: can you come up with optimum hypotheses, or equivalently the strongest version of such a result?

The following harder exercise generalizes 4.12.

Exercise 4.13. Let X, Y, Z be pure-dimensional lft k-schemes, and $f : X \to Z$ and $g : Y \to Z$ be k-morphisms. Suppose that Z is smooth over k.

(a) Show that every irreducible component of $X \times_Z Y$ has dimension at least $\dim X + \dim Y - \dim Z$.

(b) Suppose that every fiber of $f : X \to Z$ is equidimensional of dimension $\dim X - \dim Z$; this happens, for instance, when f is flat. Show that $X \times_Z Y$ is equidimensional of dimension $\dim X + \dim Y - \dim Z$.

Exercise 4.14 (Poor Man's Kleiman-Bertini)**.** (Adapted from [1, 21.6.H]) Let k be a field, G be a finite-type group scheme over k, and Z an finite-type k-scheme. Suppose we are given a k-action of G on Z which is transitive on k^{a} points, where $k \to k^{\mathrm{a}}$ is an algebraic closure of k.

(a) Show that G and Z are equidimensional. Show that if Z is generically smooth over k (e.g., if Z is geometrically reduced), then Z is smooth over k. In particular, if G is geometrically reduced, then G is smooth over k.

(b) Suppose now that Z is smooth over k. Suppose X and Y are nonempty equidimensional finite-type k-schemes, and $f : X \to Z$ and $g : Y \to Z$. Show that there is a dense open subset $V \subset G$ such for each $g \in V$, the fiber $gX \times_Z Y$ of the morphism

$$(G \times_k X) \times_Z Y \to G$$

over g is equidimensional of dimension $\dim X + \dim Y - \dim Z$ over $\kappa(g)$ (or empty).

(c) (*) Replace the hypothesis that Z is smooth over k from (b) with the hypothesis that Z is reduced, and obtain the same conclusion as in (b).

Here is some more practice with irreducible components under field extensions.

Exercise 4.15. Let k be a field, X be an lft k-scheme, and K/k a field extension.

(a) Let $d \in \mathbb{Z}_{\geq 0}$. Show that if $\dim X \leq d$, then also $\dim X_K \leq d$.
(b) Suppose that X is irreducible and K/k is algebraic. Show that every irreducible component of X_K maps surjectively onto X.
(c) Suppose that X is pure dimensional. Show that if the irreducible components of X_K are disjoint, then so are those of X. Is the converse true?

Exercise 4.16. Let k be a field and $\pi : X \to Y$ a k-morphism of irreducible smooth k-varieties of the same dimension. Show that if $Z \subset X$ is a positive-dimensional irreducible closed subset such that $\pi(Z)$ is a point, then for all $z \in Z$, the map π is not étale at z.

Exercise 4.17. Let k be a field and let $\pi : X \to Y$ be a morphism of irreducible k-varieties. Show that if π is étale, then X and Y have the same dimension, and the morphism π is generically finite and generically separable.

Exercise 4.18. Let A be a ring, $f \in A[x]$ a polynomial, and $B := A[\alpha] := A[x]/(f)$. Let $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ and $\pi : X \to Y$ the morphism corresponding to $A \to B$. Suppose $p \in X$ corresponds to the prime $\mathfrak{p} \subset B$, and let $q := \pi(p) \in Y$. Show that π is étale at p iff $f'(\alpha) \notin \mathfrak{p}$.

Chapter 5

Divisors and Line Bundles

5.1 Weil Divisors

Definition 5.1.1. Let X be a scheme.

(a) A *prime Weil divisor* on X is a codimension 1 irreducible closed subset (or equivalently codimension 1 integral closed subscheme) of X.
(b) A *Weil divisor* on X is a formal finite $\mathbb{Z}$-linear combination of prime Weil divisors on X, and the group of Weil divisors on X is denoted $\mathrm{Div}(X)$. Thus, an element of $\mathrm{Div}(X)$ is an expression of the form $D := \sum n_Z[Z]$ on X, where $Z \subset X$ are prime divisors and $n_Z \in \mathbb{Z}$ are all but finitely many zero. For such a divisor D, we define its *support* to be the closed subset $|D| := \mathrm{Supp}\, D := \bigcup_{n_Z \neq 0} Z \subset X$.
(c) A Weil divisor $D = \sum n_Z[Z]$ is said to be *effective* iff $n_Z \geq 0$ for all Z. In general, for $D, D' \in \mathrm{Div}(X)$, we write $D \geq D'$ iff $D - D'$ is effective; this defines a partial order on $\mathrm{Div}(X)$ compatible with the abelian group structure.
(d) If $U \subset X$ is an open subscheme, then for each prime Weil divisor Z, either $Z \cap U = \emptyset$, or $Z \cap U \subset U$ is a prime Weil divisor. This defines a restriction morphism $\mathrm{Div}(X) \to \mathrm{Div}(U)$, denoted by $D \mapsto D|_U$. The assignment $U \mapsto \mathrm{Div}(U)$ on X defines a separated presheaf, the sheafification of which we denote by $\underline{\mathrm{Div}}_X$; this is the *sheaf of Weil divisors*. When X is Noetherian, then in fact the presheaf above is already a sheaf and no sheafification is necessary.[1]

For the rest of this section, we adopt the

Convention 5.1.2. (in §5.1 and §5.2.2). We let X be a Noetherian, integral, and R_1 (e.g., normal) scheme with function field $K = K(X)$.

In this setting, if $Y \subset X$ is a prime Weil divisor, then $\mathcal{O}_{X,Y}$ is a discrete valuation ring of K, the discrete valuation $K^\times \to \mathbb{Z}$ associated to which we denote by ord_Z or v_Z.

Lemma/Definition 5.1.3. Let X and K be as in 5.1.2.

(a) Let $f \in K^\times$. There are only finitely many prime Weil divisors Z such that $\mathrm{ord}_Z(f) \neq 0$. We define the *divisor of* f, denoted $\mathrm{div}(f)$ or $\mathrm{div}_X(f)$, to be

$$\mathrm{div}(f) := \sum_Z \mathrm{ord}_Z(f)[Z] \in \mathrm{Div}(X).$$

(b) The map $\mathrm{div} : K^\times \to \mathrm{Div}(X)$ is a homomorphism, and we define the *class group of* X, denoted $\mathrm{Cl}(X)$ to be the cokernel of div, i.e., by the exact sequence

$$K^\times \xrightarrow{\mathrm{div}} \mathrm{Div}(X) \to \mathrm{Cl}(X) \to 0.$$

Proof. It suffices to show (a), for which it suffices to show that if A is a Noetherian domain and $f \in A \smallsetminus \{0\}$, then there are only finitely many height one primes of A containing f, but a height one prime of A containing f corresponds to some minimal prime of the Noetherian ring $A/(f)$.[2] ■

In classical terminology, the image $\mathrm{div}(K^\times) \subset \mathrm{Div}(X)$ is the subgroup of *principal Weil divisors*, two Weil divisors $D, D' \in \mathrm{Div}(X)$ are *linearly equivalent*, sometimes denoted

[1] In general, the global sections $\Gamma(X, \underline{\mathrm{Div}}_X)$ correspond to formal sums $\sum_Z n_Z[Z]$ where the collection $\{Z : n_Z \neq 0\}$ is locally finite. This is the right definition of a Weil divisor in more general contexts [TOCITE Stacks] than we will have use for.

[2] Conversely, every prime of A corresponding to a minimal prime of $A/(f)$ has height one, by Krull's theorem [TODO], although this is not needed for the proof of this result.

$D \sim D'$, iff their difference is principal, and the group $\mathrm{Cl}(X)$ is the group of Weil divisors modulo linear equivalence.

Remark 5.1.4. When X is normal, algebraic Hartogs's Lemma [TODO] tells us that the natural morphism $\mathcal{O}(X)^\times \hookrightarrow \ker \operatorname{div}$ is an isomorphism, producing the exact sequence

$$0 \to \mathcal{O}(X)^\times \to K^\times \xrightarrow{\operatorname{div}} \mathrm{Div}(X) \to \mathrm{Cl}(X) \to 0.$$

Let's now give many examples.

Example 5.1.5. Let A be a Dedekind domain and $X = \operatorname{Spec} A$. Then $\mathrm{Div}(X)$ is the free abelian group on the nonzero prime ideals of A, and $\mathrm{Cl}(X)$ is the ideal class group of A.

Example 5.1.6. Let A be a Noetherian domain and $X = \operatorname{Spec} A$. Then A is a UFD iff X is normal and $\mathrm{Cl}(X) = 0$. This follows from the fact that a Noetherian domain is a UFD iff every height 1 prime is principal. For instance, for any $n \geq 0$ and UFD A such as $A = \mathbb{Z}$ or $A = k$ a field, $\mathrm{Cl}(\mathbb{A}^n_A) = 0$.

Example 5.1.7. Let k be a field, $n \in \mathbb{Z}_{\geq 1}$, and $X = \mathbb{P}^n_k$. Then

$$0 \to k^\times \to K^\times \xrightarrow{\operatorname{div}} \mathrm{Div}(X) \xrightarrow{\deg} \mathbb{Z} \to 0,$$

so we get an isomorphism $\deg : \mathrm{Cl}(X) \xrightarrow{\sim} \mathbb{Z}$.

To produce more examples, we use

Lemma 5.1.8. Let $Z \subset X$ be a proper closed subset and $U := X \smallsetminus Z$. Write the irreducible decomposition of Z as $Z = Z_1 \cup \cdots \cup Z_r \cup Z_{r+1} \cup \cdots \cup Z_n$ for some $r, n \in \mathbb{Z}_{\geq 0}$ with $0 \leq r \leq n$, ordered so that $\operatorname{codim}_X(Z_i) = 1$ for $1 \leq i \leq r$ and $\operatorname{codim}_X(Z_i) \geq 2$ for $r+1 \leq i \leq n$. Then we have a commutative diagram with exact rows and columns as below.

$$\begin{tikzcd}
 & 0 & 0 & \\
K^\times \arrow[r, "\operatorname{div}_U"] & \mathrm{Div}(U) \arrow[u] \arrow[r] & \mathrm{Cl}(U) \arrow[u] \arrow[r] & 0 \\
K^\times \arrow[u, "="] \arrow[r, "\operatorname{div}"] & \mathrm{Div}(X) \arrow[u] \arrow[r] & \mathrm{Cl}(X) \arrow[u] \arrow[r] & 0 \\
 & \bigoplus_{i=1}^r \mathbb{Z}[Z_i] \arrow[u] \arrow[r, "="] & \bigoplus_{i=1}^r \mathbb{Z}[Z_i] \arrow[u] & \\
 & 0 \arrow[u] & &
\end{tikzcd}$$

In particular:

(a) If $\operatorname{codim}_X(Z) \geq 2$, then $\mathrm{Div}(X) \xrightarrow{\sim} \mathrm{Div}(U)$ and $\mathrm{Cl}(X) \xrightarrow{\sim} \mathrm{Cl}(U)$.
(b) If Z is irreducible of codimension 1, then the kernel of $\mathrm{Cl}(X)$ is cyclic and generated by $[Z]$.

Further, there is natural isomorphism between the kernel of $\bigoplus_{i=1}^r \mathbb{Z}[Z_i] \to \mathrm{Cl}(X)$ and $\ker \operatorname{div}_U / \ker \operatorname{div}$; in particular, if X is normal, then this gives us an exact sequence of abelian groups of the form

$$0 \to \mathcal{O}(U)^\times / \mathcal{O}(X)^\times \to \bigoplus_{i=1}^r \mathbb{Z}[Z_i] \to \mathrm{Cl}(X) \to \mathrm{Cl}(U) \to 0.$$

In principle, this gives us a way to compute the relations between any finite collection of prime divisors on X in the class group $\mathrm{Cl}(X)$, at least when X is normal.

Proof. Exercise. ∎

Example 5.1.9. Let k be a field, $n \in \mathbb{Z}_{\geq 0}$, and $U \subset \mathbb{A}^n_k$ a (nonempty) open subset. Then $\mathrm{Cl}(U) = 0$. In particular, if $U \subset \mathbb{A}^n_k$ is any (nonempty) open affine subscheme, then $\mathcal{O}(U)$ is a UFD.[3]

Example 5.1.10. Let k be a field.

(a) Let $A := k[x(x-1), x^2(x-1), xy, y] = \{f \in k[x,y] : f(0,0) = f(1,0)\} \subset k[x,y]$. Then $\operatorname{Spec} A$ is the result of collapsing the subscheme $\mathbb{V}(x^2-1, y) \subset \mathbb{A}^2_k$, and is the "plane with two points glued." Equivalently, $A \cong k[w,x,y,z]/(wy - xz, wz^2 - y^2 + yz, w^2z - xy + xd, w^3 + wx - x^2, xz^3 - y^3 + y^2z)$.

(b) Let $A := k[x^3, x^2, xy, y] = \{f \in k[x,y] : (\partial f/\partial x)|_{(0,0)} = 0\} \subset k[x,y]$. Then $\operatorname{Spec} A$ is the result of collapsing the subscheme $\mathbb{V}(x^2, y) \subset \mathbb{A}^2_k$, and is the "plane pinched at a point." Equivalently, $A \cong k[w,x,y,z]/(wz - xy, xz^2 - y^2, x^2z - wy, x^3 - w^2, wz^3 - y^3)$.

In both cases, $X = \operatorname{Spec} A$ is a two-dimensional Noetherian integral R_1 k-scheme with normalization $\mathbb{A}^2_k \to X$ given by the inclusion map $A \subset k[x,y]$. This normalization is in fact an isomorphism away from codimension two, and hence 5.1.8(a) tells us that in both cases $\mathrm{Cl}(X) = 0$. Note that in both cases, A is not normal and hence not a UFD; c.f. 5.1.6.

Example 5.1.11. Let k is a field, and $n \in \mathbb{Z}_{\geq 1}$, and $Z \subset \mathbb{P}^n_k$ be a (nonempty closed) hypersurface with irreducible components of degrees d_i, with $d := \gcd(d_i) \in \mathbb{Z}_{\geq 1}$. Then $\deg : \mathrm{Cl}(\mathbb{P}^n_k \setminus Z) \xrightarrow{\sim} \mathbb{Z}/d\mathbb{Z}$. In particular, if $d \geq 2$, then the coordinate ring of the affine scheme $\mathbb{P}^n_k \setminus Z$ is not a UFD.[4]

Example 5.1.12. We will show geometrically that in the ring $A := \mathbb{R}[x,y]/(x^2+y^2-1)$, the elements x and $1 \pm y$ are all pairwise non-associate irreducible elements, from which the factorization $x^2 = (1+y)(1-y)$ exhibits that A is not a UFD. Note that $\operatorname{div}(x) = [(0,1)] + [(0,-1)]$.

Example 5.1.13. Let k be a field and $A := k[x,y,z]/(xz - y^2)$ and $X = \operatorname{Spec} A$. This is Noetherian, integral, and normal (check!), of dimension 2. Let $Z := \mathbb{V}(x,y) \subset X$. Then $\operatorname{div}(x) = 2[Z]$ and Z is not principal, showing that $\mathrm{Cl}(X) \cong \mathbb{Z}/2$ generated by $[Z]$. In particular, A is not a UFD.

To produce even more examples, we next note

Lemma 5.1.14. If X is as in 5.1.2, then for any $n \in \mathbb{Z}_{\geq 0}$, so are $\mathbb{A}^n_X = X \times \mathbb{A}^n$ and $\mathbb{P}^n_X = X \times \mathbb{P}^n$.

Proof. The key point is showing that $X \times \mathbb{A}^1$ is R_1. Let $\pi : X \times \mathbb{A}^1 \to X$ be the projection map. If $z \in X \times \mathbb{A}^1$ is a point of codimension 1, then its image $x := \pi(z) \in X$ has codimension either 0 or 1 according to whether or not $\overline{\{z\}}$ dominates X. If $\operatorname{codim}_X(x) = 0$, then x is the generic point of X, so $\mathcal{O}_{X\times\mathbb{A}^1,z}$ is a DVR since $\mathbb{A}^1_{K(X)}$ is R_1. If $\operatorname{codim}_X(x) = 1$, then $z \in \pi^{-1}(\overline{\{x\}})$ is the generic point; then $\mathcal{O}_{X,x}$ is a DVR since X is R_1, and hence so is $\mathcal{O}_{X\times\mathbb{A}^1,z} \cong \mathcal{O}_{X,x}[t]_{\mathfrak{m}_{X,x}[t]}$. ∎

In what follows, we call a codimension one point $z \in X$ or the corresponding prime divisor $Z = \overline{\{z\}} \subset X$ of type 0 or type 1 according to whether or not Z dominates X via π respectively. Note that when z is of type 1, then a uniformizer at the projected point $x := \pi(z)$ maps to a uniformizer at z under the natural inclusion $\mathcal{O}_{X,x} \hookrightarrow \mathcal{O}_{X\times\mathbb{A}^1,z}$.

[3]This is surprising! We don't have much control over arbitrary open affine subschemes of $\mathbb{A}^n_k$.

[4]This immediately solves [1, 5.4.N].

Example 5.1.15. Let X be as in 5.1.2. Then for any $n \in \mathbb{Z}_{\geq 0}$, we have $\mathrm{Cl}(X) \xrightarrow{\sim} \mathrm{Cl}(X \times \mathbb{A}^n)$, induced by the "pullback" map. It suffices to do the case $n = 1$; fix an identification of $K(t)$ with the function field of $X \times \mathbb{A}^1$. For each prime divisor $Y \subset X$, the pullback $Y \times \mathbb{A}^1 \cong \pi^{-1}(Y) \subset X \times \mathbb{A}^1$ is a prime divisor of type 1. This gives us a commutative diagram with exact rows as below.

$$\begin{tikzcd}
K(t)^\times \arrow[r, "\mathrm{div}"] & \mathrm{Div}(X \times \mathbb{A}^1) \arrow[r] & \mathrm{Cl}(X \times \mathbb{A}^1) \arrow[r] & 0 \\
K^\times \arrow[r, "\mathrm{div}"] \arrow[u, hook] & \mathrm{Div}(X) \arrow[r] \arrow[u, "\pi^*"] & \mathrm{Cl}(X) \arrow[r] \arrow[u, dashed] & 0
\end{tikzcd}$$

We claim that the rightmost vertical map is an isomorphism. To show injectivity, it suffices to note that if $f \in K(t)^\times \smallsetminus K^\times$ is any element, then there is a type 0 prime divisor $Z \subset X \times \mathbb{A}^1$ (e.g., the vanishing locus of an irreducible factor of the numerator or denominator of f in $K[t]$) such that $\mathrm{ord}_Z(f) \neq 0$.

To show surjectivity, it suffices to show that if $Z \subset X \times \mathbb{A}^1$ is any prime divisor of type 0, then $[Z]$ is in the image of $\mathrm{Cl}(X)$. Now Z corresponds to a nonzero prime of $K[t]$, so $Z = \mathbb{V}(g)$ for some irreducible polynomial $g \in K[t]$. Then $\mathrm{div}(g)$ cannot contain any other type 0 prime divisors besides Z, i.e., $\mathrm{div}(g) = [Z] + \sum_i n_i[Y_i \times \mathbb{A}^1]$ for some prime divisors $Y_i \subset X$ and integers $n_i \in \mathbb{Z}$. This says precisely that $[Z]$ is in the image of $\mathrm{Cl}(X)$, as desired.

Example 5.1.16. Let X be a Noetherian, integral, and *normal* scheme. For any $m \in \mathbb{Z}_{\geq 1}$ and integers $n_1, \ldots, n_m \in \mathbb{Z}_{\geq 1}$, we show that

$$\mathrm{Cl}\left(X \times \prod_{i=1}^m \mathbb{P}^{n_i}\right) \cong \mathrm{Cl}(X) \oplus \bigoplus_{i=1}^m \mathbb{Z}.$$

Clearly, it suffices to do the case $m = 1$; let $n := n_1$. Applying (5.1.8) to the prime divisor $X \times \mathbb{P}^{n-1} \subset X \times \mathbb{P}^n$ gives us an exact sequence of the form

$$0 \to \mathcal{O}(X \times \mathbb{P}^{n-1})^\times / \mathcal{O}(X)^\times \to \mathbb{Z}[X \times \mathbb{P}^{n-1}] \to \mathrm{Cl}(X \times \mathbb{P}^n) \to \mathrm{Cl}(X \times \mathbb{A}^n) \to 0.$$

Clearly $\mathcal{O}(X)^\times \xrightarrow{\sim} \mathcal{O}(X \times \mathbb{P}^{n-1})^\times$, so the class $[X \times \mathbb{P}^{n-1}] \in \mathrm{Cl}(X \times \mathbb{P}^n)$ is non-torsion. Finally, the "flat pullback" $\mathrm{Cl}(X) \to \mathrm{Cl}(X \times \mathbb{P}^n)$, defined analogously to 5.1.15, composed with the restriction $\mathrm{Cl}(X \times \mathbb{P}^n) \to \mathrm{Cl}(X \times \mathbb{A}^n)$ is an isomorphism, again by 5.1.15. This gives us a splitting of the above exact sequence $\mathrm{Cl}(X \times \mathbb{P}^n) \cong \mathrm{Cl}(X) \oplus \mathbb{Z}$ as desired.

Unraveling the proof gives us explicit generators: the inclusion of the $\mathrm{Cl}(X)$ factor is again given by flat pullback, whereas a generator of the i^{th} copy of $\mathbb{Z}$ is given by the class of the prime divisor obtained by replacing $\mathbb{P}^{n_i}$ with a hyperplane $\mathbb{P}^{n_i - 1} \subset \mathbb{P}^{n_i}$ in the product $X \times \prod_{i=1}^m \mathbb{P}^{n_i}$.

Example 5.1.17. Let k be a field. Then $\mathbb{P}^2_k$ and $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$ are birational (geometrically) irreducible smooth projective surfaces over k which are not isomorphic, since their class groups are different. This is in contrast with the case of curves: birational (geometrically) irreducible smooth projective curves over k are indeed isomorphic.

Example 5.1.18. Let k be a field and $Q = \mathbb{V}(wz - xy) \subset \mathbb{P}^3_k$ be the quadric surface. Compute $\mathrm{Cl}(Q)$ and use this to show that the twisted cubic C is not (even a set-theoretic) complete intersection of Q with a surface surface $Y \subset \mathbb{P}^3_k$.

5.2 Cartier Divisors I

The notion of a Weil divisor and the divisor class group is beautiful, easy to define, and very useful, but it can only be defined on a very small class of schemes, namely those which are Noetherian, integral, and R_1. A related notion which works in more general settings is that of Cartier divisors and the Cartier class group. While this approach has the advantage of being more generally definable, it is not identical with the Weil divisor case. However, the notion of a Weil divisor is "homological", whereas that of a Cartier divisor is "cohomological"; therefore, we would expect that under exceptionally nice circumstances, these two to coincide, and that is indeed the case.

In this section, we define meromorphic functions and Cartier divisors on integral schemes, and explain the connection to line bundles (and their sections). Then, we clarify the relationship between Cartier divisors and line bundles on the one hand and Weil divisors on the other, and asking when the natural map $\mathrm{CaCl}(X) \to \mathrm{Cl}(X)$ is injective and surjective. We will explore the notion of meromorphic functions and Cartier divisors on more general (locally Noetherian) schemes in §5.3.

5.2.1 Cartier Divisors on Integral Schemes

Convention 5.2.1. In §5.2.1, we let X be an integral scheme with function field $K = K(X)$.

Definition 5.2.2. Let X and K be as in 5.2.1.

(a) We define the *sheaf of meromorphic functions* on X, denoted $\mathcal{K}_X$, to be the constant sheaf on X with value K.

This sheaf $\mathcal{K}_X$ is naturally a quasicoherent sheaf of $\mathcal{O}_X$-algebras with the structure morphism $\mathcal{O}_X \to \mathcal{K}_X$ an injection. Moreover, for each $x \in X$, the induced map on stalks $\mathcal{O}_{X,x} \to \mathcal{K}_{X,x} = K$ amounts to taking the fraction field.

Recall that if $\mathcal{R}$ is a sheaf of rings on a topological space X, then the assignment $U \mapsto \mathcal{R}(U)^\times$ defines a sheaf of abelian groups on X, denoted by $\mathcal{R}^\times$, and this construction is functorial in $\mathcal{R}$. In particular, we have a monomorphism of abelian sheaves $\mathcal{O}_X^\times \hookrightarrow \mathcal{K}_X^\times$ on X.

(b) Define the *sheaf of Cartier divisors* on X to be the quotient abelian sheaf $\mathcal{K}_X^\times/\mathcal{O}_X^\times$. We denote the global sections of this sheaf by $\mathrm{CaDiv}(X) := \Gamma(X, \mathcal{K}_X^\times/\mathcal{O}_X^\times)$ and call its elements *Cartier divisors* on X.

We therefore have the following explicit definition of $\mathrm{CaDiv}(X)$: a Cartier divisor is represented by a collection (U_α, f_α) where the U_α form an open cover of X, and for each α we have a $f_\alpha \in K^\times$ such that for all α, β we have $f_\beta \cdot f_\alpha^{-1} \in \mathcal{O}(U_\alpha \cap U_\beta)^\times$. Another such collection (V_β, g_β) represents the same Cartier divisor iff for all α, β we have $g_\beta \cdot f_\alpha^{-1} \in \mathcal{O}(U_\alpha \cap V_\beta)^\times$, and the product of two Cartier divisors represented by (U_α, f_α) and (V_β, g_β) is represented by $(U_\alpha \cap V_\beta, f_\alpha g_\beta)$.

(c) Taking global sections of the morphism $\mathcal{K}_X^\times \twoheadrightarrow \mathcal{K}_X^\times/\mathcal{O}_X^\times$ defines an abelian group morphism $K^\times \to \mathrm{CaDiv}(X)$. We define the *Cartier class group* of X, denoted $\mathrm{CaCl}(X)$, to be the cokernel of this morphism. In particular, by definition, there is a defining exact sequence

$$0 \to \mathcal{O}(X)^\times \to K^\times \to \mathrm{CaDiv}(X) \to \mathrm{CaCl}(X) \to 0.$$

(d) A Cartier divisor D is said to be *effective* iff for some (equiv. any) representative (U_α, f_α) of it, we have for all α that $f_\alpha \in \mathcal{O}(U_\alpha)$. In general, for $D, D' \in \mathrm{CaDiv}(X)$, we write $D \geq D'$ iff $D - D'$ is effective; this defines a partial order on $\mathrm{CaDiv}(X)$ compatible with the abelian group structure.

In classical terminology, the image of $K^\times$ in $\mathrm{CaDiv}(X)$ is the subgroup of *principal Cartier divisors*, two Cartier divisors $D, D' \in \mathrm{CaDiv}(X)$ are *linearly equivalent*, sometimes denoted $D \sim D'$, iff their difference is principal, and the group $\mathrm{CaCl}(X)$ is the group of Cartier divisors modulo linear equivalence.

Next, we relate Cartier divisors on an integral scheme to line bundles. For this, let $\widetilde{\mathrm{Pic}}(X)$ denote the group consisting of isomorphism classes of pairs $(\mathcal{L}, \sigma)$ where $\mathcal{L}$ is a line bundle on X and σ a nonzero rational section of $\mathcal{L}$, with group operation given by tensor product. This is an extension of $\mathrm{Pic}(X)$ by $K^\times/\mathcal{O}(X)^\times$ in the sense that there is a short exact sequence of the form

$$0 \to \mathcal{O}(X)^\times \to K^\times \to \widetilde{\mathrm{Pic}}(X) \to \mathrm{Pic}(X) \to 0.$$

This looks suspiciously similar to the exact sequence above. This is not a coincidence.

Theorem 5.2.3. Let X and K be as in 5.2.1. There is a natural isomorphism $\mathrm{CaDiv}(X) \to \widetilde{\mathrm{Pic}}(X)$ inducing an isomorphism of the short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}(X)^\times \arrow[r] \arrow[d, "="] & K^\times \arrow[r] \arrow[d, "="] & \mathrm{CaDiv}(X) \arrow[r] \arrow[d, "\cong"] & \mathrm{CaCl}(X) \arrow[r] \arrow[d, "\cong"] & 0 \\
0 \arrow[r] & \mathcal{O}(X)^\times \arrow[r] & K^\times \arrow[r] & \widetilde{\mathrm{Pic}}(X) \arrow[r] & \mathrm{Pic}(X) \arrow[r] & 0
\end{tikzcd}$$

Proof. Let D be a Cartier divisor, and pick a representative (U_α, f_α) for it. Define the corresponding line bundle $\mathcal{O}_X(D)$ to be trivial on each U_α with generating section f_α^{-1}, i.e., $\mathcal{O}_X(D)(U_\alpha) := f_\alpha^{-1}\mathcal{O}(U_\alpha) \subset K$. By the condition that for all α, β we have $f_\beta \cdot f_\alpha^{-1} \in \mathcal{O}(U_\alpha \cap U_\beta)^\times$, these agree on the overlaps and hence glue to give a line bundle on X.[5] This line bundle comes equipped with the choice of an inclusion $\mathcal{O}_X(D) \hookrightarrow \mathcal{K}_X$ of quasicoherent sheaves, and the element $1 \in K = \Gamma(X, \mathcal{K}_X)$ defines a nonzero rational section of $\mathcal{O}_X(D)$.[6] It is easy to see that the isomorphism class of the pair $(\mathcal{O}_X(D), 1)$ is independent of the choice of representative (U_α, f_α), and this construction defines a homomorphism $\mathrm{CaDiv}(X) \to \widetilde{\mathrm{Pic}}(X)$.

For injectivity, suppose we are given a $D \in \mathrm{CaDiv}(X)$, and that we can pick an isomorphism $\varphi : \mathcal{O}_X \to \mathcal{O}_X(D)$ of line bundles sending the rational (even regular) section $1 \in \mathcal{O}_X(X)$ to the rational section 1 of $\mathcal{O}_X(D)$. It would then follow that for each α that $f_\alpha \in \mathcal{O}(U_\alpha)$, and further the endomorphism of $\mathcal{O}_X(U_\alpha)$ given by multiplication by f_α is an isomorphism, i.e., that $f_\alpha \in \mathcal{O}_X(U_\alpha)^\times$. But this says exactly that the Cartier divisor D is also represented by $(X, 1)$, i.e., is trivial.

By construction, it is clear that the image of $\mathrm{CaCl}(X) \to \mathrm{Pic}(X)$ consists of the isomorphism classes of precisely those line bundles $\mathcal{L}$ which can be embedded into $\mathcal{K}_X$, i.e., admit a monomorphism of (quasicoherent) $\mathcal{O}_X$-modules $\mathcal{L} \hookrightarrow \mathcal{K}_X$. Indeed, given such an $\mathcal{L} \hookrightarrow \mathcal{K}_X$, pick a trivialization $(U_\alpha, \varphi_\alpha)$ of $\mathcal{L}$, and for each α let $f_\alpha \in K^\times$ be the image of 1 under the composite $\mathcal{O}(U_\alpha) \xrightarrow{\varphi_\alpha^{-1}} \mathcal{L}(U_\alpha) \hookrightarrow \mathcal{K}_X(U_\alpha) = K$. Then the Cartier divisor represented by (U_α, f_α) maps to (the isomorphism class of) $\mathcal{L}$. To finish the proof, therefore, it suffices to show that when X is integral, *every* line bundle $\mathcal{L}$ admits such a monomorphism. Since $\mathcal{O}_X \hookrightarrow \mathcal{K}_X$ and $\mathcal{L}$ is locally free, it suffices to show that $\mathcal{K}_X \otimes_{\mathcal{O}_X} \mathcal{L} \cong_{\mathcal{O}_X\text{-}\mathsf{Mod}} \mathcal{K}_X$.

[5] Equivalently, this is the line bundle represented by the Čech cocycle corresponding to the open cover U_α with transition functions $T_{\alpha\beta} = f_\beta \cdot f_\alpha^{-1}$.

[6] Equivalently, this is the section which on the chosen trivialization on U_α is defined by the rational function f_α; this is necessarily regular on an open dense subset.

For this, pick a trivialization $(U_\alpha, \varphi_\alpha)$ of $\mathcal{L}$ with each U_α nonempty and corresponding transition functions $T_{\alpha\beta} \in \mathcal{O}(U_\alpha \cap U_\beta)^\times \subset K^\times$, with $T_{\alpha\beta}$ corresponding to $\varphi_\beta \circ \varphi_\alpha^{-1}$. Fix an α. We define an isomorphism $\Phi : \mathcal{K}_X \otimes_{\mathcal{O}_X} \mathcal{L} \xrightarrow{\sim} \mathcal{K}_X$ of $\mathcal{O}_X$-modules as follows. For each β, we have an isomorphism $\varphi_\beta : \mathcal{L}|_{U_\beta} \xrightarrow{\sim} \mathcal{O}_{U_\beta}$, which by quasicoherence induces an isomorphism $1_{\mathcal{K}} \otimes \varphi_\beta : (\mathcal{K}_X \otimes_{\mathcal{O}_X} \mathcal{L})|_{U_\beta} \xrightarrow{\sim} \mathcal{K}_X|_{U_\beta}$. Define $\Phi|_{U_\beta} := T_{\alpha\beta}^{-1} \circ (1_{\mathcal{K}} \otimes \varphi_\beta)$, where here we think of (multiplication by) $T_{\alpha\beta}^{-1}$ as defining an isomorphism $\mathcal{K}_X|_{U_\beta} \xrightarrow{\sim} \mathcal{K}_X|_{U_\beta}$. It is easy to check–using precisely the cocycle condition on the transition functions–that this glues to define a global $\mathcal{O}_X$-module morphism $\Phi : \mathcal{K}_X \otimes_{\mathcal{O}_X} \mathcal{L} \to \mathcal{K}_X$, which is then an isomorphism because it is so locally. ■

It is clear that under this correspondence, for $D, D' \in \mathrm{CaDiv}(X)$ we have that $D \geq D'$ iff $\mathcal{O}_X(D) \supset \mathcal{O}_X(D')$ as $\mathcal{O}_X$-submodules of $\mathcal{K}_X$. In this way, we get a bijective correspondence between effective Cartier divisors D and invertible *ideal* sheaves $\mathcal{O}_X(-D)$, and so effective Cartier divisors give rise to closed subschemes $D \mapsto \mathbb{V}(\mathcal{O}_X(-D))$, often again denoted by D, and the corresponding ideal sheaf sequence

$$0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0.$$

These subschemes have the property that they are locally principal, and further the ideal sheaf is generated locally near every point by a nonzerodivisor. Conversely, if $Z \subset X$ is a closed subscheme with this property, then its ideal sheaf $\mathcal{I}_Z \subset \mathcal{O}_X$ is invertible, i.e., a line bundle. This gives us another way to think about effective Cartier divisors: these are closed subschemes such that the ideal sheaf is locally generated by a single nonzerodivisor.[7]

5.2.2 Cartier Divisors vs. Line Bundles vs. Weil Divisors

In this section, we show how to relate Cartier and Weil divisors in the setting where we have defined them both. Therefore, as in 5.1 we make invoke convention 5.1.2: X is a Noetherian, integral, R_1 scheme with function field $K = K(X)$.

Theorem 5.2.4. Let X and K be as in 5.1.2. There is a commutative diagram of abelian groups with exact rows of the form below.

$$\begin{tikzcd}
 & K^\times/\mathcal{O}(X)^\times \arrow[r, "\mathrm{div}"] & \mathrm{Div}(X) \arrow[r] & \mathrm{Cl}(X) \arrow[r] & 0 \\
0 \arrow[r] & K^\times/\mathcal{O}(X)^\times \arrow[r] \arrow[u, "="] \arrow[d, "="'] & \mathrm{CaDiv}(X) \arrow[r] \arrow[u] \arrow[d, "\cong"'] & \mathrm{CaCl}(X) \arrow[r] \arrow[u] \arrow[d, "\cong"'] & 0 \\
0 \arrow[r] & K^\times/\mathcal{O}(X)^\times \arrow[r] & \widetilde{\mathrm{Pic}}(X) \arrow[r] & \mathrm{Pic}(X) \arrow[r] & 0.
\end{tikzcd}$$

Proof. We define a group morphism $\mathrm{CaDiv}(X) \to \mathrm{Div}(X)$ making the top left square commute. For this, let $D \in \mathrm{CaDiv}(X)$ be a Cartier divisor, and $Z \subset X$ a prime Weil divisor. Pick a representative (U_α, f_α) for D, and pick a U_α intersecting Z (or equivalently containing the generic point of Z), and define $\mathrm{ord}_Z(D) := \mathrm{ord}_Z(f_\alpha)$. Then $\mathrm{ord}_Z(D)$ is independent of the choice of U_α and the representative (U_α, f_α). Further, as in 5.1.3, for a fixed $D \in \mathrm{CaDiv}(X)$, we have $\mathrm{ord}_Z(D) \neq 0$ only for finitely many Z, and we define the Weil divisor corresponding to D to be $\sum_Z \mathrm{ord}_Z(D)[Z]$. This is the required morphism. ■

[7]Of course, in the integral case, being a nonzerodivisor is the same thing as being nonzero. The reason for stating the result in this way is that this generalizes to more general (locally Noetherian schemes) in 5.3.

In general, the resulting morphism $\mathrm{CaCl}(X) \to \mathrm{Cl}(X)$ need not be either injective or surjective.

Example 5.2.5. Fix a field k. In (a) (resp. (b)) below, we construct a k-scheme X with function field K as in 5.1.2 such that the morphism $\mathrm{CaCl}(X) \to \mathrm{Cl}(X)$ of 5.2.4 is not injective (resp. surjective).

(a) Let $A := k[x^3, x^2, xy, y]$ and $X = \operatorname{Spec} A$ be as in 5.1.10(b). It suffices to show that $\mathrm{CaCl}(X) \neq 0$. Fix a $\lambda \in k$, and consider the $D_\lambda \in \mathrm{CaDiv}(X)$ represented by

$$\{(D(x^2), 1), (D(\lambda^2 x^2 - 1), 1 + \lambda x)\}.$$

We claim that if $\lambda \in k^\times$, then $0 \neq [D_\lambda] \in \mathrm{CaCl}(X)$. Indeed, if $[D_\lambda] = 0$, then there is an $f \in \mathcal{O}(D(x^2))^\times$ with $f(1 + \lambda x)^{-1} \in \mathcal{O}(D(\lambda^2 x^2 - 1))^\times$. Now $\mathcal{O}(D(x^2))^\times = k[x^{\pm 1}, y]^\times = k^\times x^{\mathbb{Z}}$, so up to scaling by an element of $k^\times$, we can then assume $f = x^n$ for some $n \in \mathbb{Z}$. Then $x^n(1 + \lambda x)^{-1} \in \mathcal{O}(D(\lambda^2 x^2 - 1))$ forces $n \geq 0$, and indeed if $\lambda \neq 0$ then $n \geq 1$. But such an element can never be invertible, since it vanishes at the pinch point (x^3, x^2, xy, y).[8] A similar example can be given with A as in 5.1.10(a) instead (5.1).

(b) Let $A := k[x, y, z]/(xz - y^2)$ and $X = \operatorname{Spec} A$ be as in 5.1.13, where we showed that $\mathrm{Cl}(X) \cong \mathbb{Z}/2$, generated by the class of $Z := \mathbb{V}(x, y)$. We show that $[Z]$ is not in the image of $\mathrm{CaCl}(X) \to \mathrm{Cl}(X)$. Indeed, if it were, then the divisor Z is locally principal at the origin, i.e., there is an open neighborhood U of the vertex $p = \mathbb{V}(x, z)$ and an $f \in K^\times$ such that $[Z]|_U = \mathrm{div}_U(f)$. But this implies in turn that the ideal $\mathcal{I}_{Z,p} \subset \mathcal{O}_{X,p}$ is principal and generated by f; but in fact, this is the ideal $(x, y) \subset A_{(x,z)}$ which is *not* principal, for instance by considerations involving the Zariski tangent space or by noting that we may use the grading of this ring to our advantage (exercise!). In fact, this observation combined with 5.2.6(a) shows that $\mathrm{Pic}(X) \cong \mathrm{CaCl}(X)$ is trivial.

We see what went wrong in the above examples: in (a), we have a Cartier divisor, not linearly equivalent to zero, which is represented by local functions (i.e., defining equations) which have no zeroes or poles; this cannot happen if X is normal. Similarly, in (b), we have a Weil divisor which is not locally principal, and this cannot happen if X is factorial. This is made precise by

Theorem 5.2.6. Let X and K be as in 5.1.2.

(a) When X is also normal, the map $\mathrm{div} : K^\times/\mathcal{O}(X)^\times \to \mathrm{Div}(X)$ is an injection, and so are the maps $\mathrm{CaDiv}(X) \to \mathrm{Div}(X)$ and $\mathrm{CaCl}(X) \to \mathrm{Cl}(X)$.

(b) If X is also factorial, then the maps $\mathrm{CaDiv}(X) \to \mathrm{Div}(X)$ and $\mathrm{CaCl}(X) \to \mathrm{Cl}(X)$ are isomorphisms. In particular, we have $\mathrm{Pic}(X) \xrightarrow{\sim} \mathrm{Cl}(X)$.

For the functoriality of the isomorphisms in (b), see 5.5.

Proof.

(a) The first part was observed in 5.1.4. The proof of the second is similar: suppose $D \in \mathrm{CaDiv}(X)$ is such that $\mathrm{ord}_Z(D) = 0$ for all prime Weil divisors $Z \subset X$. Pick some representative (U_α, f_α) of D. Then for all α, we have $\mathrm{div}_{U_\alpha}(f_\alpha) = (0)$, whence by 5.1.4 we get that $f_\alpha \in \mathcal{O}(U_\alpha)^\times$. This shows as in the proof of 5.2.3 that $D = 0$.

When X is normal, the image of $\mathrm{CaDiv}(X) \hookrightarrow \mathrm{Div}(X)$ consists of the *locally principal* Weil divisors, i.e., those $D \in \mathrm{Div}(X)$ such that for each $x \in X$, there is an open neighborhood

[8] This example was explained to me by Will Sawin, in the following but clearly equivalent way. For any $\lambda \in k$, let $M_\lambda := \{g \in k[x, y] : (\partial g/\partial x)|_{(0,0)} = \lambda g(0, 0)\} \subset k[x, y]$. This is naturally an A-module. Further, the corresponding quasicoherent sheaf $\mathcal{L}_\lambda := \widetilde{M_\lambda}$ is a line bundle on X, which is nontrivial when $\lambda \neq 0$. The same example in one dimension lower gives that $\operatorname{Pic} \operatorname{Spec} k[x^3, x^2] \cong \mathbb{G}_a(k)$.

$U \subset X$ of x and an $f \in K^\times$ such that $D|_U = \mathrm{div}_U(f)$; similarly, the map $\mathrm{CaCl}(X) \hookrightarrow \mathrm{Cl}(X)$ is an isomorphism onto the subgroup of locally principal Weil divisors modulo linear equivalence. The content of (b) is that when X is in fact factorial, every Weil divisor is locally principal. It clearly suffices to show this for prime Weil divisors; this is what we do below. The final statement then follows from this result combined with 5.2.3.

(b) Suppose that $Z \subset X$ is a prime Weil divisor and $x \in X$. We have to produce an open neighborhood $U \subset X$ of x and an $f \in K^\times$ with $[Z]|_U = \mathrm{div}_U(f)$. If $x \in X \smallsetminus Z$, we may take $U := X \smallsetminus Z$ and $f := 1$. If $x \in Z$, the subspace Z corresponds to a codimension one prime of $\mathcal{O}_{X,x}$; since $\mathcal{O}_{X,x}$ is a Noetherian UFD, this prime is principal. Pick a generator $f_Z \in \mathcal{O}_{X,x}$ of this prime, and a lift $f \in \mathcal{O}(U)$ of it to an affine open neighborhood U of x in X. Now $\mathrm{div}_U(f)$ contains $[Z]|_U$ by construction, but it may be bigger; however, none of the other prime divisors appearing in it pass through x. Therefore, by shrinking U further to avoid all these other components of $\mathrm{Supp}\,\mathrm{div}_U(f)$, we arrive at our choice of U and f.

■

Example 5.2.7. Let A be a Noetherian UFD. Then $\mathrm{Pic}(\mathrm{Spec}\, A) = 0$. For instance, $\mathrm{Pic}(\mathrm{Spec}\,\mathbb{Z}) = 0$.

Example 5.2.8. By 5.1.9 and 5.2.6, for any field k, integer $n \in \mathbb{Z}_{\geq 0}$, and open subset $U \subset \mathbb{A}^n_k$, we have $\mathrm{Pic}(U) = 0$.

Example 5.2.9. Let k be a field and $A = k[x,y,z]/(xz - y^2)$ and $X = \mathrm{Spec}\, A$ be as in 5.2.5(b), where we showed $\mathrm{Pic}(X) = 0$. If $p \in X$ is the vertex $\mathbb{V}(x,z)$, then $X \smallsetminus \{p\}$ *is* factorial, and so $\mathrm{Pic}(X \smallsetminus \{p\}) \cong \mathrm{Cl}(X \smallsetminus \{p\}) \cong \mathrm{Cl}(X) \cong \mathbb{Z}/2$. Therefore, while Cl is insensitive to removing subsets of codimension 2, the group Pic for non-factorial schemes is not.

Combining 5.2.3 with 5.2.4 tells us that when X and K are as in 5.1.2, there is a natural morphism $\widetilde{\mathrm{Pic}}(X) \to \mathrm{Div}(X)$ (resp. $\mathrm{Pic}(X) \to \mathrm{Cl}(X)$) which by 5.2.6 is injective (resp. bijective) when X is normal (resp. factorial). This admits the following clean description. Given a pair $(\mathcal{L}, \sigma)$ representing an element of $\widetilde{\mathrm{Pic}}(X)$, we can define for each prime Weil divisor $Z \subset X$ an integer $\mathrm{ord}_{Z,\mathcal{L}}(\sigma)$ as follows: pick an open $U \subset X$ meeting Z (i.e., containing the generic point of Z) on which $\mathcal{L}$ is trivial and on which σ corresponds to the rational function $f \in K^\times$, and let $\mathrm{ord}_{Z,\mathcal{L}}(\sigma) := \mathrm{ord}_Z(f)$. This is independent of the choice of U and trivialization; further, $\mathrm{ord}_{Z,\mathcal{L}}(\sigma) = 0$ for all but finitely many Z. This allows us to define the Weil divisor $\mathrm{div}(\sigma) := \sum_Z \mathrm{ord}_{Z,\mathcal{L}}(\sigma)[Z] \in \mathrm{Div}(X)$. (This definition generalizes the one 5.1.3(a), which is the case $\mathcal{L} = \mathcal{O}_X$.) The homomorphism $\widetilde{\mathrm{Pic}}(X) \to \mathrm{Div}(X)$ is given by $(\mathcal{L}, \sigma) \mapsto \mathrm{div}(\sigma)$. Similarly, $\mathrm{Pic}(X) \to \mathrm{Cl}(X)$ is given by taking an $\mathcal{L}$, pick a nonzero rational section σ of it, and sending $\mathcal{L}$ to the linear equivalence class $[\mathrm{div}(\sigma)]$.

Example 5.2.10. Let k be a field.

(a) For any $n \in \mathbb{Z}_{\geq 1}$, the group $\mathrm{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z}$ generated by the line bundle $\mathcal{O}_{\mathbb{P}^n_k}(1)$.
(b) More generally, for any $m \in \mathbb{Z}_{\geq 1}$ and integers $n_1, \ldots, n_m$, let $\mathbb{P} := \prod_{i=1}^m \mathbb{P}^{n_i}_k$. Then we have $\mathrm{Pic}(\mathbb{P}) \cong \mathbb{Z}^{\oplus m}$, generated by the line bundles $\mathcal{O}_{\mathbb{P}}(0,0,\ldots,0,1,0,\ldots,0)$

The morphism $\widetilde{\mathrm{Pic}}(X) \to \mathrm{Div}(X)$ admits a partial inverse, which when X is factorial recovers the inverse isomorphism. This is what we describe now. Let X and K be as in 5.1.2, and let $D \in \mathrm{Div}(X)$. We define the sheaf $\mathcal{O}_X(D)$ on X by the following recipe: for each nonempty open subset U, define

$$\mathcal{O}_X(D)(U) := \{f \in K^\times : \mathrm{div}_U(f) + D|_U \geq 0\} \cup \{0\} \subset K.$$

The following observations are straightforward.

Lemma 5.2.11.

(a) The above construction defines a quasicoherent $\mathcal{O}_X$-submodule $\mathcal{O}_X(D) \subset \mathcal{K}_X$.
(b) If $U \subset X$ is an open subscheme, then $\mathcal{O}_X(D)|_U = \mathcal{O}_U(D|_U)$ as $\mathcal{O}_U$-submodules of $\mathcal{K}_U = \mathcal{K}_X|_U$.
(c) The function $1 \in \Gamma(X \smallsetminus |D|, \mathcal{O}_X(D))$ defines a rational section of the quasicoherent sheaf $\mathcal{O}_X(D)$; further, it gives an $\mathcal{O}_{X\smallsetminus|D|}$-module monomorphism $\mathcal{O}_{X\smallsetminus|D|} \hookrightarrow \mathcal{O}_X(D)|_{X\smallsetminus|D|}$.
(d) If $D, D' \in \mathrm{Div}(X)$, then multiplication gives a natural isomorphism $\mathcal{O}_X(D)\otimes_{\mathcal{O}_X}\mathcal{O}_X(D') \xrightarrow{\sim} \mathcal{O}_X(D+D')$, sending the product $1 \otimes 1$ to 1.
(e) If $D' \in \mathrm{Div}(X)$ is principal, say $D' = \mathrm{div}(g)$ for $g \in K^\times$, , then multiplication by g gives a natural isomorphism $\mathcal{O}_X(D+D') \xrightarrow{\sim} \mathcal{O}_X(D)$ for any $D \in \mathrm{Div}(X)$; this takes the rational section 1 of the first to g times the rational section 1 of the second.
(f) If $D \in \mathrm{Div}(X)$ is principal, then $\mathcal{O}_X(D) \cong \mathcal{O}_X(0)$.

Proof. Exercise. ■

When X is normal, we have the strengthened

Lemma 5.2.12. Suppose that X and K are as in 5.1.2. Suppose further that X is normal.

(a) For any $D \in \mathrm{Div}(X)$, then monomorphism $\mathcal{O}_{X\smallsetminus|D|} \hookrightarrow \mathcal{O}_X(D)|_{X\smallsetminus|D|}$ of 5.2.11(c) is an isomorphism.
(b) We have that $\mathcal{O}_X \xrightarrow{\sim} \mathcal{O}_X(0)$, and the morphism $D \mapsto (\mathcal{O}_X(-), 1)$ gives a group morphism from $\mathrm{Div}(X)$ to the group of quasicoherent subsheaves of $\mathcal{K}_X$ equipped with a rational section.
(c) A divisor $D \in \mathrm{Div}(X)$ is (locally) principal iff $\mathcal{O}_X(D)$ is (locally) trivial of rank 1.
(d) Given any element of $\widetilde{\mathrm{Pic}}(X)$ represented by a pair $(\mathcal{L}, \sigma)$, we have $[\mathcal{O}_X(\mathrm{div}\,\sigma), 1] = [\mathcal{L}, \sigma] \in \widetilde{\mathrm{Pic}}(X)$. In particular, the morphism $\widetilde{\mathrm{Pic}}(X) \to \mathrm{Div}(X)$ as well as $\mathrm{Pic}(X) \to \mathrm{Cl}(X)$ of 5.2.4 is injective, with image consisting of Weil divisors D such that $\mathcal{O}_X(D)$ is a line bundle. The restriction of the morphism described in (b) to this subgroup is the inverse isomorphism to the morphism of 5.2.4.

As before, we see that if X is in fact factorial, then every Weil divisor D is locally principal, and hence $\widetilde{\mathrm{Pic}}(X) \to \mathrm{Div}(X)$ (resp. $\mathrm{Pic}(X) \to \mathrm{Cl}(X)$) are isomorphisms with inverse given by $D \mapsto [\mathcal{O}_X(D), 1]$ (resp. $[D] \mapsto [\mathcal{O}_X(D)]$).

Proof. Exercise. ■

5.3 Cartier Divisors II

In this section, we explain how to extend the notion of meromorphic or rational functions, Cartier divisors, and the Cartier class group, defined in 5.2, to an arbitrary locally Noetherian scheme. The extension to the non-locally-Noetherian case is possible, although much more care is needed; see [5, §11.10] and [13].

Convention 5.3.1. In this section, X denotes a locally Noetherian scheme.

Define the ring of meromorphic (or rational) functions on X to be the colimit of $\mathcal{O}(U)$ where U is an open subset containing all associated points of X, and denote it by $K(X)$. This is a subring of $\prod_{p\in \operatorname{Ass}(X)} \mathcal{O}_{X,p}$. The assignment $U \mapsto K(U)$ for $U \subset X$ open clearly defines a presheaf of $\mathcal{O}_X$-algebras. We denote its sheafification $\mathcal{K}_X$, and call it the *sheaf of meromorphic functions* on X; this is naturally a sheaf of $\mathcal{O}_X$-algebras.

To compute sections of $\mathcal{K}_X$ in practice, it is helpful to construct it another way, which is what we do now (5.3.3); then we show that these two constructions agree (5.3.4). For this, we start with small lemma.

Lemma 5.3.2. Let A be a ring, $X = \operatorname{Spec} A$, and $U \subset X$ an open subset. Pick an ideal $\mathfrak{a} \subset A$ such that $U = X \smallsetminus V(\mathfrak{a})$. Consider the following conditions.

(a) The ideal $\mathfrak{a}$ contains a nonzerodivisor of A.
(b) The open subset U contains a principal open subset $D(f)$ for some nonzerodivisor $f \in A$.
(c) The schematic closure of U in X is X itself.
(d) The annihilator $\operatorname{Ann}_A(\mathfrak{a})$ is zero.
(e) The open subset U contains all associated points of X, i.e., associated primes of A.

Then we have the implications (a) $\Leftrightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) $\Rightarrow$ (e). If A is Noetherian, then also (e) $\Rightarrow$ (a), so that all five conditions are equivalent, and there is a natural isomorphism $\operatorname{Quot}(A) \xrightarrow{\sim} K(\operatorname{Spec} A)$.

The equivalence of the five conditions, along with fact that the characterizations (b), (c), and (e) are intrinsic, implies that the properties (a) and (d) do not depend on the choice of ideal $\mathfrak{a}$ (which can also be shown directly).

Proof. ([5, Lemma 9.23]) Clearly, (a) $\Leftrightarrow$ (b).

(b) $\Rightarrow$ (c) Let $\mathfrak{b} \subset A$ be an ideal such that $\operatorname{Spec} A/\mathfrak{b}$ majorizes U. In particular, for all $b \in \mathfrak{b}$ we have $b|_U = 0$ and so $b|_{D(f)} = 0$, i.e., $b \in \ker(A \to A[f^{-1}])$. Since f is a nonzerodivisor, this kernel is zero, and so $b = 0$. Therefore, $\mathfrak{b} = 0$.

(c) $\Rightarrow$ (d) Let $b \in \operatorname{Ann}_A(\mathfrak{a})$. Given an $x \in U$, pick an $a \in \mathfrak{a}$ such that $a(x) \neq 0$. Now $(ab)_x = 0$ implies along with $a_x \in \mathcal{O}_{X,x}^\times$ that $b_x = 0$. Since this holds for all $x \in U$ and U is schematically dense, $b = 0$.

(d) $\Rightarrow$ (e) Suppose there is an associated prime $\mathfrak{p} \subset A$ such that $\mathfrak{p} \supset \mathfrak{a}$. Pick a $b \in A \smallsetminus \{0\}$ such that $\mathfrak{p} = \operatorname{Ann}_A(b)$. Then $b \in \operatorname{Ann}_A(\mathfrak{a}) \smallsetminus \{0\}$.

Finally, when A is a Noetherian ring, then we have

(e) $\Rightarrow$ (a) If $\mathfrak{a}$ does not contain a nonzerodivisor, then $\mathfrak{a}$ lies in the union of the finitely many associated primes of A, and hence lies in one of them by prime avoidance.

To show the last statement, note that if $f \in A$ is a nonzerodivisor, then $D(f)$ contains all associated primes of A; in particular, the natural morphism $A \to K(\operatorname{Spec} A)$ factors through $\operatorname{Quot}(A) \to K(\operatorname{Spec} A)$. This is injective, since both sides further embed into $\prod_{\mathfrak{p}\in\operatorname{Ass}(A)} A_{\mathfrak{p}}$. Surjectivity follows from the implication (e) $\Rightarrow$ (b) above. ■

The above discussion allows us to globalize the definition.

Proposition/Definition 5.3.3. The $\mathcal{O}_{X^{\mathrm{dist}}_{\mathrm{Zar}}}$-premodule $U \mapsto \operatorname{Quot} \mathcal{O}_X(U)$ is a sheaf, and hence an $\mathcal{O}_{X^{\mathrm{dist}}_{\mathrm{Zar}}}$-module. In particular, sheafification gives a unique sheaf $\mathcal{K}_X$ of $\mathcal{O}_X$-algebras, called the *sheaf of meromorphic functions* on X, such that for any affine $U \subset X$ the natural morphism $\mathcal{O}_X(U) \to \mathcal{K}_X(U)$ is given by $\mathcal{O}_X(U) \hookrightarrow \operatorname{Quot} \mathcal{O}_X(U) \xrightarrow{\sim} \mathcal{K}_X(U)$.

Proof. ([14, Lect. 9, p. 61]) We need to show that if A is a ring, $f_i \in A$ finitely many elements such that $(f_i) = (1)$, then the sequence

$$\operatorname{Quot} A \to \prod_i \operatorname{Quot} A[f_i^{-1}] \to \prod_{i,j} \operatorname{Quot} A[(f_i f_j)^{-1}]$$

is exact. Injectivity of the first map is clear from the fact that $\operatorname{Quot} A \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Ass}(A)} A_{\mathfrak{p}}$ and associated primes behave well under localization. ∎

Part of the point of the above discussion is

Lemma 5.3.4. The two constructions of $\mathcal{K}_X$ above are naturally isomorphic as sheaves of $\mathcal{O}_X$-algebras, and for $x \in X$, the natural map $\mathcal{K}_{X,x} \to \operatorname{Quot} \mathcal{O}_{X,x}$ is an isomorphism.

Proof. ∎

Remark 5.3.5. The underyling sheaf of $\mathcal{O}_X$-modules of $\mathcal{K}_X$ is not quasicoherent in general due to the existence of embedded primes; it is indeed so if X is reduced. We will not need this, and so not prove this; see [15, Exercise 3.15] or [13, p. 205].

In particular, if X is a locally Noetherian scheme, then we have a monomorphism of abelian sheaves $\mathcal{O}_X^\times \hookrightarrow \mathcal{K}_X^\times$ on X.

Definition 5.3.6.

(a) Define the *sheaf of Cartier divisors* on X to be the quotient abelian sheaf $\mathcal{K}_X^\times / \mathcal{O}_X^\times$. We denote the global sections of this sheaf by $\operatorname{CaDiv}(X) := \Gamma(X, \mathcal{K}_X^\times / \mathcal{O}_X^\times)$, and call its elements *Cartier divisors* on X.

(b) We define the *Cartier class group* to be the cokernel of the natural morphism $K(X)^\times \to \operatorname{CaDiv}(X)$ denoted by div, i.e., define it via the exact sequence

$$0 \to \mathcal{O}(X)^\times \to K(X)^\times \to \operatorname{CaDiv}(X) \to \operatorname{CaCl}(X) \to 0.$$

Super explicitly. The connecting homomorphism therefore defines an injection $\operatorname{CaCl}(X) \hookrightarrow \mathrm{H}^1(X, \mathcal{O}_X^\times) \cong \operatorname{Pic}(X)$ with image consisting of line bundles which can be embedded in the $\mathcal{O}_X$-module $\mathcal{K}_X$. In fact, there is a natural injection $\operatorname{CaDiv}(X) \to \widetilde{\operatorname{Pic}(X)}$, where $\widetilde{\operatorname{Pic}(X)}$ is the extension of $\operatorname{Pic}(X)$ by $K(X)^\times / \mathcal{O}(X)^\times$ consisting of isomorphism classes of pairs $(\mathcal{L}, \sigma)$, where $\mathcal{L}$ is a line bundle and σ a rational section giving rise to a commutative diagram of the form

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{O}(X)^\times \arrow[r] \arrow[d] & K(X)^\times \arrow[r] \arrow[d] & \operatorname{CaDiv}(X) \arrow[r] \arrow[d, hook] & \operatorname{CaCl}(X) \arrow[r] \arrow[d, hook] & 0 \\
0 \arrow[r] & \mathcal{O}(X)^\times \arrow[r] & K(X)^\times \arrow[r] & \widetilde{\operatorname{Pic}(X)} \arrow[r] & \operatorname{Pic}(X) \arrow[r] & 0.
\end{tikzcd}$$

Lemma 5.3.7. When X is integral, the above morphisms $\operatorname{CaDiv}(X) \hookrightarrow \widetilde{\operatorname{Pic}(X)}$ and $\operatorname{CaCl}(X) \hookrightarrow \operatorname{Pic}(X)$ are isomorphisms.

We give two proofs: one short and natural proof using elementary properties of sheaf and Čech cohomology, and a second direct proof following [2, Prop. II.6.15].

Proof 1. The sheaf $\mathcal{K}_X$ is the constant sheaf with value K. In particular, since X is irreducible, $\mathcal{K}_X$ is flasque, and so $\mathrm{H}^1(X, \mathcal{K}_X) = 0$, which implies the result. ■

Proof 2. ■

Remark 5.3.8. The isomorphism of 5.3.7 holds more generally if X is a Noetherian scheme that is either reduced or has an ample line bundle [TODO]. For an example where the result fails, see [TODO].

5.4 Ampleness

5.4.1 Global Generation

Theorem/Definition 5.4.1 (Globally Generated Sheaves)**.** Let X be a ringed space. The following conditions on an $\mathcal{F} \in \mathcal{O}_X\text{-}\mathsf{Mod}$ are equivalent.

(a) For each $x \in X$, the natural morphism $\mathcal{O}_{X,x} \otimes_{\mathcal{O}(X)} \mathcal{F}(X) \to \mathcal{F}_x$ is surjective.
(b) The natural $\mathcal{O}_X$-module morphism $\mathcal{O}_X \otimes_{\underline{\mathcal{O}(X)}} \underline{\mathcal{F}(X)} \to \mathcal{F}$ is an epimorphism.
(c) There is a set I and an $\mathcal{O}_X$-module epimorphism $\mathcal{O}_X^{\oplus I} \twoheadrightarrow \mathcal{F}$.

An $\mathcal{O}_X$-module $\mathcal{F}$ satisfying these equivalent conditions is said to be *globally generated.* It is said to further be *finitely globally generated* or *globally generated by a finite number of global sections* if in (c) the index set I can be taken to be finite. We further have:

(d) A quotient of a (finitely) globally generated $\mathcal{O}_X$-module is also (finitely) globally generated.
(e) If $\pi : X \to Y$ is a morphism of ringed spaces and $\mathcal{G} \in \mathcal{O}_Y\text{-}\mathsf{Mod}$ is (finitely) globally generated, then so is $\pi^*\mathcal{G}$.
(f) If $\mathcal{F}, \mathcal{G} \in \mathcal{O}_X\text{-}\mathsf{Mod}$ are (finitely) globally generated, so is $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$.

Proof. (TODO) ■

In (a) (resp. (b)), it is clear that we may equivalently ask that the morphism $\mathcal{O}_{X,x} \otimes_{\mathbb{Z}} \mathcal{F}(X)$ (resp. $\mathcal{O}_X \otimes_{\underline{\mathbb{Z}}} \underline{\mathcal{F}(X)}) \to \mathcal{F}$ be an epimorphism.

Lemma/Definition 5.4.2 (Basepoint-Free Line Bundles)**.** Let X be a ringed space and $\mathcal{L}$ a line bundle on X.

(a) We say that $\mathcal{L}$ is *basepoint-free* iff it is globally generated.

When X is a locally ringed space, this is equivalent to

(b) For each $x \in X$, there is a $\sigma \in \Gamma(X, \mathcal{L})$ with $\sigma(x) \neq 0$.

When X is a scheme, we define the *base locus* of $\mathcal{L}$ to be the closed subscheme

$$\mathrm{BL}(\mathcal{L}) := \bigcap_{\sigma \in \Gamma(X,\mathcal{L})} \mathbb{V}(\sigma) \subset X.$$

In this case, the condition that $\mathcal{L}$ be basepoint-free is further equivalent to

(c) The base locus $\mathrm{BL}(\mathcal{L})$ is empty.

Example 5.4.3. Any (finite-type) quasicoherent sheaf on an affine scheme is (finitely) globally generated.

Example 5.4.4. Let S be a graded ring which is finitely generated in degree one. Let $X = \operatorname{Proj} S$ and let $\mathcal{F} \in \mathsf{QCoh}(X)^{\mathrm{ft}}$. For all $n \gg 0$, the sheaf $\mathcal{F}(n)$ is finitely globally generated. This follows from 5.4.3 and 3.3.

Example 5.4.5. Let A be a ring, X be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$-mdoule, and $\pi : X \to \operatorname{Spec} A$ a morphism such that $\pi_*\mathcal{F} \in \mathsf{QCoh}(\operatorname{Spec} A)$.[9] Then $\mathcal{F}$ is globally generated iff the counit $\pi^*\pi_*\mathcal{F} \to \mathcal{F}$ is an epimorphism.

This motivates the following definition in the relative setting.

[9] According to 3.3.7, this happens, for instance, if $\mathcal{F} \in \mathsf{QCoh}(X)$ and π is qcqs.

Definition 5.4.6. Let $\pi : X \to Y$ be a morphism of ringed spaces and $\mathcal{F} \in \mathcal{O}_X\text{-}\mathsf{Mod}$. We say that $\mathcal{F}$ is *relatively globally generated* (relative to π) iff the counit $\pi^*\pi_*\mathcal{F} \to \mathcal{F}$ is an epimorphism of $\mathcal{O}_X$-modules.

Example 5.4.7. Let $\pi : X \to Y$ be a qcqs morphism of schemes and $\mathcal{L}$ a line bundle on X that is relatively globally generated (also known as relatively basepoint-free). Then 1.4.5 gives us a natural morphism $X \to \underline{\operatorname{Proj}}_Y \operatorname{Sym}^\bullet \pi_*\mathcal{L}$ of Y-schemes.

5.4.2 Ample Line Bundles

Theorem/Definition 5.4.8 (Ample Line Bundles)**.** The following conditions on a quasicompact scheme X and line bundle $\mathcal{L}$ on X are equivalent.

(a) The scheme X is quasiseparated, and if $\mathcal{F} \in \mathsf{QCoh}(X)^{\mathrm{ft}}$, then for all integers $n \gg 0$, the sheaf $\mathcal{L}^{\otimes n} \otimes \mathcal{F}$ is globally generated.
(b) The scheme X is quasiseparated, and if $\mathcal{I} \subset \mathcal{O}_X$ is a finite-type quasicoherent ideal sheaf, then there is an $n \in \mathbb{Z}_{>0}$ such that $\mathcal{L}^{\otimes n} \otimes \mathcal{I}$ is globally generated.
(c) For each $x \in X$ and open $U \subset X$ with $x \in U$, there is an $n \in \mathbb{Z}_{>0}$ and $f \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $x \in D(f) \subset U$.
(d) For each $x \in X$ and open $U \subset X$ with $x \in U$, there is an $n \in \mathbb{Z}_{>0}$ and $f \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $x \in D(f) \subset U$ and $D(f)$ is affine.
(e) For each $x \in X$, there is an $n \in \mathbb{Z}_{>0}$ and $f \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $x \in D(f)$ and $D(f)$ is affine.
(f) There is an $n \in \mathbb{Z}_{>0}$ and finitely many $f_i \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $X = \bigcup_i D(f_i)$ and each $D(f_i)$ is affine.
(g) Let $S := \Gamma_*(X, \mathcal{L})$. Then $\{D(f)\}_f$ for homogeneous $f \in S_+$ is an open cover of X, and the resulting morphism $\phi : X \to \operatorname{Proj} S$ is an open embedding.

We say that a line bundle $\mathcal{L}$ on a scheme X is *ample* iff X is quasicompact and $\mathcal{L}$ satisfies the equivalent conditions (a)-(e). Further:

(h) If a scheme X admits an ample line bundle, it is separated.
(i) A line bundle $\mathcal{L}$ on a scheme X is ample iff there is an $n \in \mathbb{Z}_{>0}$ such that $\mathcal{L}^{\otimes n}$ is.
(j) If $\mathcal{L}$ is an ample line bundle on a scheme X and $\rho : X' \to X$ an affine morphism, then $\rho^*\mathcal{L}$ is ample on X'.
(k) The morphism $X \to \operatorname{Proj} S$ in (g) is a quasicompact schematically dominant open embedding.
(l) If $\mathcal{L}$ is an ample line bundle on a scheme X, then for every finite $Z \subset X$ and every open neighborhood U of Z, there is an $n \in \mathbb{Z}_{>0}$ and $f \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $D(f)$ is affine and $Z \subset D(f) \subset U$.

For several other equivalent characterizations, see [3, Tag 01PR].

Proof.

(a) $\Rightarrow$ (b) Clear.
(b) $\Rightarrow$ (c) Let $\mathcal{J}$ be the quasicoherent ideal sheaf of functions vanishing on $X \smallsetminus U$, i.e., the ideal sheaf corresponding to the reduced closed subscheme supported precisely on $X \smallsetminus U$. Since $\mathcal{J}_x = \mathcal{O}_{X,x}$, by 3.3.11(b) we may replace $\mathcal{J}$ by a finite-type quasicoherent subsheaf $\mathcal{I}$ such that $\mathcal{I}_x = \mathcal{O}_{X,x}$. Let $n \in \mathbb{Z}_{>0}$ be such that $\mathcal{L}^{\otimes n} \otimes \mathcal{I}$ is globally generated, and pick an $f \in \Gamma(X, \mathcal{L}^{\otimes n} \otimes \mathcal{I}) \subset \Gamma(X, \mathcal{L}^{\otimes n})$ such that $x \in D(f)$. By construction, f vanishes on $X \smallsetminus U$, i.e., $D(f) \subset U$.
(c) $\Rightarrow$ (d) Apply (c) to an affine open neighborhood U' of x in U on which $\mathcal{L}$ is trivial.
(d) $\Rightarrow$ (e) Clear.

(e) ⇒ (f) Since X is quasicompact, there are finitely many $n_i \in \mathbb{Z}_{>0}$ and $f_i \in \Gamma(X, \mathcal{L}^{\otimes n_i})$ such that $X = \bigcup_i D(f_i)$ and each $D(f_i)$ is affine. We finish by replacing the n_i by their least common multiple and f_i by suitable powers.

(f) ⇒ (a) For each affine open $U \subset X$ and f_i, the open subscheme $D(f_i) \cap U = D(f_i|_U)$ is affine by 3.4.2(c); therefore, X is quasiseparated by 3.3.5(c).
Note that $\mathcal{L}^{\otimes n}$ is basepoint-free. If we could show that for any $\mathcal{F} \in \mathsf{QCoh}(X)^{\mathrm{ft}}$ and all $m \gg 0$, the sheaf $\mathcal{L}^{\otimes mn} \otimes \mathcal{F}$ is globally generated, then applying this observation finitely many times to $\mathcal{L}^{\otimes i} \otimes \mathcal{F}$ for $i = 0, \ldots, n-1$ shows that for all $m \gg 0$ the sheaf $\mathcal{L}^{\otimes m} \otimes \mathcal{F}$ is globally generated. Therefore, we may replace $\mathcal{L}$ by $\mathcal{L}^{\otimes n}$ to assume that $\mathcal{L}$ is globally generated and X is covered by the affine open subsets of the form $D(f)$ for $f \in \Gamma(X, \mathcal{L})$.
Suppose $\mathcal{F} \in \mathsf{QCoh}(X)^{\mathrm{ft}}$. Since X is quasicompact, by 5.4.1(f) it suffices to show that for each $x \in X$, there is an open neighborhood U of x in X and a $m \in \mathbb{Z}_{>0}$ such that $\mathcal{L}^{\otimes m} \otimes \mathcal{F}$ is globally generated at all points $y \in U$. For this, pick an $f \in \Gamma(X, \mathcal{L})$ such that $x \in D(f)$ and $D(f)$ is affine. By our hypotheses on $\mathcal{F}$, the restriction $\mathcal{F}|_{D(f)}$ is generated by a finite number of global sections, say $\sigma_i \in \Gamma(D(f), \mathcal{F})$. By 3.3(c), there is an $m \in \mathbb{Z}_{>0}$ such that each $f^m \sigma_i \in \Gamma(D(f), \mathcal{L}^{\otimes m} \otimes \mathcal{F})$ extends to a global section, i.e., an element of $\Gamma(X, \mathcal{L}^{\otimes m} \otimes \mathcal{F})$. Then the sheaf $\mathcal{L}^{\otimes m} \otimes \mathcal{F}$ is globally generated by finitely many sections at x, and hence by Geometric Nakayama [TODO] at all points y in some open neighborhood U of x as needed.

(a-f) ⇒ (g) Note that for any homogeneous $f \in S_+$, we have $\phi^{-1} D_+(f) = D(f)$. Therefore, it suffices to show that there is a cover of $\operatorname{Proj} S$ by such $D_+(f)$ such that the restriction of ϕ to each $D(f)$ is an open embedding; but $\phi|_{D(f)}$ is an isomorphism on global sections since X is qcqs (3.3) and hence an isomorphism when $D(f)$ is affine.

(g)

The statements (f) and (g) follow from (e). ■

Example 5.4.9.

(a) Every line bundle on an affine scheme is ample: combine 5.4.3 and 5.4.8(a).
(b) If S is a graded ring which is finitely generated in degree one, then the sheaf $\mathcal{O}(1)$ on $\operatorname{Proj} S$ is ample (5.4.4).

Remark 5.4.10. If a scheme X admits an ample line bundle, then it is quasicompact and separated. Indeed, by 3.6.3, it suffices to show that for any $x, y \in X$ there are affine opens U and V of X such that $x \in U$ and $y \in V$, and such that $U \cap V$ is affine and $\mathcal{O}_X(U) \otimes_{\mathbb{Z}} \mathcal{O}_X(V) \twoheadrightarrow \mathcal{O}_X(U \cap V)$. Pick n, f as in (d) so $x \in D(f) =: U$ and m, g as in (d) so $y \in D(g) := V$. By replacing f and g by suitable powers, we may assume $m = n$. Then $U \cap V$ is affine: in the affine U, it is the distinguished open corresponding to the non-vanishing of the function g/f. Finally, the morphism $\mathcal{O}_X(U) \otimes_{\mathbb{Z}} \mathcal{O}_X(V) \to \mathcal{O}_X(U \cap V) = \mathcal{O}_X(U)[(g/f)^{-1}]$ is surjective because $1 \otimes (f/g) \mapsto (g/f)^{-1}$.

The basic reason ample bundles are useful is because they are very well-behaved, and some power of them gives an embedding into projective space. This is made precise as follows.

Definition 5.4.11. Let X be a scheme and $\mathcal{L}$ a line bundle on X. Let $\pi : X \to Y$ be a morphism with Y affine. We say that $\mathcal{L}$ is *very ample relative to* π iff there is an $m \in \mathbb{Z}_{\geq 1}$ and a closed embedding $i : X \hookrightarrow \mathbb{P}^m_Y$ of Y-schemes such that $\mathcal{L} \cong i^* \mathcal{O}_{\mathbb{P}^m_Y}(1)$.

Proposition 5.4.12. Let X be a scheme and $\mathcal{L}$ a line bundle on X.

(a) If there is a morphism $\pi : X \to Y$ with Y affine and an $n \in \mathbb{Z}_{>0}$ such that $\mathcal{L}^{\otimes n}$ is very ample relative to π, then $\mathcal{L}$ is ample.
(b) If $\mathcal{L}$ is ample, then for any proper morphism $\pi : X \to Y$ with Y affine, for all $n \gg 0$, the

line bundle $\mathcal{L}^{\otimes n}$ is very ample relative to π.[10]

Proof.

(a) Note that if $\pi : X \to Y$ is a morphism with Y affine admitting a very ample line bundle, then π is projective and hence proper; in particular, X is quasicompact and separated. By 5.4.8(e), it suffices to show that if $\mathcal{L}$ is very ample relative to π, then it is ample, for which we use 5.4.8(c). Suppose that $\mathcal{L}$ is very ample relative to π, and fix an $m \in \mathbb{Z}_{\geq 1}$ and closed embedding $i : X \hookrightarrow \mathbb{P}^m_Y$ of Y-schemes, along with an isomorphism $\mathcal{L} \cong i^*\mathcal{O}(1)$. For each $x \in X$, pick an $s \in \Gamma(\mathbb{P}^m_Y, \mathcal{O}(1))$ such that $i(x) \in D_+(s)$. Let $f \in \Gamma(X, \mathcal{L})$ be the section corresponding to $i^*s \in \Gamma(X, i^*\mathcal{O}(1))$ under the given isomorphism. Then $x \in D(f)$, and $D(f) = i^{-1}(D_+(s))$ is affine.

(b) It suffices to produce a single $n \in \mathbb{Z}_{>0}$ such that $\mathcal{L}^{\otimes n}$ is very ample relative to π; then by 5.4.8(a), for $m \gg 0$, the line bundle $\mathcal{L}^{\otimes m}$ is globally generated, i.e., base-point free, and we conclude, since the tensor product of a very ample line bundle with a base-point free line bundle is again very ample [TODO].
We will show that if $\mathcal{L}$ is an ample line bundle on a scheme X and $\pi : X \to Y$ a finite-type morphism with Y affine, then there are $n, m \in \mathbb{Z}_{>0}$ such that $\mathcal{L}^{\otimes n}$ yields a (locally closed) immersion of Y-schemes into $\mathbb{P}^m_Y$. When π is further proper, this inclusion must have closed image and hence must be a closed embedding.
To show this, let $A := \mathcal{O}(Y)$. As in the proof of 5.4.8 above, pick a $p \in \mathbb{Z}_{>0}$ and finitely many $f_i \in \Gamma(X, \mathcal{L}^{\otimes p})$ such that each $D(f_i)$ is affine and $X = \bigcup_i D(f_i)$. For each i, pick finitely many generators a_{ij} of the A-algebra $\mathcal{O}(D(f_i))$. By 3.3, there is a $q \in \mathbb{Z}_{>0}$ such that for all i, j, the section $a_{ij} f_i^q \in \Gamma(D(f_i), \mathcal{L}^{\otimes pq})$ extends to a global section $b_{ij} \in \Gamma(X, \mathcal{L}^{\otimes pq})$. Take $n := pq$, and consider the linear series of the sections of $\mathcal{L}^{\otimes n}$ generated by the b_{ij} and f_i^q. As $X = \bigcup_i D(f_i^q)$, this linear series is base-point free, and hence defines an A-morphism $i : X \to \mathbb{P}^m_A$ for some $m \in \mathbb{Z}_{>0}$ with $\mathcal{L} \cong i^*\mathcal{O}(1)$; here, $\mathbb{P}^m_A$ has coordinates (i.e., an A-basis of $\Gamma(\mathbb{P}^m_A, \mathcal{O}(1))$) of the form x_{ij} and y_i such that $i^*x_{ij} = b_{ij}$ and $i^*y_i = f_i^q$ for all i, j.
Note that the morphism i factors through the open subset $\bigcup_i D_+(y_i) \subset \mathbb{P}^m_A$. To finish the proof, it suffices to show that the resulting morphism $X \to \bigcup_i D_+(y_i)$ is a closed embedding. For this, note that for each i, we have $i^{-1}D_+(y_i) = D(f_i^q) = D(f_i)$, which is affine, and the resulting A-algebra morphism $\mathcal{O}(D_+(y_i)) \to \mathcal{O}(D(f_i))$ is surjective because $x_{ij}/y_i \mapsto a_{ij}$.

■

Theorem/Definition 5.4.13 (Projective Morphisms)**.** Consider the following conditions on a morphism $\pi : X \to Y$.

(a) There is an $n \in \mathbb{Z}_{\geq 0}$ and a closed immersion $X \hookrightarrow \mathbb{P}^n_Y$ of Y-schemes.
(b) There is an $\mathcal{E} \in \mathsf{QCoh}(Y)^{\mathrm{ft}}$ and closed immersion $X \hookrightarrow \underline{\mathrm{Proj}}_Y \operatorname{Sym}^\bullet \mathcal{E}$ of Y-schemes.
(c) There is a graded quasicoherent $\mathcal{O}_Y$-algebra $\mathcal{S}$, finitely generated in degree 1, and isomorphism $X \xrightarrow{\sim} \underline{\mathrm{Proj}}_Y \mathcal{S}$ of Y-schemes.
(d) There morphism π is proper, and there is a π-ample line bundle on X.

Then:

(i) In general, we have the implications (a) $\Rightarrow$ (b) $\Leftrightarrow$ (c) $\Rightarrow$ (d).
(ii) If Y is qcqs, then we further have (d) $\Rightarrow$ (c).
(iii) If Y admits an ample line bundle, then all (a)-(d) are equivalent.

A morphism $\pi : X \to Y$ is said to be *projective* if it satisfies equivalent conditions (b) and (c).

[10] To be very clear, this means that there is an $n_0 \in \mathbb{Z}_{>0}$ (in general depending on π) such that for all integers $n > n_0$, the line bundle $\mathcal{L}^{\otimes n}$ is very ample relative to π.

A morphism π satisfying (a) is sometimes called H-projective [in Stacks, after Hartshorne]. For an example where (b) does not imply (a), see [TODO]. For an example where (d) does not imply projectivity, see [TODO].

Proof.

(a) $\Rightarrow$ (b) Take $\mathcal{E} := \mathcal{O}_Y^{n+1}$.
(b) $\Rightarrow$ (c) Follows from our understanding of the closed subschemes of $\underline{\mathrm{Proj}}_Y \operatorname{Sym}^\bullet \mathcal{E}$.
(c) $\Rightarrow$ (b) Take $\mathcal{E} := \mathcal{S}_1$ and use that $\operatorname{Sym}^\bullet \mathcal{S}_1 \twoheadrightarrow \mathcal{S}$.
(c) $\Rightarrow$ (d) Properness is local, and the line bundle $\mathcal{O}(1)$ is π-ample.

■

Theorem/Definition 5.4.14. Let $\pi : X \to Y$ be a proper morphism with Y qcqs. The following conditions on a line bundle $\mathcal{L}$ on X are equivalent:

(a) For every affine open $U \subset Y$, there is an $n \in \mathbb{Z}_{\geq 0}$ and a closed immersion $j_U : X_U = \pi^{-1}(U) \hookrightarrow \mathbb{P}^n_U$ of U-schemes with the property that $\mathcal{L}|_{X_U} \cong j_U^* \mathcal{O}_{\mathbb{P}^n_U}(1)$.
(b) There is an open cover $\{U_i\}$ of Y by affines such that for each i, there is an $n_i \in \mathbb{Z}_{\geq 0}$ and a closed immersion $j_i : X_i := \pi^{-1}(U_i) \hookrightarrow \mathbb{P}^{n_i}_{U_i}$ of U_i-schemes with the property that $\mathcal{L}|_{X_i} \cong j_i^* \mathcal{O}_{\mathbb{P}^{n_i}_{U_i}}(1)$.
(c) The line bundle $\mathcal{L}$ is relatively globally generated with respect to π (5.4.6) and the resulting Y-morphism $X \to \underline{\mathrm{Proj}}_Y \operatorname{Sym}^\bullet \pi_* \mathcal{L}$ (5.4.7) is a closed immersion.
(d) There is a finitely presented $\mathcal{O}_Y$-module $\mathcal{E}$ and a closed immersion $j : X \hookrightarrow \underline{\mathrm{Proj}}_Y \operatorname{Sym}^\bullet \mathcal{E}$ of Y-schemes with $\mathcal{L} \cong j^* \mathcal{O}(1)$.
(e) There is a graded quasicoherent $\mathcal{O}_Y$-algebra $\mathcal{S}$, finitely generated in degree 1, and an isomorphism $j : X \xrightarrow{\sim} \underline{\mathrm{Proj}}_Y \mathcal{S}$ such that $\mathcal{L} \cong j^* \mathcal{O}(1)$.

When $\pi : X \to Y$ is a proper morphism, we say that a line bundle $\mathcal{L}$ on X is *very ample* if it satisfies equivalent conditions (d) and (e).

[TODO: Clarify implications]

Proof. ([5, 13.56])

(a) $\Rightarrow$ (b) Clear.
(b) $\Rightarrow$ (c) The question is local on Y, so we may assume that $Y \xrightarrow{\sim} \operatorname{Spec} A$ is affine, and there is an $n \in \mathbb{Z}_{\geq 0}$ and closed immersion $j : X \hookrightarrow \mathbb{P}^n_A$ of A-schemes with $\mathcal{L} \cong j^* \mathcal{O}(1)$. By 1.4.5, j arises from a surjection $\mathcal{O}_X^{n+1} \twoheadrightarrow \mathcal{L}$ which by the adjunction $\pi^* \dashv \pi_*$ factors through $\pi^* \pi_* \mathcal{L} \to \mathcal{L}$, proving that $\mathcal{L}$ is relatively globally generated. The above factorization amounts to saying that j corresponds to a morphism $A[X_0, \dots, X_n] \to \bigoplus_{n \geq 0} \Gamma(X, \mathcal{L}^{\otimes n})$ of graded A-algebras, which factors through $\operatorname{Sym}_A^\bullet \Gamma(X, \mathcal{L}) \to \bigoplus_{n \geq 0} \Gamma(X, \mathcal{L}^{\otimes n})$. For each i, let $s_i \in \Gamma(X, \mathcal{L})$ be the corresponding section, and let S_i be the same section, considered as a homogeneous degree 1 element of $\operatorname{Sym}_A^\bullet \Gamma(X, \mathcal{L})$. If $\kappa : X \to \underline{\mathrm{Proj}}_Y \operatorname{Sym}^\bullet \pi_* \mathcal{L} \cong \operatorname{Proj} \operatorname{Sym}_A^\bullet \Gamma(X, \mathcal{L})$ is the canonical morphism, then $\kappa^{-1} D_+(S_i) = D(s_i)$ and we have a factorization $D(s_i) \xrightarrow{\kappa|_{D(s_i)}} D_+(S_i) \to D_+(X_i)$ of $j|_{D(s_i)}$; the Cancellation Lemma [TODO] then tells us that $\kappa|_{D(s_i)}$ is a closed immersion. Since these $D_+(S_i)$ cover $\operatorname{Proj} \operatorname{Sym}_A^\bullet \Gamma(X, \mathcal{L})$, we conclude that κ is a closed immersion as needed.
(c) $\Rightarrow$ (d) [Nontrivial, and only part that uses qcqs]
(d) $\Rightarrow$ (e) Follows from our understanding of the closed subschemes of $\underline{\mathrm{Proj}}_Y \operatorname{Sym}^\bullet \mathcal{E}$.
(e) $\Rightarrow$ (a) [Clear] The question is local on Y, so we may assume $Y \xrightarrow{\sim} \operatorname{Spec} A$ is affine.

■

5.4.3 Applications to Curves, Numerical Triviality and the Néron-Severi Group

In what follows, by a *curve* over a field k, we mean a separated finite-type k-scheme which is equidimensional of dimension 1.

Lemma 5.4.15. Let C be a proper curve over a field k. Then C is projective.

Lemma 5.4.16. Let C, C' be proper integral curves over a field k and $\varphi : C \to C'$ a surjective morphism. Then φ is finite, and for any line bundle $\mathcal{L}$ on C', we have

$$\deg_C \varphi^* \mathcal{L} = \deg f \cdot \deg_{C'} \mathcal{L}.$$

Proof. Since C is projective (5.4.15), the morphism φ is projective ([1, 17.3.C(c)]). It is also quasi-finite: the preimage of the generic point is just the generic point, whereas the preimage of a closed point of C' is a proper closed subscheme of C, hence zero-dimensional by 4.1.3(a) and hence finite by 4.2. Therefore, φ is finite by [1, 18.1.6]. For the last statement, note that

$$\deg_C \varphi^* \mathcal{L} := \chi(C, \varphi^* \mathcal{L}) - \chi(C, \mathcal{O}_C) = \chi(C', \varphi_* \varphi^* \mathcal{L}) - \chi(C', \varphi_* \mathcal{O}_C)$$

by [1, 18.2.E]. By the projection formula, we have $\varphi_* \varphi^* \mathcal{L} \cong \mathcal{L} \otimes_{\mathcal{O}_{C'}} \varphi_* \mathcal{O}_C$. By [1, 18.1.A], $\varphi_* \mathcal{O}_C$ is coherent, and so we may apply [1, 18.4.L] to conclude that this quantity is

$$\deg_{C'} \mathcal{L} \cdot \ell_{\mathcal{O}_{C',\eta}} (\varphi_* \mathcal{O}_C)_\eta,$$

where $\eta \in C'$ is the generic point. But C' is reduced, so $\mathcal{O}_{C',\eta} = K(C')$, and this last expression involving length is precisely $[K(C) : f^* K(C')] = \deg f$. ∎

Let k be a field and X a k-scheme. For any proper k-curve C and k-morphism $C \to X$, we define $\mathcal{L} \cdot C$ to be the degree of the pullback of $\mathcal{L}$ to C via $C \to X$.[11]

Theorem/Definition 5.4.17 (Nef Line Bundles)**.** Let k be a field, X a proper k-scheme, and $\mathcal{L}$ a line bundle on X. The following conditions are equivalent.

(a) For any proper curve C over k and k-morphism $C \to X$, we have $\mathcal{L} \cdot C \geq 0$ (resp. $\mathcal{L} \cdot C = 0$).
(b) For any integral regular proper k-curve C and k-morphism $C \to X$, we have $\mathcal{L} \cdot C \geq 0$ (resp $\mathcal{L} \cdot C = 0$).
(c) For any integral closed curve $C \hookrightarrow X$, we have $\mathcal{L} \cdot C \geq 0$ (resp. $\mathcal{L} \cdot C = 0$).

When these equivalent conditions hold, we say that $\mathcal{L}$ is *numerically effective* or *nef* (resp. *numerically trivial*).

(d) If $\mathcal{L}$ is nef (resp. numerically) trivial, then for any proper morphism $f : X' \to X$, so is $f^* \mathcal{L}$.
(e) The line bundle $\mathcal{L}$ is nef (resp. numerically trivial) iff it is so on each irreducible component of X.
(f) The tensor product of two nef (resp. numerically trivial) line bundles is nef (resp. numerically trivial). The dual of a numerically trivial line bundle is numerically trivial.

Of course, the dual of a nef line bundle is not nef in general; if $\mathcal{L}$ is a nef line bundle such that $\mathcal{L}^\vee$ is also nef, then $\mathcal{L}$ is numerically trivial.

Proof. Clearly, (a) implies (b)-(f), except possibly the "if" part of (e), which follows from (c).

[11] This notation is not ideal because $\mathcal{L} \cdot C$ depends on the morphism $C \to X$.

(b) $\Rightarrow$ (a) Suppose (b) holds and we are given a $C \to X$ as in (a). First note that by [1, 18.4.L, 19.2.I] we may assume that C is integral. Finally, replace C by its normalization ans use 5.4.16, noting that the normalization is a birational proper morphism.

(c) $\Rightarrow$ (b) Suppose (c) holds and that we are given a $C \to X$ as in (b). Let C' be the schematic image of C, so that $C' \subset X$ is an integral closed subscheme and we have a factorization $C \to C' \to X$ of $C \to X$. By the Cancellation Theorem and properness of C, the map $C \to C'$ is proper and surjective, so by 4.1.6(c) we have $1 = \dim C \geq \dim C'$. If $\dim C' = 0$, then C' is a closed point of X, so the pullback of $\mathcal{L}$ is trivial and hence has degree zero. If $\dim C' = 1$, then we are done by 5.4.16.

■

5.5 Blow Ups

Theorem/Definition 5.5.1. (Blow-Ups) Let X be a scheme and $Z \subset X$ a closed subscheme.

- The *blow-up of X along Z* is an X-scheme $\mathrm{Bl}_Z X \to X$ such that the pullback $Z \times_X \mathrm{Bl}_Z X \hookrightarrow \mathrm{Bl}_Z X$ of Z to it is an effective Cartier divisor, and such that $\mathrm{Bl}_Z X \to X$ is final with respect to this property.
- In this case, the effective Cartier divisor $E = Z \times_X \mathrm{Bl}_Z X \subset \mathrm{Bl}_Z X$ is called the *exceptional divisor* associated to the blow-up.

Clearly, the blow-up, if it exists, is unique; the content of the theorem, and the statement needing proof, is that it does in fact exist. There are many different ways to prove that the blow-up exists. We will give two. [TODO]

Proof 1. Blow-up closure lemma. ∎

Proof 2. Let $B := \underline{\mathrm{Proj}}_X \bigoplus_{n \geq 0} \mathcal{I}_Z^n$ with canonical projection $\pi : B \to X$; we claim that π satisfies the universal property of the blow-up. To see these, first note that $E := \pi^{-1}(Z)$ is an effective Cartier divisor corresponding to the ideal sheaf $\mathcal{O}(-E) = \mathcal{O}(1)$. Second, observe that the morphism π restricts to an isomorphism over $X \setminus Z$.[12]

Next, suppose $\phi : Y \to X$ is a morphism such that $\phi^{-1}(Z) \subset Y$ is an effective Cartier divisor corresponding to the invertible ideal sheaf $\mathcal{J} := \mathcal{I}_{\phi^{-1}(Z)/Y}$. We have to show that there is a unique morphism of X-schemes $Y \to B$. To construct such a morphism, we need to specify a triple as in 1.4.4, for which we consider the given morphism $\phi : Y \to X$, the line bundle $\mathcal{J}$, and the graded $\mathcal{O}_X$-algebra morphism $\psi : \bigoplus_{n \geq 0} \mathcal{I}_Z^n \to \bigoplus_{n \geq 0} \phi_* \mathcal{J}^{\otimes n}$, where each $\psi^n : \mathcal{I}_Z^n \to \phi_* \mathcal{J}^{\otimes n}$ is induced by the (adjoint to the n^{th} power of) the canonical morphism $\phi^* \mathcal{I}_Z \to \mathcal{J}$.[13] This triple $(\phi, \mathcal{J}, \psi)$ yields the required X-morphism $Y \to B$.

To show the uniqueness of this X-morphism, we use [1, Ex. 11.3.G] to reduce to the case where $\phi^{-1}(Z) = \emptyset$. To do this, it suffices to do the "universal case" of $\pi^{-1}(X \setminus Z)$, which follows from the fact that π is an isomorphism over $X \setminus Z$ (check!). ∎

Theorem 5.5.2. Let k be a field, and X be an integral, regular, finite-type k-scheme. Let $p \in X$ be a closed point, and let $\pi : \mathrm{Bl}_p X \to X$ be the blow-up of X at p with exceptional divisor $E \subset \mathrm{Bl}_p X$.

(a) The blow-up $\mathrm{Bl}_p X$ is also an integral, regular, finite-type k-scheme.
(b) If $D \subset X$ is an effective divisor, then its strict transform $\widetilde{D} := \overline{\beta^{-1}(D) \setminus E}$ is again an effective divisor on $\mathrm{Bl}_p X$, and further $\beta^{-1}(D) = \widetilde{D} + mE$ as closed subschemes of $\mathrm{Bl}_p X$, where $m := \mathrm{mult}_p D \in \mathbb{Z}_{\geq 0}$.

Proof.

(a) The schematic closure of an integral open subscheme is integral, and the blow-up is finite-type as shown above (it is even projective [TODO]). Regularity away from E is clear, and at E follows from the slicing criterion, since $E \cong \mathbb{P}^{n-1}_{\kappa(p)}$ is regular, where $n := \dim X$.

[12] For instance, both of these facts can be checked affine-locally on X, where they are clear. For example, for the second fact, suppose $X = \mathrm{Spec}\, A$, and that Z corresponds to the ideal $I \subset A$, so that B corresponds to $\mathrm{Proj}_B \mathrm{Bl}_I(A)$, where $\mathrm{Bl}_I(A) = \bigoplus_{n \geq 0} I^n$. To show that the canonical morphism $B \to X$ is an isomorphism on $\mathrm{Spec}\, A \setminus \mathbb{V}(I)$, we may check it affine-locally on $\mathrm{Spec}\, A \setminus \mathbb{V}(I)$. This reduces us to showing that for each $f \in I$, the morphism $\mathrm{Proj}_A A[f^{-1}] \otimes_A \mathrm{Bl}_I(A) \to \mathrm{Spec}\, A[f^{-1}]$ is an isomorphism, but of course $A[f^{-1}] \otimes_A \mathrm{Bl}_I A \cong \bigoplus_{n \geq 0} I^n A[1/f] \cong A[1/f][X]$ as graded $A[1/f]$-algebras because $IA[1/f] = (1)$.

[13] We are using, of course, that $\mathcal{J}^{\otimes n} = \mathcal{J}^n$ for all $n \geq 0$. Note that $\phi^* \mathcal{I}_Z \to \mathcal{J}$ is surjective by definition of $\mathcal{J}$.

(b) If $\dim X = 0$, there is nothing to show. Hence assume that $\dim X \geq 1$, and let $n := \dim X - 1 \in \mathbb{Z}_{\geq 0}$. The case $n = 0$ is left to the reader; hence suppose $n \geq 1$. The result is clear away from p in X (and hence E in $\mathrm{Bl}_p X$), and so it is really a local question about p. For this reason, we only have to the consider the case of $m \geq 1$, and by shrinking X, we may assume $X = \operatorname{Spec} A$ for a Noetherian domain A, with p corresponding to a maximal ideal $\mathfrak{p} \subset A$ generated by a regular sequence consisting of $\operatorname{codim}_X p = n + 1$ elements, say $x_0, x_1, \ldots, x_n \in A$, and D corresponding to $\mathbb{V}(f)$ for $0 \neq f \in \mathfrak{p}$. We may assume by shrinking X further that every permutation of x_i forms a regular sequence in A, since this is true in the local ring $A_{\mathfrak{p}}$ and A is Noetherian.

In this case, we first give an explicit description of the blow-up: we claim that $\mathrm{Bl}_p X \cong V := \mathbb{V}(x_i X_j - x_j X_i) \subset \mathbb{P}^n_A$. Indeed, we know from general principles that $\mathrm{Bl}_p X$ is certainly the schematic closure of the complement of the locally principal closed subscheme $\mathbb{P}^n_{\kappa(p)}$ in V, so it remains to show that V is integral, but that follows immediately from [1, 22.3.12].

Now the result of the problem is also local on $\mathrm{Bl}_p X$, so it suffices to work on an affine patch, say $D_+(X_0)$, where the blowup looks like $\operatorname{Spec} B$, where $B := A[x_1/x_0, \ldots, x_n/x_0]$ is a subring of $\operatorname{Frac} A$; note that $\mathfrak{p}B = x_0 B$. If $m := \operatorname{mult}_p D \in \mathbb{Z}_{\geq 1}$, then this means precisely that $f/1 \in \mathfrak{p}^m A_{\mathfrak{p}} \smallsetminus \mathfrak{p}^{m+1} A_{\mathfrak{p}}$. Since $\kappa(p)[X_0, \ldots, X_n] \xrightarrow{\sim} \operatorname{gr}_{\mathfrak{p}A_{\mathfrak{p}}}(A_{\mathfrak{p}})$ by [1, 22.3.11], there is a polynomial $g \in A[X_0, \ldots, X_n]$, homogeneous of degree m and with not all coefficients in $\mathfrak{p}$, such that in $A_{\mathfrak{p}}$ we have $f = g(x_0, \ldots, x_n) + h$ for some $h \in \mathfrak{p}^{m+1} A_{\mathfrak{p}}$. By shrinking X further, we may assume that this equality holds in fact in A for some $h \in \mathfrak{p}^{m+1}$. Treating this as an equality in B ("pulling back to the blow-up"), we get an equation of the form $f = x_0^m g(1, t_1, \ldots, t_n) + h$ where $t_i := x_i/x_0$ for $i = 1, \ldots, n$ and $h \in \mathfrak{p}^{m+1} B = (x_0^{m+1})$; write $h = x_0^{m+1}\ell$ for some $\ell \in B$, and let $q := g(1, t_1, \ldots, t_n) + x_0 \ell$, so that $f = x_0^m q$ in B.

We next claim that $q \notin \mathfrak{p}B$. Indeed, $q \in \mathfrak{p}B$ iff $g(1, t_1, \ldots, t_n) \in \mathfrak{p}B$, but again by [1, 22.3.E], we have that $B/\mathfrak{p}B \cong \kappa(p)[T_1, \ldots, T_n]$ with the t_i corresponding to T_i. Since not all the coefficients of g lie in $\mathfrak{p}$, we conclude immediately from this that $g \notin \mathfrak{p}B$ as needed.

It remains to show that in this affine patch, $\widetilde{D}$ corresponds to $\mathbb{V}(q)$. Note that, by definition, the defining ideal of $\widetilde{D}$ in B is precisely the kernel I of the morphism

$$B = A[t_1, \ldots, t_n] \to A[t_1, \ldots, t_n][x_0^{-1}] = A[x_0^{-1}] \twoheadrightarrow A[x_0^{-1}]/(f).$$

In other words, it consists of precisely those $b \in B$ such that for some $N \in \mathbb{Z}_{\geq 1}$ we have $x_0^N b \in fB$. Clearly, $q \in I$; we have to show that $I = (q)$. For that, let $b \in I$, and pick the smallest $N \in \mathbb{Z}_{\geq 1}$ such that $x_0^N b \in fB$, so that there is a $c \in B$ such that $x_0^N b = fc = x_0^m qc$. By our choice of N, we have that $x_0 \nmid c$, i.e., $c \notin \mathfrak{p}B$. If $N > m$, then we could divide out by x_0^m to conclude from $x_0^{N-m} b = qc$ and the primality of $\mathfrak{p}B$ that $q \in x_0 B = \mathfrak{p}B$, which we showed above is not true. Therefore, $N \leq m$, and then dividing out by x_0^N we conclude that $b = (x_0^{m-N} c) q$, showing that $b \in (q)$ as needed.

■

5.6 Exercises

Exercise 5.1. Let k be a field and X and A be as in 5.1.10(a). Fix a $\lambda \in k^\times$, and consider the $D_\lambda \in \operatorname{CaDiv}(X)$ represented by

$$\{(D(x^2-x),1),(D((\lambda-1)^2(x^2-x)-\lambda),1+(\lambda-1)x)\}.$$

Show that if $\lambda \neq 1$, then $0 \neq [D_\lambda] \in \operatorname{CaCl}(X)$. In particular, if k is a field different from $\mathbb{F}_2$, we have produced another example of a scheme satisfying 5.1.2 for which the natural morphism $\operatorname{CaCl}(X) \to \operatorname{Cl}(X)$ is not injective.[14]

Exercise 5.2. (Adapted from [1, 18.6.K]) Let k be a field, $n \in \mathbb{Z}_{\geq 0}$, and $X \subset \mathbb{P}^n_k$ a closed subscheme.

(a) Show that if X is a linear subspace (4.6), then $\deg X = 1$.
(b) Show that if X is equidimensional and has no embedded points, then $\deg X = 1$ implies that X is a linear subspace.
(c) Show by example that the conclusion of (b) is false if X is not equidimensional or if X has embedded points.

Exercise 5.3. Let k be a field and C be a smooth projective geometrically integral curve[15] over k.

(a) Suppose that $\mathcal{L}$ is a line bundle on C, that $n \in \mathbb{Z}_{\geq 2}$, and that $s_0, \ldots, s_n \in \mathrm{H}^0(C, \mathcal{L})$ are linearly independent sections with no common zeroes. Let $\varphi : C \to \mathbb{P}^n_k$ be the induced morphism. Show that the (schematic image), say D, of φ is a nondegenerate integral closed curve $D \subset \mathbb{P}^n_k$, say of degree $a \in \mathbb{Z}_{\geq 1}$. Show that $a \geq 2$. Show that the induced map $C \to D$ is finite, say of degree $b \in \mathbb{Z}_{\geq 1}$. Show that $\deg \mathcal{L} = a \cdot b$.

For the next two parts, suppose that $\operatorname{char} k \neq 2$.

(b) Suppose that C has genus $g \geq 1$, and $\mathcal{L}$ a degree 2 line bundle with $h^0(\mathcal{L}) \geq 2$. Show that $\mathcal{L}$ is basepoint-free, and that $h^0(\mathcal{L}) = 2$ exactly.
(c) ([1, 19.5.B]) Show that there is a degree two morphism $\pi : C \to \mathbb{P}^1_k$ iff there is a degree 2 line bundle $\mathcal{L}$ with $h^0(\mathcal{L}) = 2$.

Exercise 5.4. Let X be a Noetherian integral scheme and $D \subset X$ a closed subscheme, pure of codimension 1.

(a) Show that for each irreducible component $D_i \subset D$ of D, the local ring $\mathcal{O}_i := \mathcal{O}_{D,D_i}$ is Artinian; let $m_i := \ell_{\mathcal{O}_i}(\mathcal{O}_i) \in \mathbb{Z}_{\geq 0}$. In this way, D gives rise to the Weil divisor $\sum_i m_i D_i$ on X.
(b) Suppose further that X is factorial and that D has no embedded points. Show that the ideal sheaf of D is given by $\mathcal{I}_D = \prod_i \mathcal{I}_{D_i}^{m_i}$. Give a counterexample when D has embedded points.

Exercise 5.5. Let $\pi : X \to Y$ a morphism of schemes. Let $D \hookrightarrow Y$ be an effective Cartier divisor, considered as a locally principal closed subscheme. Suppose that $\pi^{-1}(D)$, which is a priori only a locally principal closed subscheme, is actually an effective Cartier divisor as well. Show that $\pi^*\mathcal{O}_Y(D) \cong \mathcal{O}_X(\pi^{-1}(D))$.

Exercise 5.6. Let k be a field and $Z \hookrightarrow X$ a codimension $c \in \mathbb{Z}_{\geq 1}$ closed embedding of irreducible smooth k-varieties. Let $\pi : B := \operatorname{Bl}_Z X \to X$ be the blow-up of X along Z. Show that $K_B \sim \pi^* K_X + (c-1)E$ as divisors on B.

[14] Again, this corresponds to the quasicoherent sheaf corresponding to the module $M_\lambda := \{g \in k[x,y] : g(1,0) = \lambda g(0,0)\}$. The same example in one dimension lower gives that $\operatorname{Pic}\operatorname{Spec} k[x(x-1), x^2(x-1)] \cong \mathbb{G}_m(k)$.

[15] This just means that $\dim C = 1$.

Exercise 5.7. Let k be a field of characteristic other than 2, and $a \in k^\times$. Let C_a denote the *lemniscate of Bernoulli with parameter* a, which is the quartic curve

$$C_a := \mathbb{V}((x^2+y^2)^2 - a^2(x^2-y^2)z^2) \subset \mathbb{P}^2_k.$$

(a) Calculate the arithmetic and geometric genera of C_a.
(b) When $k = \mathbb{R}$, draw a picture of the affine piece $z \neq 0$ of C_a. This appears to have one node and two "holes". According to your answer from (a), does the recipe of [1, 28.2.I] work for this picture? If yes, your answer to (a) is incorrect; try (a) again. If not, explain why the recipe doesn't work.

Chapter 6

Flatness and Theorems on Cohomology and Base-Change

6.1 Basics about Flatness

6.2 Theorems on Cohomology and Base-Change

Theorem 6.2.1 (Projective Bundles)**.** Let X be a locally Noetherian scheme. Let $\pi : E \to X$ be a morphism of schemes, and let $n \in \mathbb{Z}_{\geq 0}$. Consider the following conditions on π.

(a) There is a vector bundle $\mathcal{E}$ of rank $n+1$ on X such that $E \cong \mathbb{P}\mathcal{E}$ as X-schemes.
(b) The morphism π is projective and flat, all of its geometric fibers are isomorphic to $\mathbb{P}^n$, and there is a line bundle $\mathcal{L}$ on E restricting to $\mathcal{O}(1)$ on the geometric fibers.
(c) The morphism π is a Zariski-locally trivial projective bundle. In other words, there is an open cover $\mathcal{U}$ of X such that for each $U \in \mathcal{U}$, there is an isomorphism $\pi^{-1}(U) \cong \mathbb{P}^n \times U$ of U-schemes.
(d) The morphism π is projective and flat, all of its geometric fibers are isomorphic to $\mathbb{P}^n$, and there is a (locally finite) effective Weil divisor $D \subset E$ meeting the general geometric fiber in a hyperplane; in other words, there is a dense open subset $U \subset X$ such that for all geometric points $\overline{p} \to U$, the pullback $D_{\overline{p}} \subset E_{\overline{p}}$ is isomorphic to a hyperplane inside $E_{\overline{p}} \cong \mathbb{P}^n_{\overline{p}}$.

Then we have the following implications.

(i) We always have (a) $\Leftrightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d).
(ii) When X is Noetherian and $\dim X \leq 1$, then (c) $\Rightarrow$ (a). [TODO]
(iii) When X is Noetherian and factorial, then (d) $\Rightarrow$ (a), so all four conditions are equivalent.

Proof.

(a) $\Rightarrow$ (b) Take $\mathcal{L} := \mathcal{O}_{\mathbb{P}\mathcal{E}}(1)$.

(b) $\Rightarrow$ (a) For each geometric point $\overline{q} \to X$, we have $\mathrm{h}^1(E_{\overline{q}}, \mathcal{L}|_{E_{\overline{q}}}) = 0$, so [1, 18.2.H] tells us that for each point $q \in X$, we have $\mathrm{h}^1(E_q, \mathcal{L}|_{E_q}) = 0$. By [1, 25.1.C], we conclude that $\mathrm{R}^1\pi_*\mathcal{L} = 0$. It follows from [1, 25.1.6] applied twice to $p = 1, 0$ respectively that $\pi_*\mathcal{L}$ is locally free and commutes with arbitrary base-change; in particular, when $\overline{q} \to X$ is a geometric point, the isomorphism $\pi_*\mathcal{L}(\overline{q}) \xrightarrow{\sim} \mathrm{H}^0(E_{\overline{q}}, \mathcal{L}|_{E_{\overline{q}}})$ tells us that $\pi_*\mathcal{L}$ is locally free of rank $n+1$.
We next show that $\mathcal{L}$ is relatively basepoint-free, for which we have to show that $\pi^*\pi_*\mathcal{L} \twoheadrightarrow \mathcal{L}$. By 1.6, it suffices to show that for each geometric point $\overline{p} \to E$, the induced map $\pi^*\pi_*\mathcal{L}(\overline{p}) \to \mathcal{L}(\overline{p})$ is a surjection. Consider a geometric point $\overline{q} \to Y$ to which $\overline{p}$ maps under π; then the map $\pi^*\pi_*\mathcal{L}(\overline{p}) \to \mathcal{L}(\overline{p})$ factors as

$$\kappa(\overline{p}) \otimes_{\kappa(\overline{q})} \pi_*\mathcal{L}(\overline{q}) \xrightarrow{\sim} \kappa(\overline{p}) \otimes_{\kappa(\overline{q})} \mathrm{H}^0(E_{\overline{q}}, \mathcal{L}|_{E_{\overline{q}}}) \to \mathcal{L}|_{E_{\overline{q}}}(\overline{p}),$$

which is surjective because $\mathcal{O}_{\mathbb{P}^n}(1)$ is basepoint-free.
Now consider the X-morphism $\beta : E \to \underline{\mathrm{Proj}}_X \mathrm{Sym}^\bullet \pi_*\mathcal{L} = \mathbb{P}_X(\pi_*\mathcal{L})^\vee$ of 5.4.7. Let $\rho : \mathbb{P}_X(\pi_*\mathcal{L})^\vee \to X$ denote structure morphism. It is clear by construction for each geometric point $\overline{p} \to X$, the induced map $\beta(\overline{p}) : E_{\overline{p}} \to \mathbb{P}^n_{\kappa(\overline{p})}$ on the geometric fibers is an isomorphism, since it corresponds to the complete linear series $|\mathcal{O}(1)|$. It then follows from [1, 10.3.C] that for each $p \in X$, the induced map $\beta(p) : E_p \to \mathbb{P}^n_{\kappa(p)}$ on fibers is a bijection as well, showing that β is a bijection. From [1, 17.3.C] we know that β is projective, so that [1, 18.1.6] tells us that β is finite, and in particular affine. Using [1, 10.2.K], we then conclude that for each $p \in X$, the morphism $\beta(p)$ is an isomorphism as well.[1] By 3.12(a), we conclude that β is a surjective closed immersion. By [1, 24.6.12], the map β is also flat, and hence an open immersion (6.4), but a surjective open immersion is an isomorphism.

(a) $\Rightarrow$ (c) Clear.

(c) $\Rightarrow$ (d) We assume X is Noetherian. The first three claims are clear. To produce D, take a nonempty dense subset $U \subset X$ over which E is trivial, and consider the preimage $\mathbb{P}^n \times U \cong$

[1]Alternatively, one could also argue by using the fppf-descent of isomorphisms ([5, Prop. 14.53(2)]).

$\pi^{-1}(U) =: E_U \hookrightarrow E$; note that this last open embedding is quasicompact since $U \to X$ is. Let $H \subset \mathbb{P}^n$ be a hyperplane and let D be the schematic closure in E of $H \times U \subset \mathbb{P}^n \times U$. Then $D \cap E_U \cong H \times U$ by 3.5.12, and this D works.

(ii) This follows from $\mathrm{H}^2_{\mathrm{Zar}}(X, \mathbb{G}_m) = 0$ [TODO].

(iii) We prove (d) $\Rightarrow$ (b). Take $\mathcal{L} := \mathcal{O}_E(D)$. By hypothesis, this restricts to $\mathcal{O}(1)$ on the general fiber and hence on every fiber by the constancy of the Hilbert polynomial in flat families ([1, 24.7.2]).

■

Remark 6.2.2.

(a) In fact, the equivalence of (a) and (b) does not need X to be locally Noetherian and is true with the word "projective" in (b) replaced by "proper and finitely presented" [TODO; see Stacks 00MP. Also ZMT.]

(b) For an example of a Noetherian integral surface for which the condition (c) does not imply (a), see MSE/4013148.

(c) Just fibers isomorphic to $\mathbb{P}^n$ is not enough; see MO/213231.

(d) Brauer groups.

6.3 Castelnuovo

Example 6.3.1. Contracting on $\mathbb{F}_1 = \mathrm{Bl}_p \mathbb{P}^2$.

Example 6.3.2. Let k be a field with $\operatorname{char} k \neq 3$. Let $X := \{w^3 + x^3 + y^3 + z^3 = 0\} \subset \mathbb{P}^3_k$ be the Fermat cubic, and let $C := \{w + x = y + z = 0\}$; then C is a line on X and hence a (-1) curve. Let's try to contract C.

Following [1, 28.6.1], take $\mathcal{H} := \mathcal{O}_X(1) = -K_X$. Then $\mathrm{h}^0(\mathcal{H}) = 4$ and $\mathrm{h}^1(\mathcal{H}) = \mathrm{h}^2(\mathcal{H}) = 0$. We have $d = 1$, so that $\mathcal{L} = \mathcal{H}(C)$ works. It is easy to see that $\mathrm{h}^0(\mathcal{L}) = 5$, but in fact we can write down these five sections explicitly. If s is the defining section of $\mathcal{O}_X(C)$, then the five sections of $\mathcal{L}$ are sw, sx, sy, sz and st, where

$$t := \frac{w^2 - wx + x^2}{y + z} = -\frac{y^2 - yz + z^2}{w + x}.$$

The image Y of $\varphi_{\mathcal{L}} : X \to \mathbb{P}^4$ with coordinates $[a, b, c, d, e]$ is then the complete intersection

$$a^2 - ab + b^2 - ce - de = c^2 - cd + d^2 + ae + be = 0,$$

which is contained in the cone $a^3 + b^3 + c^3 + d^3 = 0$ over X with vertex $[0 : 0 : 0 : 0 : 1]$ to which C is contracted. This image Y is a smooth quartic del Pezzo and the projective blown down of C in X.

Observation 6.3.3. Let k be a field, X a smooth projective surface over k, and $C \subset X$ a (-1) curve. Let $\pi : X \to Y$ be the blowdown of X with $\pi(C) = \{q\}$ for some $q \in Y(k)$. Then the following diagram is a pushout and pullback square in locally ringed spaces.

$$\begin{tikzcd} C \arrow[r] \arrow[d, hook] & \{q\} \arrow[d, hook] \\ X \arrow[r, "\pi"] & Y. \end{tikzcd}$$

Example 6.3.4. The elementary transformation on $\mathbb{F}_n = \mathbb{V}(x^n v - y^n u) \subset \mathbb{P}^1 \times \mathbb{P}^2$ at the point $([0, 1], [0, 1, 0])$ is given by the map $\mathbb{F}_n \dashrightarrow \mathbb{F}_{n-1}$ given by $([x, y], [u, v, w]) \mapsto ([x, y], [yu, xv, yw])$.

6.4 Exercises

Exercise 6.1. Let $\pi : X \to Y$ be a qcqs morphism. For any moprhism $Z \to Y$, let $\pi_Z : X_Z = Z \times_Y X \to Z$ denote the pullback of π. Produce a natural morphism

$$\underline{\mathrm{Spec}}_Z \pi_{Z,*}\mathcal{O}_{X_Z} \to Z \times_Y \underline{\mathrm{Spec}}_Y \pi_*\mathcal{O}_X.$$

Show that this morphism is an isomorphism when $Z \to Y$ is flat. Produce a counterexample when the flatness hypothesis is dropped.

Exercise 6.2. (Flat Limits over Curves) Let C be an integral one-dimensional regular scheme and $0 \in C$ a regular closed point. Let $C^\circ := C \smallsetminus \{0\}$.

Let $\pi : X \to C$ be a morphism to C from a locally Noetherian scheme X, and let $X^\circ := \pi^{-1}(C^\circ)$. Suppose that $Y^\circ \hookrightarrow X^\circ$ is a closed subscheme, flat over C°. Show that there is a unique closed subscheme $Y \hookrightarrow X$ such that Y is flat over C and $Y \cap X^\circ = Y^\circ$.

In this case, the fiber Y_0 over $\{0\}$ is called the *flat limit* of the family Y° over C°.

Exercise 6.3. (Flat Limits of Arithmetic Hypersurfaces) Fix a prime p and $n, d \in \mathbb{Z}_{\geq 1}$. In 6.2, take $C = \operatorname{Spec} \mathbb{Z}_{(p)}$ and $\{0\} := \mathbb{V}(p) \cong \operatorname{Spec} \mathbb{F}_p$; take $X := \mathbb{P}^n_{\mathbb{Z}_{(p)}}$ so that $X^\circ \cong \mathbb{P}^n_{\mathbb{Q}}$. For an $f \in \mathbb{Q}[X_0, \ldots, X_n]_d$, let $Y^\circ := \mathbb{V}(f) \subset X^\circ$. What is $Y \hookrightarrow \mathbb{P}^n_{\mathbb{Z}_{(p)}}$ as in 6.2, and hence the resulting special fiber Y_0 over $\operatorname{Spec} \mathbb{F}_p$?[2]

Exercise 6.4. Let $i : Z \hookrightarrow X$ be a closed immersion. Show that if i is finitely presented and flat, then i is an open immersion. In particular, any flat closed immersion into a locally Noetherian scheme is an open immersion. Thus, closed immersion are "usually" not flat. Is the result true without the finite presentation hypotheses?

Exercise 6.5.

(a) Let Y be an integral scheme and $\pi : X \to Y$ an integral flat morphism. Show that every irreducible component of X dominates (or equivalently surjects onto) Y.
(b) Let $\pi : X \to Y$ be an integral flat morphism of schemes, and suppose that Y is equidimensional. Then so is X, and $\dim X = \dim Y$.

Exercise 6.6. Let k be a field, X be a smooth k-variety, and $\mathcal{E} \to X$ a vector bundle of rank say $r+1$ for $r \in \mathbb{Z}_{\geq 0}$. Let $\pi : \mathbb{P}\mathcal{E} \to X$ denote the projectivization of $\mathcal{E}$. Show that

$$K_{\mathbb{P}\mathcal{E}} \sim -(r+1)H + \pi^*(K_X - \det \mathcal{E})$$

as divisors on $\mathbb{P}\mathcal{E}$, where H is the divisor corresponding to the tautological bundle $\mathcal{O}_{\mathbb{P}\mathcal{E}}(1)$.

Exercise 6.7. Continuing the notation of 6.6 and [1, 20.2.10], for $n \in \mathbb{Z}_{\geq 0}$, let $X = \mathbb{P}^1$ and $\mathcal{E} = \mathcal{O} \oplus \mathcal{O}(n)$, so that $\mathbb{F}_n := \mathbb{P}\mathcal{E}$.

(a) Produce an isomorphism of $\mathbb{F}_n \to \mathbb{P}^1$ with $\mathbb{V}(x^n v - y^n u) \subset \mathbb{P}^1 \times \mathbb{P}^2 \to \mathbb{P}^1$, where the $\mathbb{P}^1$ has coordinates x, y and $\mathbb{P}^2$ has coordinates u, v, w. Show that E corresponds to the graph of the constant map $\mathbb{P}^1 \to \mathbb{P}^2$ at $[0,0,1]$ (i.e., the fiber of $\mathbb{F}_n \to \mathbb{P}^2$ over the point $[0,0,1]$); show that C corresponds to the graph of the map $\mathbb{P}^1 \to \mathbb{P}^2$ given by $[x : y] \mapsto [x^n : y^n : 0]$.
(b) Show that $H = E$ and hence $K_{\mathbb{F}_n} \sim -2E-(n+2)F \sim -2F-E-C$. Check that $K^2_{\mathbb{F}_n} = 8$.
(c) Show that for any $a, b \in \mathbb{Z}$, any curve in the class $aE + bF$ has arithmetic genus

$$(a-1)(b-1) - n\binom{a}{2}.$$

[2] The same example can also be given by $(\mathbb{Z}_{(p)}, \mathbb{Q})$ replaced by the completion $(\mathbb{Z}_p, \mathbb{Q}_p)$.

(d) Show that $\mathsf{R}^{>0}\pi_*\mathcal{O}_{\mathbb{F}_n} = 0$, and conclude that $\chi(\mathbb{F}_n, \mathcal{O}) = 1$.

(e) Show that for any $a, b \in \mathbb{Z}$, we have

$$\chi(\mathbb{F}_n, aE + bF) = -n\binom{a+1}{2} + (a+1)(b+1).$$

(f) Show that for $n > 0$, we have $\mathrm{h}^j(\mathbb{F}_n, E)$ for $j = 0, 1, 2$ is $1, n-1, 0$; in particular, E is the unique reduced curve in $|E|$. Show that $\mathrm{h}^j(\mathbb{F}_0, E)$ for $j = 0, 1, 2$ is $2, 0, 0$.

(g) Suppose now that $n > 0$. Let $D \subset \mathbb{F}_n$ be an irreducible effective divisor (ECD) such that $D^2 < 0$. Show that there is an $a \in \mathbb{Z}_{\geq 1}$ such that $D = aE$. In particular, if D is an irreducible ECD such that $D^2 = -n$, then $D = E$.

(h) Continue to assume that $n > 0$. Let D be an effective divisor with irreducible components C_i; say $D = \sum_i a_i C_i$ for some $a_i \geq 0$. Show that D is reduced iff each $a_i = 1$. Show that if this is the case, then $D^2 \geq -n$. When does equality hold?

Note that the example of $2E + 3F$ on $\mathbb{F}_4$ shows that, in general, E is not the unique curve on $\mathbb{F}_n$ such that $E^2 = -n$ (as [1, 20.2.R] erroneously claims); it is the unique *irreducible* curve (and unique *reduced* curve) with this property. For any $n \in \mathbb{Z}_{\geq 0}$, can you find all curves D on $\mathbb{F}_n$ such that $D^2 = -n$?

Exercise 6.8. Let k be a field, Y a smooth projective surface over k, and $q \in Y(k)$. Let $\pi : \mathrm{Bl}_q(Y) \to Y$ be the blow-up with exceptional divisor E. Show that $\pi_*\mathcal{I}_{E/\mathrm{Bl}_q Y} \cong \mathcal{I}_q$.

6.4.1 PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 6.9. Let k be a field, X and Y be lft k-schemes and $\pi : X \to Y$ be a k-morphism. Let $m, n \in \mathbb{Z}_{\geq 0}$. Suppose that Y is equidimensional of dimension n, and for each $y \in Y$ the fiber X_y is equidimensional of dimension m over $\kappa(y)$. Then X is equidimensional of dimension $m + n$.

Chapter 7

Intersection Theory

7.1 Numerical Intersection Theory

Lemma 7.1.1. Let $\pi : X \to Y$ be a quasicompact separated morphism of schemes with Y quasicompact. There is an $n \in \mathbb{Z}_{\geq 0}$ such that $\mathsf{R}^{>n}\pi_* = 0$ as functors $\mathsf{QCoh}(X) \to \mathsf{QCoh}(Y)$.

Proof. Pick a finite cover V_i of Y by affine opens; for each i, the preimage $\pi^{-1}(V_i)$ has an open cover by a finite number, say n_i, of affine opens. Taking $n := \max\{n_i\}$ works. ∎

Lemma 7.1.2. Let k be a field, X and Y be proper k-schemes, and $\pi : X \to Y$ a k-morphism. For every $\mathcal{F} \in \mathsf{Coh}(X)$, we have

$$\chi(X, \mathcal{F}) = \sum_{p \geq 0} (-1)^p \chi(Y, \mathsf{R}^p\pi_*\mathcal{F}).$$

Proof. Follows immediately from the existence of the Leray spectral sequence $\mathrm{H}^q(Y, \mathsf{R}^p\pi_*\mathcal{F}) \Rightarrow \mathrm{H}^{p+q}(X, \mathcal{F})$, which is supported only on finitely many rows and columns thanks to 7.1.1, and Grothendieck's coherence theorem. ∎

Theorem 7.1.3. Let k be a field and X, Y be projective k-schemes. Let $\pi : X \to Y$ be a k-morphism, $n \in \mathbb{Z}_{\geq 0}$, and $\mathcal{L}_1, \ldots, \mathcal{L}_n$ be line bundles on Y. For every $\mathcal{F} \in \mathsf{Coh}(X)$, we have that

$$(\pi^*\mathcal{L}_1 \cdot \pi^*\mathcal{L}_2 \cdots \pi^*\mathcal{L}_n \cdot \mathcal{F}) = \sum_{p \geq 0} (-1)^p (\mathcal{L}_1 \cdot \mathcal{L}_2 \cdots \mathcal{L}_n \cdot \mathsf{R}^p\pi_*\mathcal{F}).$$

Proof. Follows from 7.1.2 and the projection formula for higher pushforwrads ([1, 18.7.E(b)]). ∎

7.2 Exercises

Exercise 7.1. Let B be a locally Noetherian scheme and $\pi : X \to B$ a proper smooth morphism of relative dimension $n \in \mathbb{Z}_{\geq 0}$. Let $D_1, \ldots, D_n \subset X$ be ECDs that are also relative ECDs, i.e., such that for all $b \in B$, they restrict to ECDs $D_{1,b}, \ldots, D_{n,b}$ on the fiber X_b. Show that the function $B \to \mathbb{Z}$ given by $b \mapsto (D_{1,b} \cdot D_{2,b} \cdots D_{n,b})_{X_b}$ is locally constant.

Exercise 7.2. Let k be a field, $n \in \mathbb{Z}$ with $0 \leq n \leq 8$, and let $p_1, \ldots, p_n \in \mathbb{P}^2_k$ be k-valued points, no three collinear, no six on a conic, and if $n = 8$ then not all on a singular cubic with one at the singularity. Show that the surface S_n obtained by blowing up $\mathbb{P}^2_k$ at these n points is del Pezzo.

Exercise 7.3. Let k be a field, Y a smooth projective k-surface, $q \in Y(k)$, and $\pi : B = \mathrm{Bl}_q(Y) \to Y$ the blow-up with exceptional divisor $E \subset B$. Suppose $C \subset B$ is a smooth integral k-curve such that C intersects E transversely at one k-point. Show that $\pi(C) \subset Y$ is a smooth integral k-curve passing through q, and the map $C \to \pi(C)$ is an isomorphism.

Chapter 8

Appendices

8.1 Generic Freeness

Theorem 8.1.1 (Geometric Nakayama)**.** Let X be a locally Noetherian scheme, $\mathscr{F} \in \mathsf{Coh}(X)$ and $x \in X$. Let $n \in \mathbb{Z}_{\geq 0}$.

(a) Given an epimorphism $\psi(x) : \kappa(x)^{\oplus n} \twoheadrightarrow \mathscr{F}(x)$ of $\kappa(x)$-vector spaces, there is an open subset $V \subset X$ containing x and an $\mathcal{O}_V$-module epimorphism $\psi_V : \mathcal{O}_V^{\oplus n} \twoheadrightarrow \mathscr{F}|_V$ lifting $\psi(x)$. Further, V can be chosen to be affine and such that $\psi(V) : \mathcal{O}_X(V)^{\oplus n} \to \mathscr{F}(V)$ is an $\mathcal{O}_X(V)$-module epimorphism.

(b) Given an isomorphism $\psi_x : \mathcal{O}_{X,x}^{\oplus n} \xrightarrow{\sim} \mathscr{F}_x$ of $\mathcal{O}_{X,x}$-modules, then there is an open subset $W \subset X$ containing x and an isomorphism $\psi_W : \mathcal{O}_W^{\oplus n} \to \mathscr{F}|_W$ of $\mathcal{O}_W$-modules lifting ψ_x.

In particular, $\mathscr{F}$ is flat iff it is locally free of finite rank, i.e. a vector bundle.

Proof.

(a) Pick an open subset $U \subset X$ containing x and elements $a_1, \ldots, a_n \in \mathscr{F}(U)$ such that for $i = 1, \ldots, n$, the map $\psi(x)$ sends the i^{th} basis vector to $a_i(x)$. Without loss of generality, we may assume $U = \operatorname{Spec} A$ for a Noetherian ring A and $\mathscr{F} = \widetilde{M}$ for a finite A-module M; then $\mathscr{F}(U) = M$. Let x correspond to the prime ideal $\mathfrak{p} \subset A$. The hypothesis when combined with Nakayama's Lemma tells us that (the classes of) $a_1, \ldots, a_n$ generate $M_{\mathfrak{p}}$ as an $A_{\mathfrak{p}}$-module. Let $M' \subset M$ be the submodule of M generated by $a_1, \ldots, a_n$; since M is finite over A, there is an $s \in A \setminus \mathfrak{p}$ such that $sM \subset M'$. It suffices to take $V := D(s)$.

(b) By (the proof of) (a), there is an affine open $V \subset U$ and a lifting $\psi_V : \mathcal{O}_V^{\oplus n} \twoheadrightarrow \mathscr{F}|_V$ of ψ_x to an $\mathcal{O}_V$-module epimorphism. Then $\ker \psi_V \in \mathsf{Coh}(V)$; taking $W := V \setminus \operatorname{supp} \ker \psi_V$ works.

The last statement follows from (b) and Lemma 8.2.1. ■

Remark 8.1.2. For an abitrary scheme, the correct statement is that vector bundles, i.e. locally free sheaves of finite rank, are exactly the flat modules of finite presentation.

Corollary 8.1.3 (Generic Freeness)**.** Let X be a locally Noetherian integral scheme and $\mathscr{F} \in \mathsf{Coh}(X)$.

(a) Let $\eta \in X$ denote the generic point. If $\mathscr{F}_\eta = 0$, then there is a dense open $U \subset X$ with $\mathscr{F}|_U = 0$.

(b) There is a dense open $U \subset X$ such that $\mathscr{F}|_U$ is locally free of finite rank, i.e. a vector bundle.

8.2 Flatness

Lemma 8.2.1. . Let A be a ring, and M be a finitely presented flat module. (For instance, this holds if A is Noetherian and M finite.) Then M is locally free, i.e. projective.

Proof. Localizing, we may assume that $(A, \mathfrak{m}, k)$ is a local ring; then we have to show that a finitely presented flat A-module M is free. Since M is finitely generated, $k \otimes_A M$ is a finite-dimensional k-vector space, of dimension say $r \in \mathbb{Z}_{\geq 0}$. Pick $m_1, \ldots, m_r \in M$ such that the images of these in $k \otimes_A M$ constitute a k-basis; we show that the corresponding map $A^{\oplus r} \to M$ is an isomorphism. Indeed, it is surjective by another application of Nakayama's Lemma. For injectivity, let I denote the kernel, so that we have a short exact sequence $0 \to I \to A^{\oplus r} \to M \to 0$. Note that I is a finitely generated A-module.[1] Since B is flat, we have $\mathrm{Tor}_1^A(k, B) = 0$, and hence tensoring with k yields the exact sequence $0 \to k \otimes_A I \to k^{\oplus r} \to k \otimes_A B \to 0$. Since the corresponding map $k^{\oplus r} \to k \otimes_A B$ is an injection, we conclude that $k \otimes_A I = 0$, and so by a final application of Nakayama's Lemma we get $I = 0$. ∎

Lemma 8.2.2. Let A be a Noetherian domain with fraction field $A \to K$, let B an irreducible Noetherian ring (i.e. a Noetherian ring such that $\gamma := \mathrm{Nil}(B)$ is a prime ideal), and $\varphi : A \hookrightarrow B$ be a finite flat injective morphism. The natural map $K \otimes_A B \cong (A \smallsetminus \{0\})^{-1} B \to B_\gamma$ is an isomorphism of K-vector spaces.

Proof. The idea is that in this setting, the nilpotent elements of $K \otimes_A B$ are precisely the zero-divisors. Injectivity then comes from the fact that the localization map obtained by inverting non-zero-divisors is injective. Surjectivity comes from checking that [TODO]. ∎

[1] This is due to the "finitely presented implies always finitely presented" property.

8.3 Counterexamples in Algebraic Geometry

Example 8.3.1. A locally ringed space that is not a scheme. Take X to be a point with local ring that has more than one prime ideal.

Example 8.3.2. Open subscheme of an affine scheme that is not affine.

Example 8.3.3. Scheme with no closed point.

Example 8.3.4. An affine open subscheme of an affine scheme that is not a distinguished affine open. (Elliptic curve - non-torsion)

Example 8.3.5. A non-Noetherian scheme whose underlying space is Noetherian.

Example 8.3.6. A Noetherian scheme X and open subset $U \subset X$ such that $\mathcal{O}(U)$ is not a Noetherian ring. Form U by gluing $\mathbb{A}^2_k$ and $\mathbb{P}^2_k \smallsetminus \{0\}$ along the closed subscheme $\mathbb{A}^1_k = \mathbb{P}^1_k \smallsetminus \{0\}$. Then X embeds into the union of two planes in $\mathbb{P}^3_k$, but its coordinate ring is not Noetherian.

8.4 Possible Hints to Selected Exercises

1.8. This is long but straightforward. Construct Z as a topological space by taking the quotient of $\coprod_U Z_U$ by the equivalence relation that $z_U \sim z_{U'}$ iff there is an affine open $V \subset U \cap U'$, simultaneously distinguished in both, and an element $z_V \in Z_V$ such that $\rho_V^U(z_V) = z_U$ and $\rho_V^{U'}(z_V) = z_{U'}$. (Check that this defines an equivalence relation!) Define continuous maps π and ι_U such that Z_U maps homeomorphically to $\pi^{-1}(U)$. Finally, define the structure sheaf $\mathcal{O}_Z$ by the following recipe: for an open subset $W \subset Z$, let

$$\mathcal{O}_Z(W) := \left\{ (f_U) \in \prod_U \mathcal{O}_{Z_U}(\iota_U^{-1} W) : (\forall \text{ dist. } V \subset U)\, f_V = (\rho_V^U)^\sharp(\iota_U^{-1}(W))(f_U) \right\}.$$

Upgrade ι_U and π to morphisms of ringed spaces. The longest part is then checking that each ι_U is an open embedding of ringed spaces.

2.6(b). Show that if R is a ring and M, N finitely generated R-modules with N finitely presented, and $\varphi : M \to N$ an R-module homomorphism, then for each $r \in \mathbb{Z}_{\geq 0}$, the locus $\{p \in \operatorname{Spec} R : \operatorname{rank} \varphi(p) \leq r\}$ is a finite disjoint union of locally closed subsets. To do this, pick generators for M to get the exact sequence $R^{\oplus a} \to M \to 0$, and a presentation for N to get the exact sequence $R^{\oplus b} \xrightarrow{\psi} R^{\oplus c} \to N \to 0$ for some $a, b, c \in \mathbb{Z}_{\geq 0}$ and $c \times b$ matrix ψ with entries in R. Lift the map φ to a morphism $R^{\oplus a} \xrightarrow{\pi} R^{\oplus c}$. Show that for each p, we have $\operatorname{Im} \varphi(p) \cong (\operatorname{Im} \psi(p) + \operatorname{Im} \pi(p)) / \operatorname{Im} \psi(p)$. Let $\tau : R^{\oplus a+b} \to R^{\oplus c}$ be the sum of ψ and π. Conclude that $\operatorname{rank} \varphi(p) = \operatorname{rank} \tau(p) - \operatorname{rank} \psi(p)$, so that the required locus is $\coprod_{s \geq 0} X_s$ where $X_s := \{p \in \operatorname{Spec} R : \operatorname{rank} \psi(p) = s \text{ and } \operatorname{rank} \tau(p) \leq r + s\}$. Show that each X_s is locally closed, and $X_s = \emptyset$ for $s \gg 0$.

3.11. Represent $\mathcal{L}$ by a cocycle consisting of open sets which are either contained in $X \smallsetminus X_i \cong X_{3-i} \smallsetminus Z$, or which are affine. Compare two trivializing sections $(s_{\alpha,i})$; restricted to Z, these must differ by a nonzero constant. After scaling as needed, use these to glue these to produce a trivialization (s_α) of $\mathcal{L}$.

3.15. The statement is true for separated and universally closed morphisms (say by the valuative criterion), and if π is proper then $\pi \circ \iota$ is, with the converse holding when π is locally of finite type. For a counterexample otherwise, take k to be a field, $Y = \operatorname{Spec} k$, and $X := \operatorname{Spec} k[\varepsilon_1, \varepsilon_2, \dots]/(\varepsilon_i^2)$.

3.12 For (a), reduce to showing that if $(A, \mathfrak{m})$ is a local ring and $A \to B$ a finite ring morphism such that $A/\mathfrak{m} \twoheadrightarrow B/\mathfrak{m}B$, then $A \twoheadrightarrow B$; this is just Nakayama's Lemma. For (b), use 3.8.

4.6(e). This is can be done by hand using (d), but another approach is to invoke [1, 18.2.H] for $\mathcal{O}_Y(1)$ and use induction.

4.9.

(a) Use Krull's theorems [TODO].
(b) Combine (a) for $W = \overline{\{y\}}$ with 2.1.16, 4.1.6(a), and 4.1.8(a).
(c) Use Noether normalization [TODO] (and a little more) to show that for some nonempty open subset $V \subset Y$, the map $X_V = \pi^{-1}(V) \to V$ factors as $X_V \to \mathbb{A}_V^r \to V$, where the first map is finite surjective.
(d) Combine (c) with induction on $\dim Y$.

4.10. Replace Y by Y^{red} and X by X^{red} to assume that X and Y are reduced. Next, show that it suffices to shrink Y to assume that Y is integral and finite-type over k. At this point, use 2.2.7(c) and 4.1.3(d) to further assume that X is integral. If π is not dominant, we are vacuously done. If π is dominant, invoke 4.9(c).

4.11. Show that if Y is an irreducible topological space, $r, n \in \mathbb{Z}_{\geq 1}$, and $f_1, f_2, \dots, f_r : Y \to \mathbb{N}$

upper semicontinuous functions such that $\max_i f_i \equiv n$, then there is an i such that $f_i \equiv n$.

4.12. For (a), use 2.1.13 to show the first statement. For the second statement, reduce to the case of affine X, Y, Z and surjective f, g; you might need Chevalley's Theorem along with 2.3.4(b). First use Krull's Theorems [TODO] to prove the result when $Z = \mathbb{A}^n_k$ for some $n \in \mathbb{N}$, and in fact solve the second part of (b). For the general case, use Noether normalization combined with the diagonal-base-change diagram ([1, 1.2.S]), noting that if $Z \to \mathbb{A}^n_k$ is finite surjective, then the diagonal morphism $Z \to Z \times_{\mathbb{A}^n_k} Z$ is the inclusion of an irreducible component for dimensional reasons (this might need 4.1.8). Since f, g are surjective, so is $X \times_{\mathbb{A}^n_k} Y \to Z \times_{\mathbb{A}^n_k} Z$, and so this map is dominant by 3.3.3; in particular, there is an irreducible component of $X \times_{\mathbb{A}^n_k} Y$ dominating $\Delta(Z)$, and this must also be an irreducible component of $X \times_Z Y$. For (b), for both examples take $X = Y$ and $f = g$; for the first one, consider the blow-up of $\mathbb{A}^3_k$ at the origin, and for the second one the normalization of the nodal cubic.

4.13. For (a), show that a product of smooth k-schemes is smooth, and hence that the diagonal $\Delta : Z \to Z \times_k Z$ is a regular locally closed embedding of codimension $\dim Z$. Next, using Krull's Theorems, prove that if A, B, C are locally Noetherian schemes and $B \hookrightarrow C$ a regular locally closed embdding of codimension say $d \in \mathbb{N}$ and $p : A \to C$ any morphism, then every irreducible component of $p^{-1}(B)$ has codimension at most d in A. For (b), have a peek at 6.9.

4.14.

(a) Use 4.3.11 and the transitivity of the action.
(b) Use 4.1.12 to conclude that $G \times_k X$ is equidimensional of dimension $\dim G + \dim Z$. Using 4.10 and G-equivariance, show that $G \times_k X \to Z$ is surjective, with fibers pure of dimension $\dim G + \dim X - \dim Z$. Conclude using 4.13(b) that $(G \times_k X) \times_Z Y$ is equidimensional of dimension $\dim G + \dim X + \dim Y - \dim Z$. Apply 4.10 once more.
(c) Here's how the only proof I know of this goes: Reduce to the case of k algebraically closed. Show using generic flatness (before passing to the algebraic closure!) that $G \times_k X \to Z$ is faithfully flat of relative dimension $\dim G + \dim X - \dim Z$. Now proceed as in (b).[2]

4.15.

(a) Use 4.1.3 to reduce to the affine case; then use 4.1.5.
(b) Using 4.1.9, reduce to the case of affine integral X. Now use 4.1.10(a) along with the fact that integral morphisms are closed [TOCITE].
(c) Let $\pi : X_K \to X$ denote the projection. If $x \in X$ lies in an irreducible component $Z \subset X$, then $\pi^{-1}(x) \subset Z_K$. If this happens for two distinct Z, say Z_1 and Z_2, then pick any point $\tilde{x} \in \pi^{-1}(x)$ and, for $i = 1, 2$, an irreducible component $Z'_i \subset Z_i$ containing $\tilde{x}$. Use 4.1.9 to show that $Z'_1, Z'_2 \subset X_K$ are irreducible components which intersect.

4.4(b). Use 2.3.3(b) and 4.1.3(d) to reduce to the case of Y locally closed. Now use 2.3.1(g), 2.2.12, 4.1.3(b), and 4.1.6(b).

4.17. [1, 21.2.Q].

5.2.

(b) First, use [4, §9.2.1][3] and 4.6 to reduce to the case of k algebraically closed. Second, proceed by induction on $d := \dim X$, with the case $d = 0$ being clear. Suppose now that $d \geq 1$, and show that X is irreducible and contains $d+1$ distinct closed points $p_0, \ldots, p_d \in X$ in linear general position. Use 3.8 to reduce to showing that every hyperplane H containing $p_0, \ldots, p_d$ contains X. If H is a hyperplane containing $p_0, \ldots, p_d$ but not X, then by [1, 18.6.J] and dimension theory [TODO], the intersection $X \cap H$ is a equidimensional closed subscheme of dimension $d-1$ of degree 1. Use [1, 18.6.E(a)] to show that for any nonempty

[2] If you know or find a different proof that does not use generic flatness, please write to me!
[3] Unstable citation.

closed subscheme $Z \subset \mathbb{P}^n_k$ we have $1 \leq \deg Z^{\mathrm{red}} \leq \deg Z$; apply this to $Z = X \cap H$ to conclude that $\deg(X \cap H)^{\mathrm{red}} = 1$. The induction hypothesis applied to $(X \cap H)^{\mathrm{red}}$ tells us that $(X \cap H)^{\mathrm{red}}$ is a linear subspace of dimension $d-1$, but such a space cannot contain $d+1$ points in linear general position.

(c) Consider $X = \mathbb{V}(XY, Y(sZ - tY)) \subset \mathbb{P}^3_k$ for $[s:t] = [1:0], [0:1]$.

5.3.

(a) The schematic image D is an integral closed subscheme which has dimension 0 or 1 by 4.1.6. If $\dim D = 0$, then D is a closed point and then $\mathcal{L}$ is trivial, but the trivial line bundle does not have $n+1$ linearly independent sections. Therefore, D has dimension 1. Show that D is nondegenerate and use 5.2 to conclude that $\deg D \geq 2$. Next, the morphism $C \to D$ is projective and quasifinite and hence finite. Finally, use 5.4.16.

(b) Reduce to the case in which k is algebraically closed. Show that $\mathcal{L}$ is basepoint-free using [1, 19.4.2]. If $h^0(\mathcal{L}) \geq 3$, use (a) to produce a morphism $C \to \mathbb{P}^2_k$, the schematic image D of which is a smooth conic and such that the induced morphism $C \to D$ is birational and hence an isomorphism, contradicting our assumption on the genus.

5.6. This is a local question, so we work locally around closed $p \in Z$. Given such a p, replace X by a small affine open neighborhood of p in X to assume that $X = \operatorname{Spec} A$ is affine, with A a Noetherian domain, such that Z corresponds to a prime ideal $I \subset A$ of the form $I = (x_1, \ldots, x_c)$ for some regular sequence $x_1, \ldots, x_c$ stable under permutations, and such that it can be completed to a regular sequence $x_1, \ldots, x_c, x_{c+1}, \ldots, x_d$, also stable under permutations, where $d := \dim X$. At this point, $\mathrm{Bl}_Z X = \mathbb{V}(x_i X_j - x_j X_i)_{1 \leq i < j \leq c} \subset \mathbb{P}^{c-1}_A$. Work in the affine patch $D_+(X_1)$, so the blowup looks like $\operatorname{Spec} B$ where $B := A[t_2, \ldots, t_c]$, where $t_i = x_i/x_1$ for $i = 2, \ldots, c$, so that $IB = (x_1)$ and $B/IB \cong (A/I)[T_2, \ldots, T_c]$. Note in particular that $x_1, t_2, \ldots, t_c, x_{c+1}, \ldots, x_d$ is a regular sequence in B, that $\Omega_{X/k}$ is freely generated by $\mathrm{d}x_1, \ldots, \mathrm{d}x_d$ near p, and hence that K_X is has generator $\mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_d$ near p. Now explictly compute the pullback:

$$\mathrm{d}x_1 \wedge \mathrm{d}(t_2 x_1) \wedge \cdots \wedge \mathrm{d}(t_c x_1) \wedge \mathrm{d}x_{c+1} \wedge \cdots \wedge \mathrm{d}x_d = x_1^{c-1} \mathrm{d}x_1 \wedge \mathrm{d}t_2 \wedge \cdots \wedge \mathrm{d}t_c \wedge \mathrm{d}x_{c+1} \wedge \cdots \wedge \mathrm{d}x_d.$$

6.9. Consider the example when X is the disjoint union of $\mathbb{A}^1_k$ and a point, and Y is $\mathbb{A}^1_k$. One good salvage is: X has dimension $m+n$, and if π is flat, then X is equidimensional of dimension $m+n$. Note that flatness of π is sufficient but not necessary for equidimensionality of X.

6.7.

(b) What is the pullback of $\mathcal{O}(1)$ under the section $\sigma : \mathbb{P}^1 \to \mathbb{F}_n$ whose image is E?

(c) Use cohomology and base-change.

(d) Combine (a) and (b) with Riemann-Roch ([1, 20.2.B(b)]).

(e) Show that $\mathsf{R}^{>0}\pi_* \mathcal{O}(1) = 0$, and use the Leray spectral sequence.

(f) There are unique $a, b \in \mathbb{Z}$ such that $D \sim aE + bF$; by effectivity, we know $a, b \geq 0$ ([1, 20.2.M, 20.2.P]). If $a = 0$, then $D^2 = 0$; therefore, $a > 0$. If D is not supported precisely on E, then $D \cdot E \geq 0$, which implies that $b \geq an > 0$, and so $D^2 = -na^2 + 2ab = a(2b - an) > 0$, which is a contradiction. Therefore, D is supported precisely on E, and then $D \sim aE$ implies $D = aE$. In this case, $D^2 = -a^2 n$, so if this is $-n$, then $a = 1$.

(g) The first statement follows from the fact that D is Cohen-Macaulay and so has no embedded points ([1, 26.2.B, 26.2.6]). For the second, again let $a, b \in \mathbb{Z}_{\geq 0}$ be the unique integers so that $D \sim aE + bF$, so that $D^2 = a(2b - an)$. If E is not one of these components C_i, then $D^2 = \sum_i C_i^2 + 2\sum_{i<j} C_i \cdot C_j$. Since each C_i is irreducible and distinct from E, part (e) tells us that each $C_i^2 \geq 0$. Also, trivially $C_i \cdot C_j \geq 0$ for $i \neq j$. Therefore, in this case we must have $D^2 \geq 0 > -n$. If E is one of these components, write $D = E + D'$ for some reduced effective D' not containing E as a component. Then $D^2 = -n + 2D' \cdot E + (D')^2$. By the part already established, we know $(D')^2 \geq 0$. Since E is not one of the components of D', we also conclude that $2D' \cdot E \geq 0$. It follows that $D^2 \geq -n$.

7.2. Use the Nakai-Moishezon criterion ([1, 20.4.1]) to reduce to showing that if $C \subset \mathbb{P}^2_k$ is an integral curve of degree $d \in \mathbb{Z}_{\geq 1}$ with multiplicity $m_i \in \mathbb{Z}_{\geq 0}$ at p_i for $i = 1, \ldots, n$, then $3d > S := \sum_{i=1}^n m_i$. Check $d = 1$ by hand. When $d \geq 2$, if $S \geq 3d \geq 6 \geq n/2$, then

$$\frac{1}{n}(3d)^2 - 3d \leq \frac{1}{n}S^2 - S \leq \sum_{i=1}^n (m_i^2 - m_i) \leq (d-1)(d-2) \Rightarrow d \leq \sqrt{\frac{2n}{9-n}} \leq 4.$$

Check $d = 2, 3, 4$ by hand. One interesting special case is when $d = 4$ and C is a rational quartic with three nodes, with three of the eight p_i located at the nodes; then $S = 11$ while $3d = 12$.

Bibliography

[1] R. Vakil, *The Rising Sea: Foundations of Algebraic Geometry.* Princeton University Press, 2025.

[2] R. Hartshorne, *Algebraic Geometry.* No. 52 in Graduate Texts in Mathematics, Springer, 1977.

[3] The Stacks Project Authors, "*Stacks Project.*" `https://stacks.math.columbia.edu`, 2018.

[4] D. Goel, "Commutative Algebra." `https://gdmgoel.github.io/notes/CA_Notes.pdf`.

[5] U. Görtz and T. Wedhorn, *Algebraic Geometry I: Schemes.* Springer Studium Mathematik - Master, Springer Spektrum, second ed., 2010.

[6] Q. Liu, *Algebraic Geometry and Arithmetic Curves.* No. 6 in Oxford Graduate Texts in Mathematics, Oxford University Press, 2002.

[7] T. Wedhorn, *Manifolds, Sheaves, and Cohomology.* Springer Studium Mathematik - Master, Springer Spektrum, 2016.

[8] M. S. Davis, "Quasiseparated if finitely covered by affines in appropriate way." Mathematics Stack Exchange. URL: https://math.stackexchange.com/q/4919351 (version: 2024-06-14).

[9] M. Escardó, "Intersections of compactly many open sets are open." `https://martinescardo.github.io/papers/compactness-submitted.pdf`.

[10] B. Poonen, "Is a universally closed morphism of schemes quasi-compact?." MathOverflow. URL: https://mathoverflow.net/q/23528 (version: 2010-05-05).

[11] B. Poonen, *Rational Points on Varieties*, vol. 186 of *Graduate Studies in Mathematics.* American Mathematical Society, 2017.

[12] J. Harris, *Algebraic Geometry: A First Course*, vol. 133 of *Graduate Texts in Mathematics.* Springer, 1992.

[13] S. L. Kleiman, "Misconceptions about K_X," *L'Enseignment Mathématique*, vol. 25, no. II, pp. 203–206, 1979.

[14] D. Mumford, *Lectures on Curves on an Algebraic Surface.* Princeton University Press, 1996.

[15] D. Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, vol. 150 of *Graduate Texts in Mathematics.* Springer, 1995.